

Model Analysis

Table of contents

1	Data analysis /Dataframe import and filters	3
1.1	DataFrame import and filters	3
1.2	Metadata	5
2	Model(s) analysis	6
2.1	Explanation legend tables	6
2.2	Repartition of train and test	7
2.2.1	External Split: X_train vs X_test	7
2.2.2	Internal CV Folds	7
2.3	Model : Linear	10
2.3.1	Gridsearch — Group: 1	10
2.3.2	Scatter	12
2.3.3	Scatter residuals	13
2.3.4	Learning curve — Group: 1	13
2.3.5	Gridsearch — Group: 2	14
2.3.6	Scatter	16
2.3.7	Scatter residuals	17
2.3.8	Learning curve — Group: 2	17
2.3.9	Gridsearch — Group: 3	18
2.3.10	Scatter	20
2.3.11	Scatter residuals	21
2.3.12	Learning curve — Group: 3	21
2.3.13	Gridsearch — Group: 4	22
2.3.14	Scatter	24
2.3.15	Scatter residuals	25
2.3.16	Learning curve — Group: 4	25
2.3.17	Gridsearch — Group: 6	26
2.3.18	Scatter	28
2.3.19	Scatter residuals	29

2.3.20	Learning curve — Group: 6	29
2.4	Model : LGBM	30
2.4.1	Gridsearch — Group: 1	30
2.4.2	Scatter	35
2.4.3	Scatter residuals	36
2.4.4	Learning curve — Group: 1	36
2.4.5	Gridsearch — Group: 2	37
2.4.6	Scatter	43
2.4.7	Scatter residuals	44
2.4.8	Learning curve — Group: 2	44
2.4.9	Gridsearch — Group: 3	45
2.4.10	Scatter	50
2.4.11	Scatter residuals	51
2.4.12	Learning curve — Group: 3	51
2.4.13	Gridsearch — Group: 4	52
2.4.14	Scatter	56
2.4.15	Scatter residuals	57
2.4.16	Learning curve — Group: 4	57
2.4.17	Gridsearch — Group: 6	58
2.4.18	Scatter	62
2.4.19	Scatter residuals	63
2.4.20	Learning curve — Group: 6	63
2.5	Model : CatBoost	64
2.5.1	Gridsearch — Group: 1	64
2.5.2	Scatter	70
2.5.3	Scatter residuals	71
2.5.4	Learning curve — Group: 1	71
2.5.5	Gridsearch — Group: 2	72
2.5.6	Scatter	78
2.5.7	Scatter residuals	79
2.5.8	Learning curve — Group: 2	79
2.5.9	Gridsearch — Group: 3	80
2.5.10	Scatter	87
2.5.11	Scatter residuals	88
2.5.12	Learning curve — Group: 3	88
2.5.13	Gridsearch — Group: 4	89
2.5.14	Scatter	95
2.5.15	Scatter residuals	96
2.5.16	Learning curve — Group: 4	96
2.5.17	Gridsearch — Group: 6	97
2.5.18	Scatter	103
2.5.19	Scatter residuals	104
2.5.20	Learning curve — Group: 6	104

2.6	Sumup-table	105
3	Plot	106
3.1	Standalone plots	106
3.1.1	Group: campaign: 2025_Jun,id_rabbit: 1,id_channel: 3	106
3.1.2	Group: campaign: 2025_Jun,id_rabbit: 2,id_channel: 11	109
3.1.3	Group: campaign: 2025_Jun,id_rabbit: 4,id_channel: 5	112
3.1.4	Group: campaign: 2025_Jun,id_rabbit: 6,id_channel: 4	115
3.1.5	Group: campaign: 2025_Sep,id_rabbit: 2,id_channel: 11	118
3.1.6	Group: campaign: 2025_Sep,id_rabbit: 3,id_channel: 4	121
3.1.7	Group: campaign: 2025_Sep,id_rabbit: 4,id_channel: 5	124
3.1.8	Group: campaign: 2025_Sep,id_rabbit: 6,id_channel: 3	127
3.2	Compare plots	130
3.2.1	Group: campaign: 2025_Jun,id_rabbit: 1,id_channel: 3	130
3.2.2	Group: campaign: 2025_Jun,id_rabbit: 2,id_channel: 11	131
3.2.3	Group: campaign: 2025_Jun,id_rabbit: 4,id_channel: 5	133
3.2.4	Group: campaign: 2025_Jun,id_rabbit: 6,id_channel: 4	134
3.2.5	Group: campaign: 2025_Sep,id_rabbit: 2,id_channel: 11	135
3.2.6	Group: campaign: 2025_Sep,id_rabbit: 3,id_channel: 4	137
3.2.7	Group: campaign: 2025_Sep,id_rabbit: 4,id_channel: 5	138
3.2.8	Group: campaign: 2025_Sep,id_rabbit: 6,id_channel: 3	139

1 Data analysis /Dataframe import and filters

1.1 DataFrame import and filters

The data has been imported with the following(s) function(s):

```
def build_df():
    if flag_IR:
        df, features_engineered = import_data_IR()
        if flag_IR_maskP:
            df = mask_period_IR(df)
        if flag_IR_balR:
            df = mask_balanceHC_random_IR(df, n_samples=None)
    if flag_ER:
        df, features_engineered = import_data_ER()

    cols_to_check = target_col + features_X
    mask = np.isfinite(df[cols_to_check]).all(axis=1)
    df = df[mask]
```

```

cols = list(dict.fromkeys(target_col + features_X +
                          feats_to_balance + feats_grouping +
                          feats_reference+ feats_postproc))

df = df[cols]
#df['campaign'] = "ALL"

return df

```

```

def import_data_IR():
    if target_col[0] == "i_NH3":
        #file_path = "csv/innova/switched/df_signal_decomposition_NH3_current_db2.csv"
        file_path = "csv/innova/notswitched/df_signal_decomposition_current.csv"
    elif target_col[0] == "i_CO2":
        file_path = "csv/innova/switched/df_signal_decomposition_CO2_current.csv"
    else:
        raise ValueError("Target is not supported")

    df = pd.read_csv(file_path, sep=";")
    df["datetime"] = pd.to_datetime(df["datetime"])
    df.sort_values("datetime").reset_index(drop=True)

    if target_col[0] == "i_NH3":
        df["campaign"] = df["datetime"].apply(assign_campaign)
        df = df[df["campaign"] != "other"].copy()
        df["innova_point"] = df.apply(assign_innova_point, axis=1)
        df = df[df["innova_point"] != "no"].copy()
        with open("csv/innova/notswitched/features_decomposition_list.json", "r") as f:
            dict_features = json.load(f)
    elif target_col[0] == "i_CO2":
        df["campaign"] = df["datetime"].apply(assign_campaign_iCO2)
        df = df[df["campaign"] != "other"].copy()
        df["innova_point"] = df.apply(assign_innova_point_iCO2, axis=1)
        df = df[df["innova_point"] != "no"].copy()
        with open("csv/innova/switched/features_decomposition_CO2_list.json", "r") as f:
            dict_features = json.load(f)
    else:
        raise ValueError("The target col do not match any existing function")

    df["hot"] = df.apply(assign_hot_season,axis=1)
    df["umidity_range"] = df.apply(assign_humidity, axis = 1)

```

```
#Removing of rabbit 4
df["column"] = df.apply(assign_colonna, axis = 1)
df['height'] = df.apply(assign_altezza, axis = 1)

return df,dict_features
```

```
def assign_innova_point(row):
    camp = row["campaign"]
    rid = row["id_rabbit"]

    # if camp == "2025_Jun":
    #     mapping = {2: "sl11", 4: "sl5", 6: "sl4", 1: "sl3", 5: "no"}
    # elif camp == "2025_Sep":
    #     mapping = {2: "sl11", 4: "sl5", 3: "sl4", 6: "sl3", 5: "no"}
    if camp == "2025_Apr":
        #mapping = {1: "no", 2: "no", 4: "no", 6: "sl5"}
        #mapping = {1: "no", 2: "sl5", 4: "no", 6: "sl5"}
        mapping = {1: "no", 2: "no", 4: "no", 6: "no"}
    elif camp == "2025_Jun":
        mapping = {2: "sl11", 4: "sl5", 1: "sl3", 6: "sl4", 5: "no"}
        #mapping = {2: "sl11", 4: "no", 1: "sl4", 6: "sl3", 5: "no"}
    elif camp == "2025_Sep":
        mapping = {2: "sl11", 4: "sl5", 6: "sl3", 3: "sl4", 5: "no"}
        #mapping = {2: "sl11", 4: "no", 6: "sl4", 3: "sl3", 5: "no"}
    elif camp == "2026_Gen":
        #mapping = {1: "no", 4: "no", 5: "sl11", 6: "no"}
        #mapping = {1: "sl11", 4: "no", 5: "sl11", 6: "sl11"}
        mapping = {1: "no", 4: "no", 5: "no", 6: "no"}
    else:
        mapping = {}

    return mapping.get(rid, "no")
```

Function print :

1.2 Metadata

Nombre de lignes df

8168

Nombre de lignes df clean	8168
Nombre de features	5
% de NaN	0.0
Taille dataset entraînement	5717
Taille dataset test	2450
Métrique utilisée	r2

Column target : i_NH3

Features_X : r_NH3, r_temperature, r_umidity, r_CO2, r_THI

feats_to_balance :

feats_grouping : id_rabbit

feats_postproc : datetime, id_channel, id_rabbit, campaign

feats_ref :

id_cols : campaign

Grouping : id_rabbit

2 Model(s) analysis

2.1 Explanation legend tables

Columns

- rank: ranking regarding cross validation test
- train_cv: mean score train of the CV
- test_cv: mean score test of the CV
- test: score on the testing dataset
- diff: test-test_cv
- r_test: test/test_cv
- r_train: train_cv/test_cv

Colors:

[Orange] $r_{\text{test}} < 0.95$ (under performance of the test)

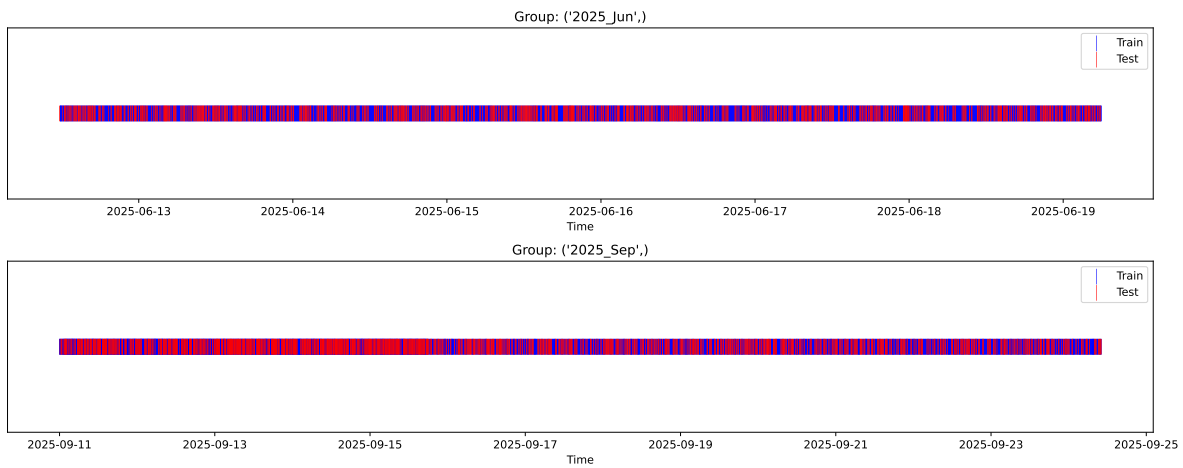
[Green] $r_{\text{test}} > 1.05$ (over performance of the test)

[Red] r_{train} not in $[0.95, 1.05]$ (gap between train/test)

2.2 Repartition of train and test

2.2.1 External Split: X_{train} vs X_{test}

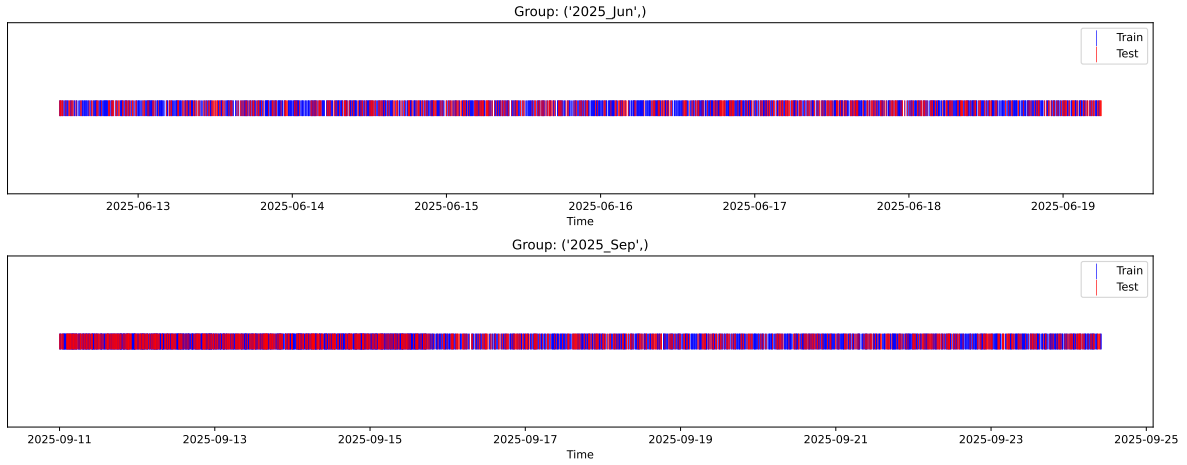
External Split: Train: 5717 samples Test: 2451 samples



2.2.2 Internal CV Folds

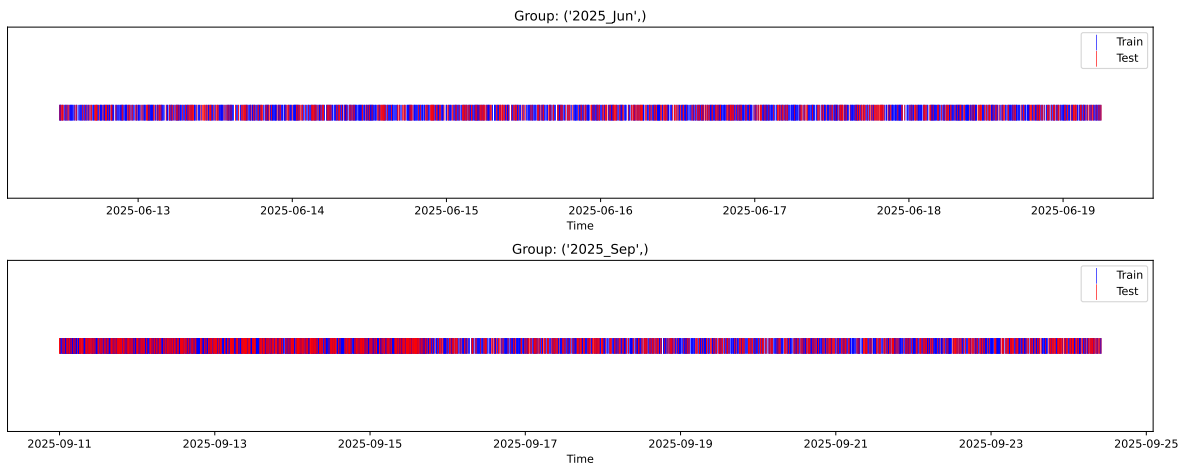
Fold 1/5

Train: 4001 samples Test: 1716 samples Excluded: 0 samples Window exclusion: 0h



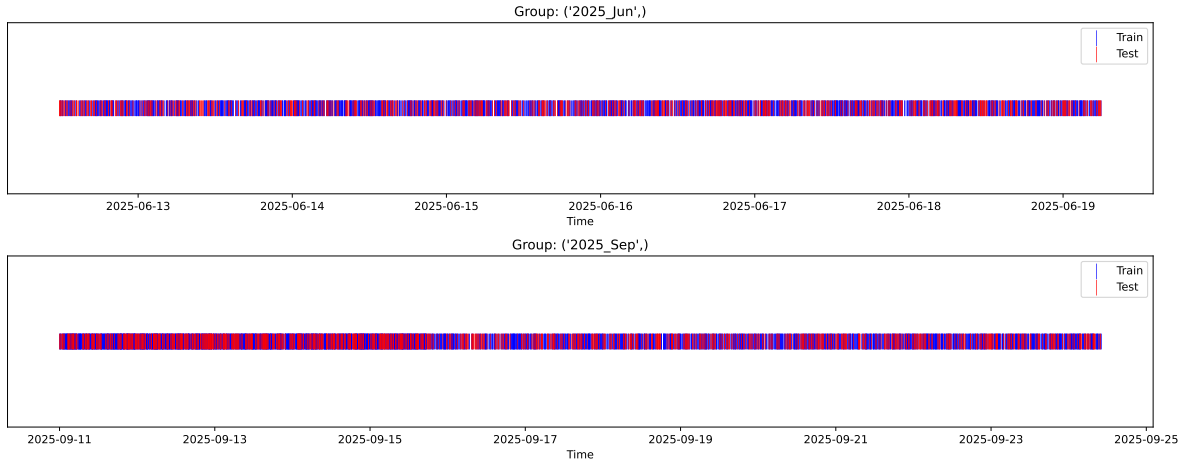
Fold 2/5

Train: 4001 samples Test: 1716 samples Excluded: 0 samples Window exclusion: 0h



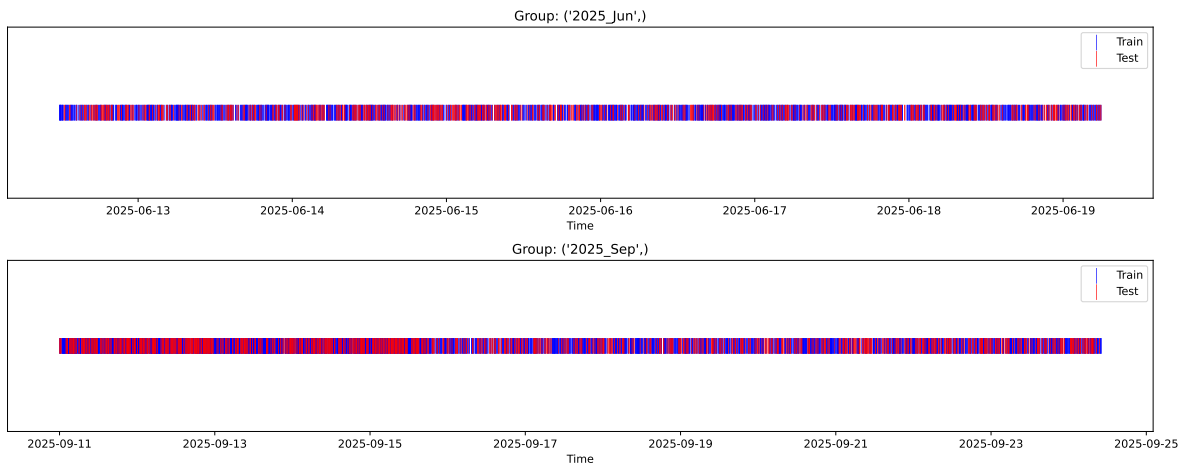
Fold 3/5

Train: 4001 samples Test: 1716 samples Excluded: 0 samples Window exclusion: 0h



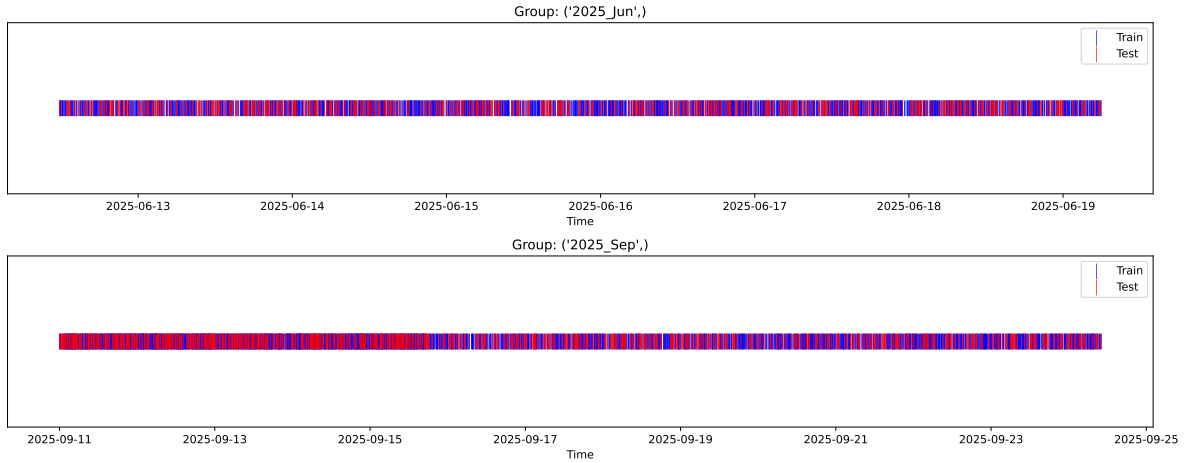
Fold 4/5

Train: 4001 samples Test: 1716 samples Excluded: 0 samples Window exclusion: 0h



Fold 5/5

Train: 4001 samples Test: 1716 samples Excluded: 0 samples Window exclusion: 0h



2.3 Model : Linear

2.3.1 Gridsearch — Group: 1

Len(group) : 795, Len(training) : 557, Len(test) : 238

Distribution y_train vs y_test: y_train: mean=2.597, std=0.579, min=1.437, max=4.986
y_test: mean=2.492, std=0.625, min=1.466, max=5.447

Fold 0 — y_test: mean=2.57, std=0.62, y_train: mean=2.61, std=0.56

Fold 1 — y_test: mean=2.53, std=0.55, y_train: mean=2.63, std=0.59

Fold 2 — y_test: mean=2.65, std=0.55, y_train: mean=2.58, std=0.59

Fold 3 — y_test: mean=2.64, std=0.59, y_train: mean=2.58, std=0.57

Fold 4 — y_test: mean=2.56, std=0.54, y_train: mean=2.61, std=0.59

train scores CV: [0.56730715 0.5334636 0.56784944 0.59353423 0.55417991]

test scores CV: [0.5431311 0.6179077 0.53621024 0.4830173 0.57408525]

rank	mean_train_cv	mean_test_cv	std_test_cv	test_score_cv	test	diff	r_test	r_train
1.0	0.5633	0.5509	0.0445	0.4979	0.5003	-0.0505	0.9082	1.0225

Coefficients:

$\text{coef_r_CO2} = -0.0079$

$\text{coef_r_NH3} = 0.0524$

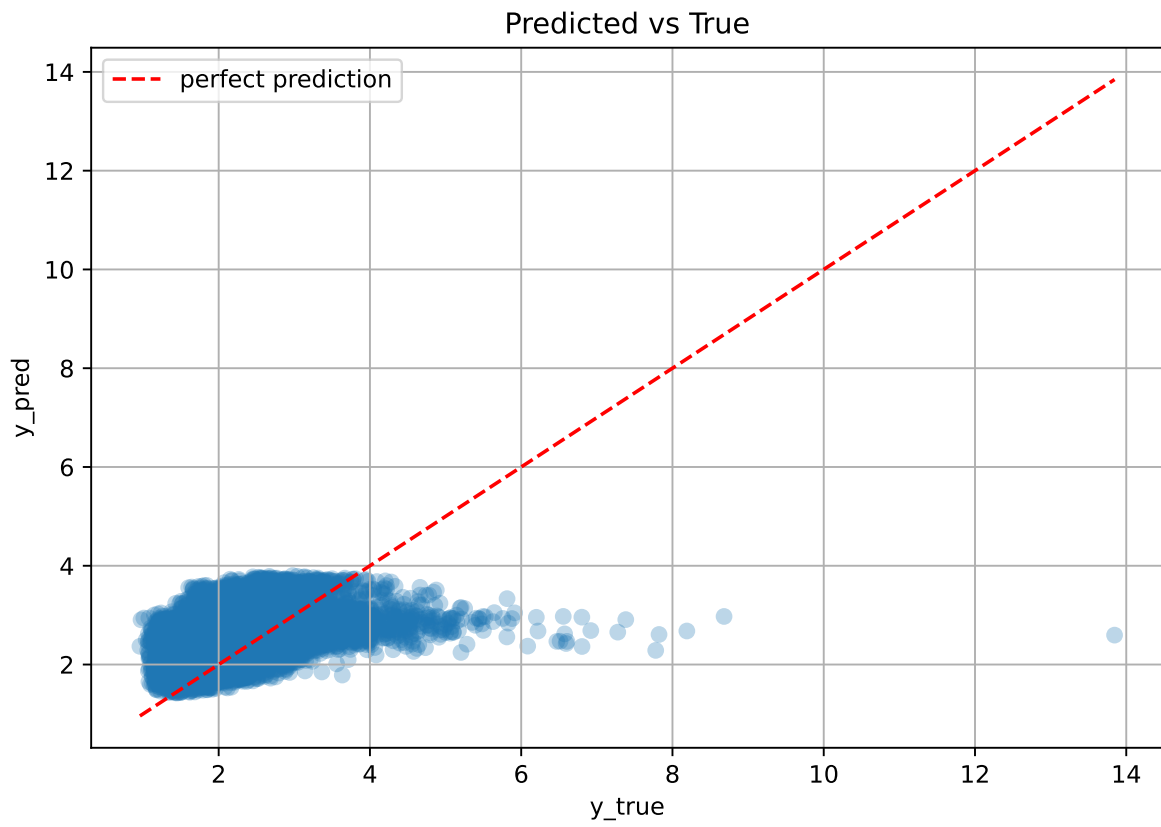
$\text{coef_r_THI} = 0.5946$

$\text{coef_r_temperature} = -0.3581$

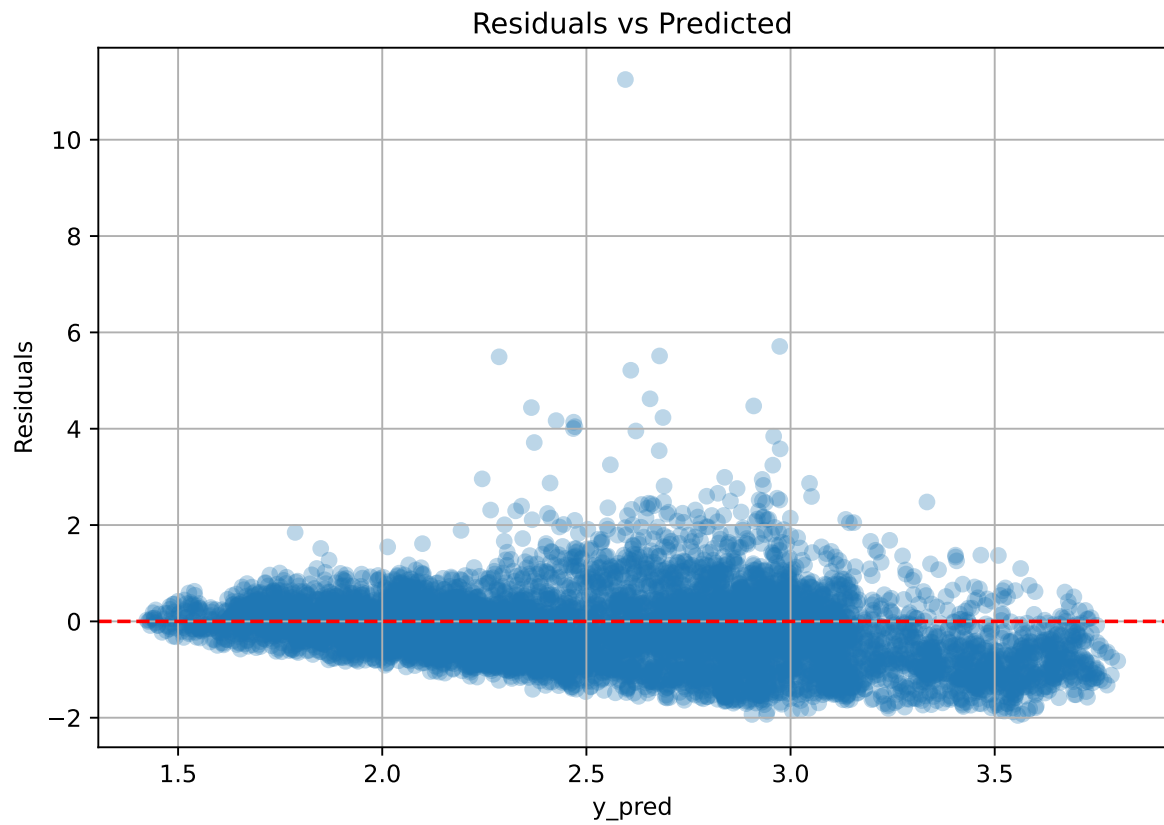
$\text{coef_r_umidity} = -0.2680$

$\text{intercept} = 2.5975$

2.3.2 Scatter



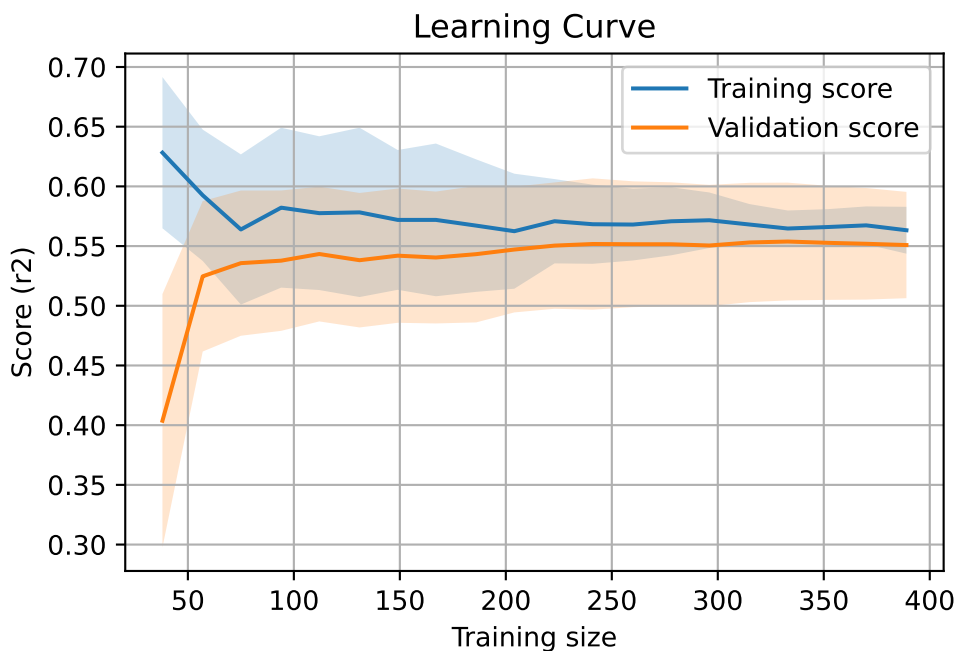
2.3.3 Scatter residuals



2.3.4 Learning curve — Group: 1

`Len(group)` : 795, `Len(training)` : 557,

`Len(training_real)` : 390, `Len(test_of_training)` : 167, `Len(test)` : 238



2.3.5 Gridsearch — Group: 2

Len(group) : 2726, Len(training) : 1909, Len(test) : 817

Distribution y_train vs y_test: y_train: mean=2.540, std=0.859, min=1.150, max=8.682
y_test: mean=2.561, std=0.947, min=1.120, max=13.845

Fold 0 — y_test: mean=2.58, std=0.86, y_train: mean=2.52, std=0.86

Fold 1 — y_test: mean=2.56, std=0.84, y_train: mean=2.53, std=0.87

Fold 2 — y_test: mean=2.50, std=0.84, y_train: mean=2.55, std=0.87

Fold 3 — y_test: mean=2.53, std=0.87, y_train: mean=2.55, std=0.85

Fold 4 — y_test: mean=2.53, std=0.82, y_train: mean=2.54, std=0.87

train scores CV: [0.46497694 0.46580479 0.46495948 0.46844345 0.45091902]

test scores CV: [0.45757013 0.45504757 0.45666765 0.44978339 0.49647352]

rank	mean_train_cv	mean_test_cv	std_test_cv	test_score_cv	test	diff	r_test	r_train
1.0	0.463	0.4631	0.0169	0.3942	0.3945	-0.0686	0.8519	0.9998

Coefficients:

coef_r_CO2 = 0.0009

coef_r_NH3 = 0.0485

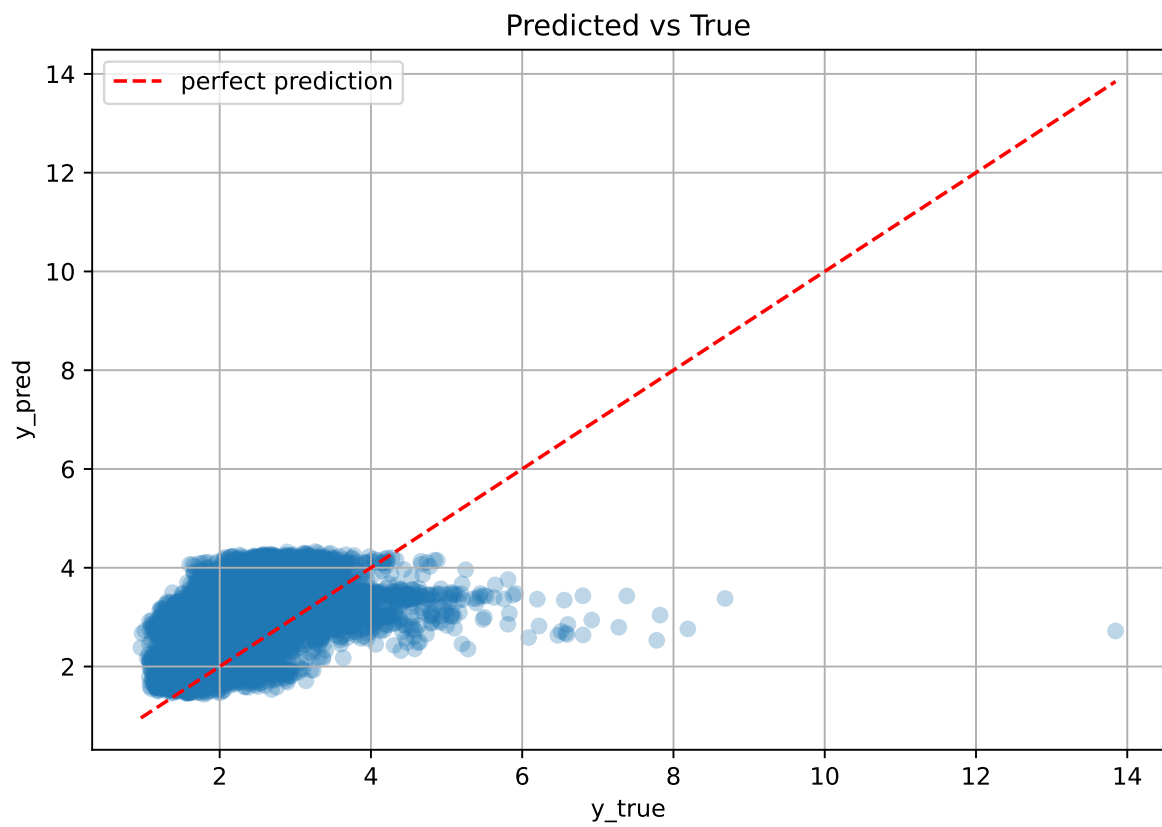
coef_r_THI = 0.8323

coef_r_temperature = -0.9731

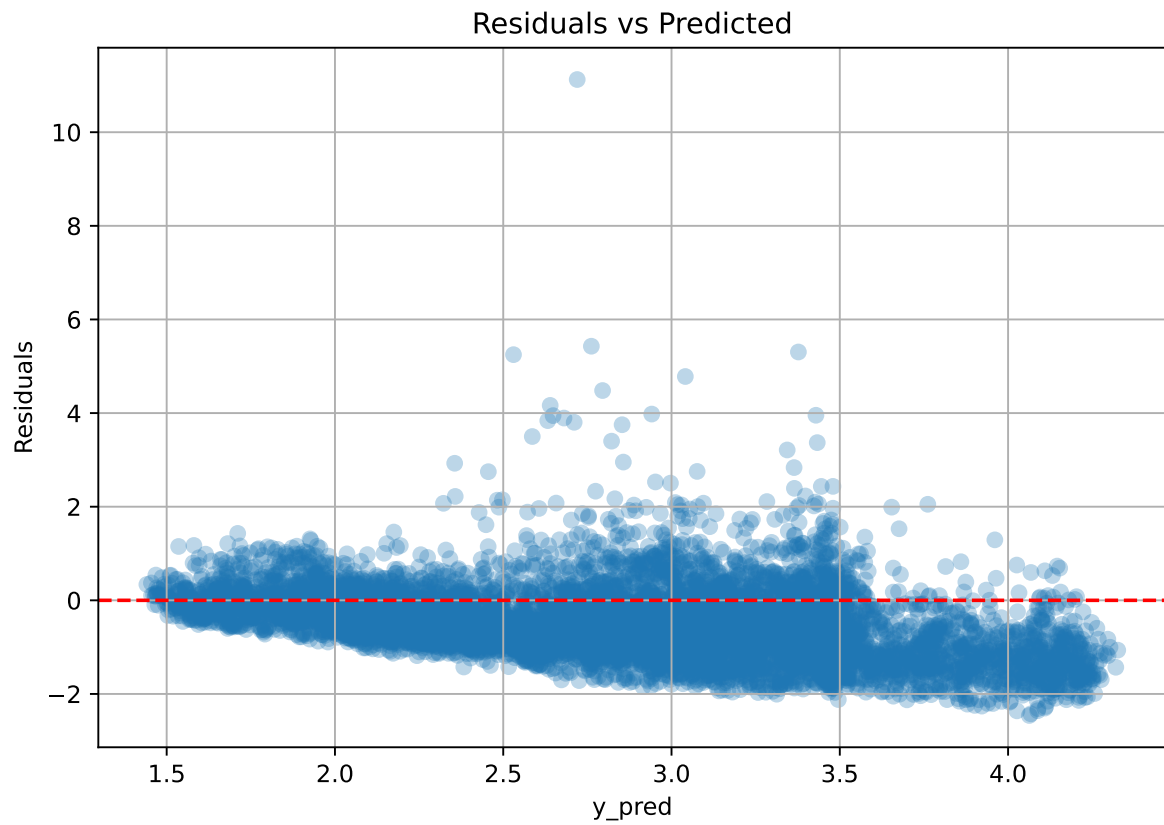
coef_r_umidity = -0.7852

intercept = 2.5398

2.3.6 Scatter



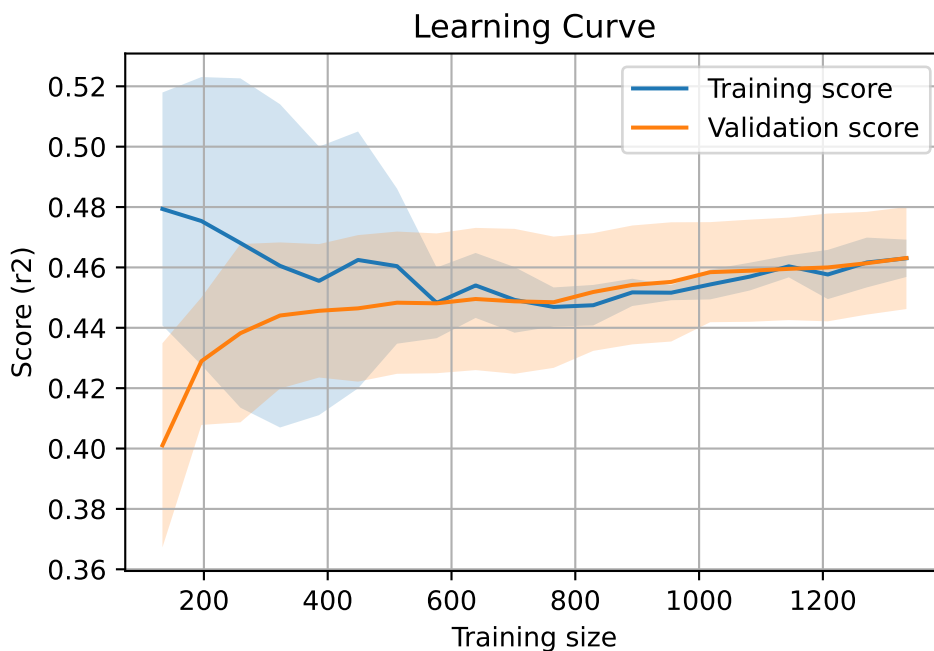
2.3.7 Scatter residuals



2.3.8 Learning curve — Group: 2

`Len(group)` : 2726, `Len(training)` : 1909,

`Len(training_real)` : 1337, `Len(test_of_training)` : 572, `Len(test)` : 817



2.3.9 Gridsearch — Group: 3

Len(group) : 568, Len(training) : 398, Len(test) : 170

Distribution y_train vs y_test: y_train: mean=1.928, std=0.816, min=0.959, max=6.804
y_test: mean=1.833, std=0.729, min=1.098, max=7.822

Fold 0 — y_test: mean=1.93, std=0.75, y_train: mean=1.93, std=0.84

Fold 1 — y_test: mean=1.85, std=0.67, y_train: mean=1.96, std=0.87

Fold 2 — y_test: mean=2.00, std=0.92, y_train: mean=1.90, std=0.76

Fold 3 — y_test: mean=1.92, std=0.88, y_train: mean=1.93, std=0.78

Fold 4 — y_test: mean=1.92, std=0.86, y_train: mean=1.93, std=0.80

train scores CV: [0.3155034 0.29364485 0.29854522 0.32073505 0.30855581]

test scores CV: [0.17929864 0.27024886 0.27717852 0.24665152 0.26532812]

rank	mean_train_cv	mean_test_cv	std_test_cv	test_score_cv	test	diff	r_test	r_train
1.0	0.3074	0.2477	0.0357	0.1869	0.2027	-0.045	0.8182	1.2408

Coefficients:

$\text{coef_r_CO2} = 0.1066$

$\text{coef_r_NH3} = -0.2625$

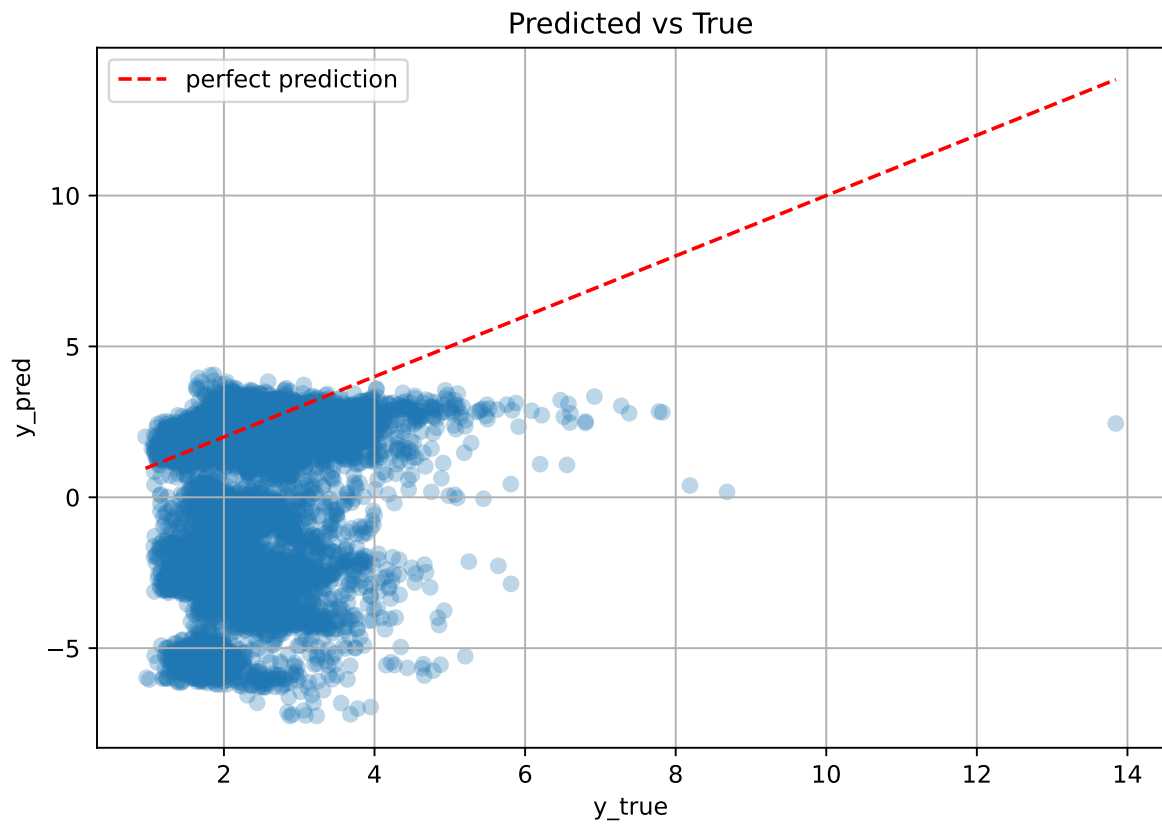
$\text{coef_r_THI} = 2.6048$

$\text{coef_r_temperature} = -3.7070$

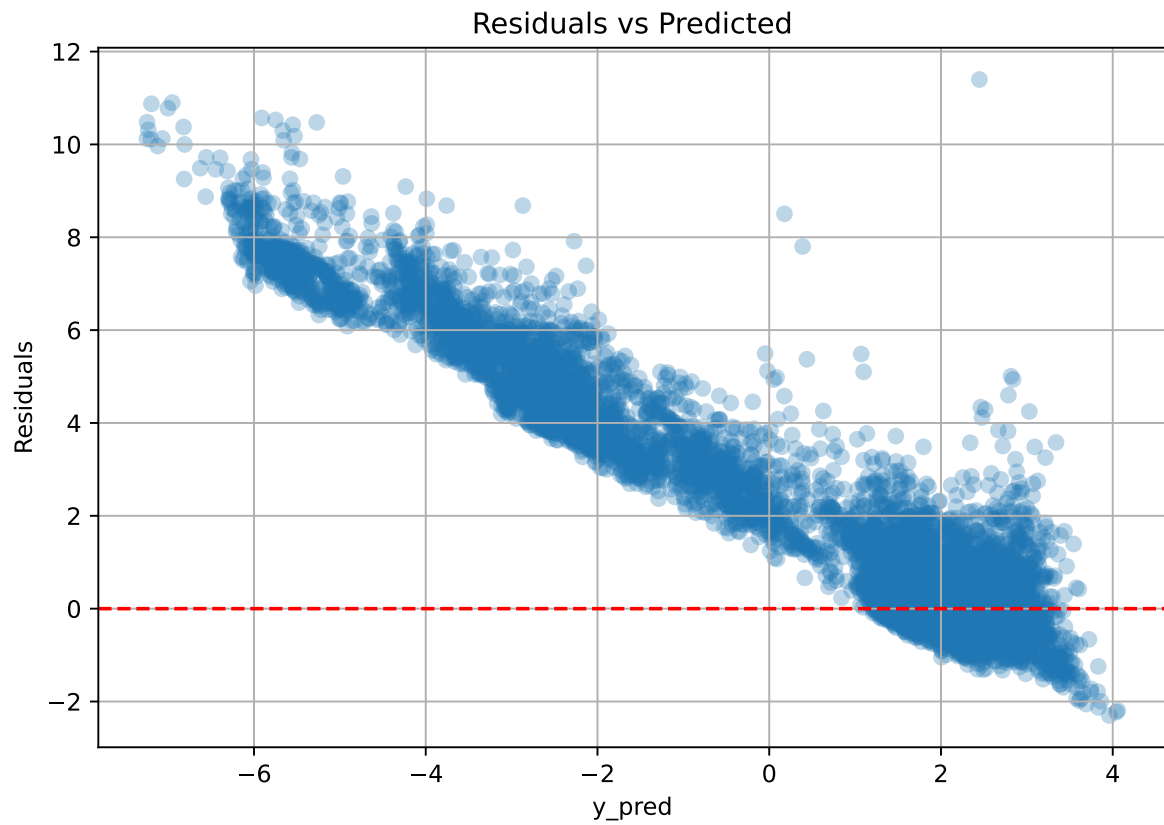
$\text{coef_r_umidity} = -1.4591$

$\text{intercept} = 1.9278$

2.3.10 Scatter



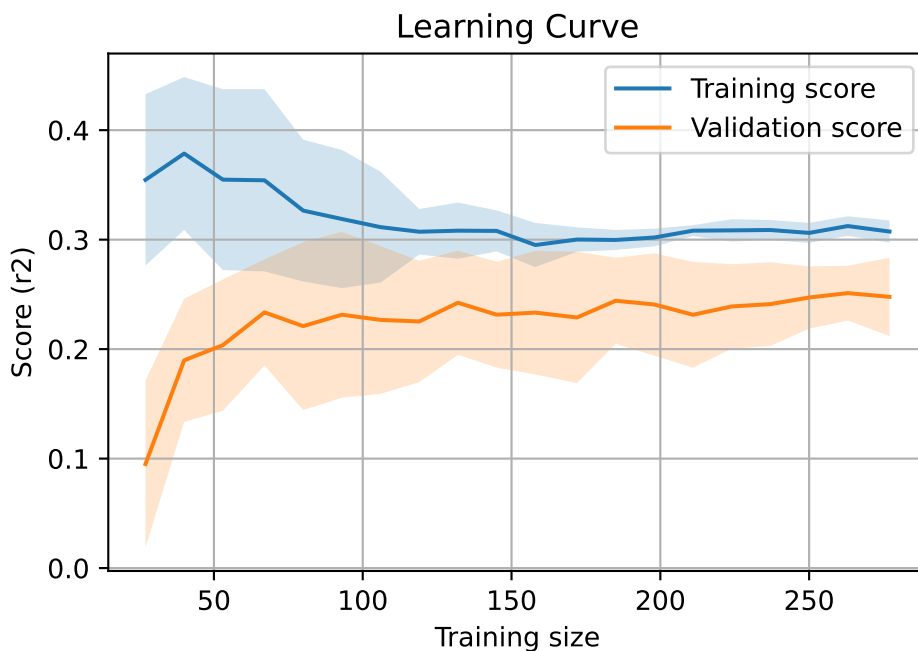
2.3.11 Scatter residuals



2.3.12 Learning curve — Group: 3

`Len(group)` : 568, `Len(training)` : 398,

`Len(training_real)` : 279, `Len(test_of_training)` : 119, `Len(test)` : 170



2.3.13 Gridsearch — Group: 4

Len(group) : 2726, Len(training) : 1909, Len(test) : 817

Distribution y_train vs y_test: y_train: mean=2.238, std=0.681, min=0.973, max=5.813
y_test: mean=2.212, std=0.659, min=1.071, max=4.858

Fold 0 — y_test: mean=2.27, std=0.68, y_train: mean=2.22, std=0.68

Fold 1 — y_test: mean=2.25, std=0.71, y_train: mean=2.23, std=0.67

Fold 2 — y_test: mean=2.20, std=0.65, y_train: mean=2.25, std=0.69

Fold 3 — y_test: mean=2.21, std=0.65, y_train: mean=2.25, std=0.70

Fold 4 — y_test: mean=2.22, std=0.63, y_train: mean=2.24, std=0.70

train scores CV: [0.4277706 0.44021497 0.41172833 0.43714743 0.44005866]

test scores CV: [0.41299685 0.39069188 0.45040747 0.3878951 0.37549212]

rank	mean_train_cv	mean_test_cv	std_test_cv	test_score_cv	test	diff	r_test	r_train
1.0	0.4314	0.4035	0.0264	0.4566	0.4572	0.0537	1.1332	1.0691

Coefficients:

coef_r_CO2 = -0.0952

coef_r_NH3 = -0.1254

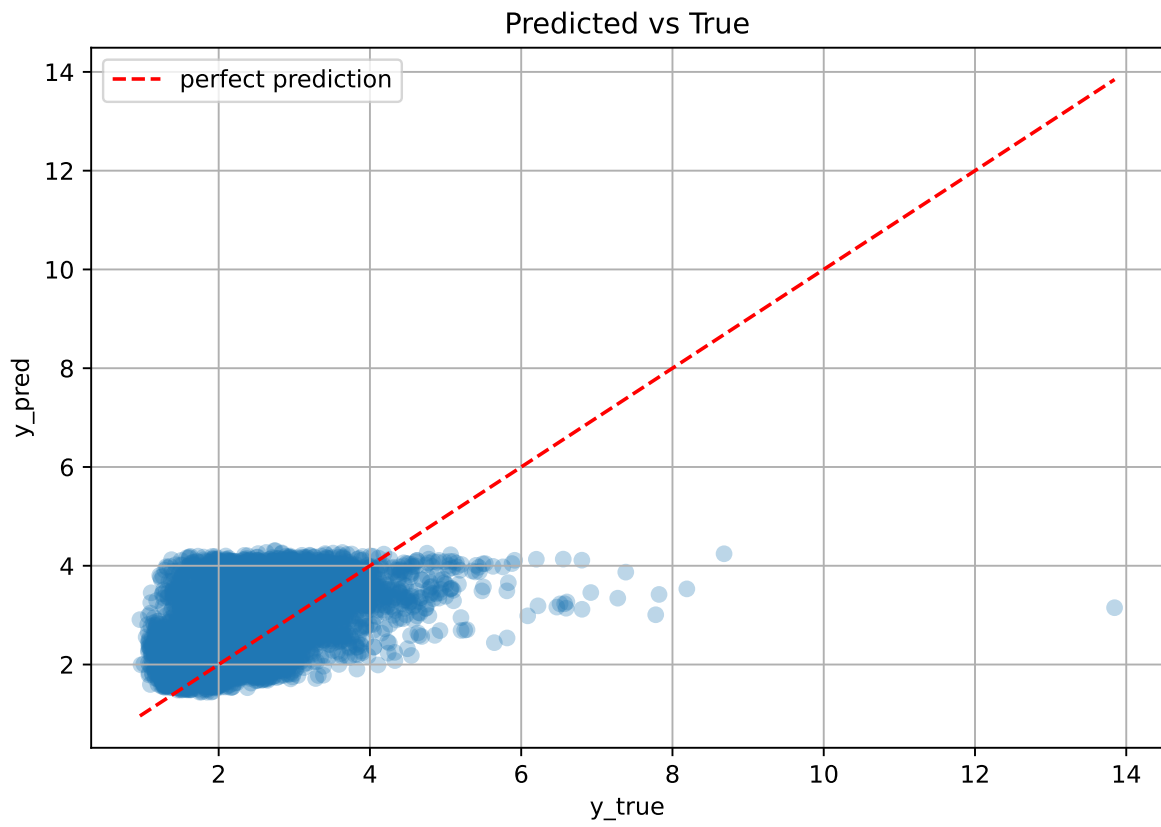
coef_r_THI = 0.0994

coef_r_temperature = 0.0362

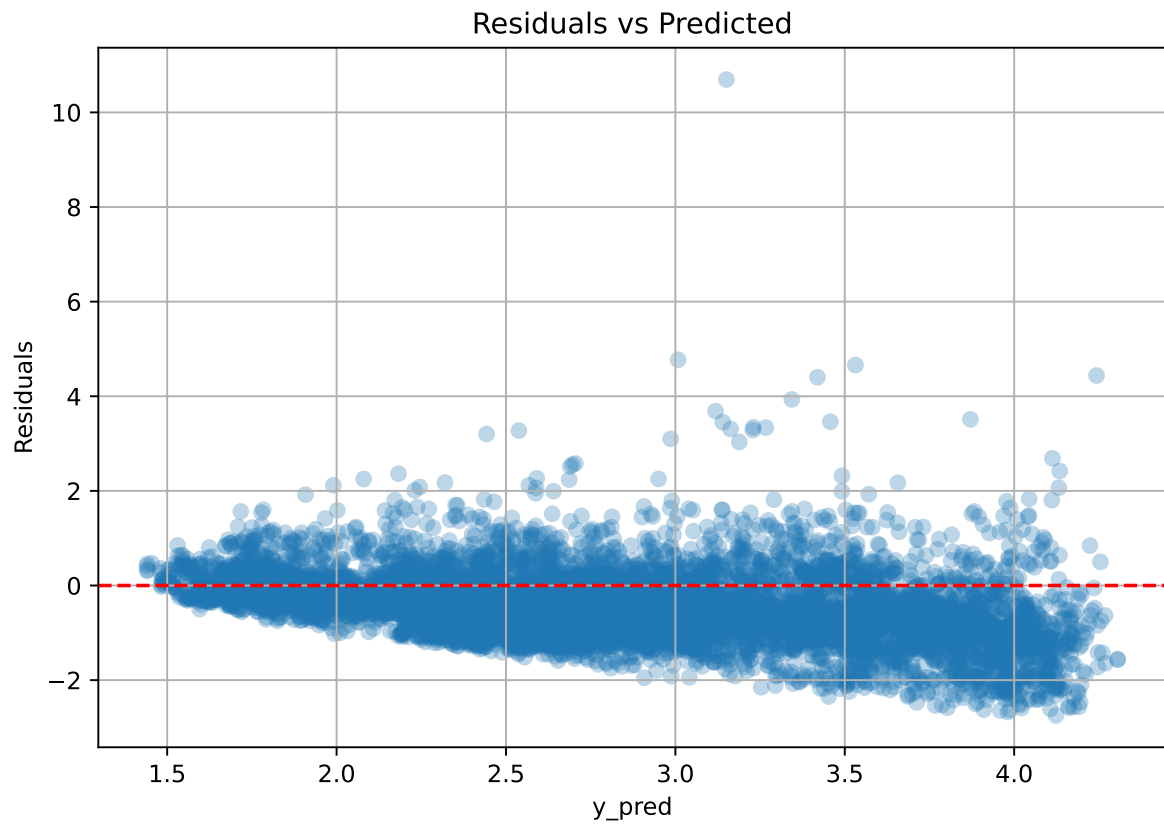
coef_r_umidity = -0.4502

intercept = 2.2379

2.3.14 Scatter



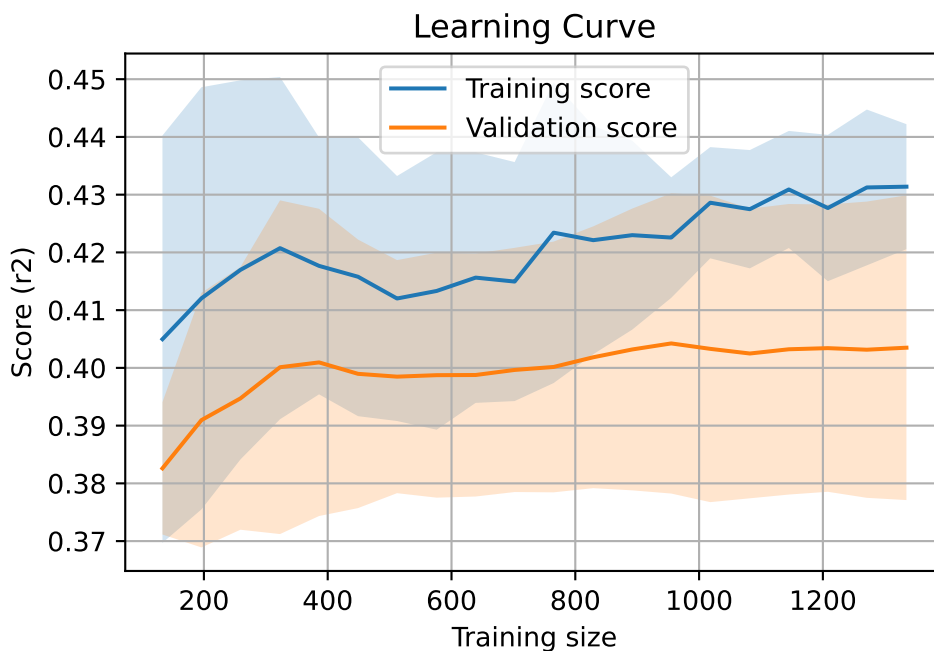
2.3.15 Scatter residuals



2.3.16 Learning curve — Group: 4

`Len(group)` : 2726, `Len(training)` : 1909,

`Len(training_real)` : 1337, `Len(test_of_training)` : 572, `Len(test)` : 817



2.3.17 Gridsearch — Group: 6

Len(group) : 1353, Len(training) : 948, Len(test) : 405

Distribution y_train vs y_test: y_train: mean=2.286, std=0.802, min=1.041, max=7.383
y_test: mean=2.187, std=0.733, min=1.078, max=5.525

Fold 0 — y_test: mean=2.28, std=0.82, y_train: mean=2.29, std=0.79

Fold 1 — y_test: mean=2.30, std=0.83, y_train: mean=2.28, std=0.79

Fold 2 — y_test: mean=2.25, std=0.73, y_train: mean=2.30, std=0.83

Fold 3 — y_test: mean=2.29, std=0.80, y_train: mean=2.28, std=0.80

Fold 4 — y_test: mean=2.28, std=0.76, y_train: mean=2.29, std=0.82

train scores CV: [0.35655782 0.39427008 0.3362054 0.36466187 0.35206257]

test scores CV: [0.37929847 0.30341745 0.44634569 0.36397919 0.3946844]

rank	mean_train_cv	mean_test_cv	std_test_cv	test_score_cv	test	diff	r_test	r_train
1.0	0.3608	0.3775	0.0463	0.4129	0.4124	0.0348	1.0923	0.9555

Coefficients:

coef_r_CO2 = -0.1485

coef_r_NH3 = -0.0059

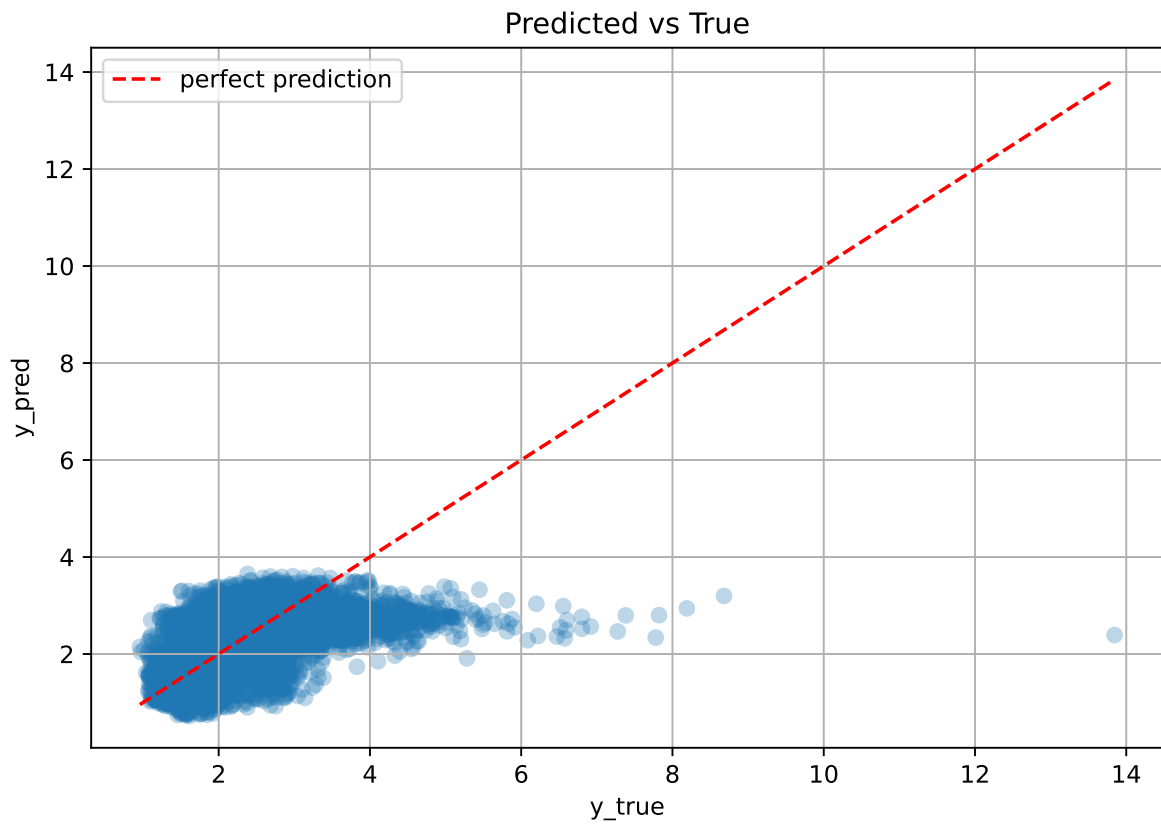
coef_r_THI = 2.0119

coef_r_temperature = -2.8461

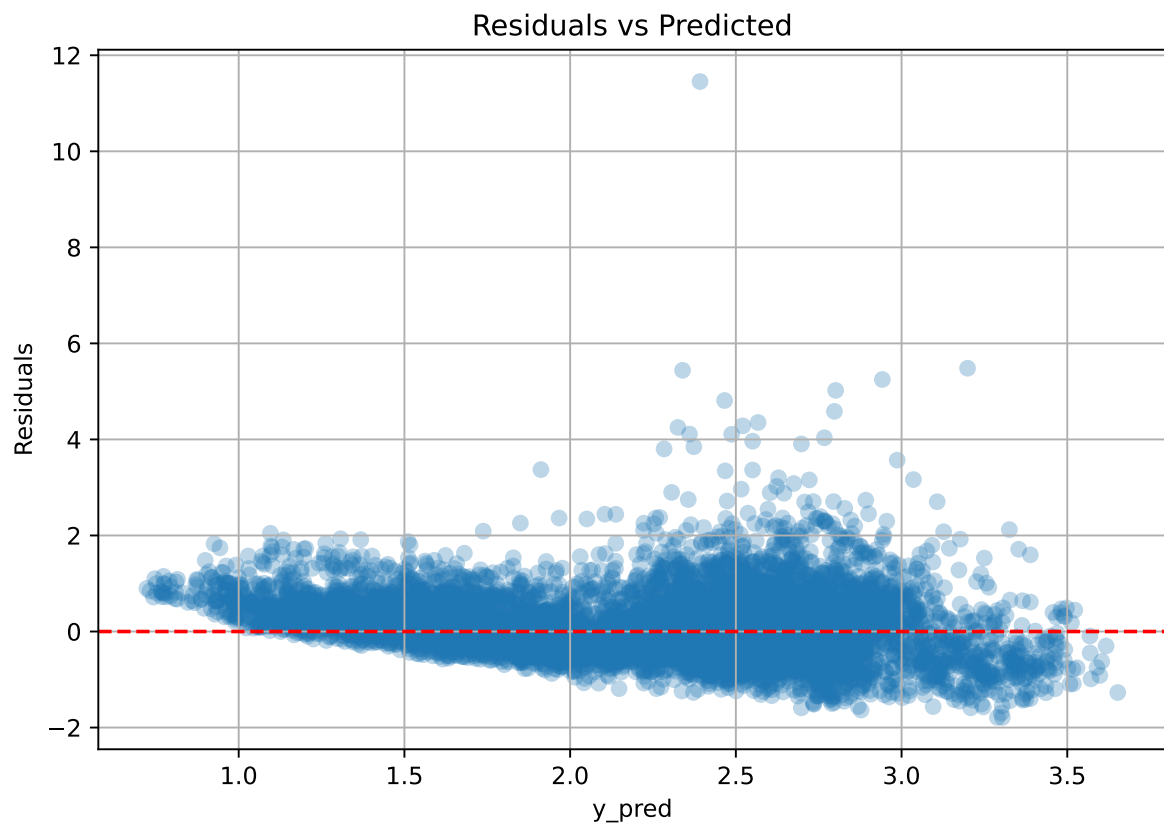
coef_r_umidity = -1.4147

intercept = 2.2858

2.3.18 Scatter



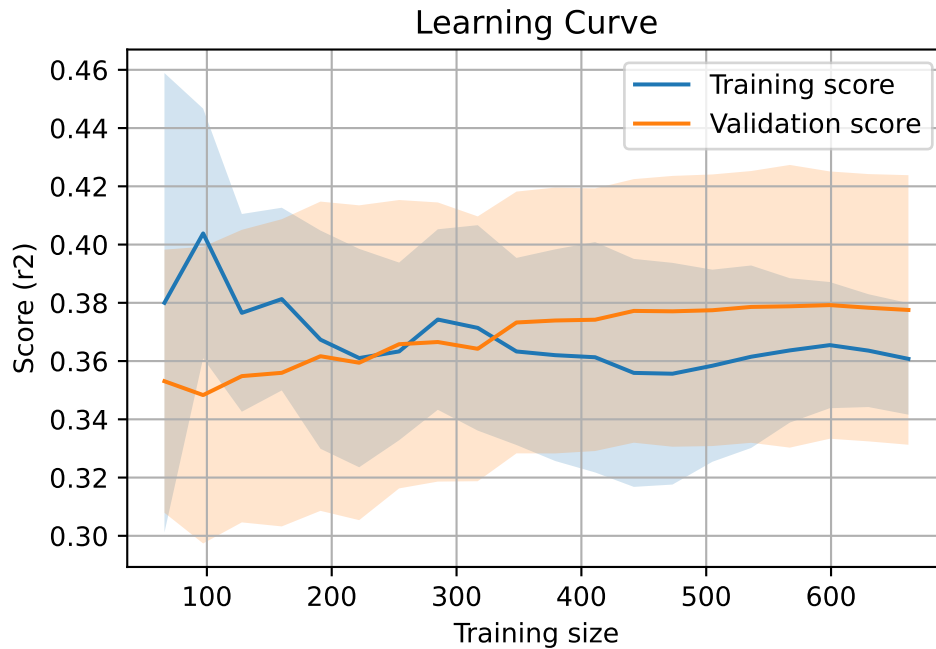
2.3.19 Scatter residuals



2.3.20 Learning curve — Group: 6

`Len(group)` : 1353, `Len(training)` : 948,

`Len(training_real)` : 664, `Len(test_of_training)` : 284, `Len(test)` : 405



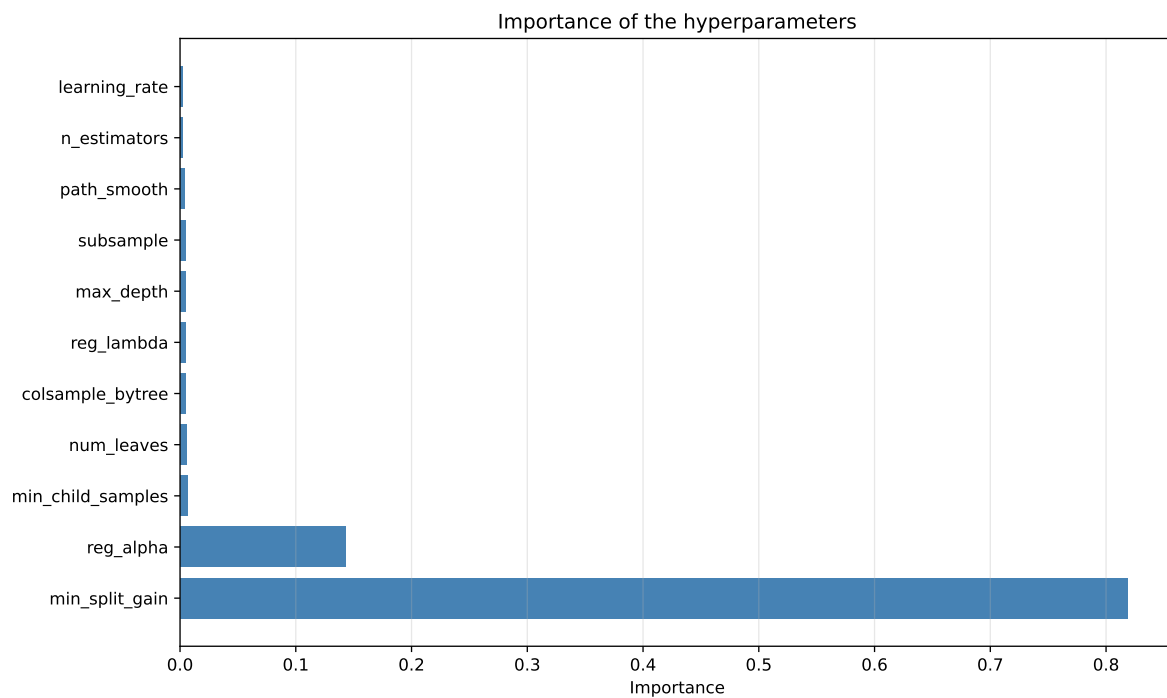
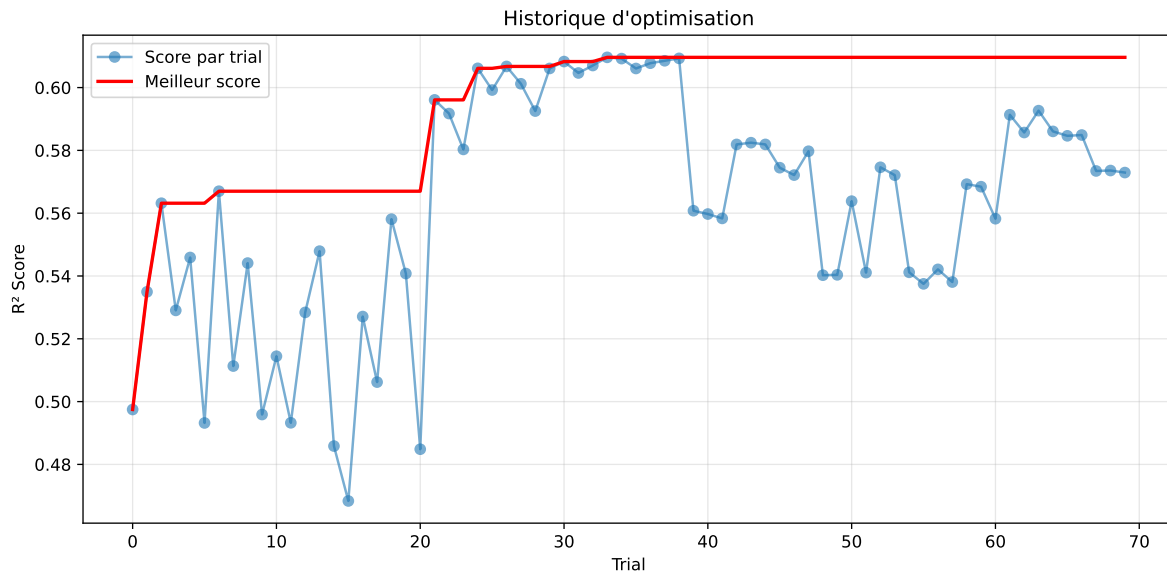
2.4 Model : LGBM

2.4.1 Gridsearch — Group: 1

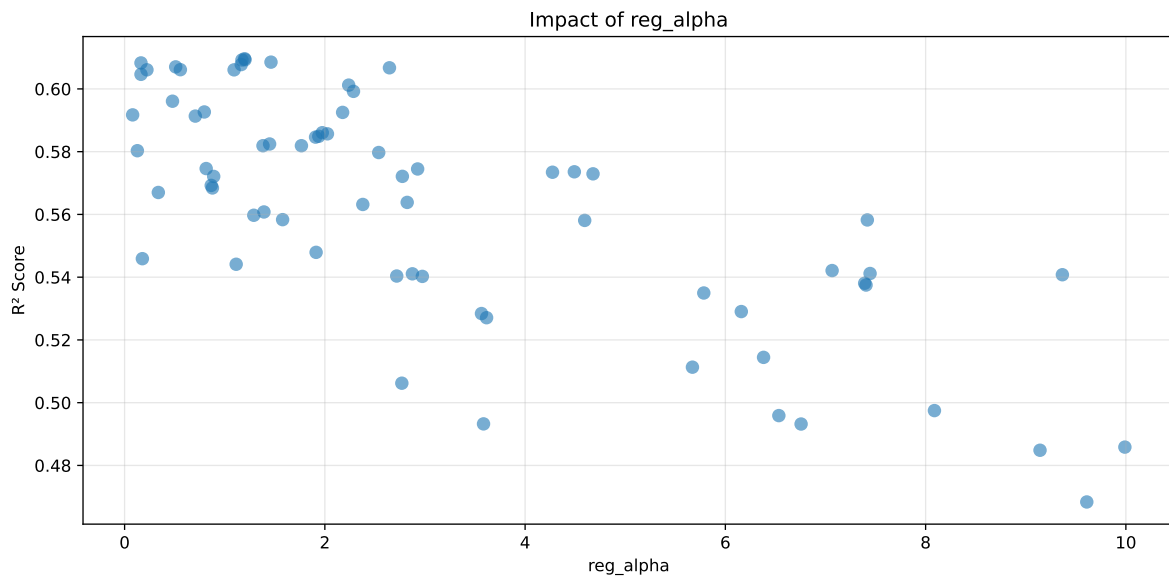
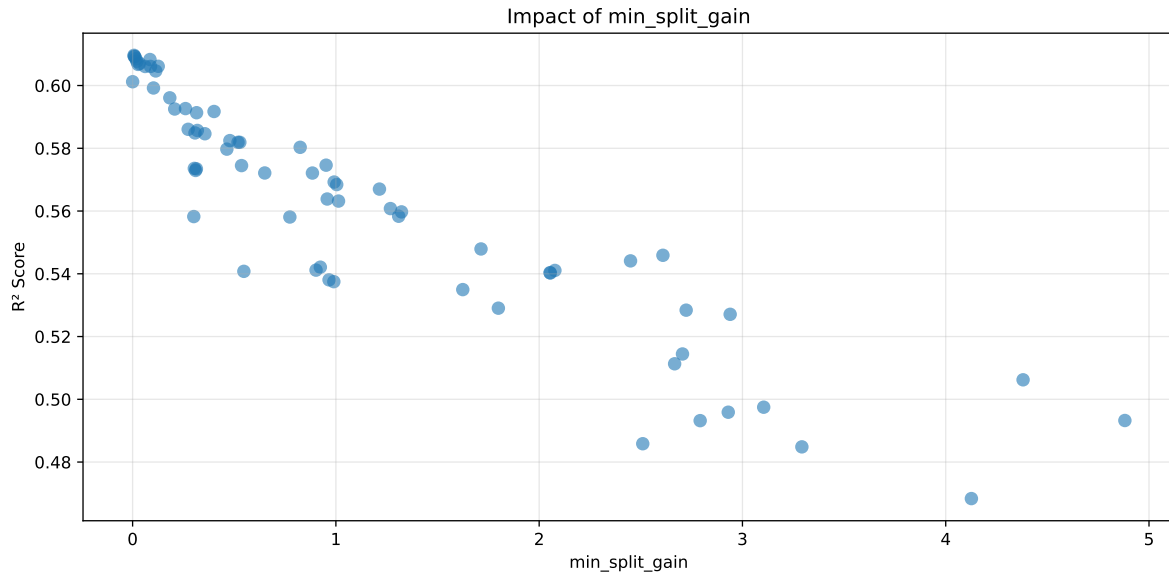
Len(group) : 795, Len(training) : 557, Len(test) : 238

Distribution y_train vs y_test: y_train: mean=2.597, std=0.579, min=1.437, max=4.986
y_test: mean=2.492, std=0.625, min=1.466, max=5.447

===== TOP Importance FEATURES =====
feature importance r_temperature 66 r_umidity 38 r_CO2 25 r_THI 13 r_NH3 4

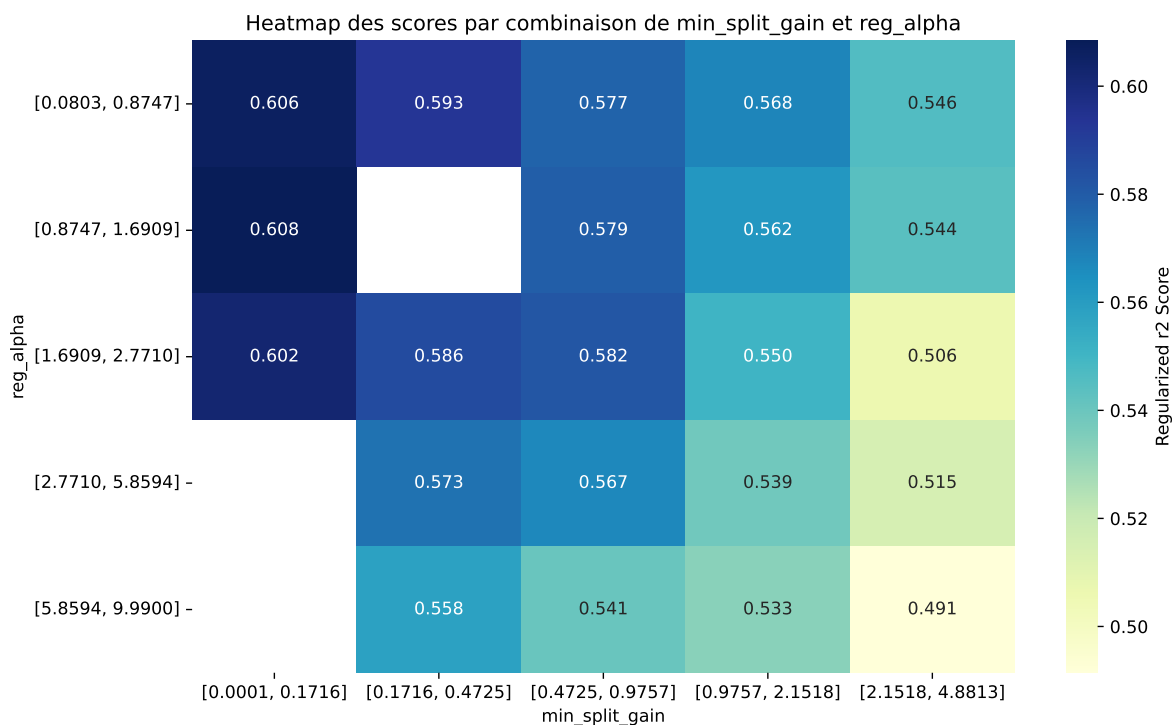


Paramètres avec >5% d'importance : ['min_split_gain', 'reg_alpha']



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The default value of observed=False is deprecated and will change to observed=True in a future



rank	mean_train_cv	mean_test_cv	test_score_cv	std_test_cv	penalized_score	test	diff	r_test	r_train
1	0.611	0.5721	0.5401	0.0432	0.4555	0.5573	-0.0148	0.9741	1.0679
2	0.6048	0.567	0.5325	0.0437	0.4537	0.552	-0.0149	0.9736	1.0666
3	0.6083	0.5692	0.5373	0.0428	0.4519	0.556	-0.0132	0.9768	1.0687
4	0.6004	0.5632	0.5273	0.0437	0.4516	0.5481	-0.0151	0.9733	1.066
5	0.6017	0.5638	0.5312	0.0436	0.4502	0.5499	-0.0139	0.9753	1.0672
6	0.6079	0.5684	0.5373	0.0424	0.4502	0.5537	-0.0147	0.9741	1.0694
7	0.5984	0.5608	0.5271	0.0427	0.4481	0.5435	-0.0173	0.9692	1.067
8	0.597	0.5597	0.526	0.0437	0.4479	0.5417	-0.018	0.9679	1.0666
9	0.614	0.5721	0.5417	0.0431	0.4466	0.5614	-0.0108	0.9812	1.0732
10	0.6174	0.5746	0.5476	0.0446	0.4462	0.5627	-0.0119	0.9793	1.0745
---	---	---	---	---	---	---	---	---	---
70	0.7227	0.6096	0.6293	0.0496	0.2705	0.6462	0.0366	1.06	1.1854

Hyperparameters:

Rank 1: {'reg__n_estimators': 200, 'reg__max_depth': 4, 'reg__learning_rate': 0.051750513617113636, 'reg__num_leaves': 57, 'reg__subsample': 0.5462752052434899, 'reg__colsample_bytree': 0.5925859055627839, 'reg__min_child_samples': 38, 'reg__reg_alpha': 0.8899250846641266, 'reg__reg_lambda': 49.876957327363314, 'reg__min_split_gain': 0.8845687324452858, 'reg__path_smooth': 2.6869128749164997}

Rank 2: {'reg__n_estimators': 700, 'reg__max_depth': 7, 'reg__learning_rate': 0.010766912154577894, 'reg__num_leaves': 49, 'reg__subsample': 0.5606159443248334, 'reg__colsample_bytree': 0.7021801955651611, 'reg__min_child_samples': 10, 'reg__reg_alpha': 0.3369047862818553, 'reg__reg_lambda': 48.23612685698684, 'reg__min_split_gain': 1.2142937173168045, 'reg__path_smooth': 1.2339186164699418}

Rank 3: {'reg__n_estimators': 300, 'reg__max_depth': 3, 'reg__learning_rate': 0.08755895034386586, 'reg__num_leaves': 55, 'reg__subsample': 0.5205319021611581, 'reg__colsample_bytree': 0.6048489589482763, 'reg__min_child_samples': 47, 'reg__reg_alpha': 0.8651040296865689, 'reg__reg_lambda': 44.61846228965172, 'reg__min_split_gain': 0.9914573152558626, 'reg__path_smooth': 2.7596119956255465}

Rank 4: {'reg__n_estimators': 700, 'reg__max_depth': 4, 'reg__learning_rate': 0.011187644021755615, 'reg__num_leaves': 41, 'reg__subsample': 0.6814919313368393, 'reg__colsample_bytree': 0.8303594099235947, 'reg__min_child_samples': 26, 'reg__reg_alpha': 2.379104469957184, 'reg__reg_lambda': 36.746142305151594, 'reg__min_split_gain': 1.0129128508460155, 'reg__path_smooth': 3.608098193257778}

Rank 5: {'reg__n_estimators': 200, 'reg__max_depth': 4, 'reg__learning_rate': 0.04998919665007327, 'reg__num_leaves': 43, 'reg__subsample': 0.5410157976834473, 'reg__colsample_bytree': 0.5448868689047832, 'reg__min_child_samples': 36, 'reg__reg_alpha': 2.822362932315896, 'reg__reg_lambda': 23.429677040378333, 'reg__min_split_gain': 0.9572635722455161, 'reg__path_smooth': 1.7197960949298863}

Rank 6: {'reg__n_estimators': 300, 'reg__max_depth': 3, 'reg__learning_rate': 0.08405602385261439, 'reg__num_leaves': 40, 'reg__subsample': 0.6362863100213314, 'reg__colsample_bytree': 0.5933940364572954, 'reg__min_child_samples': 40, 'reg__reg_alpha': 0.8770967944392825, 'reg__reg_lambda': 45.5053929107392, 'reg__min_split_gain': 1.0040590324162753, 'reg__path_smooth': 2.7439580294702903}

Rank 7: {'reg__n_estimators': 300, 'reg__max_depth': 3, 'reg__learning_rate': 0.0608596001627862, 'reg__num_leaves': 48, 'reg__subsample': 0.5567309914283638, 'reg__colsample_bytree': 0.6532409873959962, 'reg__min_child_samples': 42, 'reg__reg_alpha': 1.3916709393992868, 'reg__reg_lambda': 40.92567086422165, 'reg__min_split_gain': 1.2682885855266295, 'reg__path_smooth': 3.0892311724247854}

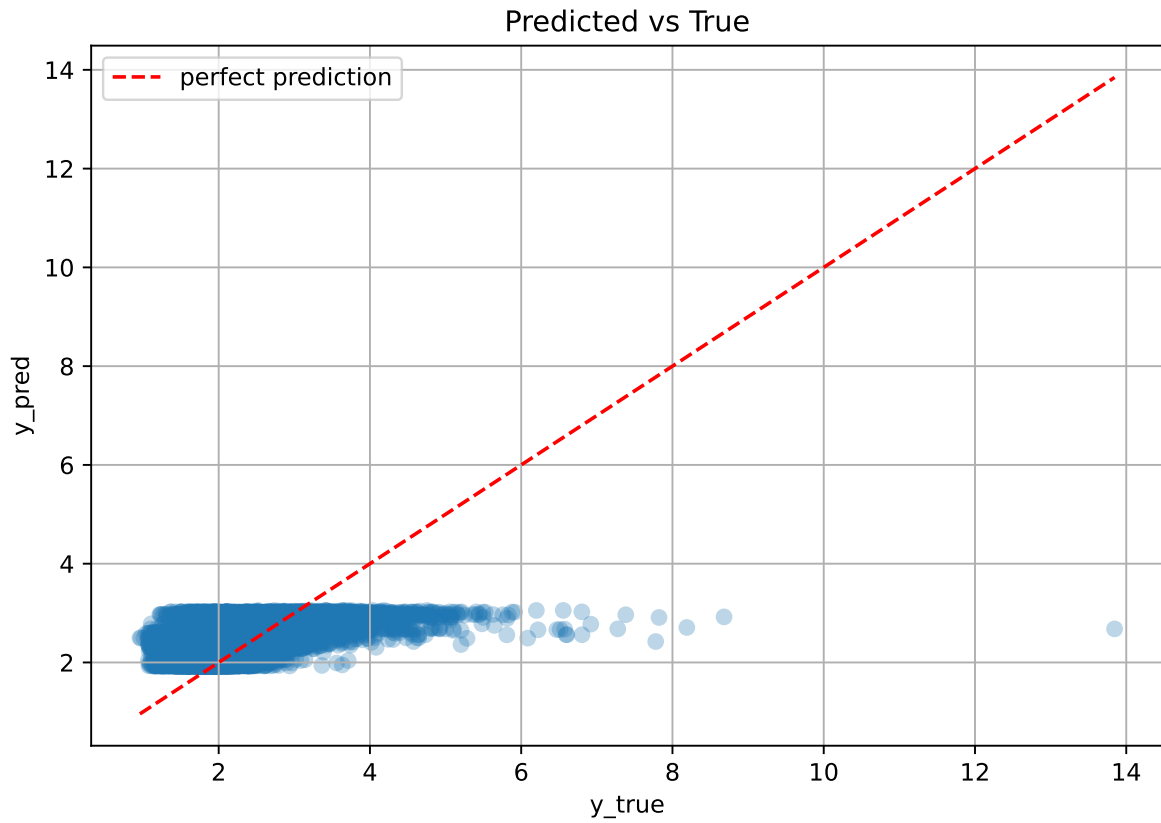
Rank 8: {'reg__n_estimators': 300, 'reg__max_depth': 3, 'reg__learning_rate': 0.058595686972028486, 'reg__num_leaves': 63, 'reg__subsample': 0.5796966448198317, 'reg__colsample_bytree': 0.6430013789732342, 'reg__min_child_samples': 42, 'reg__reg_alpha': 1.2901925846499505, 'reg__reg_lambda': 41.95059112390522, 'reg__min_split_gain': 1.3227646896972765, 'reg__path_smooth': 2.216644100275203}

Rank 9: {'reg__n_estimators': 200, 'reg__max_depth': 4, 'reg__learning_rate': 0.07299662796652993, 'reg__num_leaves': 44, 'reg__subsample': 0.5951763492393698,

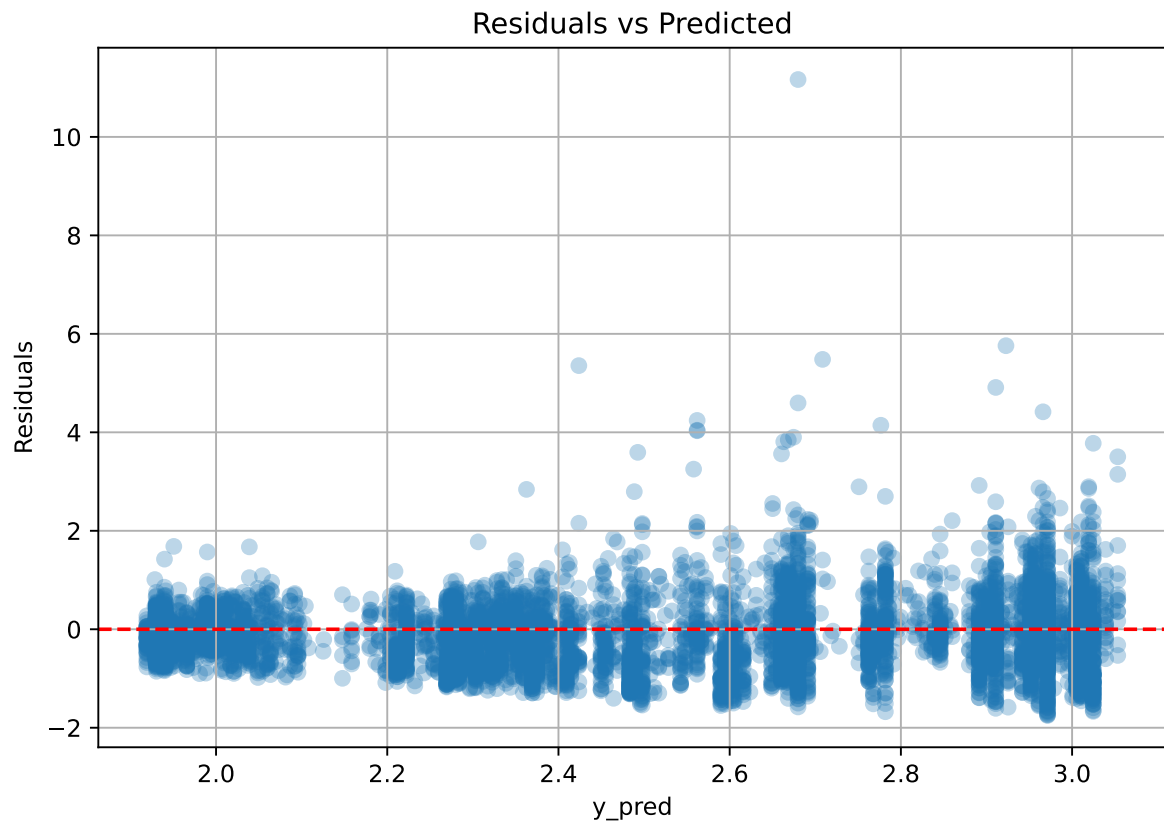
'reg__colsample_bytree': 0.5578994585356548, 'reg__min_child_samples': 36, 'reg__reg_alpha': 2.7741460128395214, 'reg__reg_lambda': 28.300530906460413, 'reg__min_split_gain': 0.6499907977530865, 'reg__path_smooth': 1.635896546808449}

Rank 10: {'reg__n_estimators': 200, 'reg__max_depth': 3, 'reg__learning_rate': 0.05289229605528271, 'reg__num_leaves': 57, 'reg__subsample': 0.5496186946030459, 'reg__colsample_bytree': 0.6029239441489181, 'reg__min_child_samples': 38, 'reg__reg_alpha': 0.8147312423473332, 'reg__reg_lambda': 24.737440124500207, 'reg__min_split_gain': 0.9518675111736494, 'reg__path_smooth': 0.011441260148454122}

2.4.2 Scatter



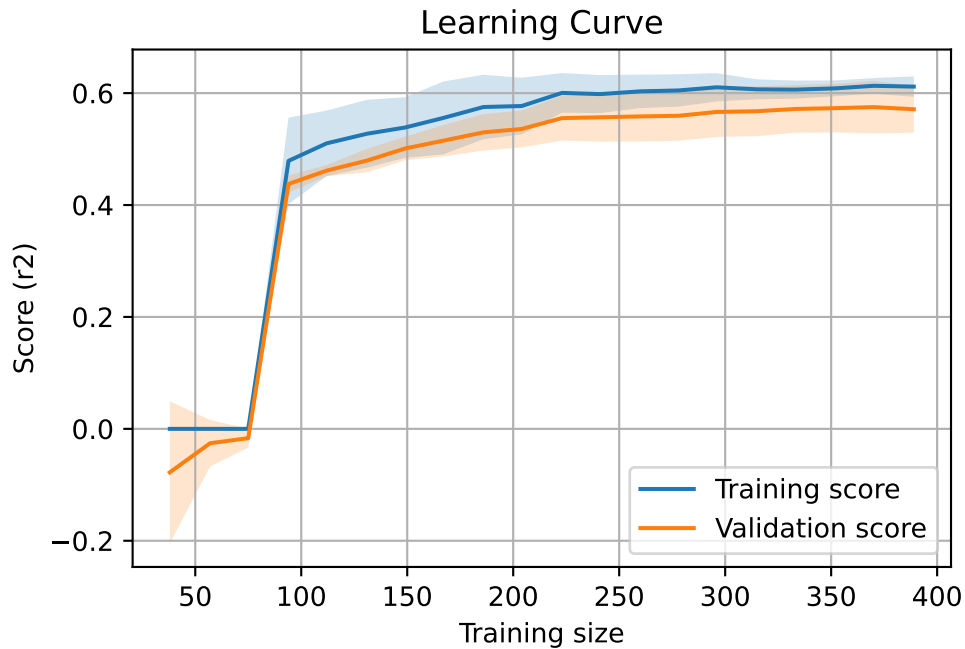
2.4.3 Scatter residuals



2.4.4 Learning curve — Group: 1

`Len(group)` : 795, `Len(training)` : 557,

`Len(training_real)` : 390, `Len(test_of_training)` : 167, `Len(test)` : 238

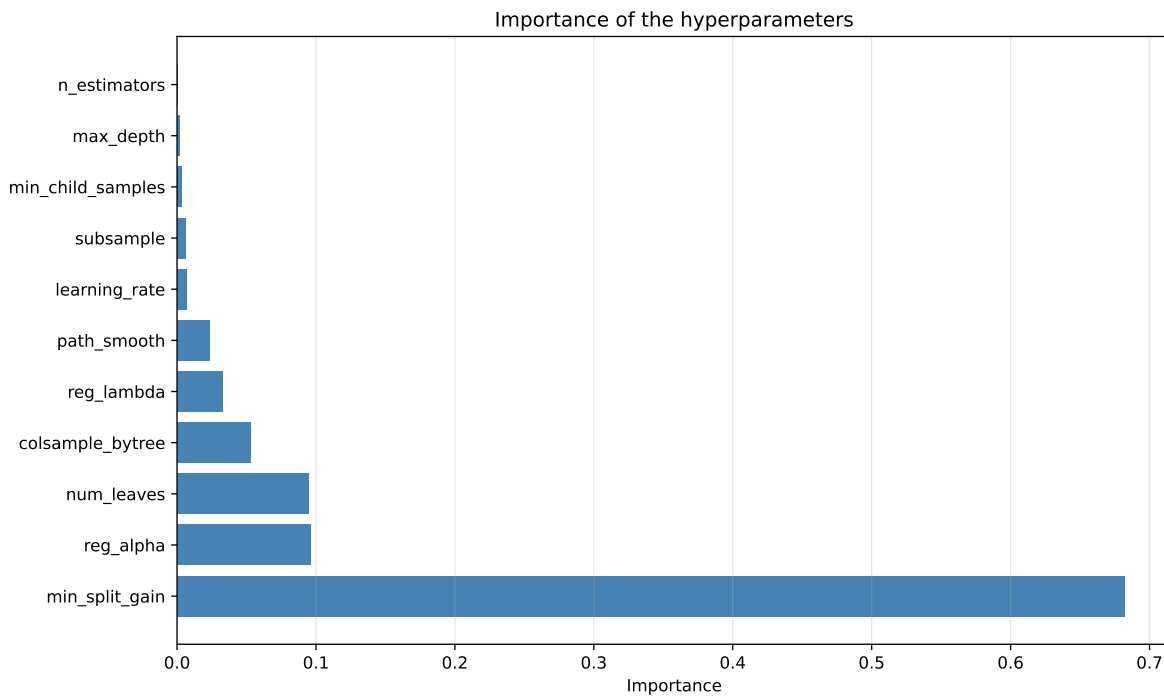
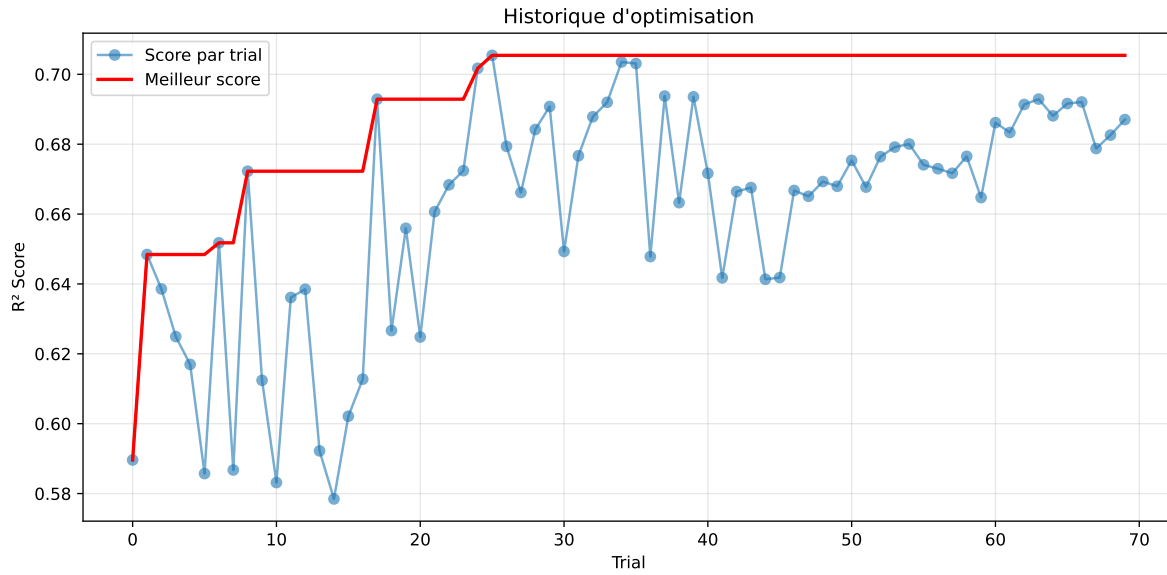


2.4.5 Gridsearch — Group: 2

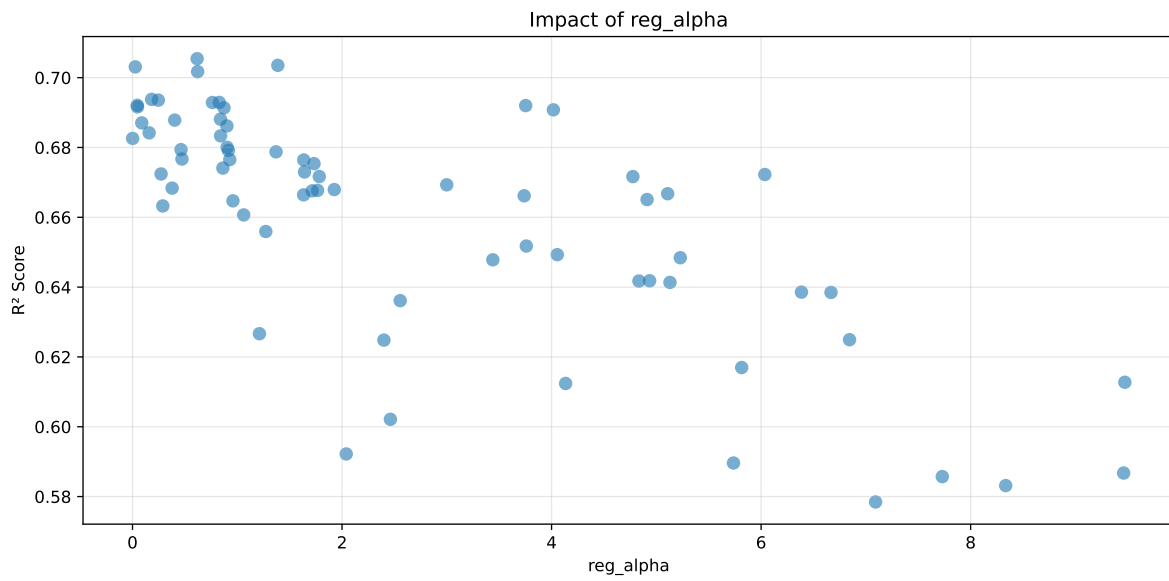
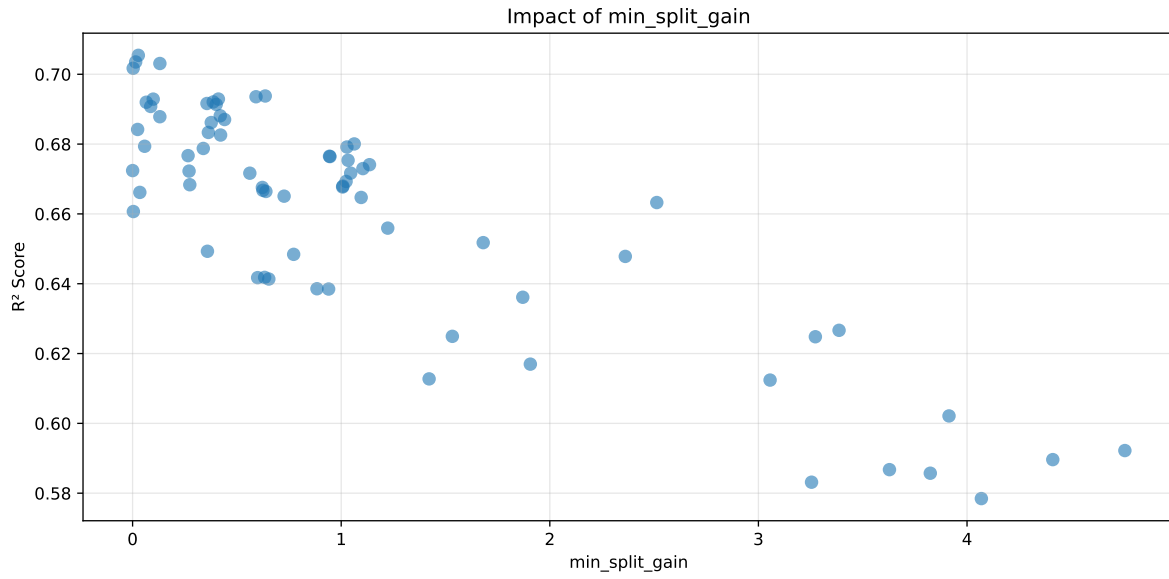
Len(group) : 2726, Len(training) : 1909, Len(test) : 817

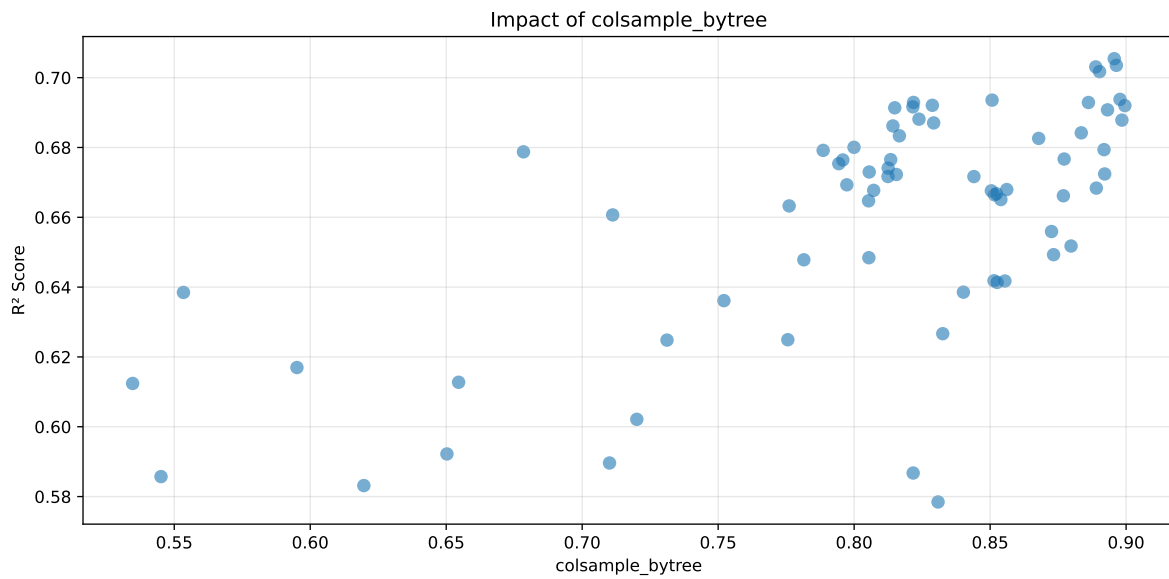
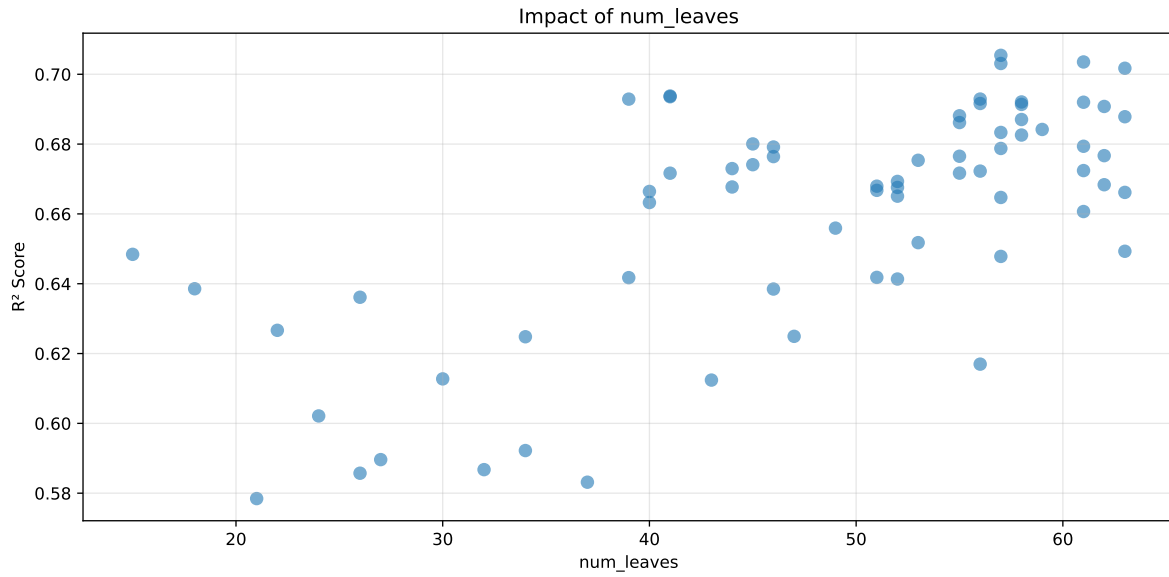
Distribution y_train vs y_test: y_train: mean=2.540, std=0.859, min=1.150, max=8.682
y_test: mean=2.561, std=0.947, min=1.120, max=13.845

===== TOP Importance FEATURES =====
feature importance r_NH3 131 r_umidity 115 r_CO2 85 r_temperature 36 r_THI 33



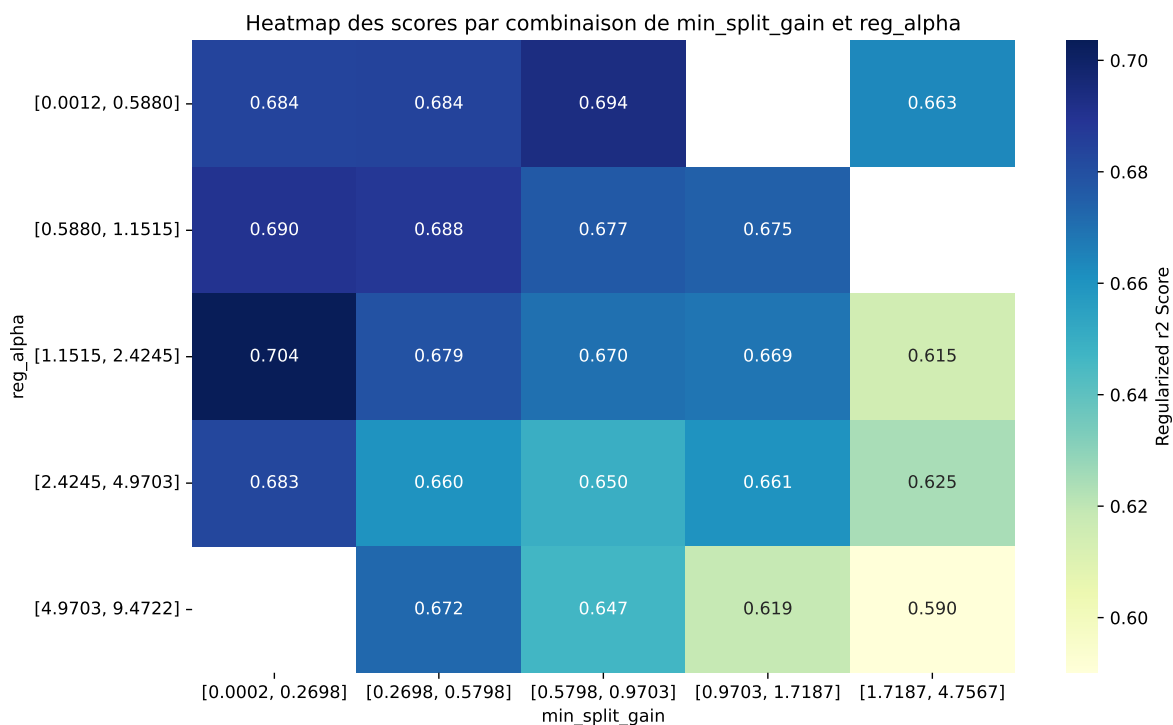
Paramètres avec >5% d'importance : ['min_split_gain', 'reg_alpha', 'num_leaves', 'colsample_bytree']





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The default value of observed=False is deprecated and will change to observed=True in a future



rank	mean_train_cv	mean_test_cv	test_score_cv	std_test_cv	penalized_score	test	diff	r_test	r_train
1	0.6762	0.6484	0.5493	0.0134	0.6484	0.5627	-0.0858	0.8677	1.0428
2	0.6765	0.6478	0.5501	0.0136	0.6478	0.5697	-0.0782	0.8794	1.0443
3	0.6651	0.6386	0.5449	0.0118	0.6386	0.5597	-0.0789	0.8765	1.0416
4	0.6667	0.6385	0.5452	0.0138	0.6385	0.5633	-0.0752	0.8822	1.0442
5	0.661	0.6361	0.5432	0.0133	0.6361	0.5606	-0.0755	0.8813	1.0391
6	0.6454	0.6267	0.5305	0.0101	0.6267	0.5481	-0.0786	0.8746	1.0299
7	0.6438	0.6249	0.532	0.0127	0.6249	0.5484	-0.0765	0.8775	1.0301
8	0.6433	0.6248	0.5277	0.015	0.6248	0.5484	-0.0764	0.8777	1.0295
9	0.6357	0.617	0.5277	0.0126	0.617	0.5459	-0.0711	0.8848	1.0304
10	0.6305	0.6127	0.524	0.0109	0.6127	0.5462	-0.0665	0.8915	1.029
---	---	---	---	---	---	---	---	---	---
70	0.8619	0.7054	0.5889	0.0113	0.236	0.6095	-0.0959	0.864	1.2218

Hyperparameters:

Rank 1: {'reg__n_estimators': 800, 'reg__max_depth': 4, 'reg__learning_rate': 0.08158817590158551, 'reg__num_leaves': 15, 'reg__subsample': 0.8087298338899738, 'reg__colsample_bytree': 0.8054140308514781, 'reg__min_child_samples': 48, 'reg__reg_alpha': 5.228686777044728, 'reg__reg_lambda': 10.825923482356709, 'reg__min_split_gain': 0.7720766589540806, 'reg__path_smooth': 4.612287430308158}

Rank 2: {'reg__n_estimators': 500, 'reg__max_depth': 7, 'reg__learning_rate': 0.014986441890469422, 'reg__num_leaves': 57, 'reg__subsample': 0.6686716279736793, 'reg__colsample_bytree': 0.7815011632024715, 'reg__min_child_samples': 17, 'reg__reg_alpha': 3.4392362191536145, 'reg__reg_lambda': 1.2635554209923006, 'reg__min_split_gain': 2.3618946814543573, 'reg__path_smooth': 3.418175225362707}

Rank 3: {'reg__n_estimators': 300, 'reg__max_depth': 6, 'reg__learning_rate': 0.01987261242042138, 'reg__num_leaves': 18, 'reg__subsample': 0.5056972315491545, 'reg__colsample_bytree': 0.8401747159103179, 'reg__min_child_samples': 37, 'reg__reg_alpha': 6.385237567415763, 'reg__reg_lambda': 30.074742134437727, 'reg__min_split_gain': 0.8839063237635597, 'reg__path_smooth': 4.942450668834553}

Rank 4: {'reg__n_estimators': 700, 'reg__max_depth': 7, 'reg__learning_rate': 0.08365961546250332, 'reg__num_leaves': 46, 'reg__subsample': 0.6716582657888797, 'reg__colsample_bytree': 0.5533814531066673, 'reg__min_child_samples': 10, 'reg__reg_alpha': 6.667192596662747, 'reg__reg_lambda': 9.004990692441222, 'reg__min_split_gain': 0.9397324072898033, 'reg__path_smooth': 2.8147427131675853}

Rank 5: {'reg__n_estimators': 800, 'reg__max_depth': 5, 'reg__learning_rate': 0.09560743504331584, 'reg__num_leaves': 26, 'reg__subsample': 0.6166376947177616, 'reg__colsample_bytree': 0.7521120651920146, 'reg__min_child_samples': 24, 'reg__reg_alpha': 2.5552084834229682, 'reg__reg_lambda': 34.09097668954873, 'reg__min_split_gain': 1.8703555734615307, 'reg__path_smooth': 4.841548412504906}

Rank 6: {'reg__n_estimators': 600, 'reg__max_depth': 4, 'reg__learning_rate': 0.058069922634249135, 'reg__num_leaves': 22, 'reg__subsample': 0.5409985394876538, 'reg__colsample_bytree': 0.8325814384009926, 'reg__min_child_samples': 37, 'reg__reg_alpha': 1.2116138246949903, 'reg__reg_lambda': 14.78648531703653, 'reg__min_split_gain': 3.3865587570337574, 'reg__path_smooth': 1.4951411576225504}

Rank 7: {'reg__n_estimators': 200, 'reg__max_depth': 6, 'reg__learning_rate': 0.024670724806889012, 'reg__num_leaves': 47, 'reg__subsample': 0.8528613368265534, 'reg__colsample_bytree': 0.775608652059298, 'reg__min_child_samples': 45, 'reg__reg_alpha': 6.844237635818999, 'reg__reg_lambda': 27.408230809844287, 'reg__min_split_gain': 1.532792196372494, 'reg__path_smooth': 3.1309326562017272}

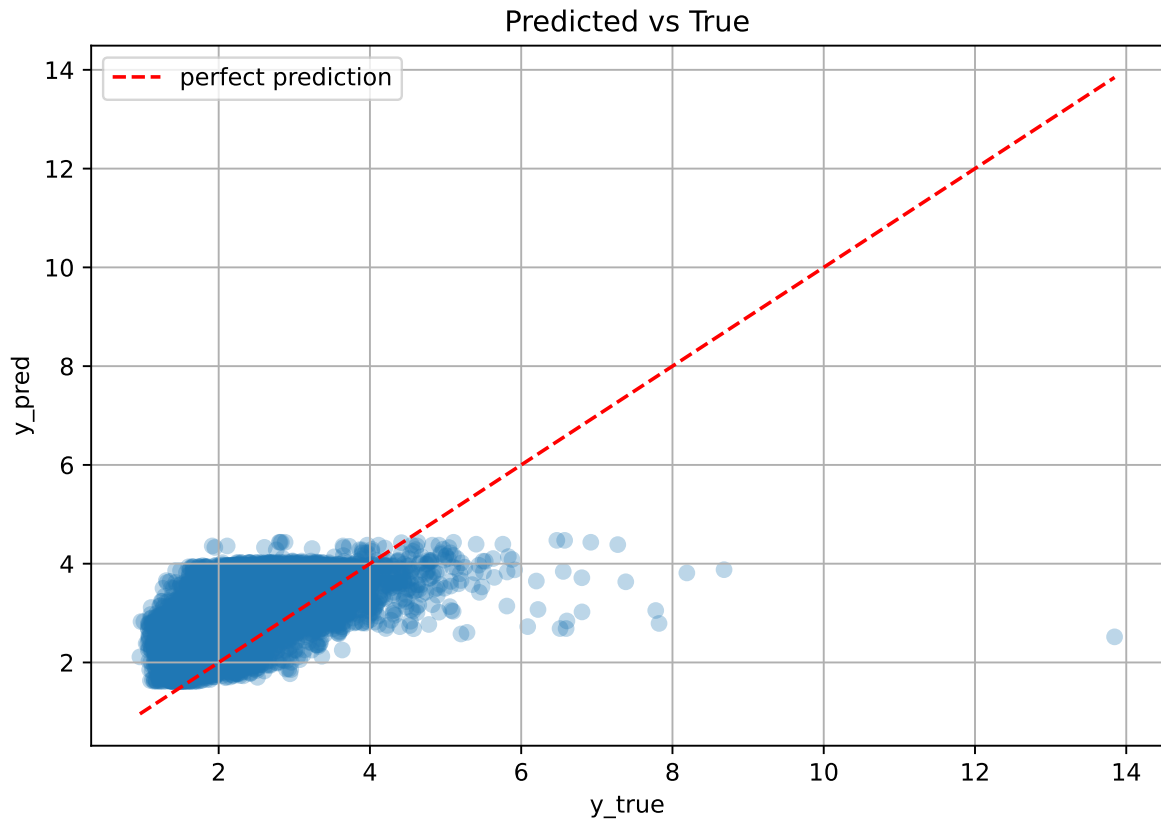
Rank 8: {'reg__n_estimators': 500, 'reg__max_depth': 5, 'reg__learning_rate': 0.026094886489246794, 'reg__num_leaves': 34, 'reg__subsample': 0.5777384107497926, 'reg__colsample_bytree': 0.7311828411915828, 'reg__min_child_samples': 46, 'reg__reg_alpha': 2.399761094485208, 'reg__reg_lambda': 6.750542154560208, 'reg__min_split_gain': 3.2729897901198295, 'reg__path_smooth': 2.5155538674760662}

Rank 9: {'reg__n_estimators': 700, 'reg__max_depth': 4, 'reg__learning_rate': 0.05161595750337681, 'reg__num_leaves': 56, 'reg__subsample': 0.7939058707080835,

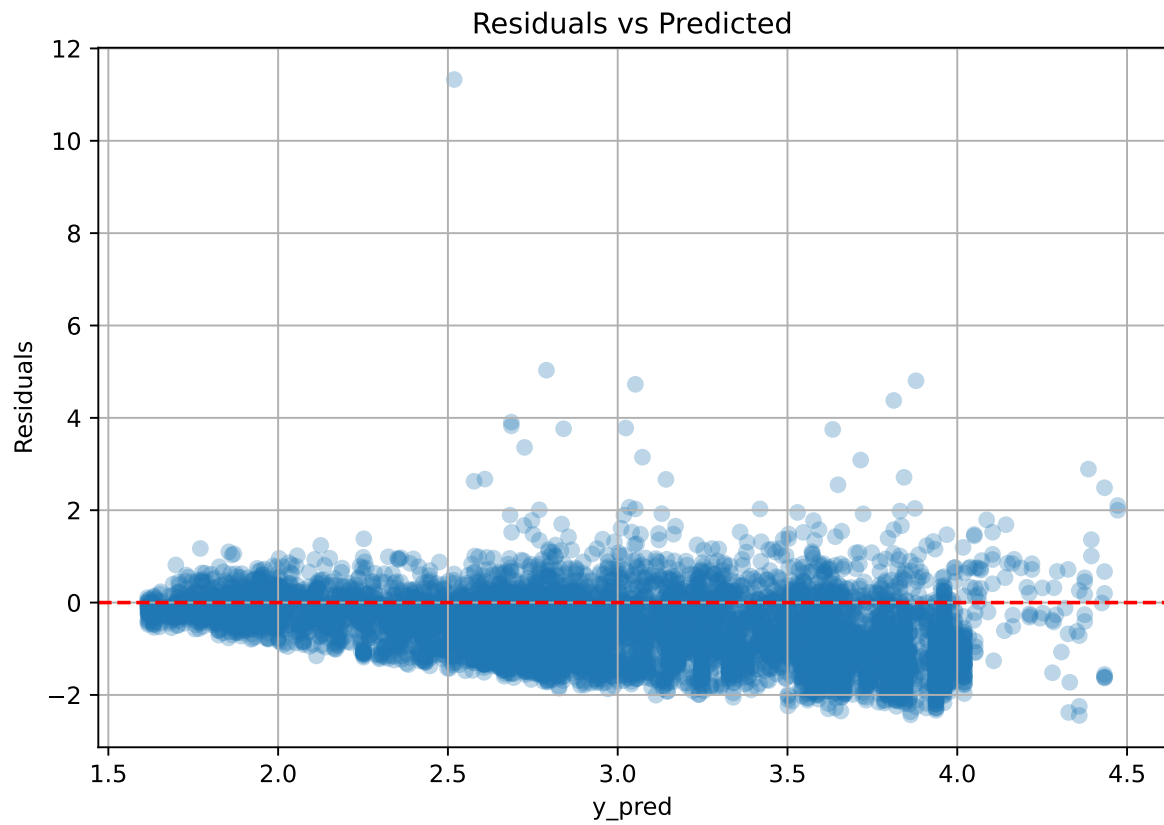
'reg__colsample_bytree': 0.5951120100690239, 'reg__min_child_samples': 38, 'reg__reg_alpha': 5.814640166791946, 'reg__reg_lambda': 18.976531101930718, 'reg__min_split_gain': 1.9066567772346787, 'reg__path_smooth': 2.580973363550557}

Rank 10: {'reg__n_estimators': 300, 'reg__max_depth': 5, 'reg__learning_rate': 0.021163926878151496, 'reg__num_leaves': 30, 'reg__subsample': 0.569283091682576, 'reg__colsample_bytree': 0.6545621318948052, 'reg__min_child_samples': 12, 'reg__reg_alpha': 9.472208136877377, 'reg__reg_lambda': 19.675248129870976, 'reg__min_split_gain': 1.42138739055879, 'reg__path_smooth': 1.9859193897004723}

2.4.6 Scatter



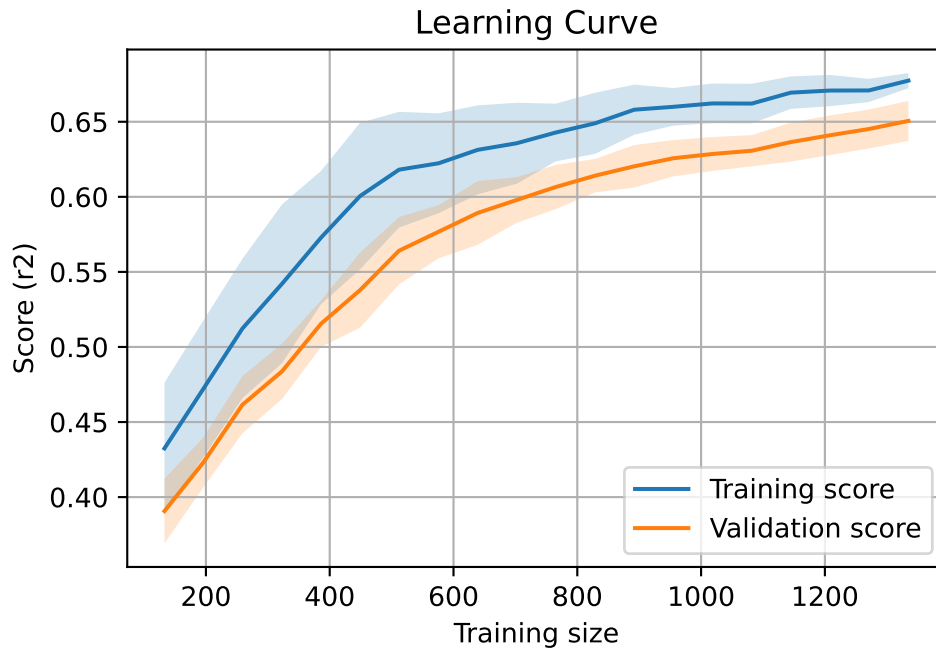
2.4.7 Scatter residuals



2.4.8 Learning curve — Group: 2

`Len(group)` : 2726, `Len(training)` : 1909,

`Len(training_real)` : 1337, `Len(test_of_training)` : 572, `Len(test)` : 817

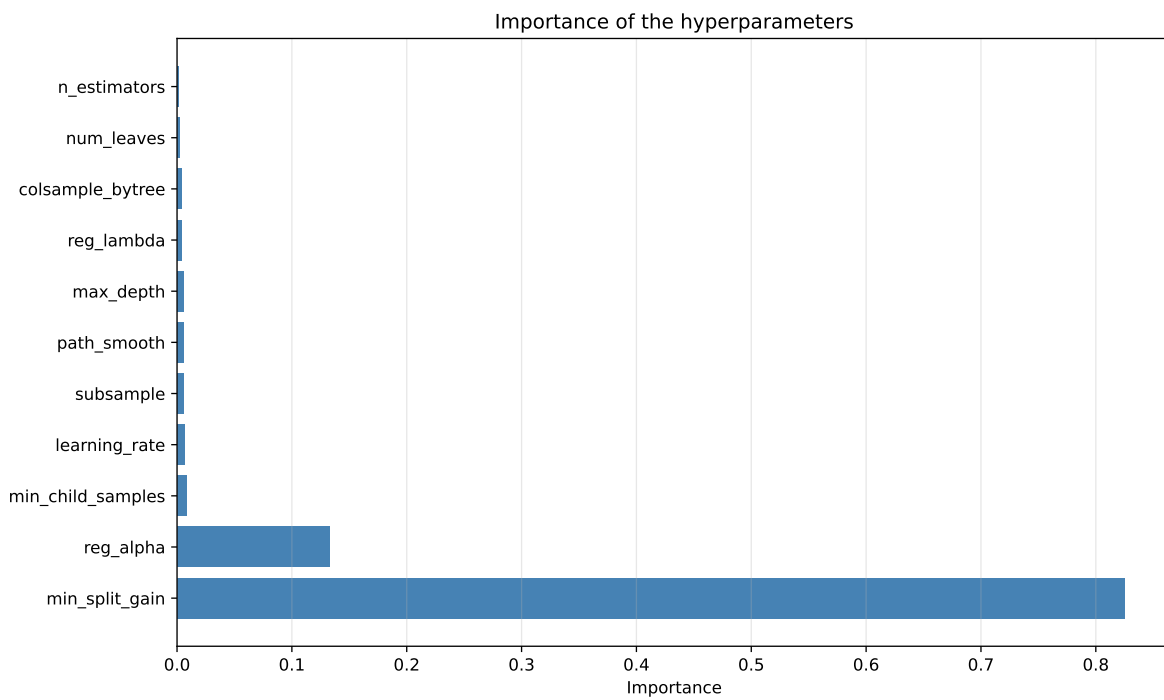
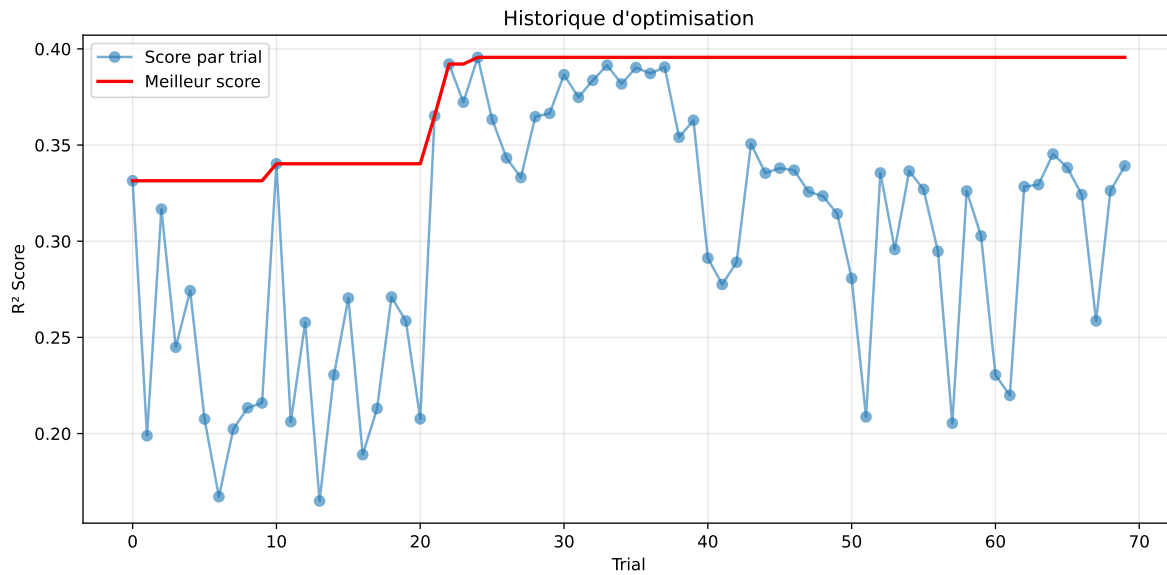


2.4.9 Gridsearch — Group: 3

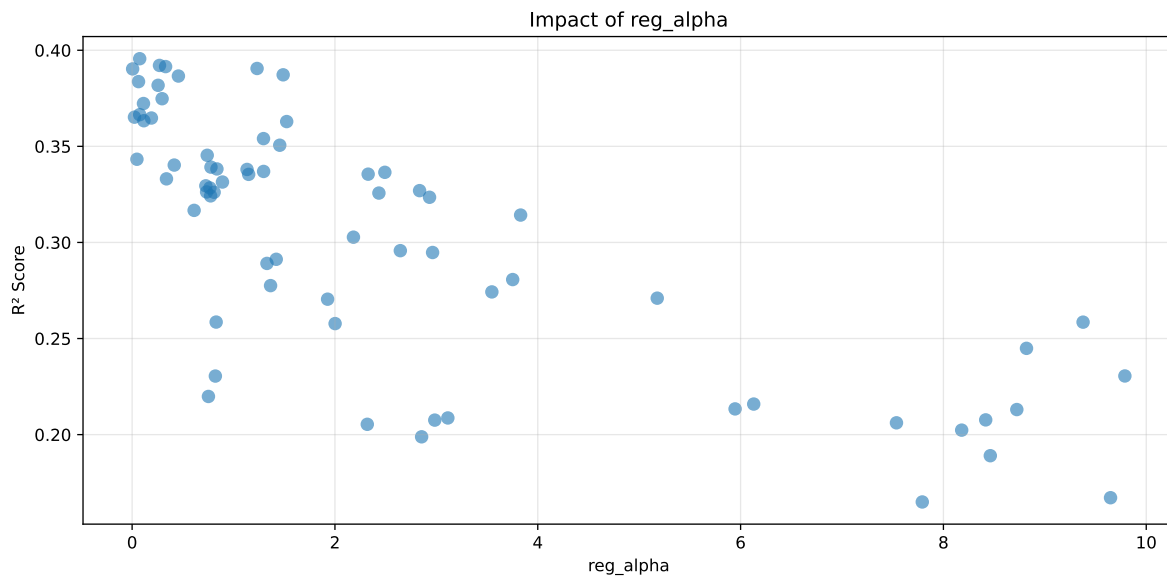
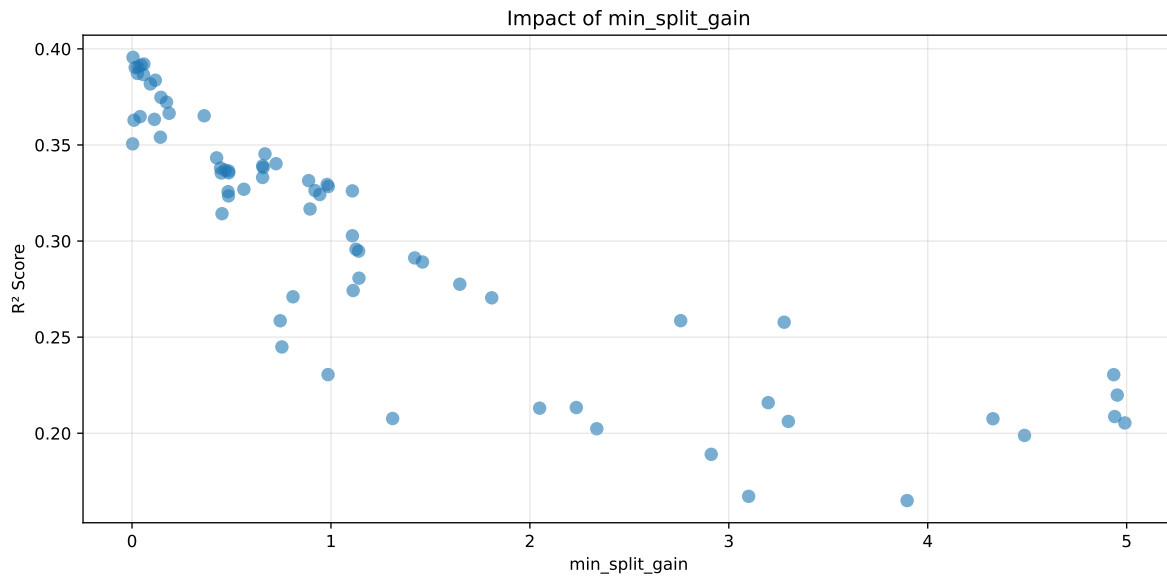
Len(group) : 568, Len(training) : 398, Len(test) : 170

Distribution y_train vs y_test: y_train: mean=1.928, std=0.816, min=0.959, max=6.804
y_test: mean=1.833, std=0.729, min=1.098, max=7.822

===== TOP Importance FEATURES =====
feature importance r_umidity 69 r_THI 26 r_NH3 25 r_CO2 13 r_temperature 7

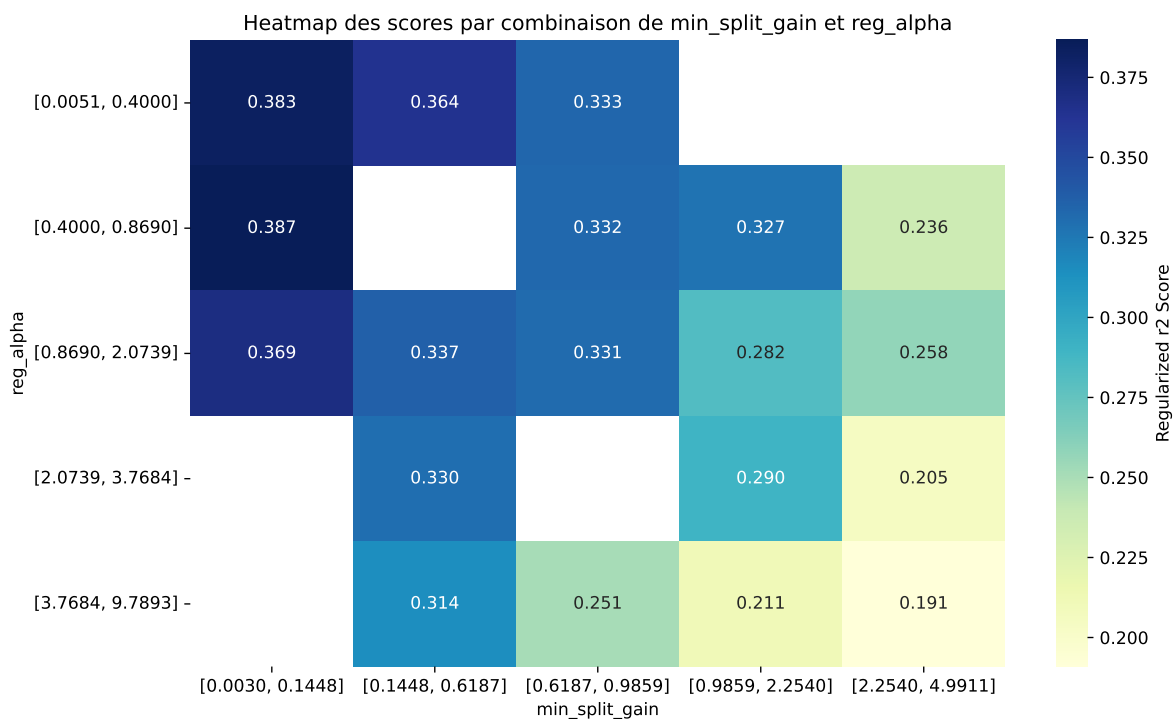


Paramètres avec >5% d'importance : ['min_split_gain', 'reg_alpha']



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The default value of observed=False is deprecated and will change to observed=True in a future



rank	mean_train_cv	mean_test_cv	test_score_cv	std_test_cv	penalized_score	test	diff	r_test	r_train
1	0.2648	0.2305	0.0945	0.0385	0.1278	0.1331	-0.0974	0.5774	1.1485
2	0.2319	0.2023	0.0856	0.0366	0.1136	0.1209	-0.0814	0.5977	1.1462
3	0.2532	0.2159	0.0918	0.0343	0.1039	0.1294	-0.0865	0.5993	1.1729
4	0.2308	0.1989	0.084	0.0373	0.1032	0.1213	-0.0776	0.61	1.1604
5	0.2187	0.189	0.0799	0.0368	0.1001	0.1171	-0.072	0.6193	1.1568
6	0.2416	0.2054	0.0913	0.0342	0.0967	0.1287	-0.0767	0.6265	1.1763
7	0.2457	0.2075	0.0894	0.0364	0.0932	0.1247	-0.0829	0.6007	1.1837
8	0.1912	0.165	0.0655	0.0344	0.0862	0.0957	-0.0692	0.5804	1.1592
9	0.2646	0.2199	0.1008	0.0314	0.0855	0.1354	-0.0845	0.6159	1.2037
10	0.3166	0.2585	0.1075	0.0311	0.0842	0.1447	-0.1138	0.5598	1.2248
---	---	---	---	---	---	---	---	---	---
70	0.7218	0.3506	0.1995	0.0969	-0.7632	0.2681	-0.0824	0.7649	2.059

Hyperparameters:

Rank 1: {'reg__n_estimators': 500, 'reg__max_depth': 6, 'reg__learning_rate': 0.037018232343808515, 'reg__num_leaves': 32, 'reg__subsample': 0.8386913365557007, 'reg__colsample_bytree': 0.8266344257345388, 'reg__min_child_samples': 46, 'reg__reg_alpha': 9.789286935136115, 'reg__reg_lambda': 49.89946878442561, 'reg__min_split_gain': 0.9852504891343355, 'reg__path_smooth': 3.1053701269576846}

Rank 2: {'reg__n_estimators': 400, 'reg__max_depth': 6, 'reg__learning_rate': 0.014177513401691344, 'reg__num_leaves': 35, 'reg__subsample': 0.6167484598097068, 'reg__colsample_bytree': 0.7585376400385104, 'reg__min_child_samples': 36, 'reg__reg_alpha': 8.180039623943534, 'reg__reg_lambda': 44.88305689003038, 'reg__min_split_gain': 2.336309945922772, 'reg__path_smooth': 3.224319655299128}

Rank 3: {'reg__n_estimators': 200, 'reg__max_depth': 8, 'reg__learning_rate': 0.036883371741996955, 'reg__num_leaves': 25, 'reg__subsample': 0.6094087741425129, 'reg__colsample_bytree': 0.829383511687974, 'reg__min_child_samples': 44, 'reg__reg_alpha': 6.12939786856548, 'reg__reg_lambda': 12.685873990595958, 'reg__min_split_gain': 3.198158293371332, 'reg__path_smooth': 1.9581566305687987}

Rank 4: {'reg__n_estimators': 300, 'reg__max_depth': 8, 'reg__learning_rate': 0.06460328460286349, 'reg__num_leaves': 34, 'reg__subsample': 0.7544329785872839, 'reg__colsample_bytree': 0.8173778094517747, 'reg__min_child_samples': 37, 'reg__reg_alpha': 2.854661040104225, 'reg__reg_lambda': 43.577494123348146, 'reg__min_split_gain': 4.486520349261418, 'reg__path_smooth': 2.1799852691003947}

Rank 5: {'reg__n_estimators': 500, 'reg__max_depth': 3, 'reg__learning_rate': 0.010139978399584, 'reg__num_leaves': 22, 'reg__subsample': 0.5571706482474004, 'reg__colsample_bytree': 0.7262521985432205, 'reg__min_child_samples': 28, 'reg__reg_alpha': 8.462409329455792, 'reg__reg_lambda': 34.98520848218031, 'reg__min_split_gain': 2.911612363072009, 'reg__path_smooth': 3.447205666913823}

Rank 6: {'reg__n_estimators': 800, 'reg__max_depth': 4, 'reg__learning_rate': 0.014011166937924762, 'reg__num_leaves': 46, 'reg__subsample': 0.8441926655673195, 'reg__colsample_bytree': 0.8517956120548169, 'reg__min_child_samples': 13, 'reg__reg_alpha': 2.318940259712998, 'reg__reg_lambda': 27.747829757939027, 'reg__min_split_gain': 4.991058891562551, 'reg__path_smooth': 4.440785544752881}

Rank 7: {'reg__n_estimators': 200, 'reg__max_depth': 4, 'reg__learning_rate': 0.04167057293584181, 'reg__num_leaves': 16, 'reg__subsample': 0.6099677757266987, 'reg__colsample_bytree': 0.7413880229482669, 'reg__min_child_samples': 15, 'reg__reg_alpha': 2.9843444518894935, 'reg__reg_lambda': 30.76708366606255, 'reg__min_split_gain': 4.327813799052393, 'reg__path_smooth': 2.493256625877289}

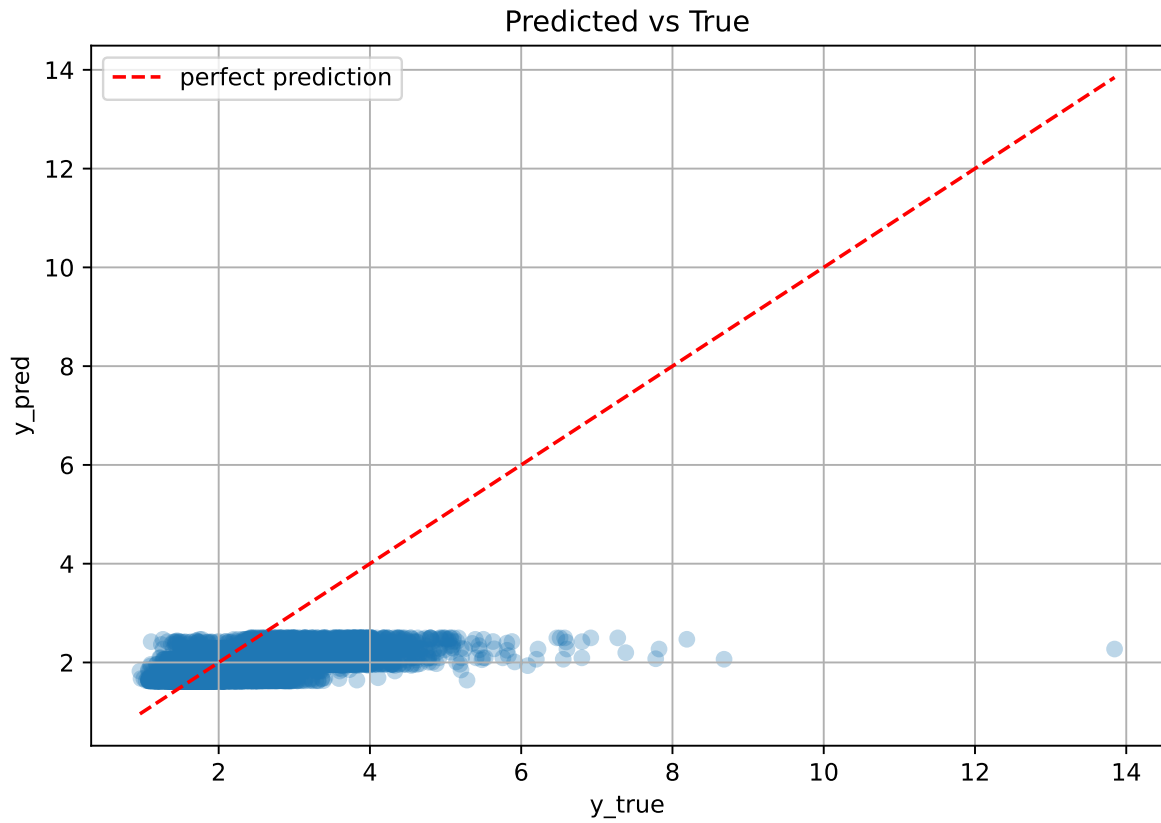
Rank 8: {'reg__n_estimators': 700, 'reg__max_depth': 4, 'reg__learning_rate': 0.051048302945539734, 'reg__num_leaves': 44, 'reg__subsample': 0.5068989486220639, 'reg__colsample_bytree': 0.8072145758297662, 'reg__min_child_samples': 16, 'reg__reg_alpha': 7.791566342080555, 'reg__reg_lambda': 43.08093632642597, 'reg__min_split_gain': 3.8964157601960796, 'reg__path_smooth': 3.1300422971797297}

Rank 9: {'reg__n_estimators': 800, 'reg__max_depth': 6, 'reg__learning_rate': 0.014400154273612367, 'reg__num_leaves': 60, 'reg__subsample': 0.6955242904772269,

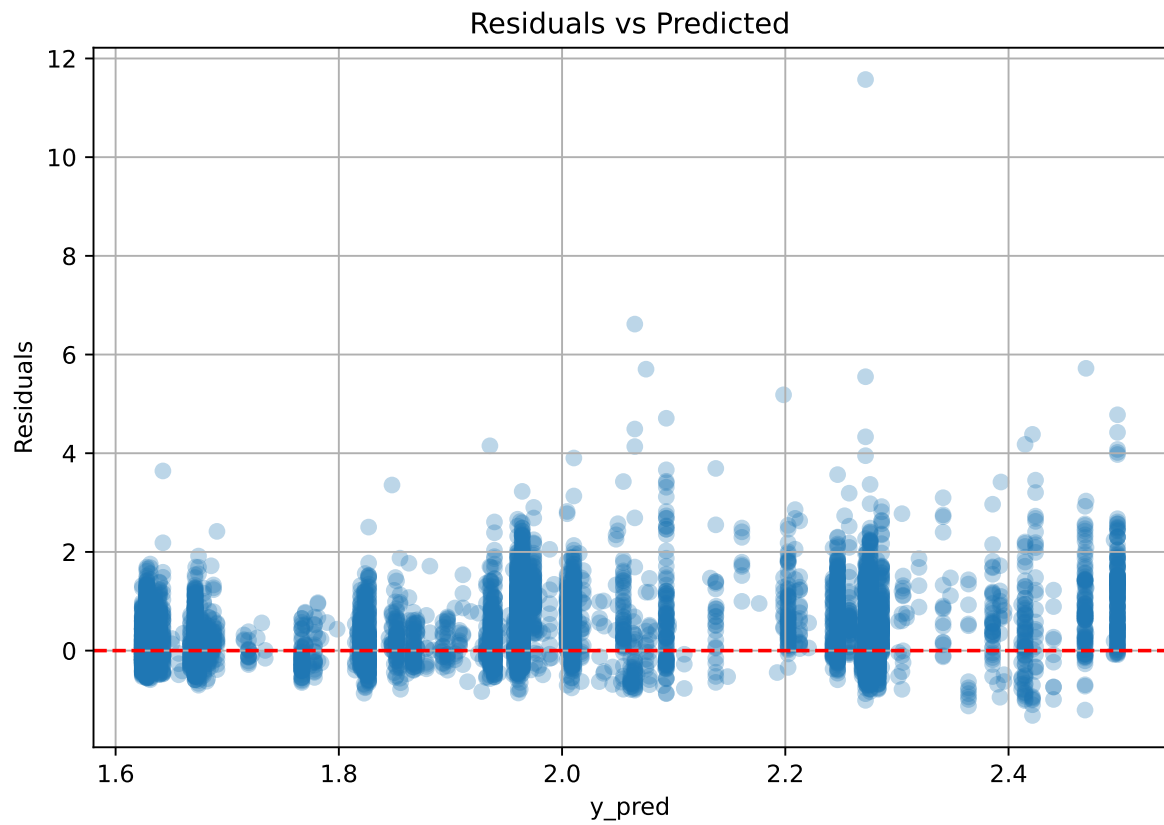
'reg__colsample_bytree': 0.8659158871037155, 'reg__min_child_samples': 16, 'reg__reg_alpha': 0.7525084051767913, 'reg__reg_lambda': 28.225554847133793, 'reg__min_split_gain': 4.952750319299865, 'reg__path_smooth': 4.791097716862886}

Rank 10: {'reg__n_estimators': 200, 'reg__max_depth': 5, 'reg__learning_rate': 0.05257957033575301, 'reg__num_leaves': 33, 'reg__subsample': 0.6364016718541099, 'reg__colsample_bytree': 0.8100941031998953, 'reg__min_child_samples': 46, 'reg__reg_alpha': 9.377093449419089, 'reg__reg_lambda': 7.563807168087122, 'reg__min_split_gain': 0.7448773314176638, 'reg__path_smooth': 0.3401026644650573}

2.4.10 Scatter



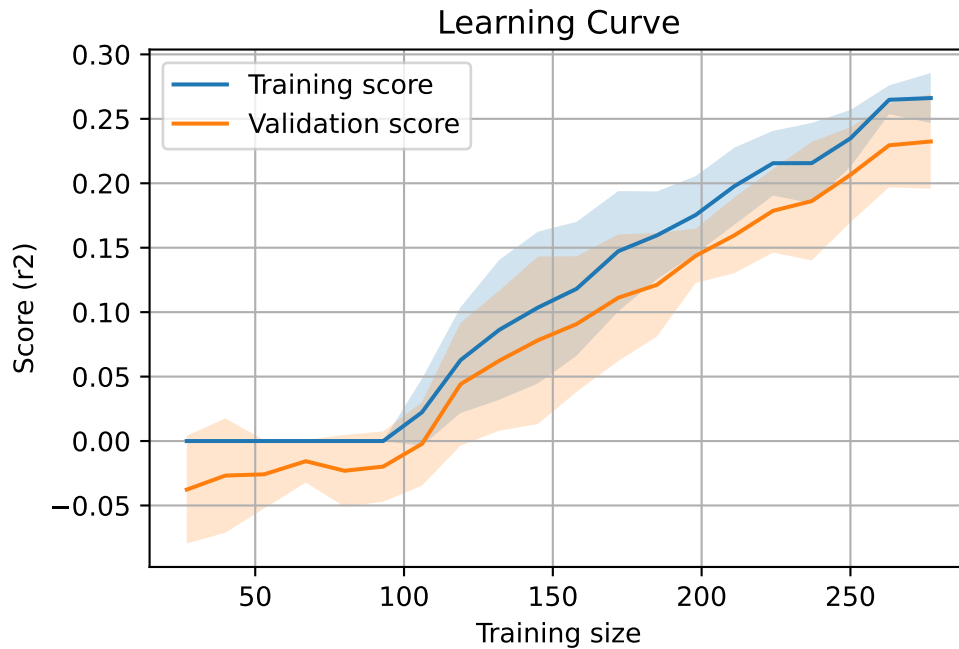
2.4.11 Scatter residuals



2.4.12 Learning curve — Group: 3

`Len(group)` : 568, `Len(training)` : 398,

`Len(training_real)` : 279, `Len(test_of_training)` : 119, `Len(test)` : 170

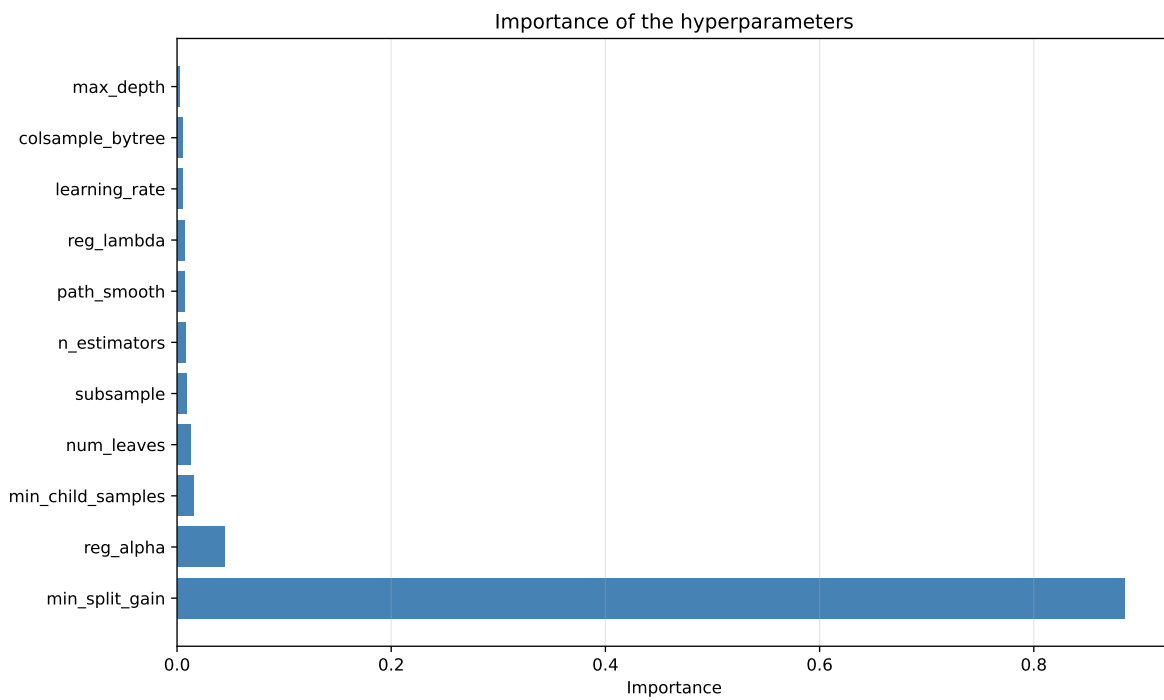
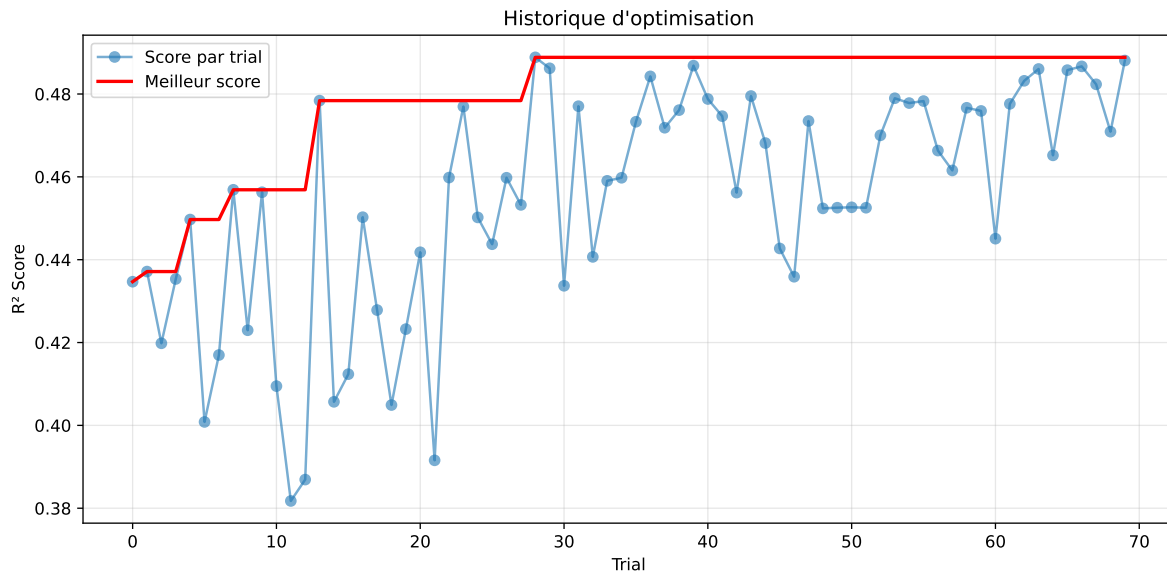


2.4.13 Gridsearch — Group: 4

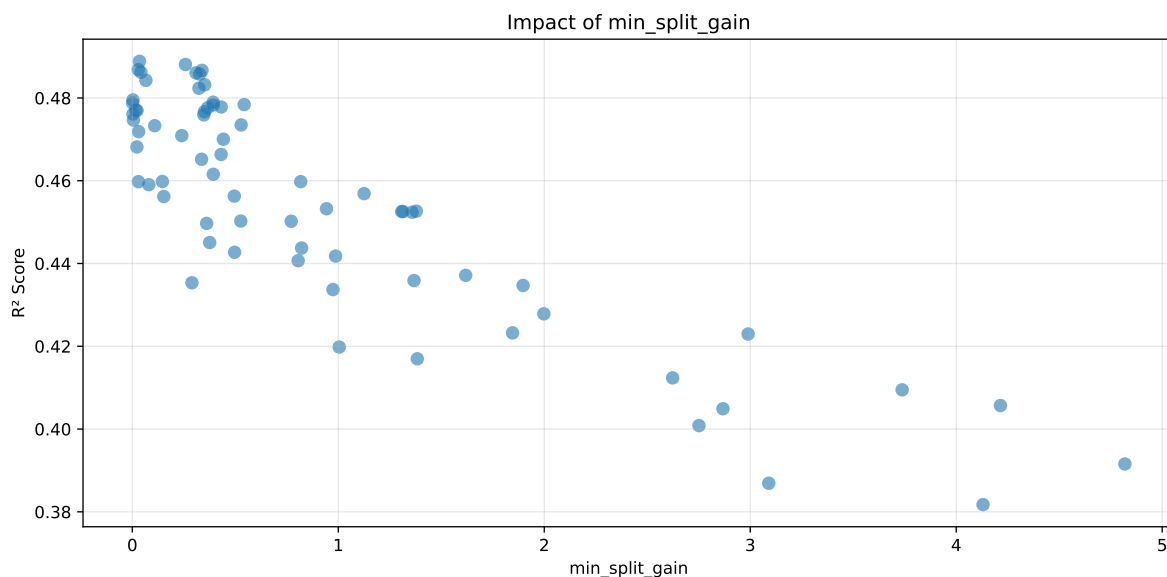
Len(group) : 2726, Len(training) : 1909, Len(test) : 817

Distribution y_train vs y_test: y_train: mean=2.238, std=0.681, min=0.973, max=5.813
y_test: mean=2.212, std=0.659, min=1.071, max=4.858

===== TOP Importance FEATURES =====
feature importance r_umidity 118 r_NH3 48 r_CO2 45 r_temperature 27 r_THI 17



Paramètres avec >5% d'importance : ['min_split_gain']



Pas assez de paramètres avec >5% d'importance pour une heatmap 2D.

rank	mean_train_cv	mean_test_cv	test_score_cv	std_test_cv	penalized_score	test	diff	r_test	r_train
1	0.4621	0.4198	0.4785	0.0256	0.2931	0.498	0.0781	1.1861	1.1006
2	0.4369	0.4008	0.4591	0.0249	0.2926	0.4824	0.0815	1.2034	1.09
3	0.4119	0.3818	0.4379	0.0229	0.2914	0.4594	0.0777	1.2035	1.0789
4	0.4599	0.417	0.4756	0.0248	0.2882	0.498	0.081	1.1943	1.1029
5	0.4439	0.4049	0.4648	0.0259	0.2881	0.4849	0.08	1.1977	1.0962
6	0.4203	0.3869	0.4484	0.0235	0.2868	0.4696	0.0827	1.2138	1.0863
7	0.4558	0.4124	0.4757	0.0238	0.2822	0.4946	0.0823	1.1995	1.1052
8	0.4528	0.4095	0.4749	0.0247	0.2796	0.4919	0.0824	1.2013	1.1058
9	0.4294	0.3916	0.4569	0.0231	0.2782	0.4711	0.0796	1.2032	1.0965
10	0.4489	0.4057	0.4702	0.0252	0.2762	0.4848	0.0791	1.195	1.1064
---	---	---	---	---	---	---	---	---	---
70	0.6362	0.4862	0.5601	0.0287	0.0364	0.5804	0.0942	1.1938	1.3084

Hyperparameters:

Rank 1: {'reg__n_estimators': 600, 'reg__max_depth': 3, 'reg__learning_rate': 0.0511930697402105, 'reg__num_leaves': 33, 'reg__subsample': 0.859090463418694, 'reg__colsample_bytree': 0.8024763205459262, 'reg__min_child_samples': 18, 'reg__reg_alpha': 9.9600187773909, 'reg__reg_lambda': 48.13006916221096, 'reg__min_split_gain': 1.0044886322974622, 'reg__path_smooth': 4.635337667099243}

Rank 2: {'reg__n_estimators': 500, 'reg__max_depth': 4, 'reg__learning_rate': 0.01261531648410241, 'reg__num_leaves': 58, 'reg__subsample': 0.728449389798368,

'reg__colsample_bytree': 0.7172391964602778, 'reg__min_child_samples': 14, 'reg__reg_alpha': 9.175032965685867, 'reg__reg_lambda': 22.39834942340438, 'reg__min_split_gain': 2.7511318600254016, 'reg__path_smooth': 0.8937694065031082}

Rank 3: {'reg__n_estimators': 300, 'reg__max_depth': 8, 'reg__learning_rate': 0.027970681550497804, 'reg__num_leaves': 24, 'reg__subsample': 0.7260162012201833, 'reg__colsample_bytree': 0.8692328392556994, 'reg__min_child_samples': 47, 'reg__reg_alpha': 9.794236405410047, 'reg__reg_lambda': 34.55623331277966, 'reg__min_split_gain': 4.130132988376919, 'reg__path_smooth': 1.6814141886591987}

Rank 4: {'reg__n_estimators': 700, 'reg__max_depth': 7, 'reg__learning_rate': 0.030673311104309185, 'reg__num_leaves': 42, 'reg__subsample': 0.5789354749297123, 'reg__colsample_bytree': 0.8200397215686749, 'reg__min_child_samples': 47, 'reg__reg_alpha': 9.574968380971935, 'reg__reg_lambda': 31.793110981662785, 'reg__min_split_gain': 1.3836701968834224, 'reg__path_smooth': 3.376982791290879}

Rank 5: {'reg__n_estimators': 800, 'reg__max_depth': 3, 'reg__learning_rate': 0.01599232025454497, 'reg__num_leaves': 23, 'reg__subsample': 0.5467221669481785, 'reg__colsample_bytree': 0.8004157278485162, 'reg__min_child_samples': 11, 'reg__reg_alpha': 9.662261393168148, 'reg__reg_lambda': 2.1308438767521585, 'reg__min_split_gain': 2.8676220890984494, 'reg__path_smooth': 0.33373274351221727}

Rank 6: {'reg__n_estimators': 400, 'reg__max_depth': 6, 'reg__learning_rate': 0.01361527434184657, 'reg__num_leaves': 22, 'reg__subsample': 0.644909447916742, 'reg__colsample_bytree': 0.5229772136068935, 'reg__min_child_samples': 47, 'reg__reg_alpha': 9.029666366169408, 'reg__reg_lambda': 42.20695891496128, 'reg__min_split_gain': 3.0900111895309283, 'reg__path_smooth': 0.9560668322911359}

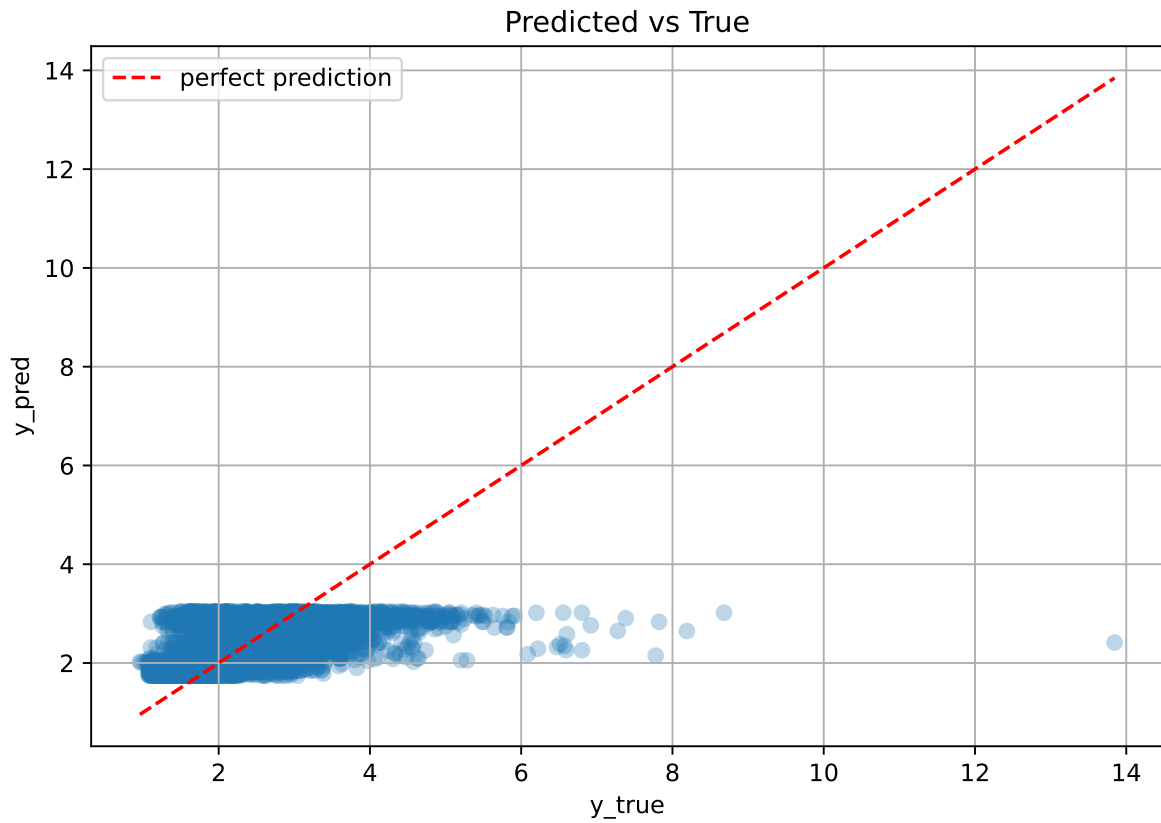
Rank 7: {'reg__n_estimators': 400, 'reg__max_depth': 8, 'reg__learning_rate': 0.03502621327468407, 'reg__num_leaves': 26, 'reg__subsample': 0.7458827175391596, 'reg__colsample_bytree': 0.5522475002203716, 'reg__min_child_samples': 17, 'reg__reg_alpha': 4.276428874985141, 'reg__reg_lambda': 31.63378190386114, 'reg__min_split_gain': 2.623197189453786, 'reg__path_smooth': 3.051333921624626}

Rank 8: {'reg__n_estimators': 800, 'reg__max_depth': 7, 'reg__learning_rate': 0.06067756664432863, 'reg__num_leaves': 24, 'reg__subsample': 0.6149834729615102, 'reg__colsample_bytree': 0.5943779921622502, 'reg__min_child_samples': 33, 'reg__reg_alpha': 0.5824369890409076, 'reg__reg_lambda': 36.31128196886962, 'reg__min_split_gain': 3.7374583451592738, 'reg__path_smooth': 3.650054984875908}

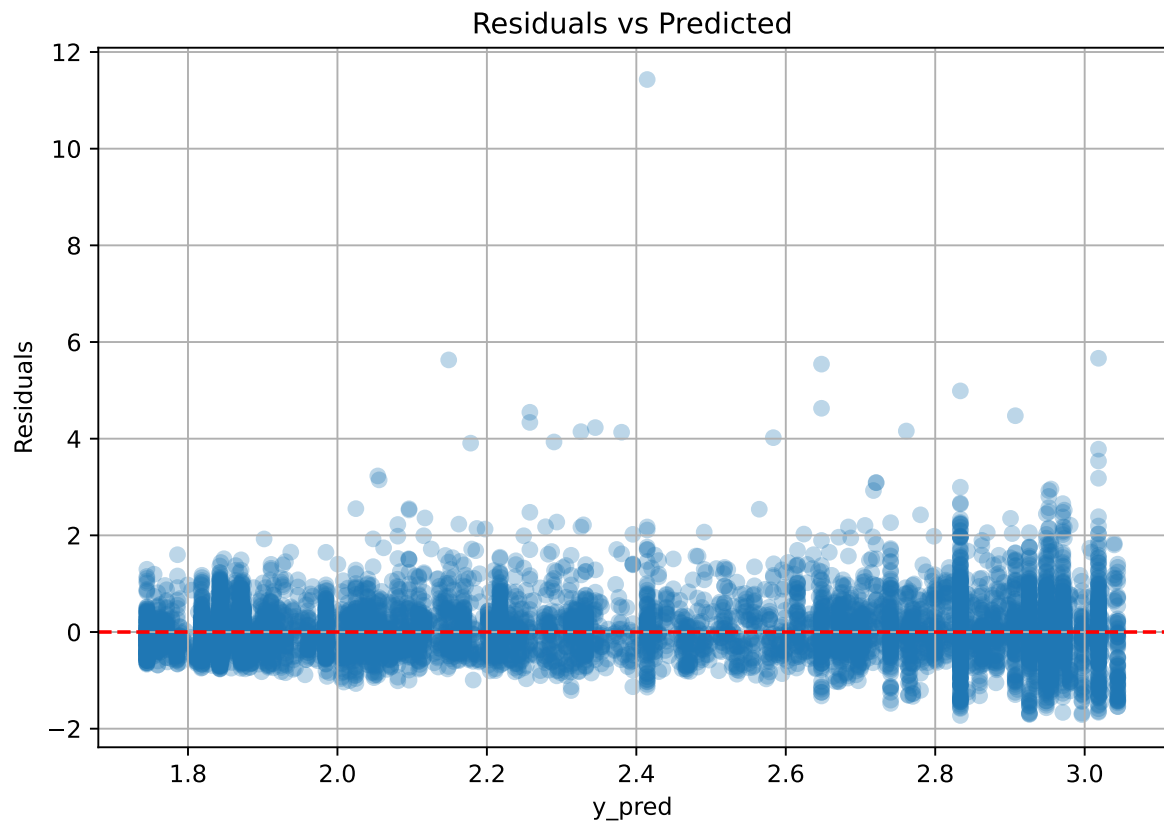
Rank 9: {'reg__n_estimators': 400, 'reg__max_depth': 5, 'reg__learning_rate': 0.09528703923469947, 'reg__num_leaves': 41, 'reg__subsample': 0.88265997765506, 'reg__colsample_bytree': 0.5049760254747703, 'reg__min_child_samples': 27, 'reg__reg_alpha': 5.063062304462754, 'reg__reg_lambda': 11.375627398413984, 'reg__min_split_gain': 4.818763506146314, 'reg__path_smooth': 2.586600613128515}

Rank 10: {'reg__n_estimators': 600, 'reg__max_depth': 3, 'reg__learning_rate': 0.01088474552137028, 'reg__num_leaves': 50, 'reg__subsample': 0.6449447153664816, 'reg__colsample_bytree': 0.531921696033234, 'reg__min_child_samples': 26, 'reg__reg_alpha': 1.7813592920742172, 'reg__reg_lambda': 15.991548073870993, 'reg__min_split_gain': 4.214618765029444, 'reg__path_smooth': 4.261590478121888}

2.4.14 Scatter



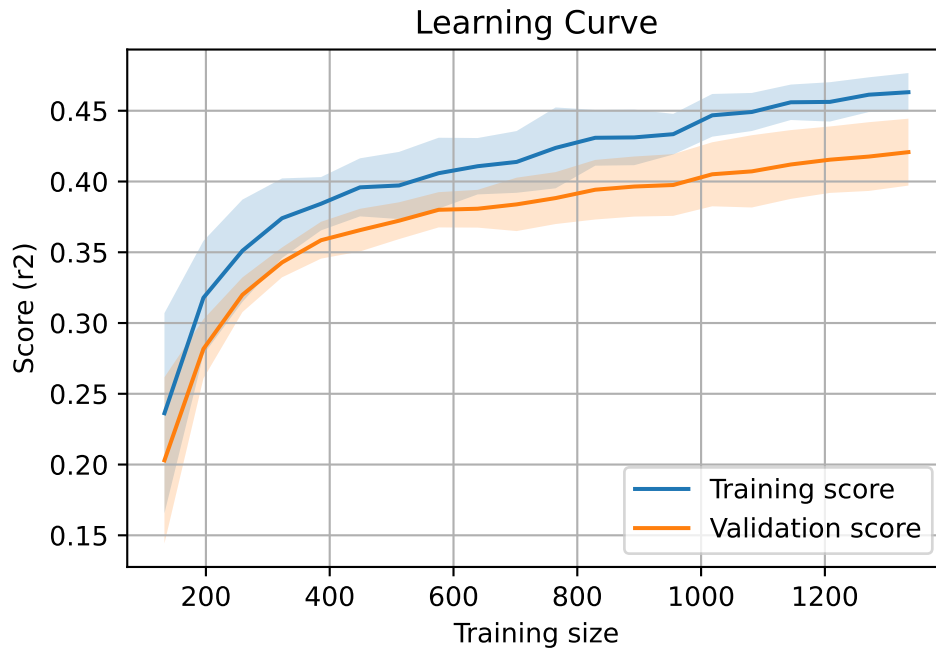
2.4.15 Scatter residuals



2.4.16 Learning curve — Group: 4

`Len(group)` : 2726, `Len(training)` : 1909,

`Len(training_real)` : 1337, `Len(test_of_training)` : 572, `Len(test)` : 817

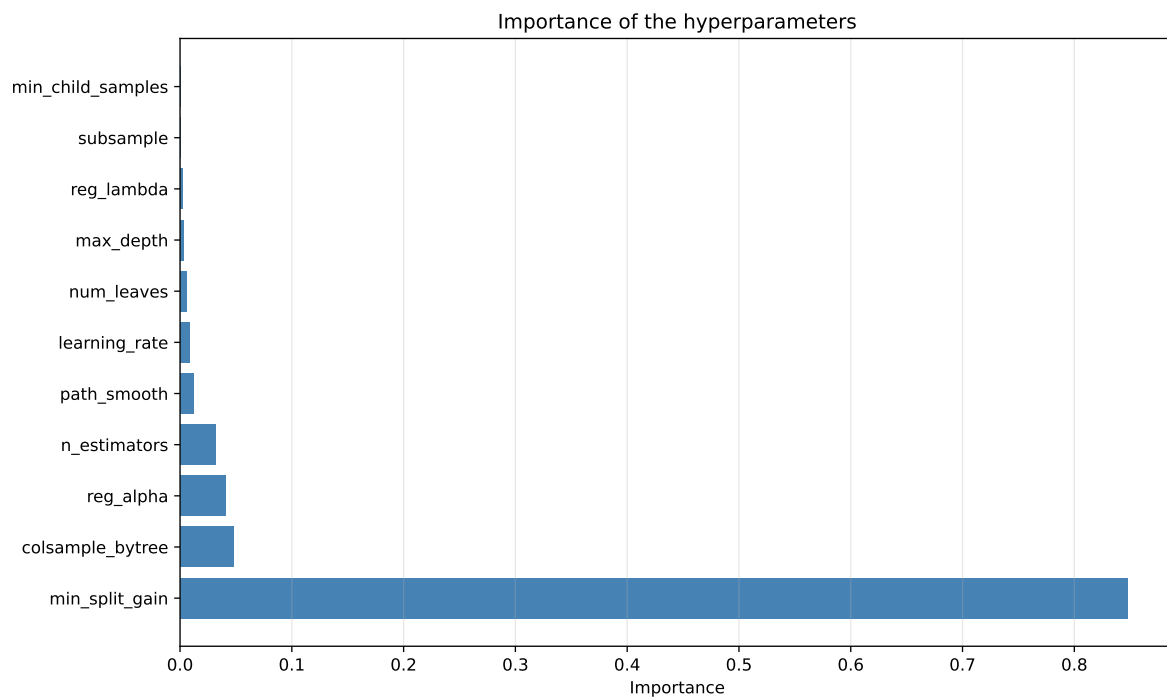
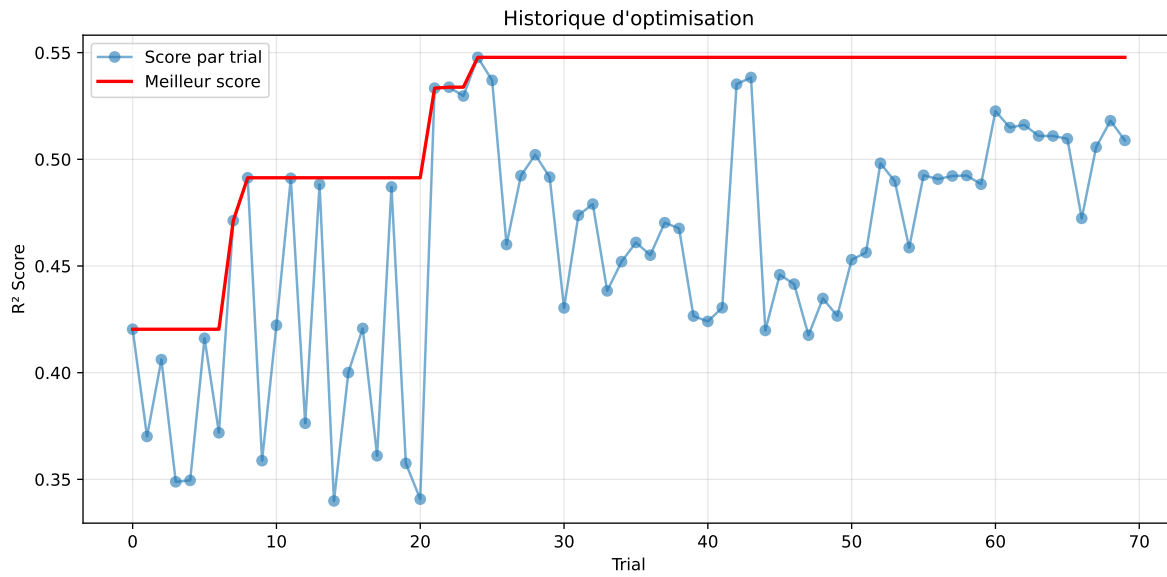


2.4.17 Gridsearch — Group: 6

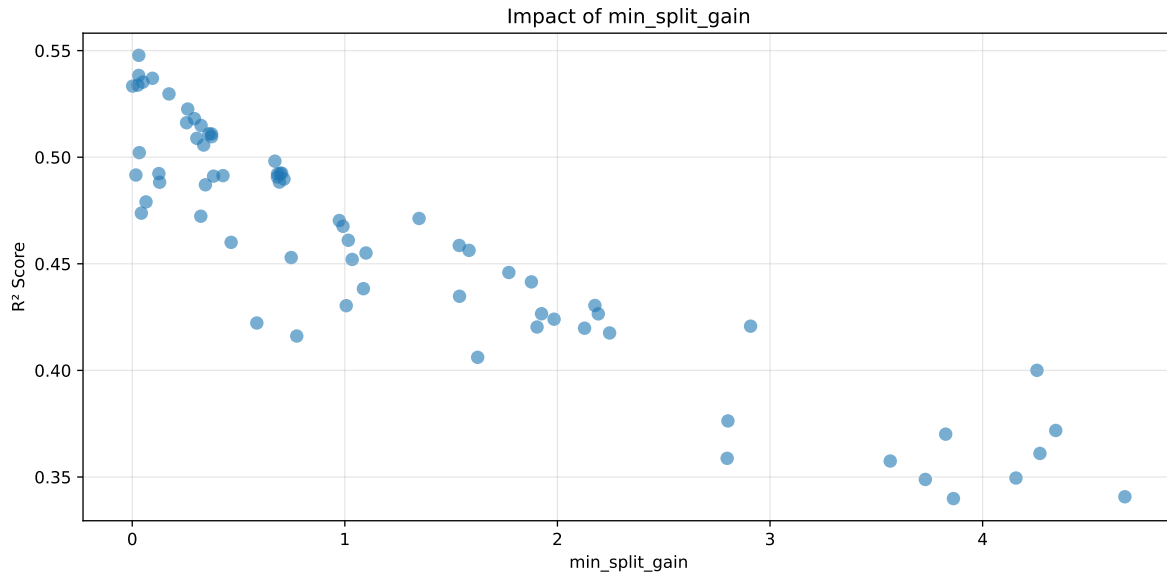
Len(group) : 1353, Len(training) : 948, Len(test) : 405

Distribution y_train vs y_test: y_train: mean=2.286, std=0.802, min=1.041, max=7.383
y_test: mean=2.187, std=0.733, min=1.078, max=5.525

===== TOP Importance FEATURES =====
feature importance r_umidity 120 r_NH3 85 r_temperature 45 r_THI 42 r_CO2 0



Paramètres avec >5% d'importance : ['min_split_gain']



Pas assez de paramètres avec >5% d'importance pour une heatmap 2D.

rank	mean_train_cv	mean_test_cv	test_score_cv	std_test_cv	penalized_score	test	diff	r_test	r_train
1	0.4077	0.3763	0.451	0.0531	0.282	0.4859	0.1096	1.2912	1.0836
2	0.4023	0.3718	0.4458	0.0543	0.2803	0.4858	0.1139	1.3064	1.0821
3	0.3858	0.3588	0.4321	0.0525	0.2777	0.4622	0.1035	1.2885	1.0753
4	0.402	0.3701	0.4464	0.0538	0.2742	0.4746	0.1045	1.2822	1.0863
5	0.3754	0.3495	0.4223	0.0515	0.2718	0.4556	0.106	1.3034	1.0741
6	0.3913	0.361	0.4373	0.0527	0.2702	0.4709	0.1098	1.3042	1.0838
7	0.3866	0.3575	0.433	0.0531	0.2701	0.4606	0.1031	1.2884	1.0815
8	0.365	0.3407	0.4082	0.0497	0.2679	0.4362	0.0955	1.2803	1.0712
9	0.376	0.3489	0.4238	0.052	0.2675	0.4567	0.1078	1.3091	1.0777
10	0.4664	0.4161	0.5101	0.0555	0.2653	0.5516	0.1355	1.3256	1.1208
---	---	---	---	---	---	---	---	---	---
70	0.8111	0.5478	0.6922	0.072	-0.2423	0.7239	0.1761	1.3214	1.4808

Hyperparameters:

Rank 1: {'reg__n_estimators': 800, 'reg__max_depth': 7, 'reg__learning_rate': 0.02630817088998247, 'reg__num_leaves': 49, 'reg__subsample': 0.72939689473056, 'reg__colsample_bytree': 0.7337062864512567, 'reg__min_child_samples': 30, 'reg__reg_alpha': 7.496327079182899, 'reg__reg_lambda': 30.128085161731043, 'reg__min_split_gain': 2.802037650255599, 'reg__path_smooth': 3.5726888362901263}

Rank 2: {'reg__n_estimators': 800, 'reg__max_depth': 7, 'reg__learning_rate': 0.03199238326723125, 'reg__num_leaves': 45, 'reg__subsample': 0.8013793538767562,

'reg__colsample_bytree': 0.8529961710430085, 'reg__min_child_samples': 46, 'reg__reg_alpha': 5.607717132800431, 'reg__reg_lambda': 9.928665496508426, 'reg__min_split_gain': 4.3439786936602856, 'reg__path_smooth': 2.465201421567655}

Rank 3: {'reg__n_estimators': 500, 'reg__max_depth': 4, 'reg__learning_rate': 0.011877133081090724, 'reg__num_leaves': 44, 'reg__subsample': 0.6068175202307904, 'reg__colsample_bytree': 0.6260110038699248, 'reg__min_child_samples': 41, 'reg__reg_alpha': 8.896012950815454, 'reg__reg_lambda': 23.000250282134004, 'reg__min_split_gain': 2.798366327430473, 'reg__path_smooth': 2.0691415324112485}

Rank 4: {'reg__n_estimators': 400, 'reg__max_depth': 8, 'reg__learning_rate': 0.015049341433659819, 'reg__num_leaves': 48, 'reg__subsample': 0.824034541198864, 'reg__colsample_bytree': 0.581015063356159, 'reg__min_child_samples': 17, 'reg__reg_alpha': 1.6435605504485595, 'reg__reg_lambda': 44.08780508593309, 'reg__min_split_gain': 3.8262474078426996, 'reg__path_smooth': 2.306067869630848}

Rank 5: {'reg__n_estimators': 800, 'reg__max_depth': 6, 'reg__learning_rate': 0.024481163917463973, 'reg__num_leaves': 48, 'reg__subsample': 0.6152119538528098, 'reg__colsample_bytree': 0.5334890105872678, 'reg__min_child_samples': 50, 'reg__reg_alpha': 8.12031991415833, 'reg__reg_lambda': 2.2322892042934974, 'reg__min_split_gain': 4.156866781199433, 'reg__path_smooth': 2.167637817472075}

Rank 6: {'reg__n_estimators': 500, 'reg__max_depth': 3, 'reg__learning_rate': 0.06663831650160562, 'reg__num_leaves': 28, 'reg__subsample': 0.5770055192884933, 'reg__colsample_bytree': 0.6797292543635425, 'reg__min_child_samples': 31, 'reg__reg_alpha': 3.1570921920011363, 'reg__reg_lambda': 26.290692042386887, 'reg__min_split_gain': 4.2695810958301985, 'reg__path_smooth': 0.43773894875931196}

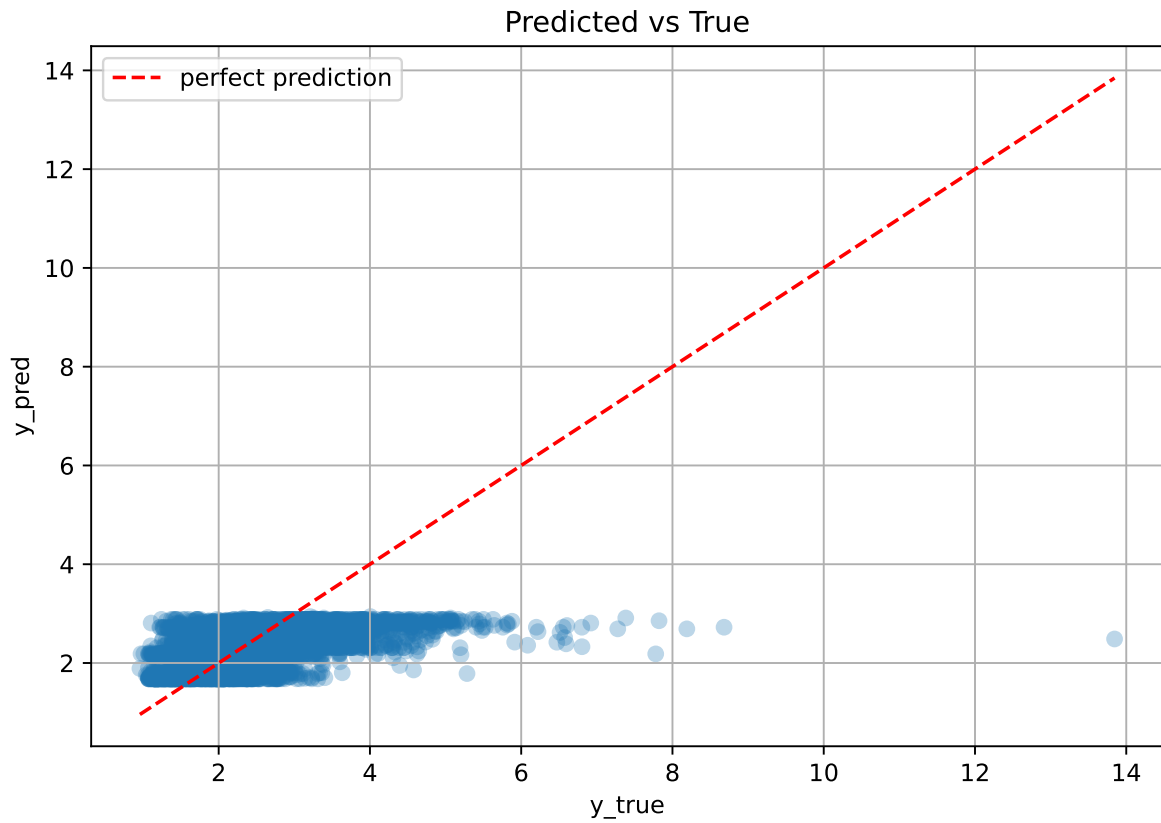
Rank 7: {'reg__n_estimators': 800, 'reg__max_depth': 4, 'reg__learning_rate': 0.09570872950658552, 'reg__num_leaves': 59, 'reg__subsample': 0.8125019024335458, 'reg__colsample_bytree': 0.5889110729917606, 'reg__min_child_samples': 29, 'reg__reg_alpha': 4.512258662176959, 'reg__reg_lambda': 43.366149757783965, 'reg__min_split_gain': 3.565431486881704, 'reg__path_smooth': 2.6717722600740452}

Rank 8: {'reg__n_estimators': 700, 'reg__max_depth': 5, 'reg__learning_rate': 0.08636273484947656, 'reg__num_leaves': 39, 'reg__subsample': 0.8659148504901351, 'reg__colsample_bytree': 0.5536062829170126, 'reg__min_child_samples': 21, 'reg__reg_alpha': 7.292240383127108, 'reg__reg_lambda': 39.0335754251581, 'reg__min_split_gain': 4.669119258423563, 'reg__path_smooth': 1.9842029280502111}

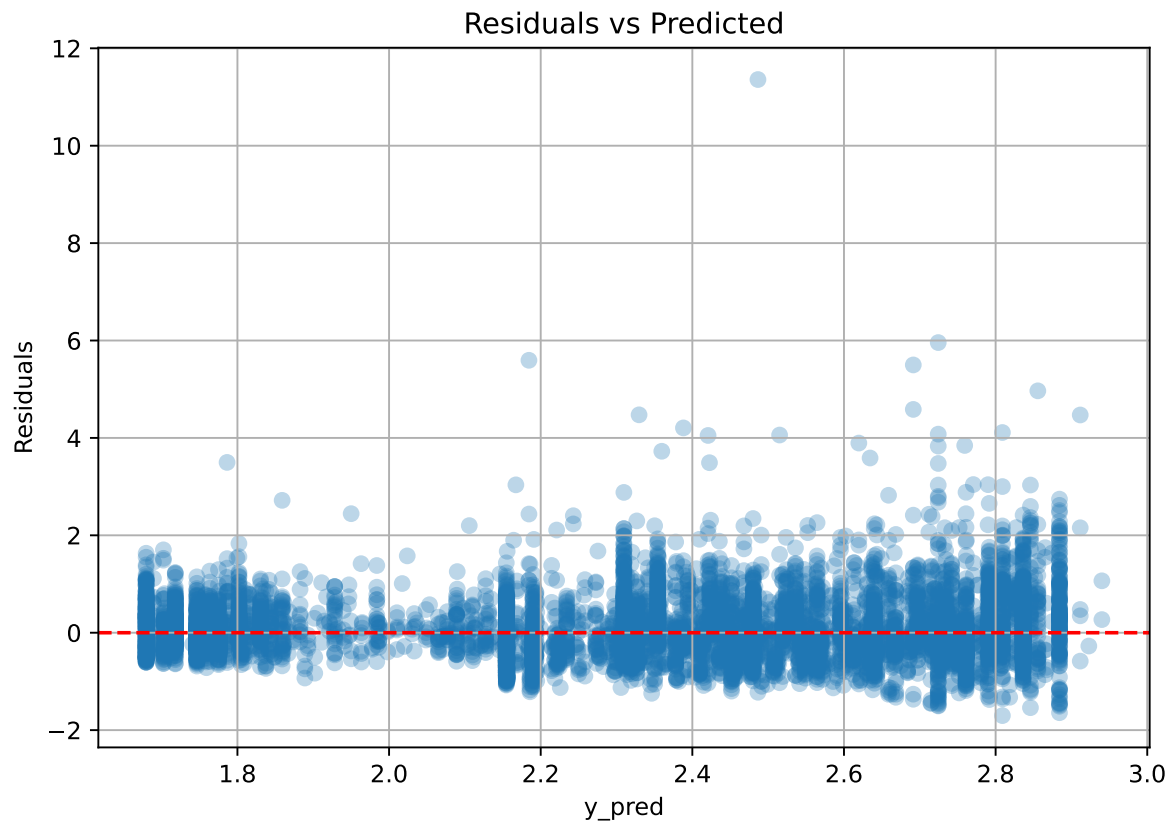
Rank 9: {'reg__n_estimators': 500, 'reg__max_depth': 7, 'reg__learning_rate': 0.06118560325109706, 'reg__num_leaves': 23, 'reg__subsample': 0.7273554817646177, 'reg__colsample_bytree': 0.6530064387323227, 'reg__min_child_samples': 47, 'reg__reg_alpha': 6.426488999782201, 'reg__reg_lambda': 31.357311832445443, 'reg__min_split_gain': 3.730522110279961, 'reg__path_smooth': 4.318891131004653}

Rank 10: {'reg__n_estimators': 800, 'reg__max_depth': 8, 'reg__learning_rate': 0.0254113050638219, 'reg__num_leaves': 39, 'reg__subsample': 0.5796653667992269, 'reg__colsample_bytree': 0.8849726516905565, 'reg__min_child_samples': 41, 'reg__reg_alpha': 8.986740946469876, 'reg__reg_lambda': 12.5347223870577, 'reg__min_split_gain': 0.7738090240280987, 'reg__path_smooth': 2.0899272758530154}

2.4.18 Scatter



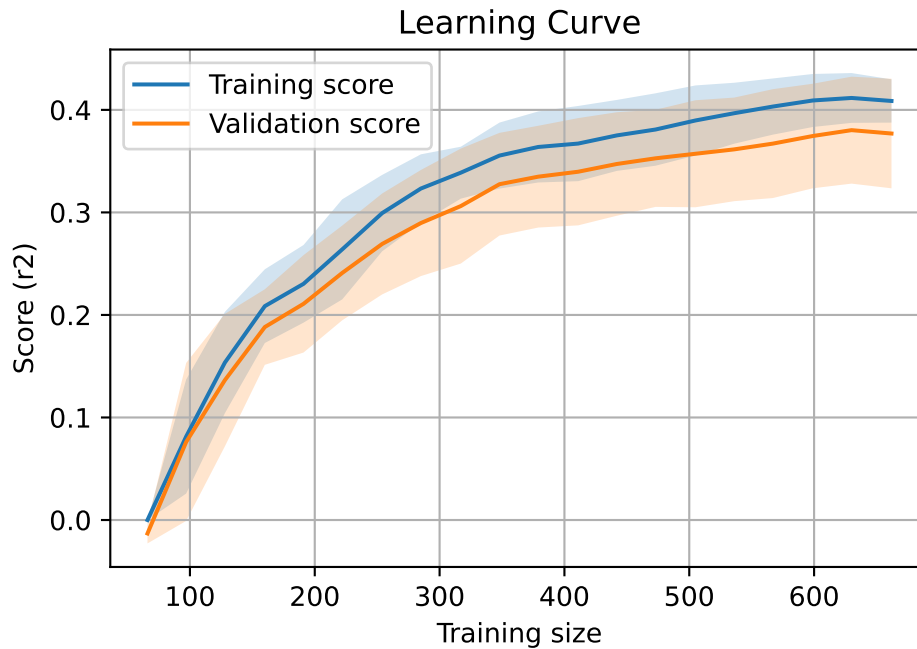
2.4.19 Scatter residuals



2.4.20 Learning curve — Group: 6

`Len(group)` : 1353, `Len(training)` : 948,

`Len(training_real)` : 664, `Len(test_of_training)` : 284, `Len(test)` : 405



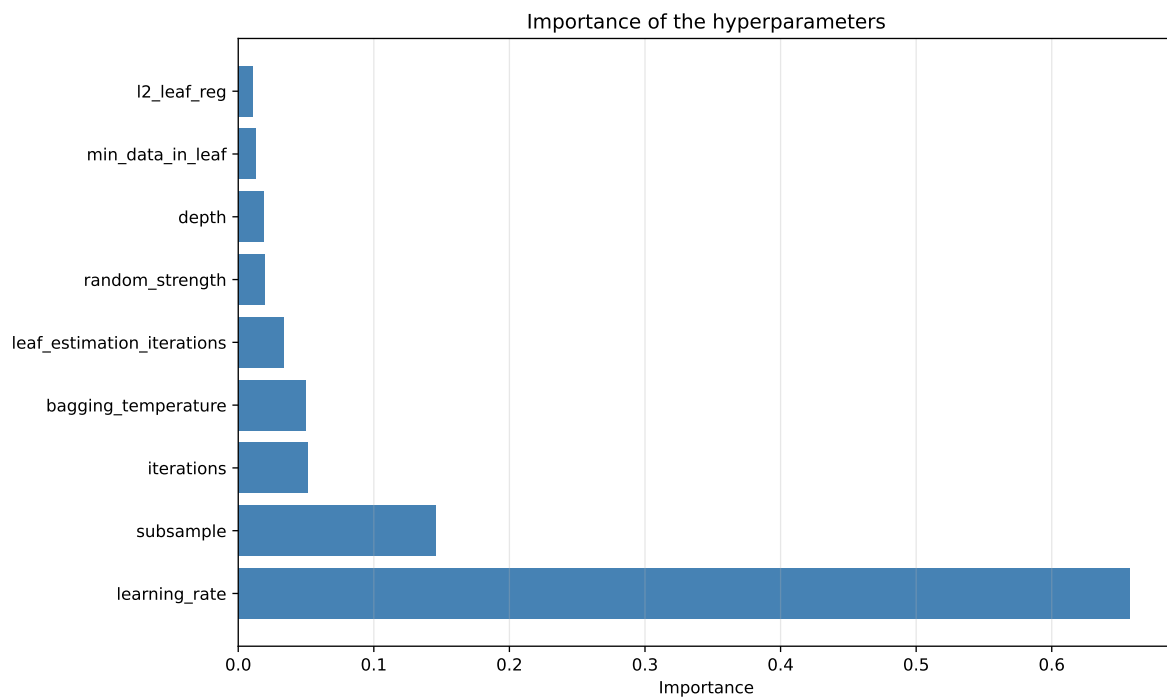
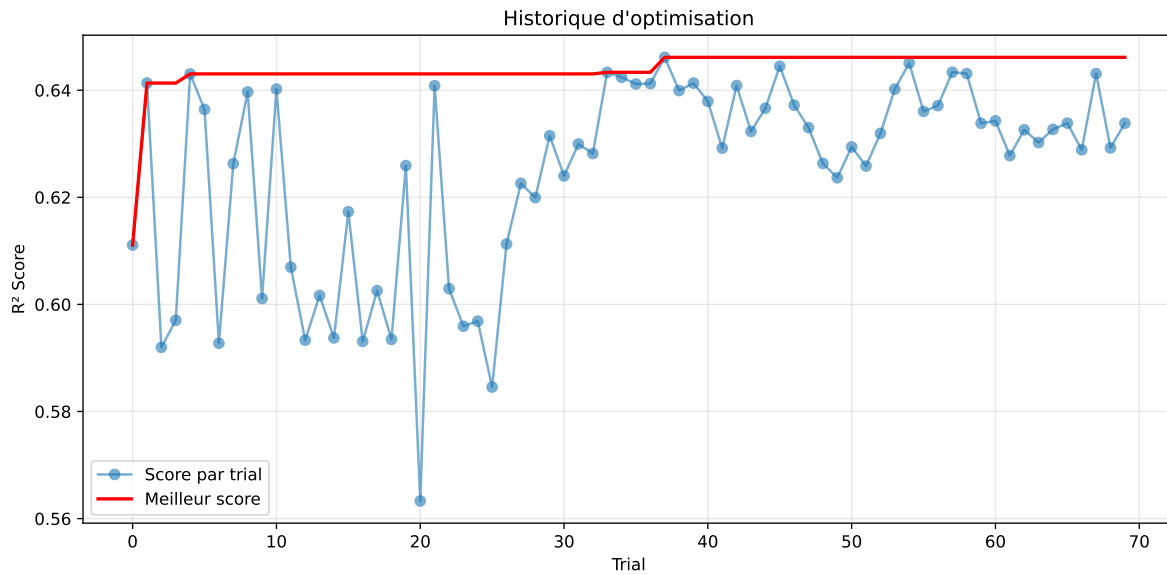
2.5 Model : CatBoost

2.5.1 Gridsearch — Group: 1

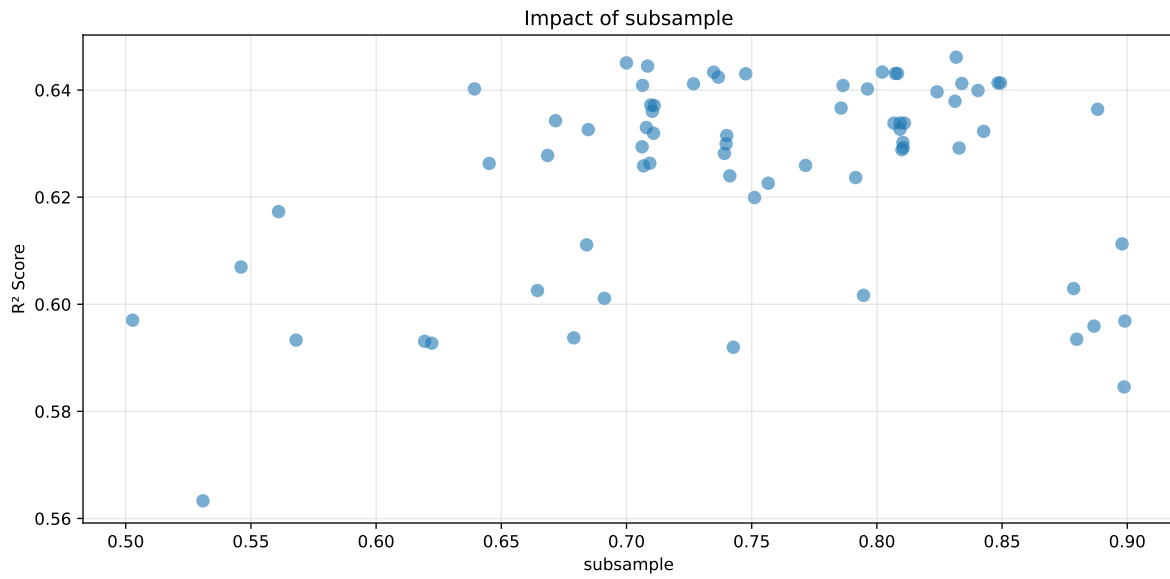
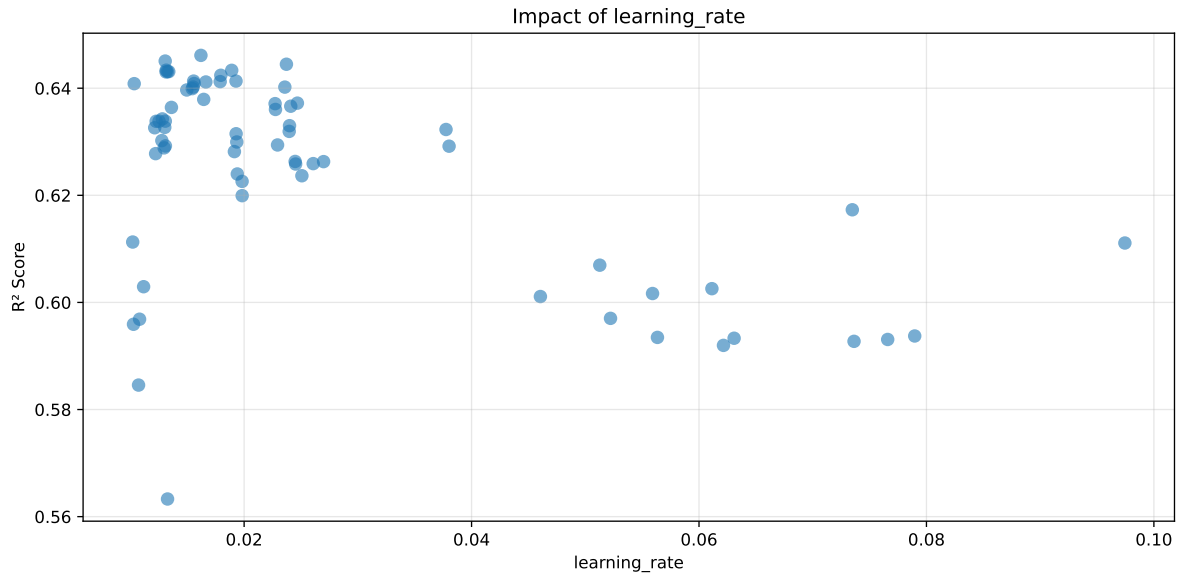
Len(group) : 795, Len(training) : 557, Len(test) : 238

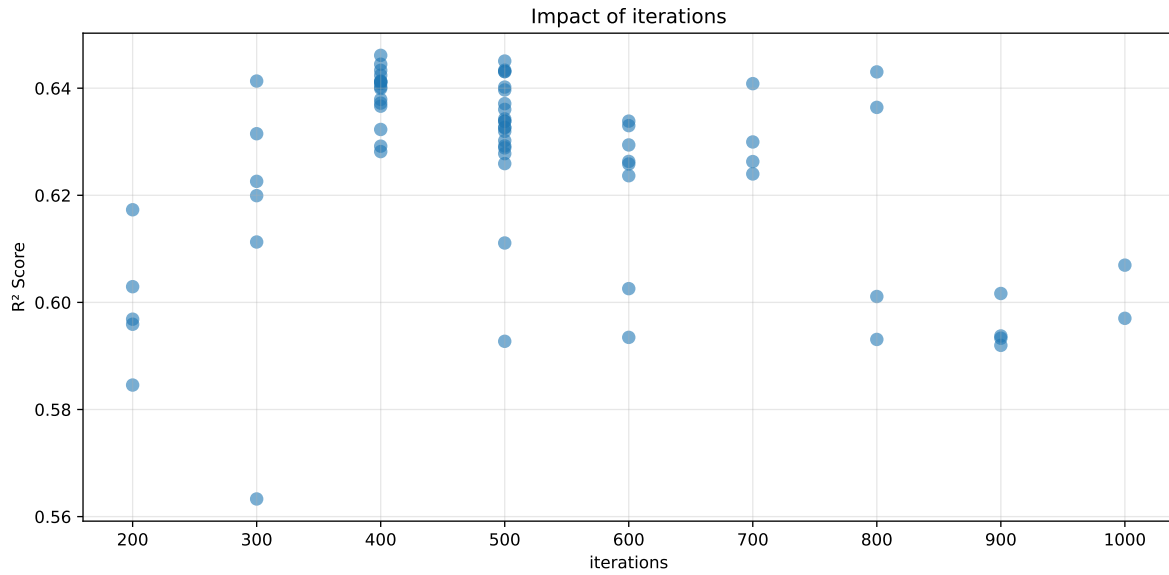
Distribution y_train vs y_test: y_train: mean=2.597, std=0.579, min=1.437, max=4.986
y_test: mean=2.492, std=0.625, min=1.466, max=5.447

===== TOP Importance FEATURES =====
feature importance r_temperature 25.819917 r_umidity 24.821217 r_CO2 18.326253 r_THI
17.496483 r_NH3 13.536131



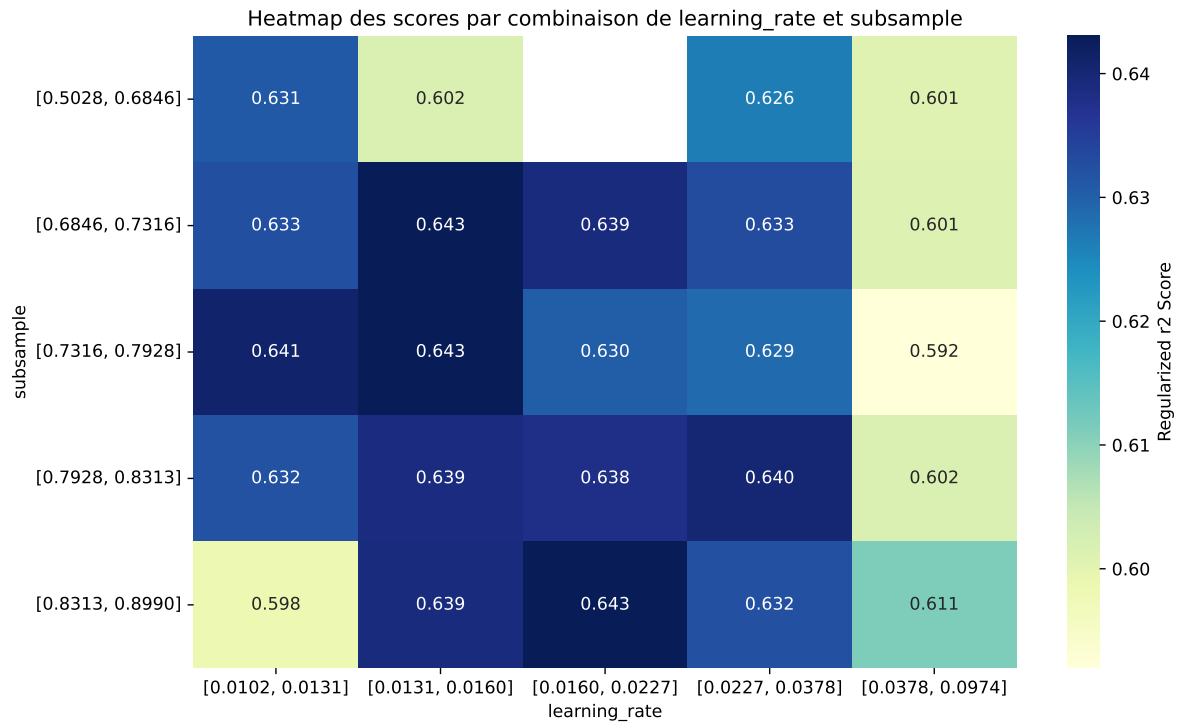
Paramètres avec >5% d'importance : ['learning_rate', 'subsample', 'iterations']





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The default value of observed=False is deprecated and will change to observed=True in a future



rank	mean_train_cv	mean_test_cv	test_score_cv	std_test_cv	penalized_score	test	diff	r_test	r_train
1	0.6075	0.5633	0.5244	0.0433	0.4307	0.5551	-0.0082	0.9855	1.0784
2	0.6664	0.6029	0.5741	0.0462	0.4126	0.5891	-0.0138	0.9771	1.1052
3	0.6794	0.6113	0.5889	0.0451	0.407	0.6015	-0.0097	0.9841	1.1114
4	0.6603	0.5969	0.571	0.0459	0.4066	0.5831	-0.0138	0.9769	1.1063
5	0.6595	0.5959	0.5692	0.0446	0.4051	0.581	-0.0149	0.975	1.1067
6	0.7032	0.6278	0.6195	0.0466	0.4015	0.6254	-0.0024	0.9962	1.1201
7	0.7064	0.6292	0.6229	0.0482	0.3978	0.6298	0.0006	1.0009	1.1226
8	0.7063	0.6289	0.6225	0.0487	0.3965	0.6292	0.0003	1.0005	1.1232
9	0.7088	0.6302	0.6237	0.0472	0.3945	0.6271	-0.0032	0.995	1.1247
10	0.723	0.6343	0.6276	0.0471	0.3682	0.6346	0.0004	1.0006	1.1398
---	---	---	---	---	---	---	---	---	---
70	0.9988	0.592	0.6434	0.0491	-0.6286	0.6421	0.0501	1.0847	1.6873

Hyperparameters:

Rank 1: {'reg__iterations': 300, 'reg__depth': 8, 'reg__learning_rate': 0.013266217272809317, 'reg__l2_leaf_reg': 46.76316118750011, 'reg__bagging_temperature': 2.3689235210059234, 'reg__random_strength': 4.14947483955967, 'reg__subsample': 0.5308606950075669, 'reg__min_data_in_leaf': 39, 'reg__leaf_estimation_iterations': 1}

Rank 2: {'reg__iterations': 200, 'reg__depth': 8, 'reg__learning_rate': 0.01116019282581263, 'reg__l2_leaf_reg': 16.903149751153798, 'reg__bagging_temperature': 4.729276626414115, 'reg__random_strength': 4.891079455367005, 'reg__subsample': 0.8785919690068992, 'reg__min_data_in_leaf': 22, 'reg__leaf_estimation_iterations': 3}

Rank 3: {'reg__iterations': 300, 'reg__depth': 8, 'reg__learning_rate': 0.010202128013761882, 'reg__l2_leaf_reg': 18.404703600136905, 'reg__bagging_temperature': 0.08707462535578347, 'reg__random_strength': 4.957712651148741, 'reg__subsample': 0.8979162455041443, 'reg__min_data_in_leaf': 23, 'reg__leaf_estimation_iterations': 2}

Rank 4: {'reg__iterations': 200, 'reg__depth': 8, 'reg__learning_rate': 0.01080578072408986, 'reg__l2_leaf_reg': 24.84581644993106, 'reg__bagging_temperature': 0.10398675436344251, 'reg__random_strength': 4.78498440758327, 'reg__subsample': 0.8990433757619186, 'reg__min_data_in_leaf': 23, 'reg__leaf_estimation_iterations': 4}

Rank 5: {'reg__iterations': 200, 'reg__depth': 8, 'reg__learning_rate': 0.010272766571399196, 'reg__l2_leaf_reg': 23.72640894414117, 'reg__bagging_temperature': 0.43304840375181053, 'reg__random_strength': 3.2920934248089755, 'reg__subsample': 0.8867140596843783, 'reg__min_data_in_leaf': 20, 'reg__leaf_estimation_iterations': 4}

Rank 6: {'reg__iterations': 500, 'reg__depth': 4, 'reg__learning_rate': 0.012217121016614369, 'reg__l2_leaf_reg': 5.582304219120873, 'reg__bagging_temperature': 2.061491237053404, 'reg__random_strength': 4.450126186751236, 'reg__subsample': 0.668494342044993, 'reg__min_data_in_leaf': 41, 'reg__leaf_estimation_iterations': 3}

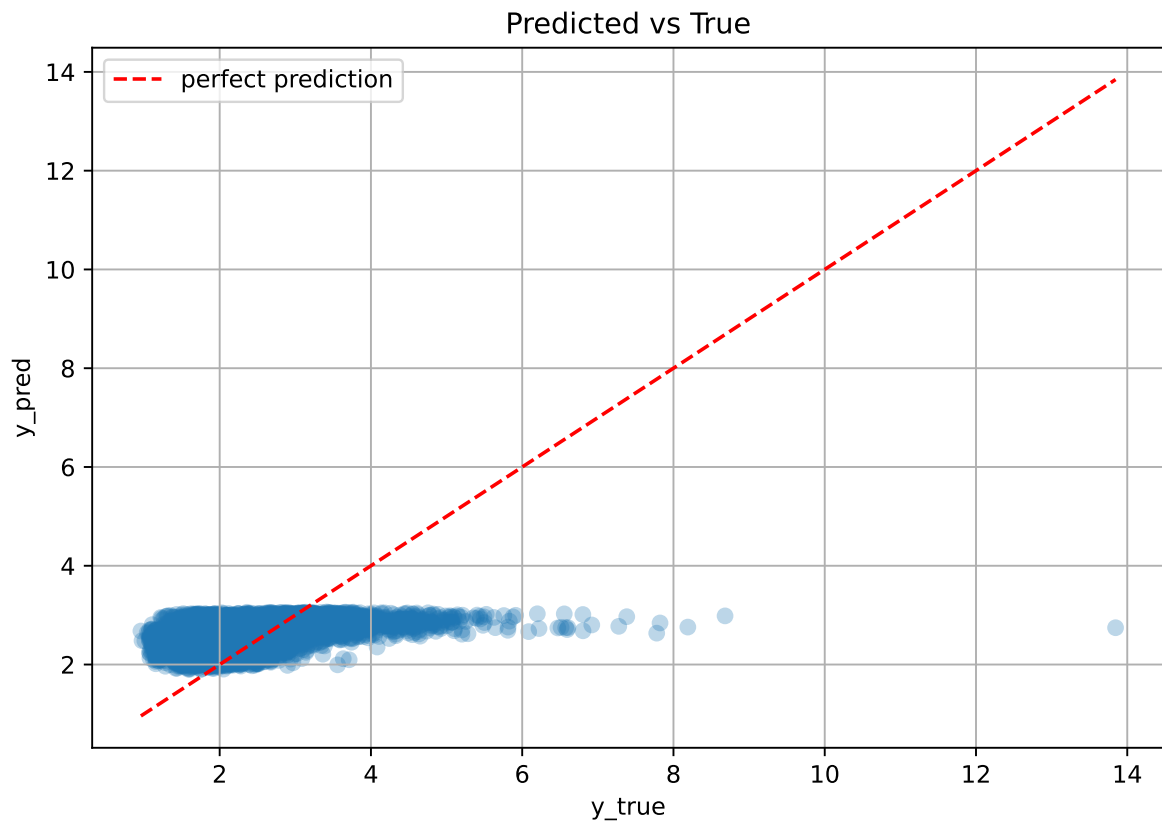
Rank 7: {'reg__iterations': 500, 'reg__depth': 4, 'reg__learning_rate': 0.013077575273826669, 'reg__l2_leaf_reg': 15.426111792660844, 'reg__bagging_temperature': 2.1013116279932285, 'reg__random_strength': 3.1077893488105746, 'reg__subsample': 0.8105053873786144, 'reg__min_data_in_leaf': 45, 'reg__leaf_estimation_iterations': 3}

Rank 8: {'reg__iterations': 500, 'reg__depth': 4, 'reg__learning_rate': 0.012968680992692695, 'reg__l2_leaf_reg': 14.614164690240173, 'reg__bagging_temperature': 1.8782670674047761, 'reg__random_strength': 3.1148740141142484, 'reg__subsample': 0.8100151835582026, 'reg__min_data_in_leaf': 46, 'reg__leaf_estimation_iterations': 3}

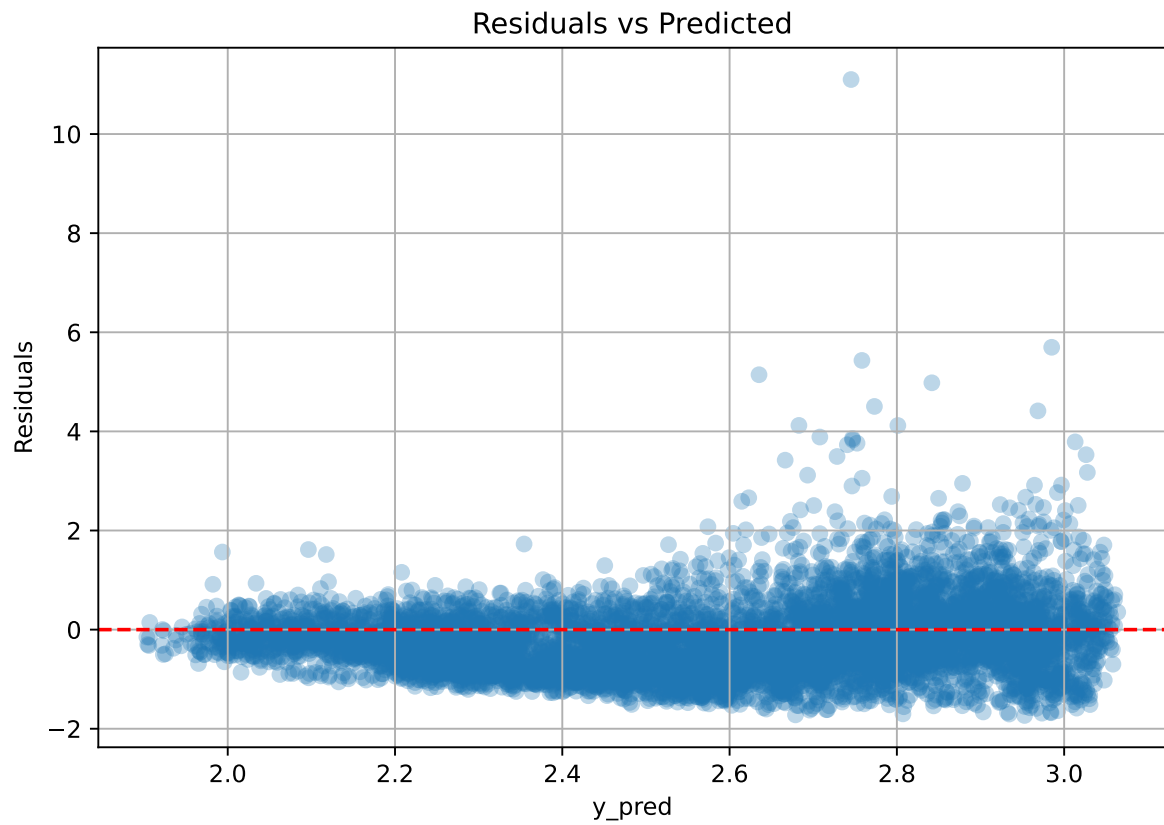
Rank 9: {'reg__iterations': 500, 'reg__depth': 4, 'reg__learning_rate': 0.012766077235814874, 'reg__l2_leaf_reg': 4.886438925453988, 'reg__bagging_temperature': 3.790999688555327, 'reg__random_strength': 4.45062178403556, 'reg__subsample': 0.8104083048975365, 'reg__min_data_in_leaf': 45, 'reg__leaf_estimation_iterations': 3}

Rank 10: {'reg__iterations': 500, 'reg__depth': 4, 'reg__learning_rate': 0.012807302339601182, 'reg__l2_leaf_reg': 4.877200835995702, 'reg__bagging_temperature': 2.056865073343191, 'reg__random_strength': 2.7474927962245794, 'reg__subsample': 0.6716658064301303, 'reg__min_data_in_leaf': 44, 'reg__leaf_estimation_iterations': 3}

2.5.2 Scatter



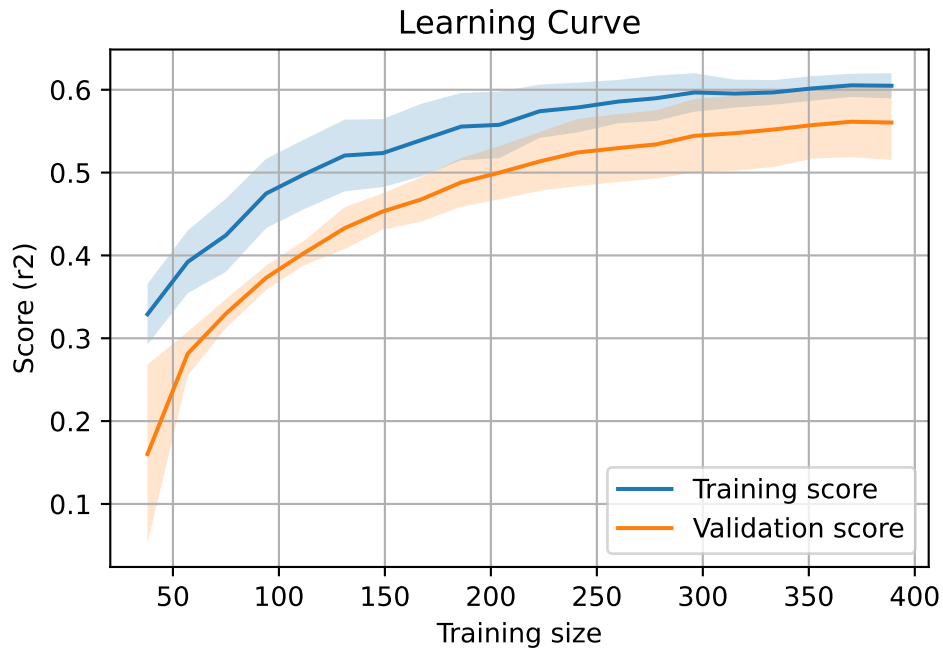
2.5.3 Scatter residuals



2.5.4 Learning curve — Group: 1

`Len(group)` : 795, `Len(training)` : 557,

`Len(training_real)` : 390, `Len(test_of_training)` : 167, `Len(test)` : 238

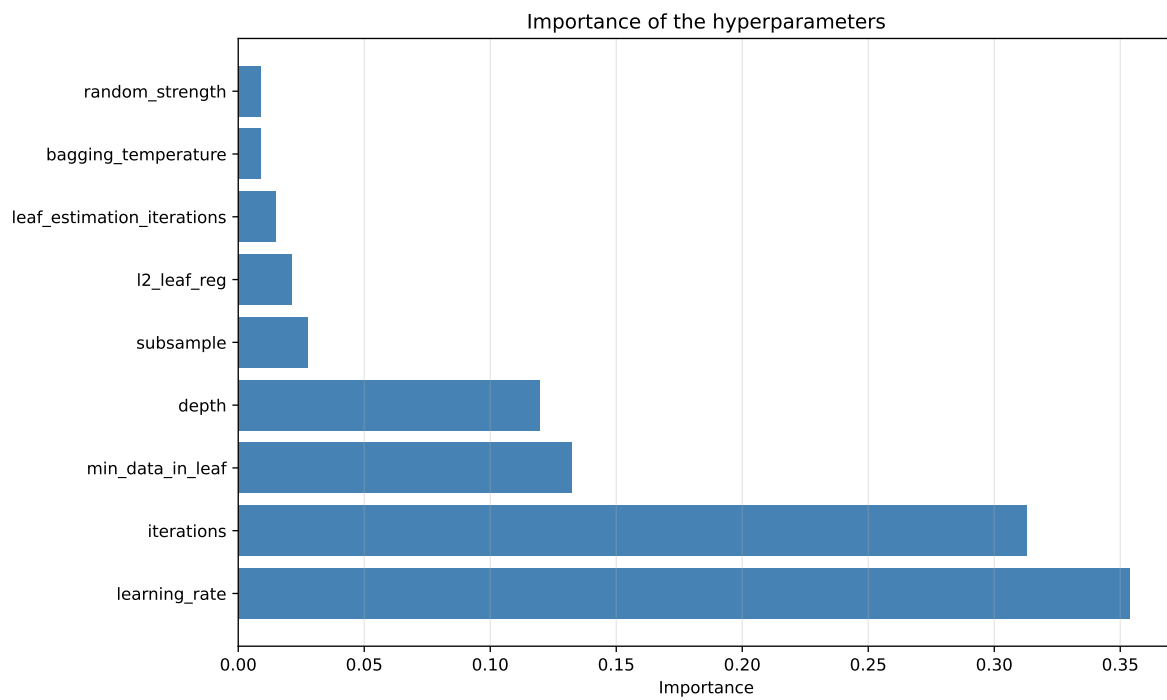
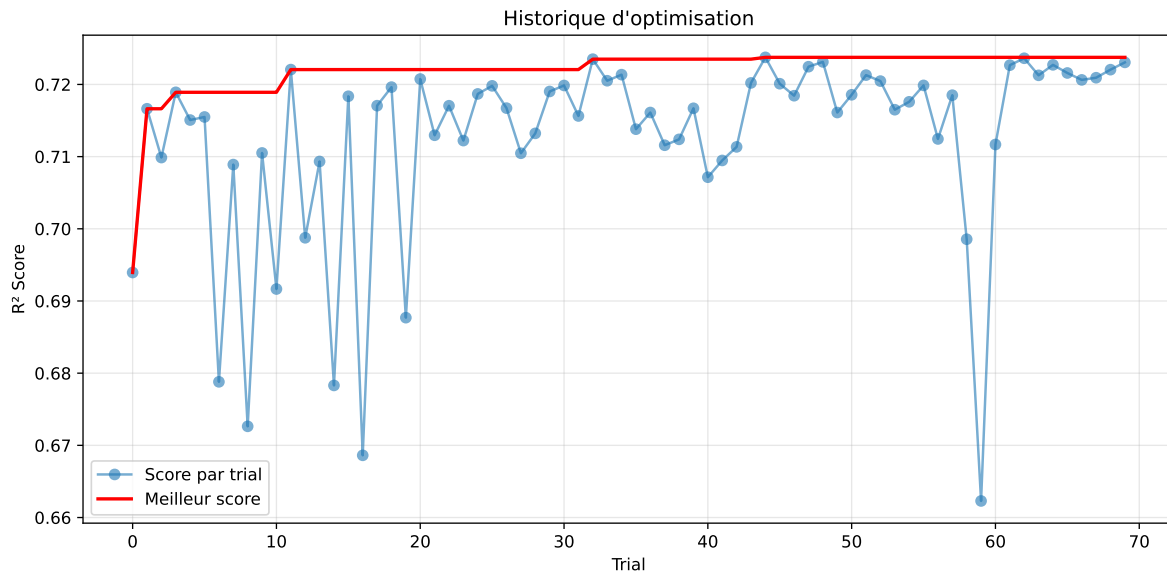


2.5.5 Gridsearch — Group: 2

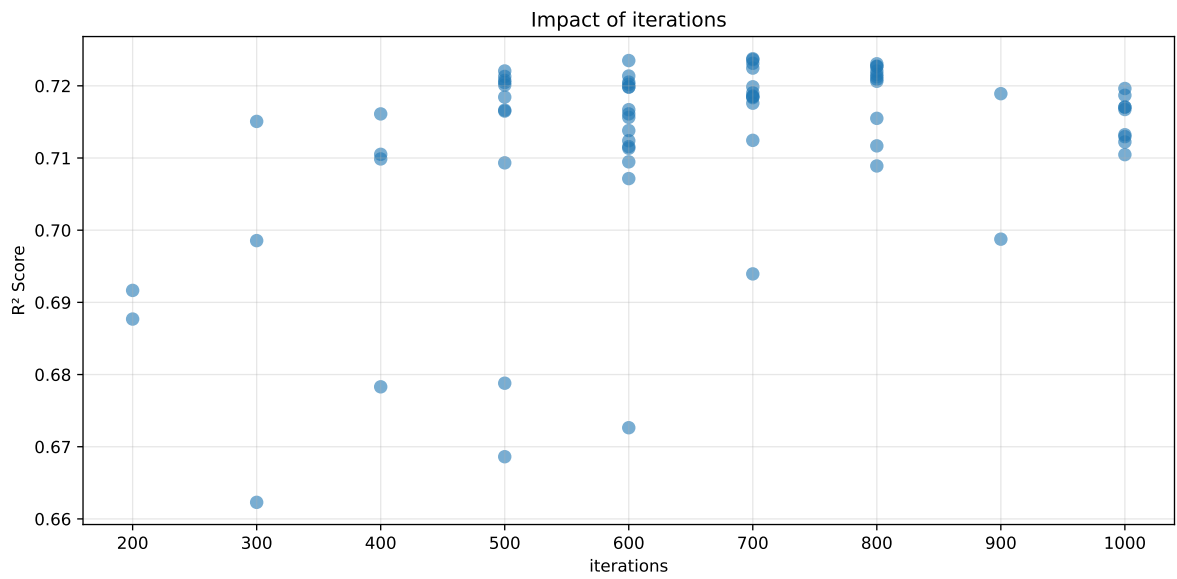
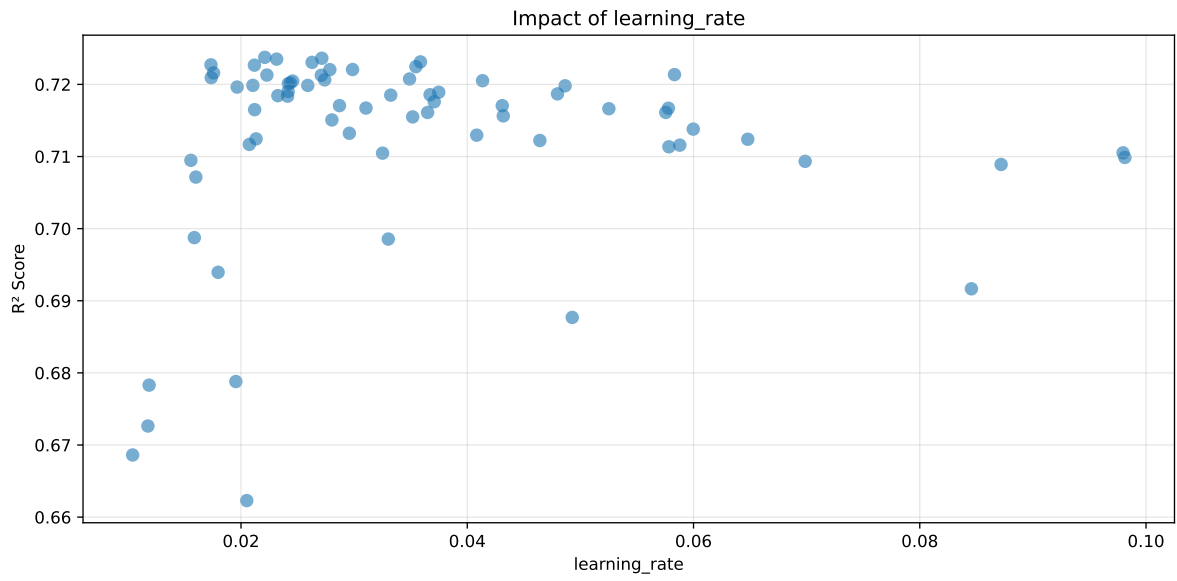
Len(group) : 2726, Len(training) : 1909, Len(test) : 817

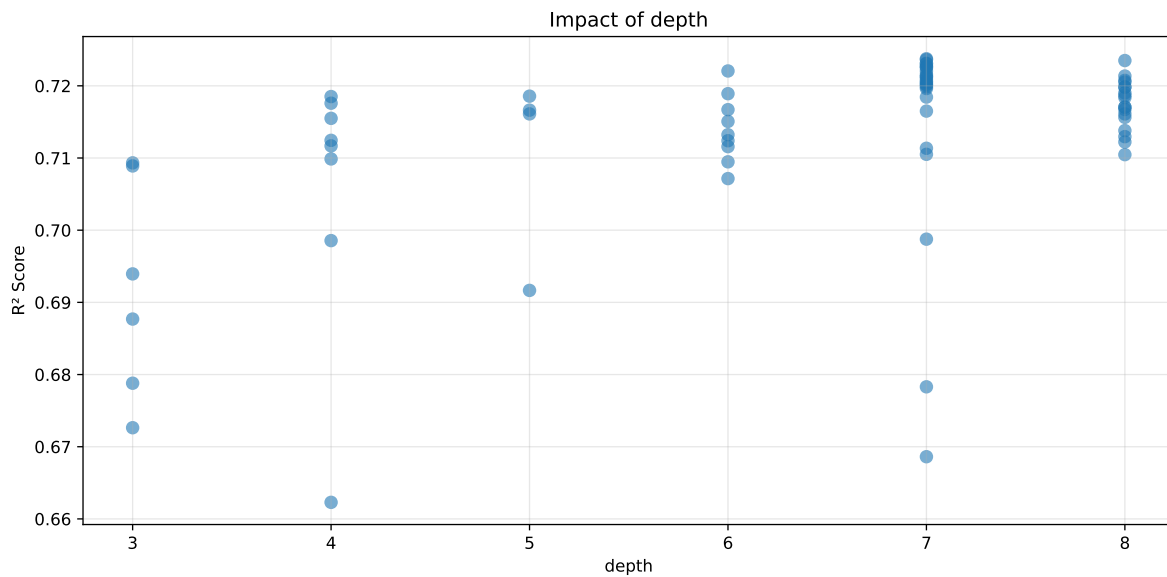
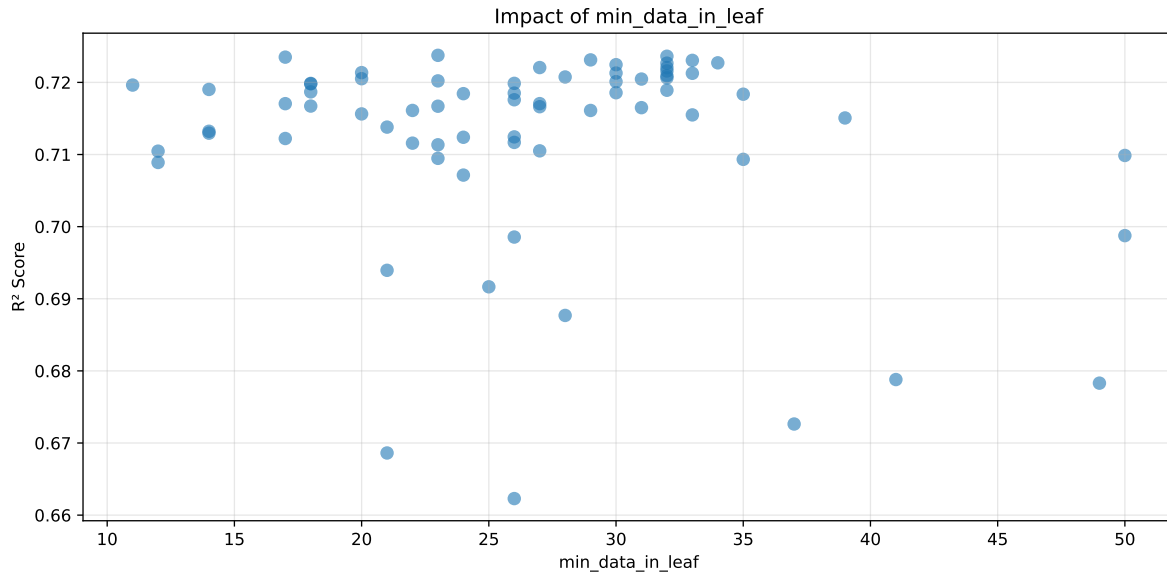
Distribution y_train vs y_test: y_train: mean=2.540, std=0.859, min=1.150, max=8.682
y_test: mean=2.561, std=0.947, min=1.120, max=13.845

===== TOP Importance FEATURES =====
feature importance r_umidity 35.637288 r_NH3 32.163400 r_CO2 13.701506 r_THI 9.730547
r_temperature 8.767259



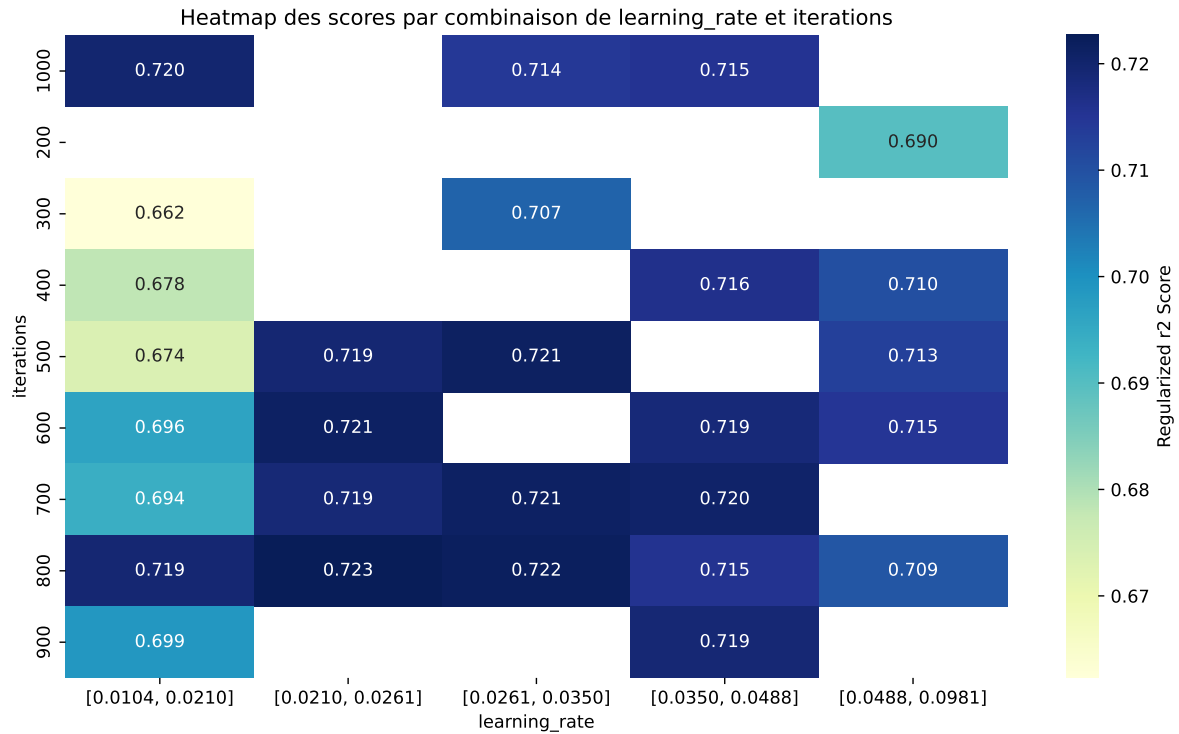
Paramètres avec >5% d'importance : ['learning_rate', 'iterations', 'min_data_in_leaf', 'depth']





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The default value of observed=False is deprecated and will change to observed=True in a future



rank	mean_train_cv	mean_test_cv	test_score_cv	std_test_cv	penalized_score	test	diff	r_test	r_train
1	0.7332	0.6986	0.6013	0.0169	0.6986	0.6045	-0.0941	0.8654	1.0496
2	0.7165	0.6877	0.5924	0.0207	0.6877	0.5931	-0.0946	0.8625	1.0419
3	0.7027	0.6788	0.5892	0.0185	0.6788	0.591	-0.0878	0.8707	1.0352
4	0.7028	0.6726	0.5829	0.0176	0.6726	0.5848	-0.0878	0.8695	1.0449
5	0.6981	0.6686	0.5745	0.0137	0.6686	0.5862	-0.0824	0.8768	1.0441
6	0.6781	0.6623	0.5741	0.0144	0.6623	0.5808	-0.0815	0.8769	1.0239
7	0.7288	0.6939	0.5981	0.0196	0.5893	0.5965	-0.0975	0.8596	1.0503
8	0.7274	0.6917	0.5914	0.0165	0.5843	0.6028	-0.0889	0.8715	1.0517
9	0.7163	0.6783	0.586	0.0133	0.5643	0.5927	-0.0856	0.8738	1.056
10	0.7482	0.6988	0.5997	0.0144	0.5505	0.6073	-0.0915	0.8691	1.0707
---	---	---	---	---	---	---	---	---	---
70	0.9784	0.7122	0.5892	0.016	-0.0863	0.6007	-0.1115	0.8434	1.3737

Hyperparameters:

Rank 1: {'reg__iterations': 300, 'reg__depth': 4, 'reg__learning_rate': 0.033020368862118336, 'reg__l2_leaf_reg': 25.952105795534358, 'reg__bagging_temperature': 4.110717551071699, 'reg__random_strength': 3.8318488960477826, 'reg__subsample': 0.7151697461887936, 'reg__min_data_in_leaf': 26, 'reg__leaf_estimation_iterations': 5}

Rank 2: {'reg__iterations': 200, 'reg__depth': 3, 'reg__learning_rate': 0.04928101356373144, 'reg__l2_leaf_reg': 11.573589014408537, 'reg__bagging_temperature': 0.694381971172614, 'reg__random_strength': 2.9885840324586423, 'reg__subsample': 0.7494027537807475, 'reg__min_data_in_leaf': 28, 'reg__leaf_estimation_iterations': 8}

Rank 3: {'reg__iterations': 500, 'reg__depth': 3, 'reg__learning_rate': 0.019546627993675975, 'reg__l2_leaf_reg': 28.31922449252573, 'reg__bagging_temperature': 1.4553530036202167, 'reg__random_strength': 4.0106999383762, 'reg__subsample': 0.7449953474126356, 'reg__min_data_in_leaf': 41, 'reg__leaf_estimation_iterations': 10}

Rank 4: {'reg__iterations': 600, 'reg__depth': 3, 'reg__learning_rate': 0.01177079788440557, 'reg__l2_leaf_reg': 45.73000979017463, 'reg__bagging_temperature': 0.8734077577220312, 'reg__random_strength': 0.5554802371810103, 'reg__subsample': 0.6150738737165502, 'reg__min_data_in_leaf': 37, 'reg__leaf_estimation_iterations': 9}

Rank 5: {'reg__iterations': 500, 'reg__depth': 7, 'reg__learning_rate': 0.010420587769968502, 'reg__l2_leaf_reg': 38.821906254434325, 'reg__bagging_temperature': 1.3023949443258283, 'reg__random_strength': 4.616865773854239, 'reg__subsample': 0.5093739991242933, 'reg__min_data_in_leaf': 21, 'reg__leaf_estimation_iterations': 3}

Rank 6: {'reg__iterations': 300, 'reg__depth': 4, 'reg__learning_rate': 0.02051006829447814, 'reg__l2_leaf_reg': 26.360884079803625, 'reg__bagging_temperature': 4.205919646816266, 'reg__random_strength': 4.081936890914042, 'reg__subsample': 0.7256586100820033, 'reg__min_data_in_leaf': 26, 'reg__leaf_estimation_iterations': 5}

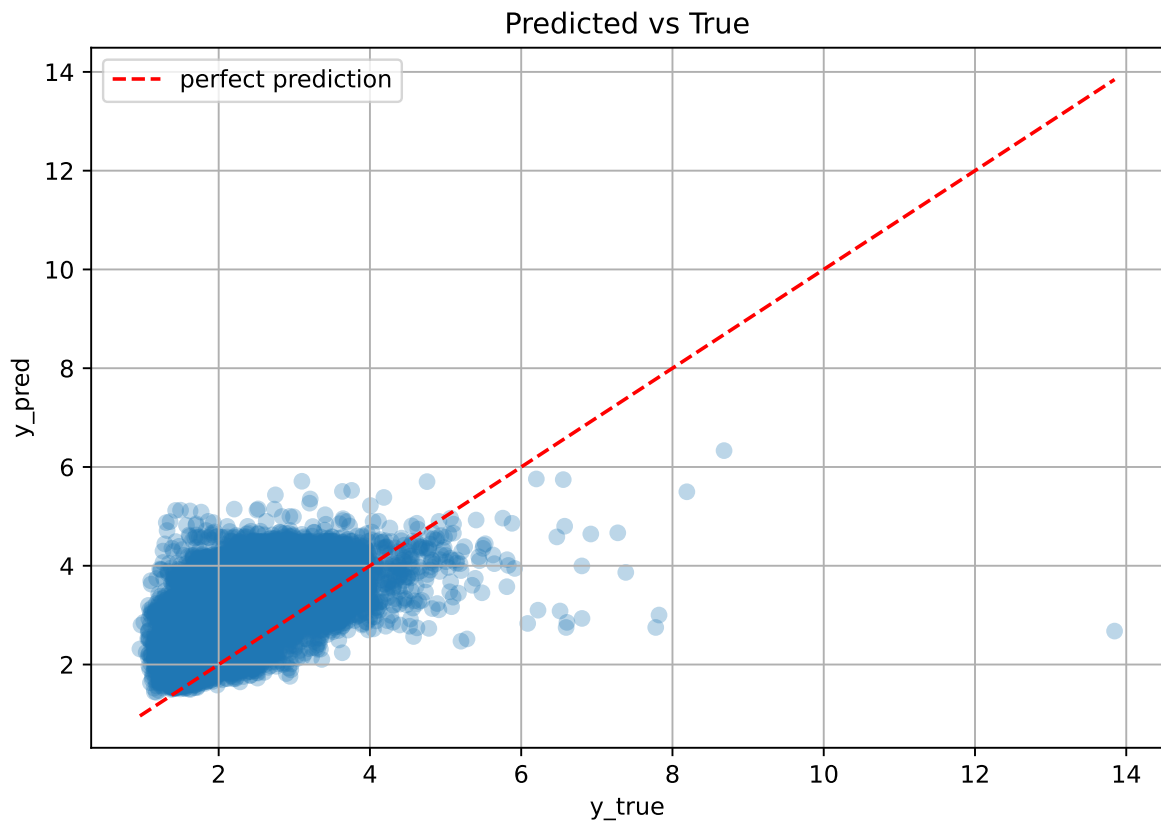
Rank 7: {'reg__iterations': 700, 'reg__depth': 3, 'reg__learning_rate': 0.01797561191945123, 'reg__l2_leaf_reg': 31.904979834851805, 'reg__bagging_temperature': 0.43452306726666723, 'reg__random_strength': 4.757899124600992, 'reg__subsample': 0.5619159239885716, 'reg__min_data_in_leaf': 21, 'reg__leaf_estimation_iterations': 9}

Rank 8: {'reg__iterations': 200, 'reg__depth': 5, 'reg__learning_rate': 0.08456528355510316, 'reg__l2_leaf_reg': 27.359141097169875, 'reg__bagging_temperature': 3.730768237753904, 'reg__random_strength': 3.674539382982131, 'reg__subsample': 0.893683561833204, 'reg__min_data_in_leaf': 25, 'reg__leaf_estimation_iterations': 1}

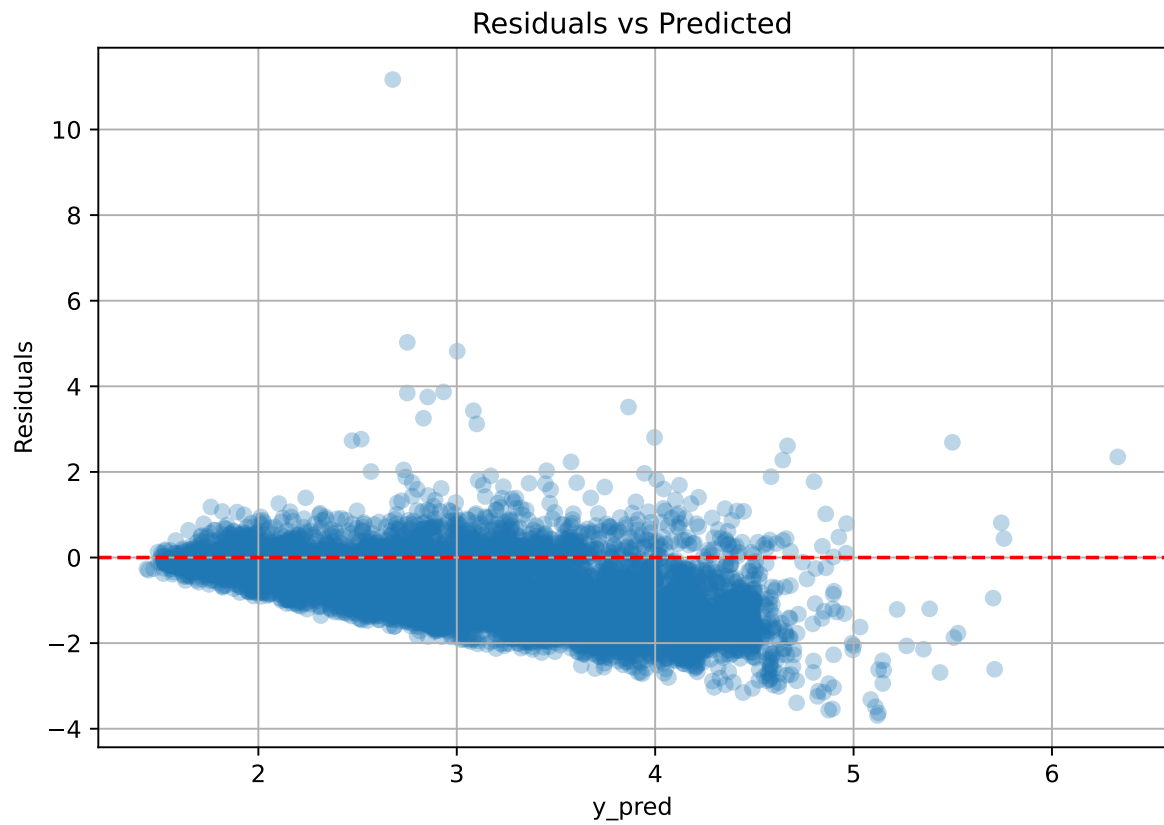
Rank 9: {'reg__iterations': 400, 'reg__depth': 7, 'reg__learning_rate': 0.011876361271480926, 'reg__l2_leaf_reg': 28.85731820267632, 'reg__bagging_temperature': 0.9734533681480434, 'reg__random_strength': 4.7119460659313175, 'reg__subsample': 0.6790670628043722, 'reg__min_data_in_leaf': 49, 'reg__leaf_estimation_iterations': 6}

Rank 10: {'reg__iterations': 900, 'reg__depth': 7, 'reg__learning_rate': 0.015874363118254978, 'reg__l2_leaf_reg': 49.492080298803344, 'reg__bagging_temperature': 2.1947565405454172, 'reg__random_strength': 4.192609651701013, 'reg__subsample': 0.6875878435410835, 'reg__min_data_in_leaf': 50, 'reg__leaf_estimation_iterations': 2}

2.5.6 Scatter



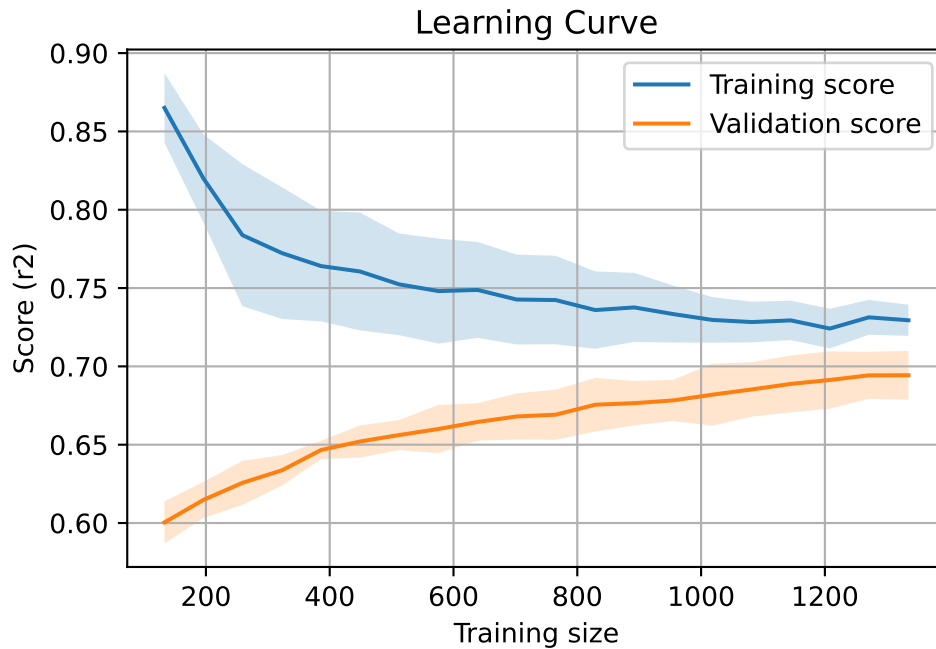
2.5.7 Scatter residuals



2.5.8 Learning curve — Group: 2

`Len(group)` : 2726, `Len(training)` : 1909,

`Len(training_real)` : 1337, `Len(test_of_training)` : 572, `Len(test)` : 817

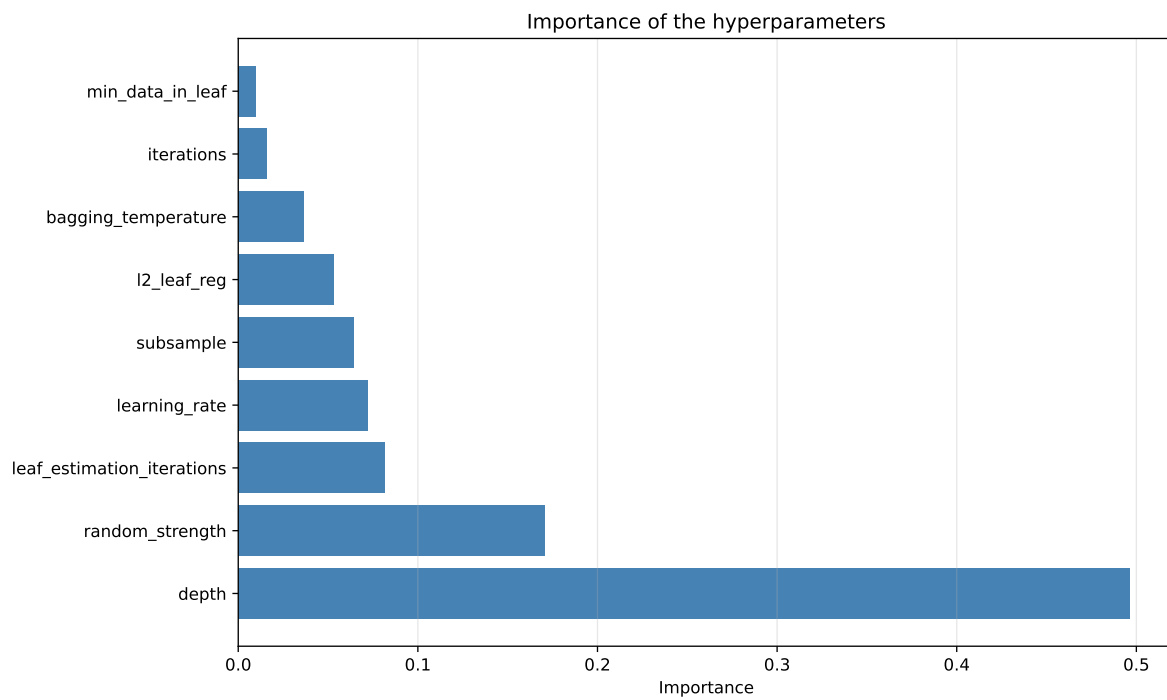
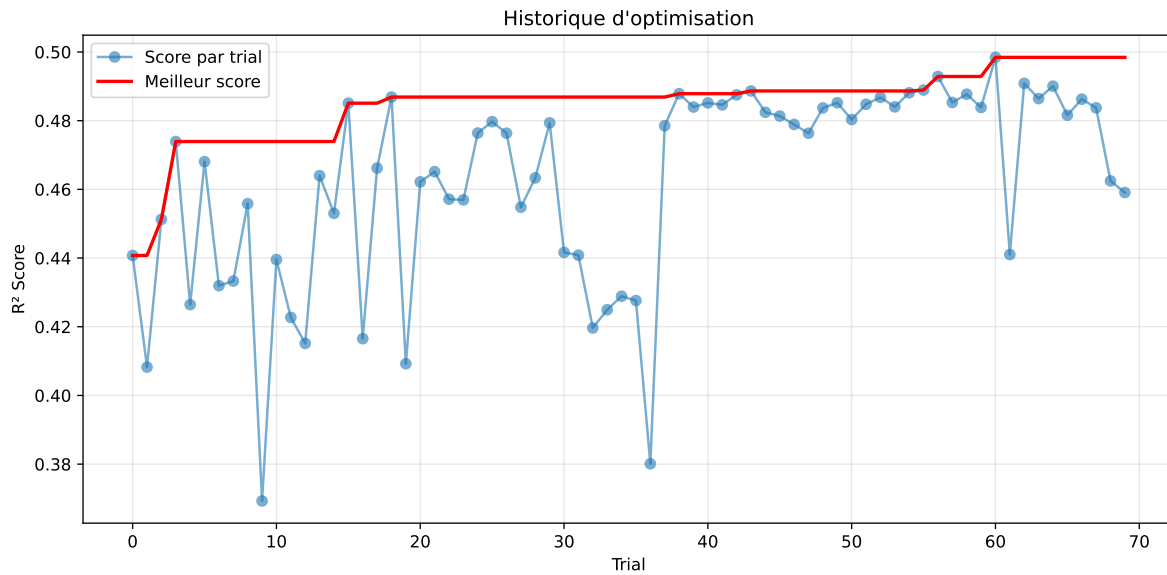


2.5.9 Gridsearch — Group: 3

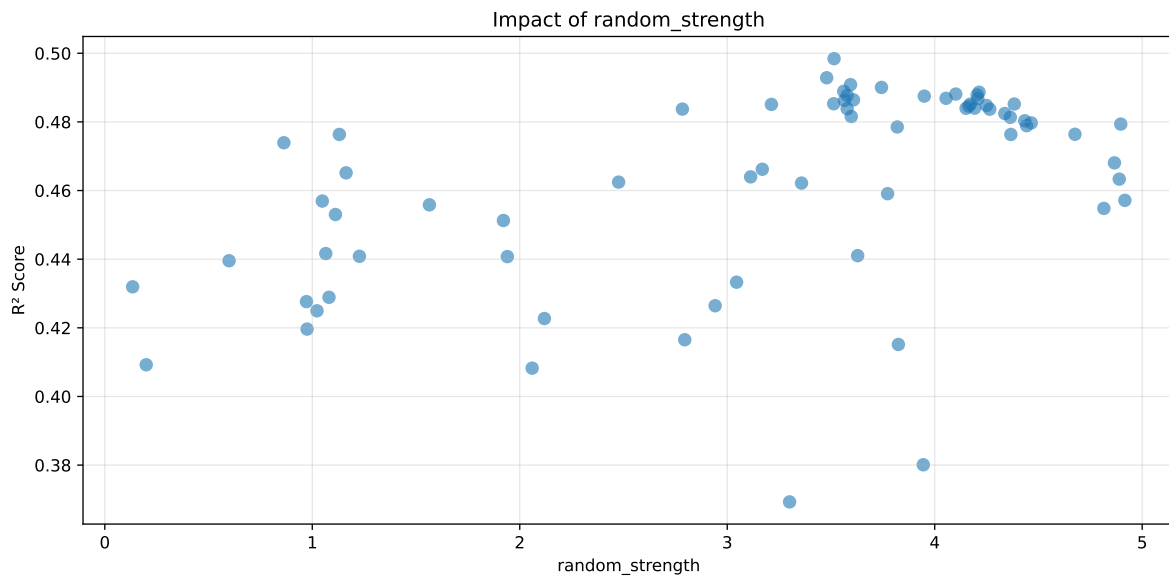
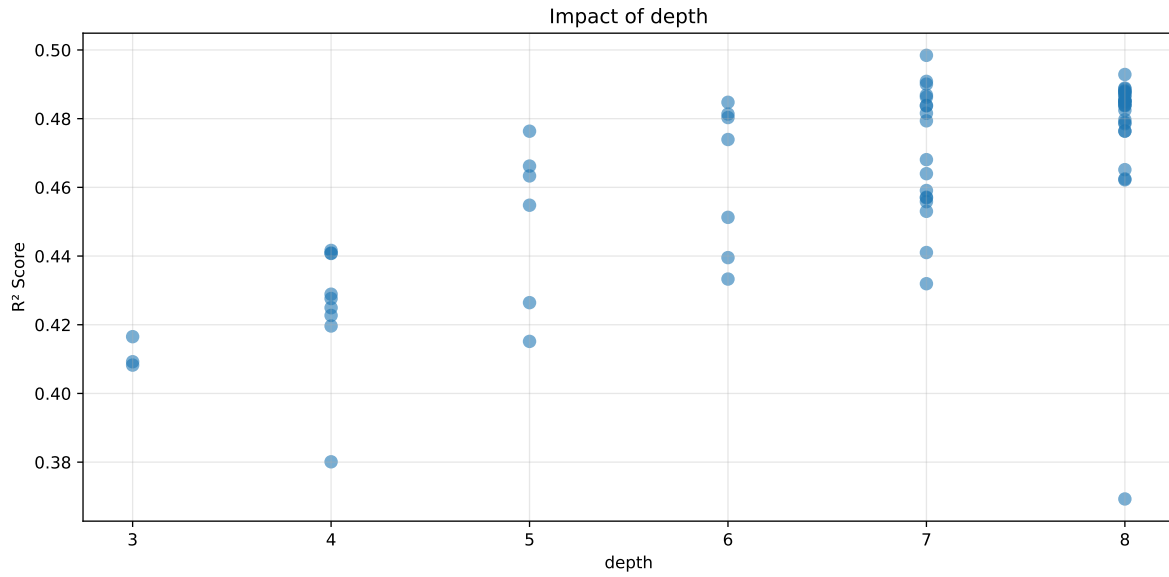
Len(group) : 568, Len(training) : 398, Len(test) : 170

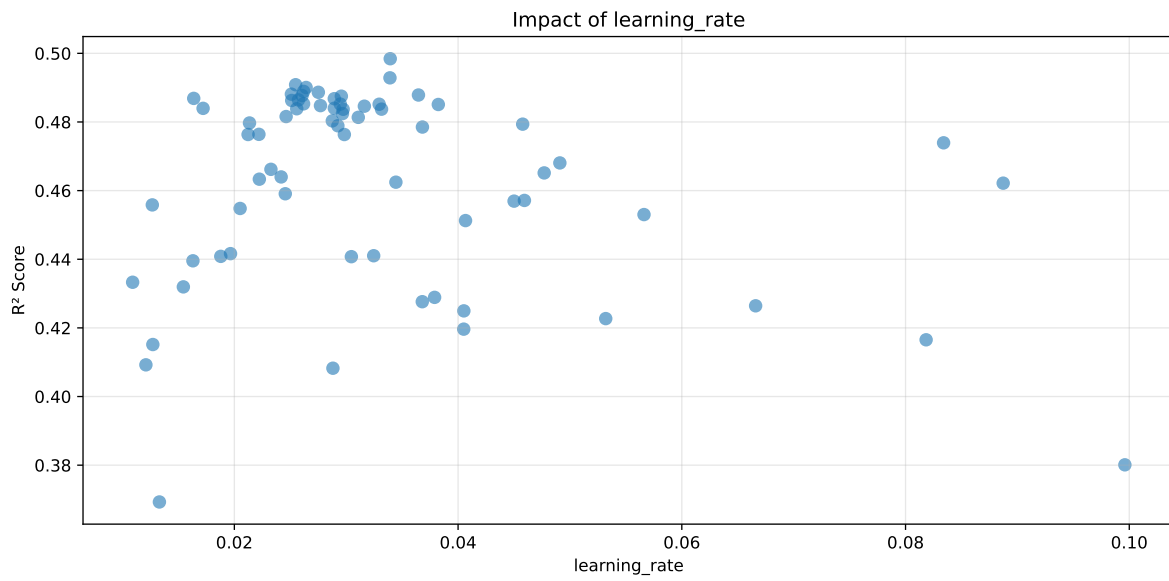
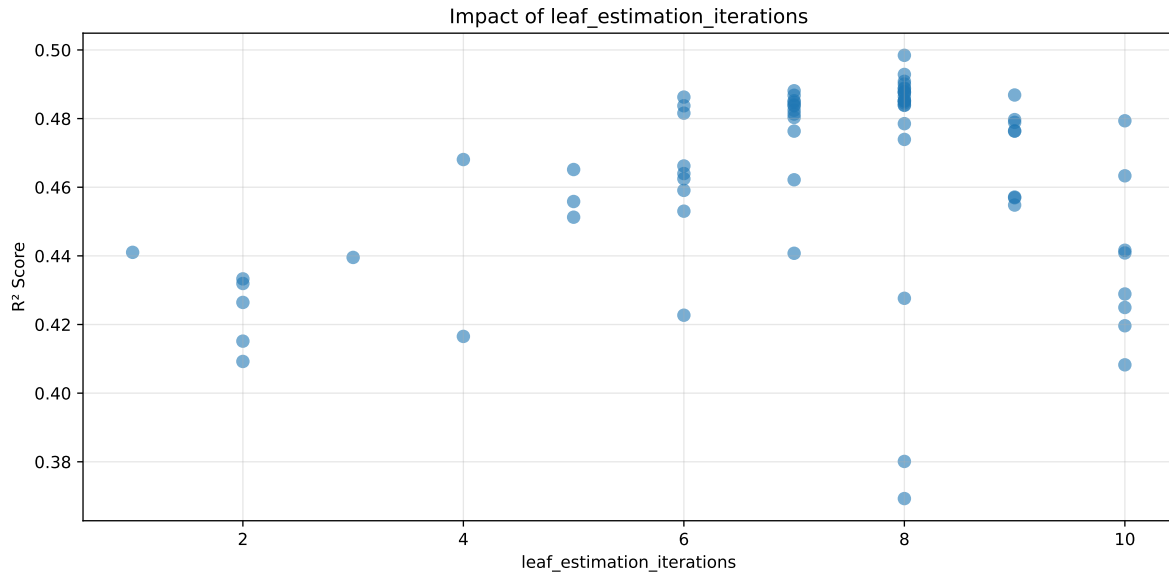
Distribution y_train vs y_test: y_train: mean=1.928, std=0.816, min=0.959, max=6.804
y_test: mean=1.833, std=0.729, min=1.098, max=7.822

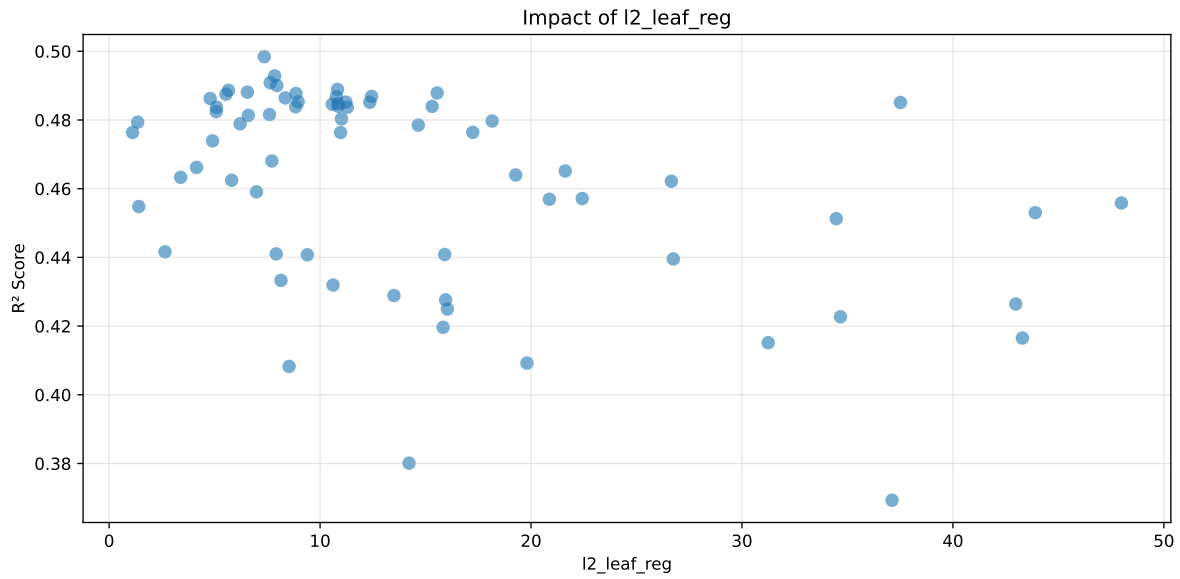
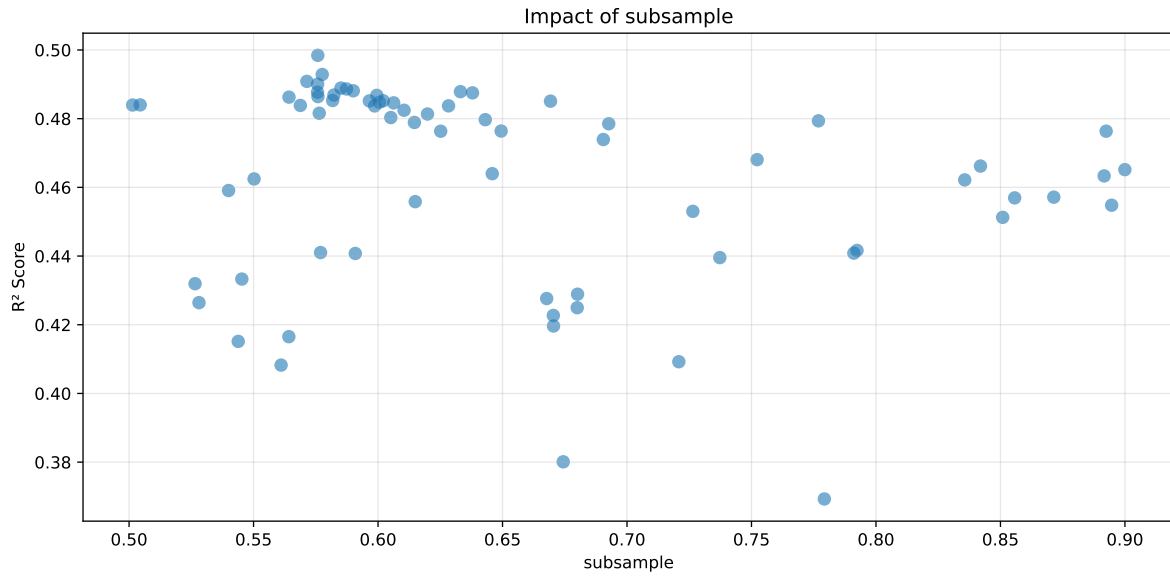
===== TOP Importance FEATURES =====
feature importance r_NH3 28.416292 r_THI 24.603932 r_umidity 23.312160 r_temperature
11.916169 r_CO2 11.751447



Paramètres avec >5% d'importance : ['depth', 'random_strength', 'leaf_estimation_iterations', 'learning_rate', 'subsample', 'l2_leaf_reg']

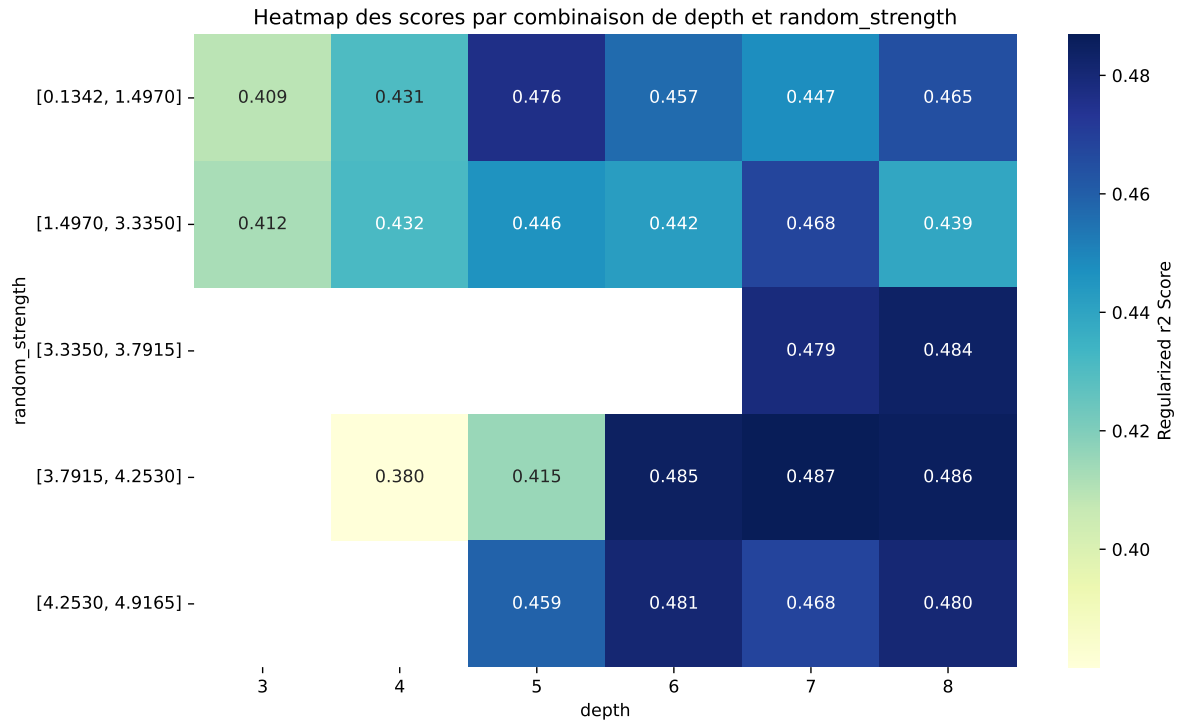






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The default value of observed=False is deprecated and will change to observed=True in a future



rank	mean_train_cv	mean_test_cv	test_score_cv	std_test_cv	penalized_score	test	diff	r_test	r_train
1	0.6183	0.3693	0.2191	0.0533	-0.3777	0.2593	-0.11	0.7022	1.6742
2	0.6845	0.4152	0.2501	0.0816	-0.3928	0.3011	-0.114	0.7253	1.6487
3	0.7337	0.4333	0.2519	0.0815	-0.468	0.3115	-0.1218	0.719	1.6933
4	0.8015	0.4591	0.2848	0.0832	-0.5682	0.3149	-0.1442	0.686	1.7459
5	0.7734	0.4165	0.263	0.1167	-0.654	0.3017	-0.1148	0.7244	1.8567
6	0.7723	0.4092	0.2019	0.108	-0.68	0.2705	-0.1387	0.6611	1.8872
7	0.856	0.464	0.2782	0.0945	-0.712	0.3404	-0.1236	0.7337	1.8448
8	0.8327	0.441	0.2773	0.1053	-0.734	0.3371	-0.104	0.7643	1.8881
9	0.8921	0.4816	0.289	0.1022	-0.75	0.3483	-0.1333	0.7232	1.8525
10	0.907	0.4909	0.2824	0.0981	-0.7576	0.341	-0.1498	0.6948	1.8478
---	---	---	---	---	---	---	---	---	---
70	0.9922	0.3801	0.206	0.1225	-1.4563	0.3602	-0.0199	0.9477	2.6104

Hyperparameters:

Rank 1: {'reg__iterations': 200, 'reg__depth': 8, 'reg__learning_rate': 0.013300724146281579, 'reg__l2_leaf_reg': 37.114117883667944, 'reg__bagging_temperature': 0.5169484787364492, 'reg__random_strength': 3.3005003913639857, 'reg__subsample': 0.7792632082083053, 'reg__min_data_in_leaf': 30, 'reg__leaf_estimation_iterations': 8}

Rank 2: {'reg__iterations': 700, 'reg__depth': 5, 'reg__learning_rate': 0.012713622461795646,

'reg__l2_leaf_reg': 31.24389206796481, 'reg__bagging_temperature': 3.430591574649947, 'reg__random_strength': 3.824863348115545, 'reg__subsample': 0.543838615928368, 'reg__min_data_in_leaf': 40, 'reg__leaf_estimation_iterations': 2}

Rank 3: {'reg__iterations': 500, 'reg__depth': 6, 'reg__learning_rate': 0.010907435031916265, 'reg__l2_leaf_reg': 8.147521890117401, 'reg__bagging_temperature': 0.5509817776282466, 'reg__random_strength': 3.045033742565032, 'reg__subsample': 0.5452951145420353, 'reg__min_data_in_leaf': 25, 'reg__leaf_estimation_iterations': 2}

Rank 4: {'reg__iterations': 200, 'reg__depth': 7, 'reg__learning_rate': 0.024553608410101704, 'reg__l2_leaf_reg': 6.986992000335173, 'reg__bagging_temperature': 1.1576762334721984, 'reg__random_strength': 3.7728821937240853, 'reg__subsample': 0.5399711878257502, 'reg__min_data_in_leaf': 40, 'reg__leaf_estimation_iterations': 6}

Rank 5: {'reg__iterations': 200, 'reg__depth': 3, 'reg__learning_rate': 0.08183942860307107, 'reg__l2_leaf_reg': 43.292478715182746, 'reg__bagging_temperature': 1.512140662422397, 'reg__random_strength': 2.7952201479896797, 'reg__subsample': 0.564182186012457, 'reg__min_data_in_leaf': 16, 'reg__leaf_estimation_iterations': 4}

Rank 6: {'reg__iterations': 1000, 'reg__depth': 3, 'reg__learning_rate': 0.012093253712849054, 'reg__l2_leaf_reg': 19.807487219343443, 'reg__bagging_temperature': 0.9365930628980862, 'reg__random_strength': 0.19964316355323142, 'reg__subsample': 0.7208173569442988, 'reg__min_data_in_leaf': 47, 'reg__leaf_estimation_iterations': 2}

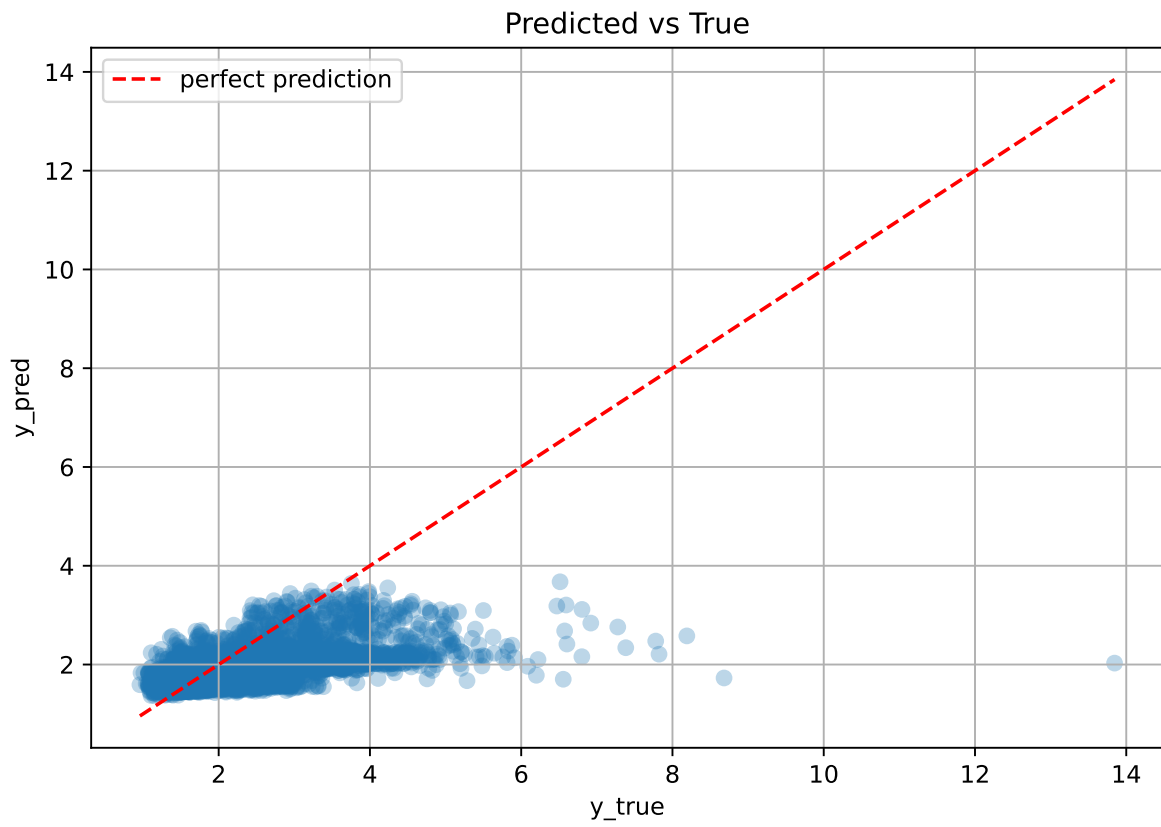
Rank 7: {'reg__iterations': 300, 'reg__depth': 7, 'reg__learning_rate': 0.024184126322391503, 'reg__l2_leaf_reg': 19.27222367990267, 'reg__bagging_temperature': 1.3815181924926607, 'reg__random_strength': 3.112790055196464, 'reg__subsample': 0.6458747382796449, 'reg__min_data_in_leaf': 27, 'reg__leaf_estimation_iterations': 6}

Rank 8: {'reg__iterations': 300, 'reg__depth': 7, 'reg__learning_rate': 0.03245550822977189, 'reg__l2_leaf_reg': 7.92310831461208, 'reg__bagging_temperature': 0.05616644346272526, 'reg__random_strength': 3.6286145251969617, 'reg__subsample': 0.576918980886432, 'reg__min_data_in_leaf': 38, 'reg__leaf_estimation_iterations': 1}

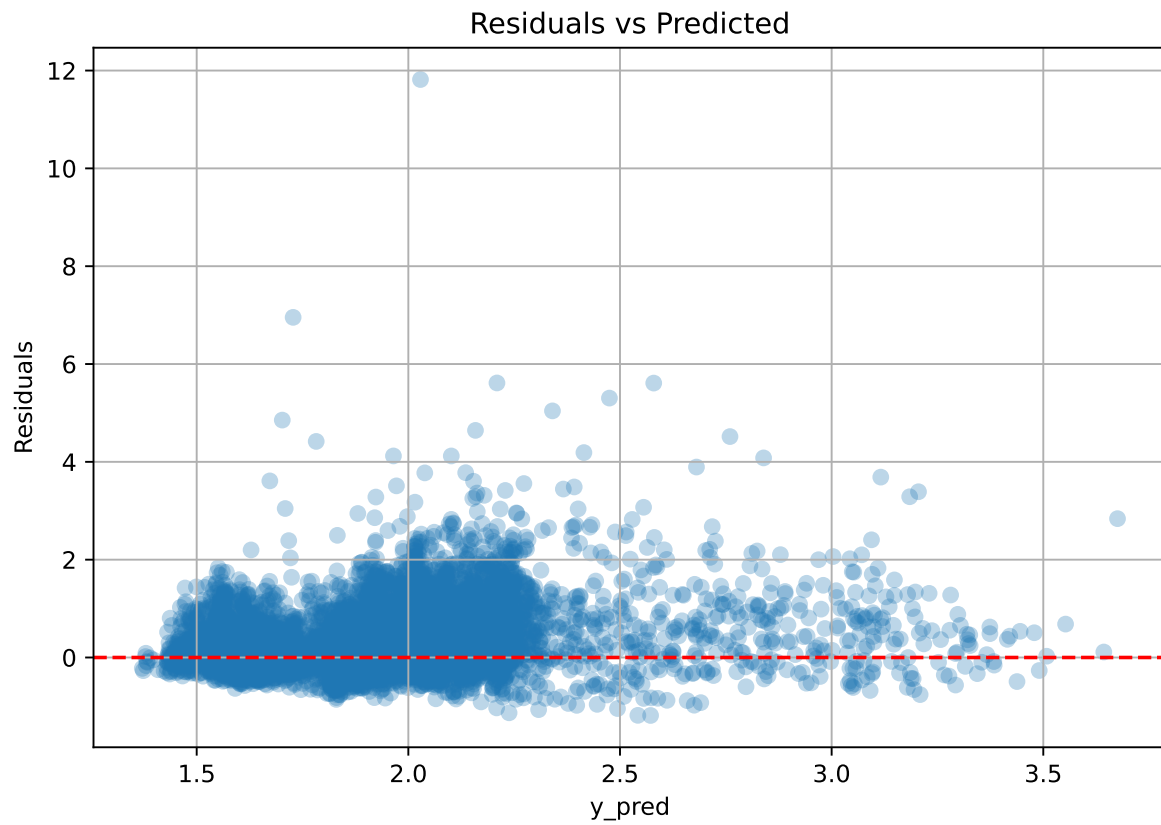
Rank 9: {'reg__iterations': 300, 'reg__depth': 7, 'reg__learning_rate': 0.024628073972209487, 'reg__l2_leaf_reg': 7.603612162737981, 'reg__bagging_temperature': 4.350800373303566, 'reg__random_strength': 3.5977739756212874, 'reg__subsample': 0.5763880002553572, 'reg__min_data_in_leaf': 38, 'reg__leaf_estimation_iterations': 6}

Rank 10: {'reg__iterations': 300, 'reg__depth': 7, 'reg__learning_rate': 0.02547768778479791, 'reg__l2_leaf_reg': 7.636859993191249, 'reg__bagging_temperature': 4.356462086644634, 'reg__random_strength': 3.5944168503002896, 'reg__subsample': 0.5714128488531178, 'reg__min_data_in_leaf': 38, 'reg__leaf_estimation_iterations': 8}

2.5.10 Scatter



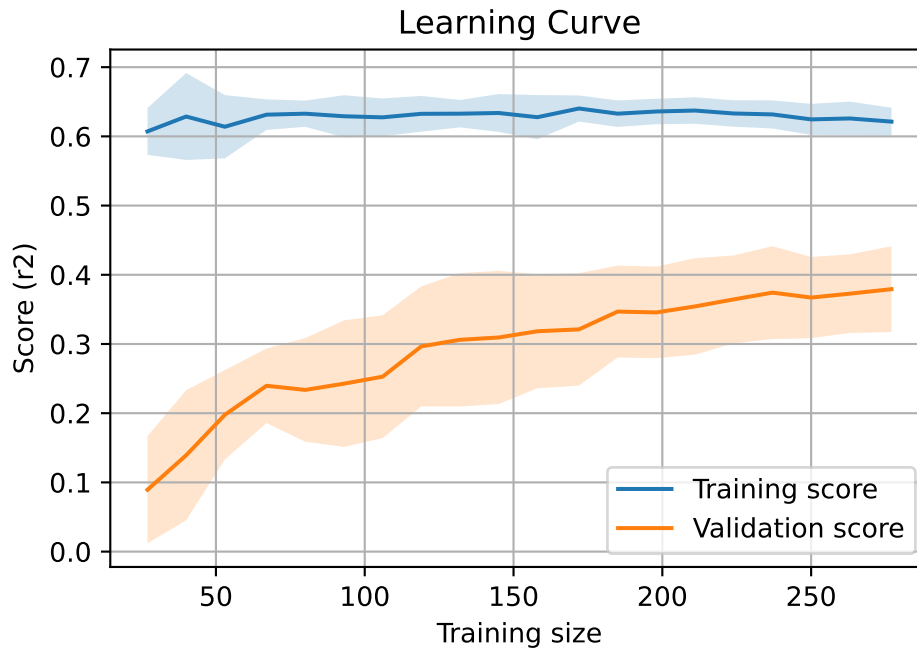
2.5.11 Scatter residuals



2.5.12 Learning curve — Group: 3

`Len(group)` : 568, `Len(training)` : 398,

`Len(training_real)` : 279, `Len(test_of_training)` : 119, `Len(test)` : 170

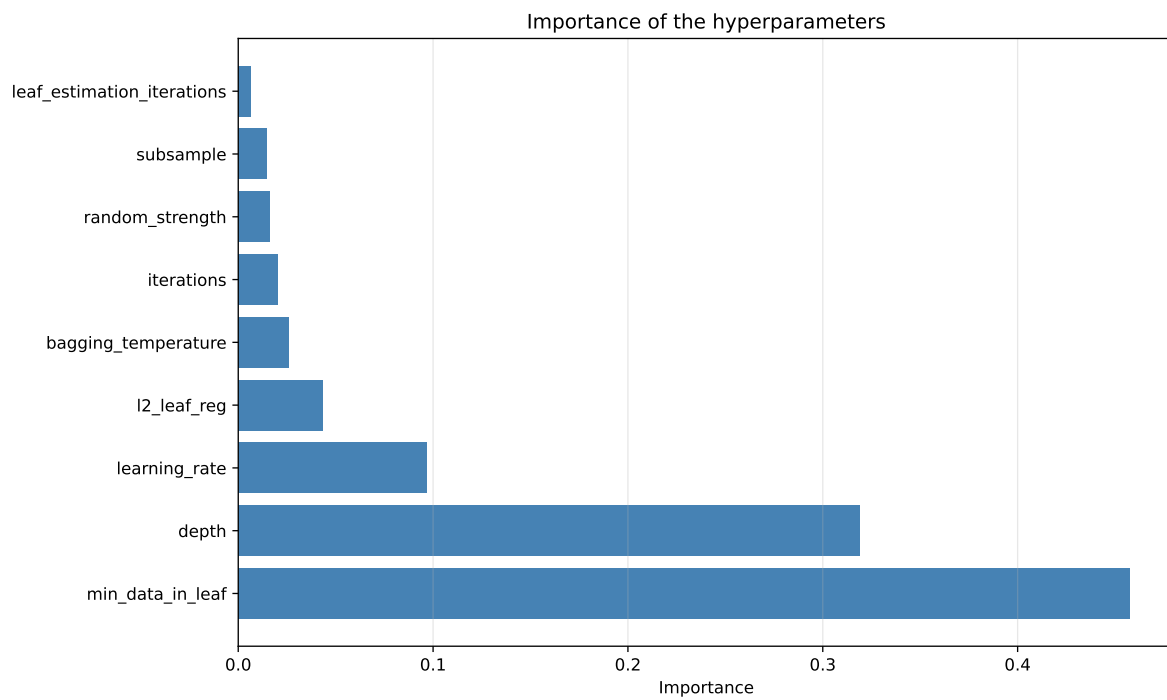
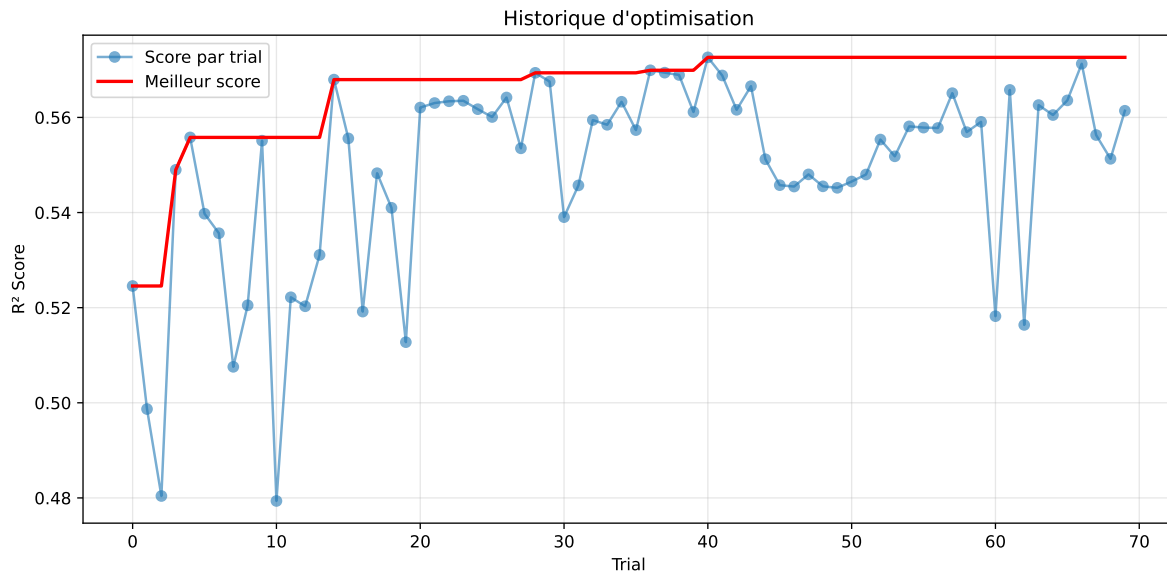


2.5.13 Gridsearch — Group: 4

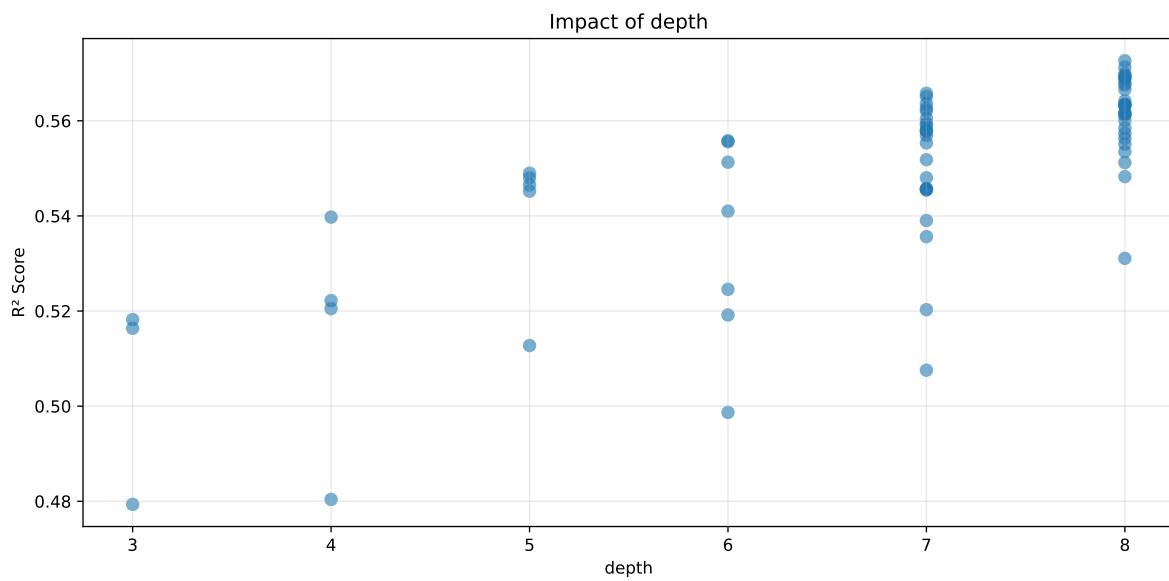
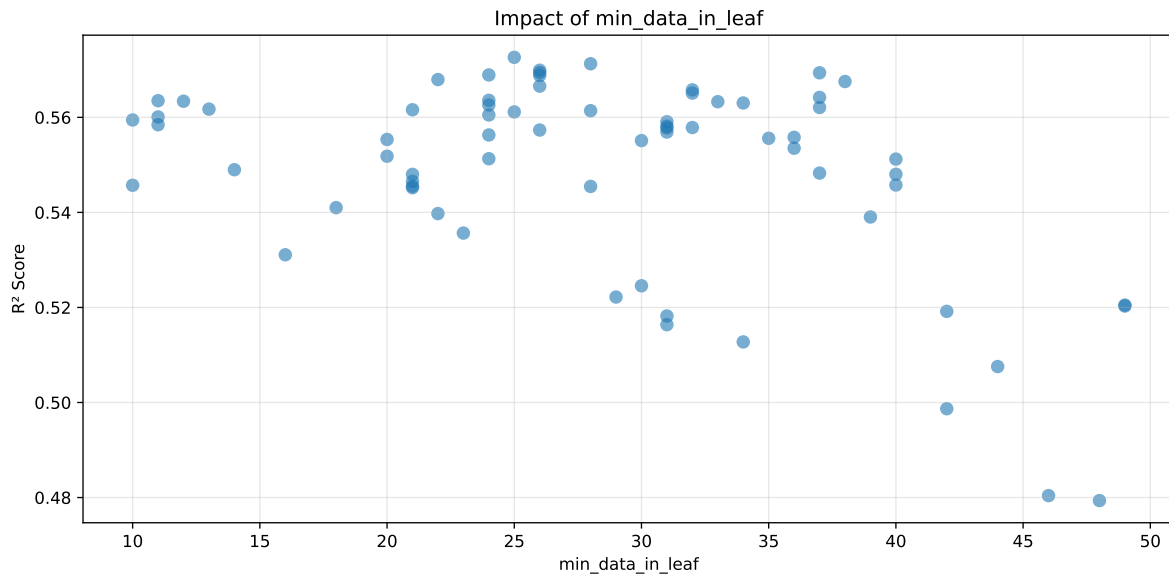
Len(group) : 2726, Len(training) : 1909, Len(test) : 817

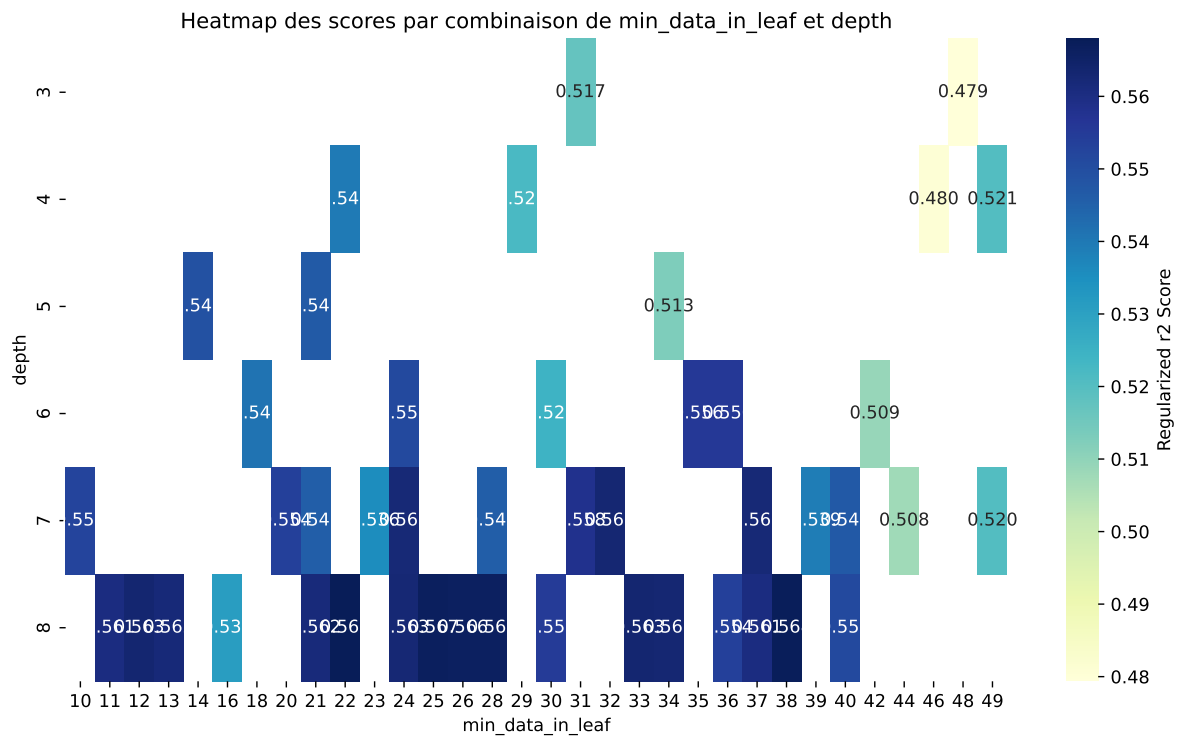
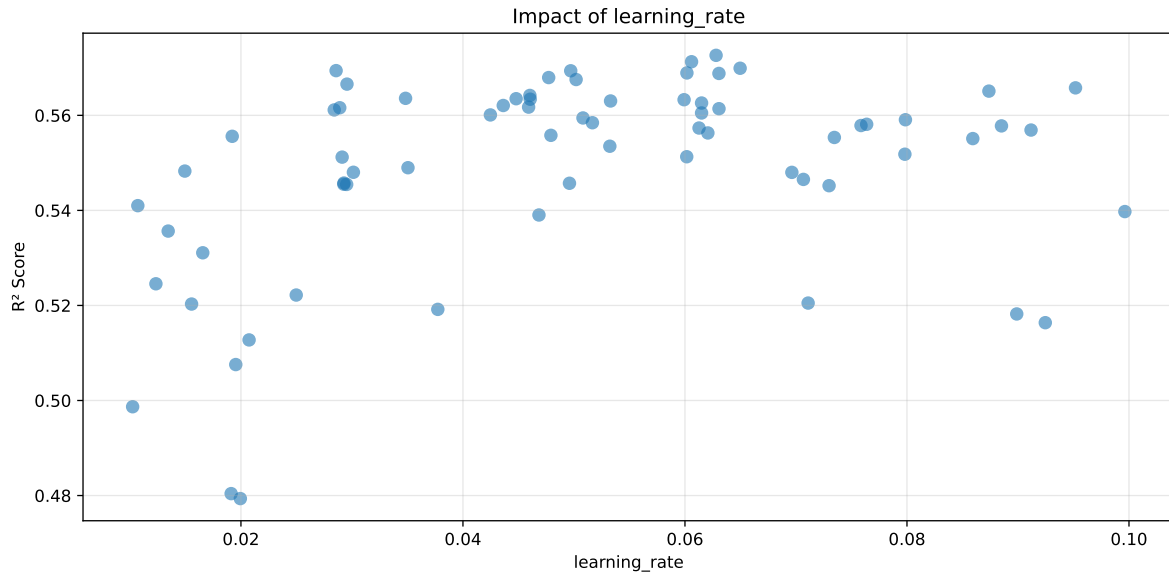
Distribution y_train vs y_test: y_train: mean=2.238, std=0.681, min=0.973, max=5.813
y_test: mean=2.212, std=0.659, min=1.071, max=4.858

===== TOP Importance FEATURES =====
feature importance r_umidity 51.873338 r_NH3 15.031610 r_temperature 13.254769 r_CO2
12.333820 r_THI 7.506463



Paramètres avec >5% d'importance : ['min_data_in_leaf', 'depth', 'learning_rate']





rank	mean_train_cv	mean_test_cv	test_score_cv	std_test_cv	penalized_score	test	diff	r_test	r_train
1	0.5376	0.4804	0.5457	0.0226	0.3087	0.5489	0.0685	1.1426	1.1192
2	0.5718	0.4987	0.5611	0.0232	0.2793	0.5688	0.0701	1.1406	1.1466
3	0.5865	0.5076	0.5683	0.0239	0.2708	0.5764	0.0689	1.1357	1.1555
4	0.5512	0.4794	0.5472	0.0259	0.264	0.5524	0.073	1.1524	1.1498
5	0.6018	0.5127	0.5725	0.0231	0.2456	0.5814	0.0686	1.1339	1.1737
6	0.6184	0.5203	0.5817	0.0208	0.2261	0.5926	0.0723	1.1389	1.1885
7	0.6185	0.5192	0.5792	0.0202	0.2212	0.5899	0.0708	1.1363	1.1913
8	0.6312	0.5246	0.5842	0.0226	0.2046	0.5951	0.0705	1.1344	1.2033
9	0.649	0.5311	0.5903	0.0195	0.1774	0.6035	0.0724	1.1363	1.222
10	0.6659	0.5356	0.597	0.0218	0.1448	0.6125	0.0769	1.1435	1.2432
---	---	---	---	---	---	---	---	---	---
70	0.9681	0.5535	0.6042	0.0295	-0.6903	0.6288	0.0753	1.136	1.7491

Hyperparameters:

Rank 1: {'reg__iterations': 300, 'reg__depth': 4, 'reg__learning_rate': 0.019118974360241518, 'reg__l2_leaf_reg': 4.086275587467578, 'reg__bagging_temperature': 4.760244067161578, 'reg__random_strength': 4.5922365772110805, 'reg__subsample': 0.584742178962738, 'reg__min_data_in_leaf': 46, 'reg__leaf_estimation_iterations': 10}

Rank 2: {'reg__iterations': 600, 'reg__depth': 6, 'reg__learning_rate': 0.01024311713208336, 'reg__l2_leaf_reg': 40.68863156651481, 'reg__bagging_temperature': 1.5568444267141328, 'reg__random_strength': 3.7033534641672237, 'reg__subsample': 0.8249711825661572, 'reg__min_data_in_leaf': 42, 'reg__leaf_estimation_iterations': 3}

Rank 3: {'reg__iterations': 300, 'reg__depth': 7, 'reg__learning_rate': 0.019533473955794134, 'reg__l2_leaf_reg': 36.780247353776254, 'reg__bagging_temperature': 3.9821278281466457, 'reg__random_strength': 3.6621392677173947, 'reg__subsample': 0.7038966966051566, 'reg__min_data_in_leaf': 44, 'reg__leaf_estimation_iterations': 3}

Rank 4: {'reg__iterations': 400, 'reg__depth': 3, 'reg__learning_rate': 0.019945523931558646, 'reg__l2_leaf_reg': 27.890787019661797, 'reg__bagging_temperature': 4.584169272530097, 'reg__random_strength': 1.2629980379432166, 'reg__subsample': 0.8538204139383028, 'reg__min_data_in_leaf': 48, 'reg__leaf_estimation_iterations': 2}

Rank 5: {'reg__iterations': 400, 'reg__depth': 5, 'reg__learning_rate': 0.020733949107563727, 'reg__l2_leaf_reg': 39.844754358753136, 'reg__bagging_temperature': 0.9934962864443336, 'reg__random_strength': 2.989050346572881, 'reg__subsample': 0.5338235149880033, 'reg__min_data_in_leaf': 34, 'reg__leaf_estimation_iterations': 10}

Rank 6: {'reg__iterations': 500, 'reg__depth': 7, 'reg__learning_rate': 0.015552987971021578, 'reg__l2_leaf_reg': 35.83042769094813, 'reg__bagging_temperature': 1.8832013367797455, 'reg__random_strength': 2.9672624247258557, 'reg__subsample': 0.5015260716349518, 'reg__min_data_in_leaf': 49, 'reg__leaf_estimation_iterations': 3}

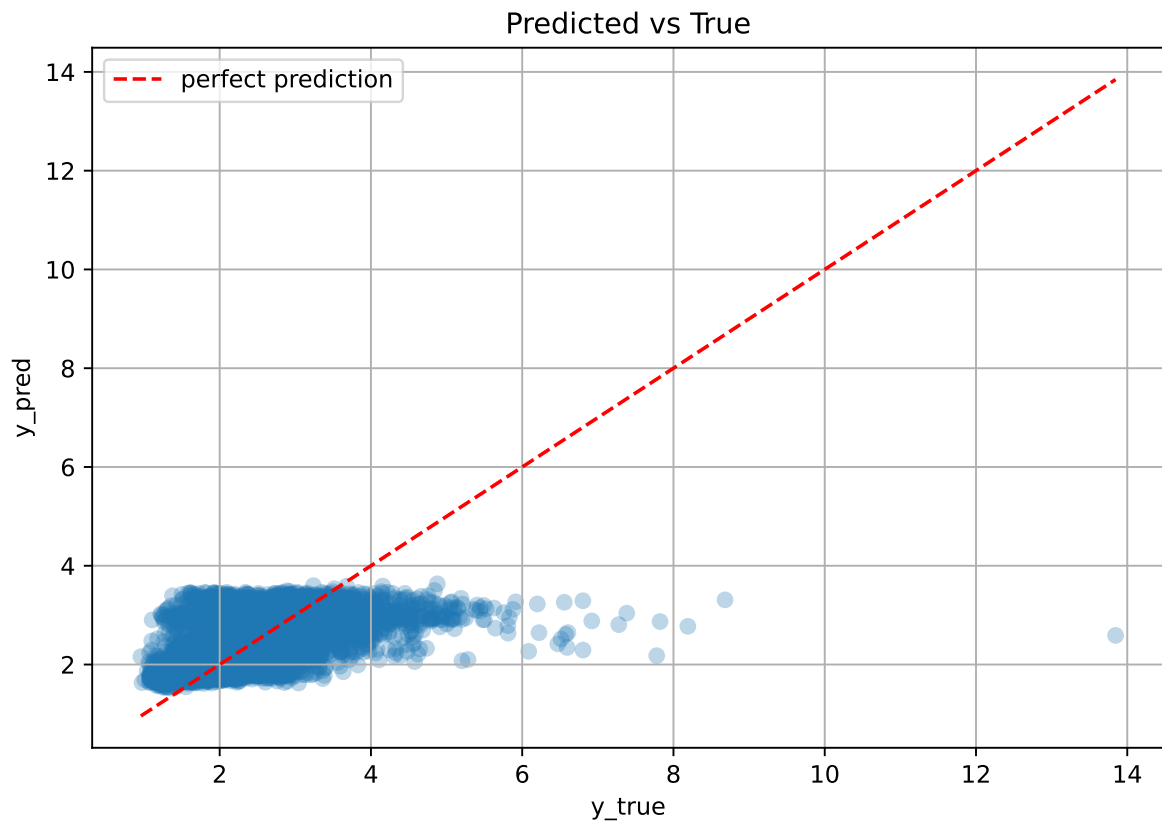
Rank 7: {'reg__iterations': 300, 'reg__depth': 6, 'reg__learning_rate': 0.037735378685962474, 'reg__l2_leaf_reg': 42.14326693759851, 'reg__bagging_temperature': 3.196674967154774, 'reg__random_strength': 3.766198612145714, 'reg__subsample': 0.610945697346306, 'reg__min_data_in_leaf': 42, 'reg__leaf_estimation_iterations': 2}

Rank 8: {'reg__iterations': 700, 'reg__depth': 6, 'reg__learning_rate': 0.012339764533360558, 'reg__l2_leaf_reg': 43.99897799590888, 'reg__bagging_temperature': 1.894034287675448, 'reg__random_strength': 2.6321032101760107, 'reg__subsample': 0.706405237809012, 'reg__min_data_in_leaf': 30, 'reg__leaf_estimation_iterations': 7}

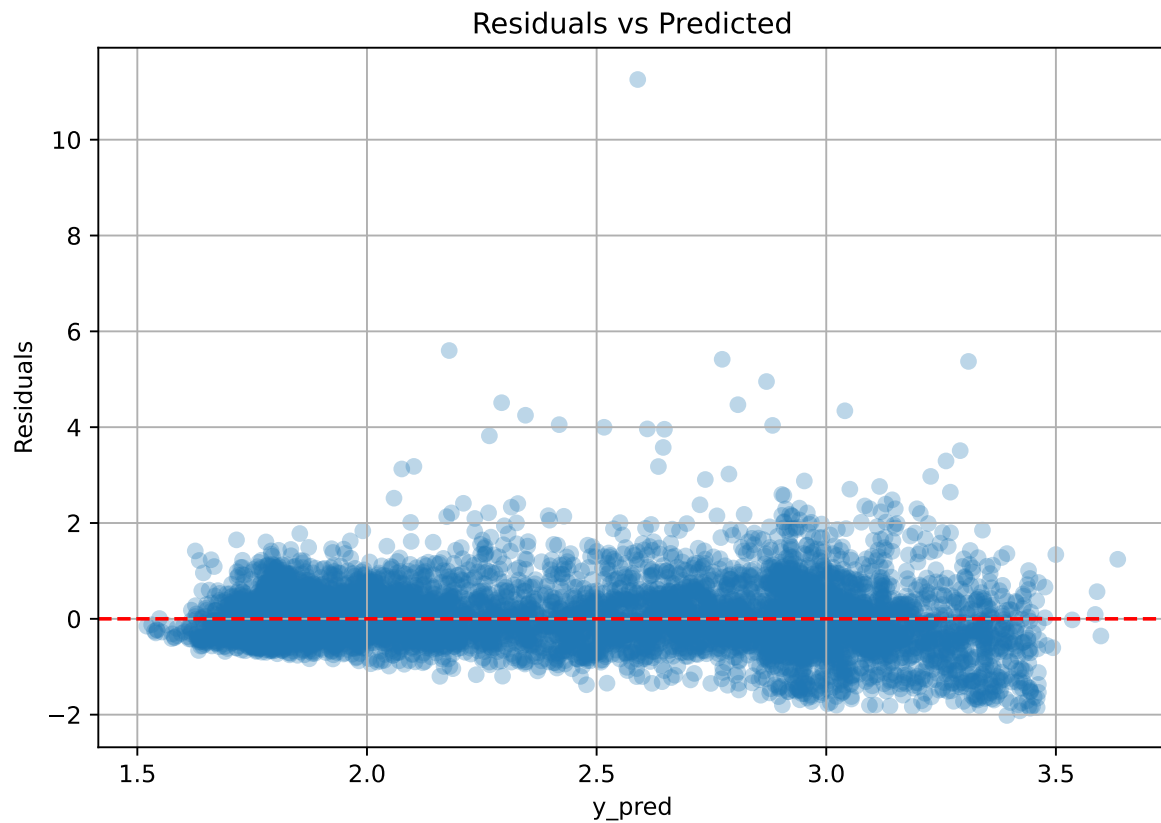
Rank 9: {'reg__iterations': 700, 'reg__depth': 8, 'reg__learning_rate': 0.016561037470580176, 'reg__l2_leaf_reg': 40.964096376067616, 'reg__bagging_temperature': 2.833288335330133, 'reg__random_strength': 4.88978489188793, 'reg__subsample': 0.8234219119437027, 'reg__min_data_in_leaf': 16, 'reg__leaf_estimation_iterations': 2}

Rank 10: {'reg__iterations': 800, 'reg__depth': 7, 'reg__learning_rate': 0.01344564244209363, 'reg__l2_leaf_reg': 42.20304317705934, 'reg__bagging_temperature': 3.074066553387137, 'reg__random_strength': 3.5026832028158883, 'reg__subsample': 0.7736508650154317, 'reg__min_data_in_leaf': 23, 'reg__leaf_estimation_iterations': 5}

2.5.14 Scatter



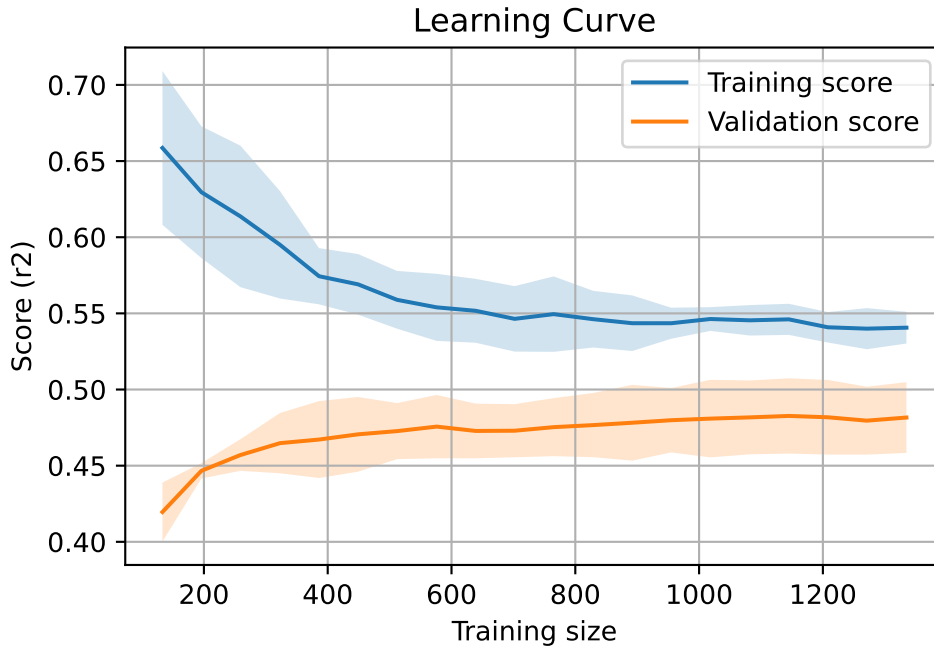
2.5.15 Scatter residuals



2.5.16 Learning curve — Group: 4

`Len(group)` : 2726, `Len(training)` : 1909,

`Len(training_real)` : 1337, `Len(test_of_training)` : 572, `Len(test)` : 817



2.5.17 Gridsearch — Group: 6

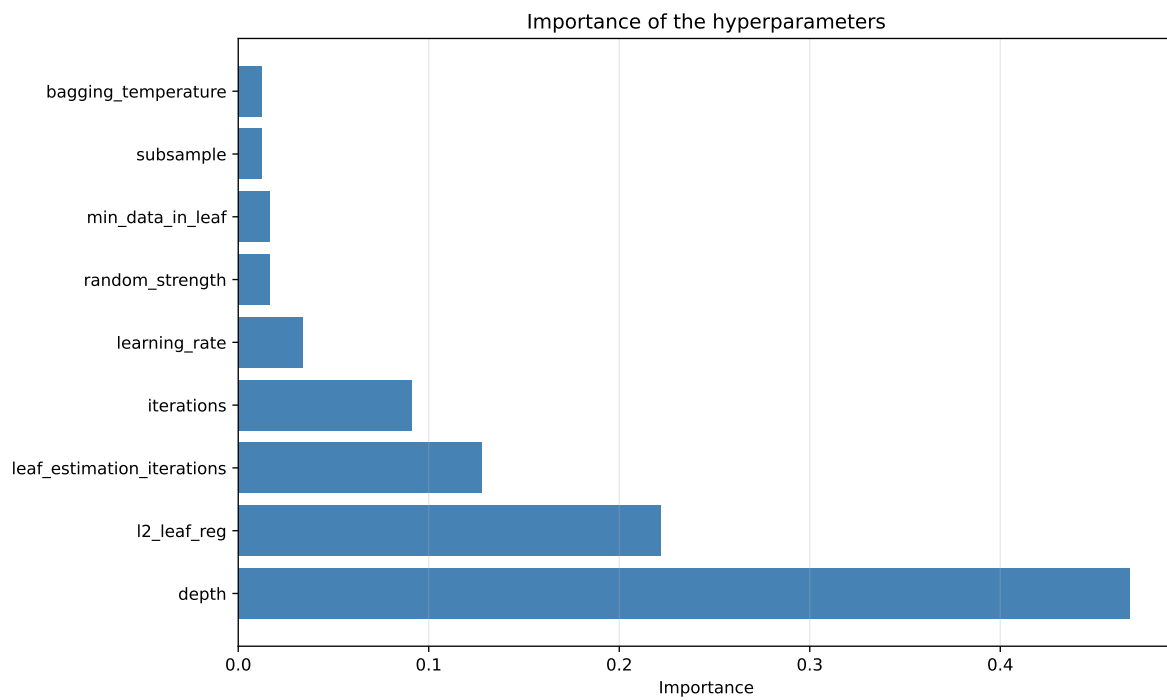
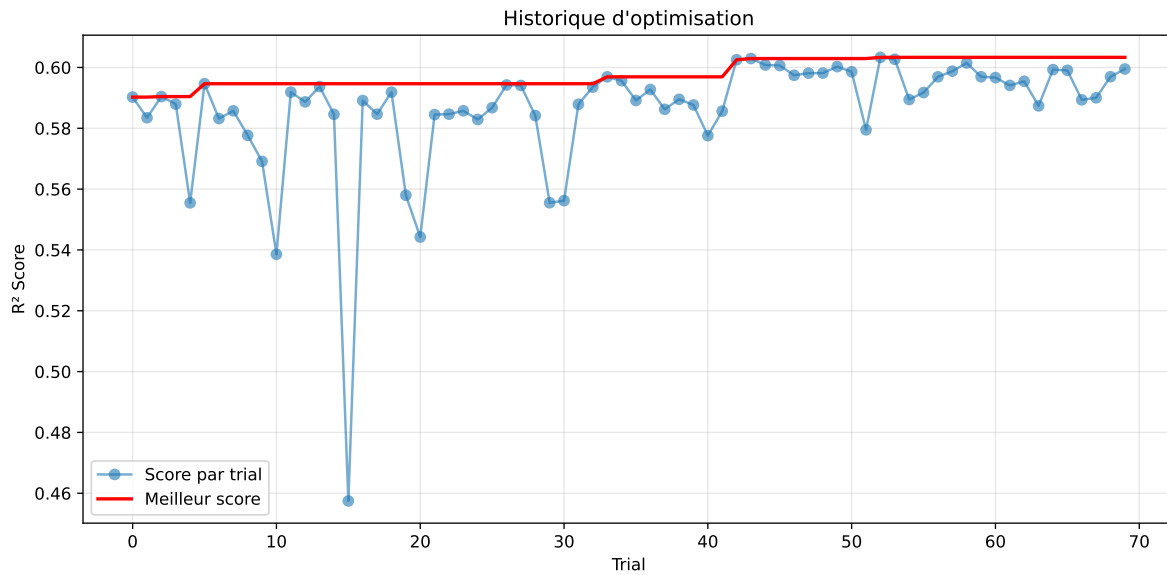
Len(group) : 1353, Len(training) : 948, Len(test) : 405

Distribution y_train vs y_test: y_train: mean=2.286, std=0.802, min=1.041, max=7.383
y_test: mean=2.187, std=0.733, min=1.078, max=5.525

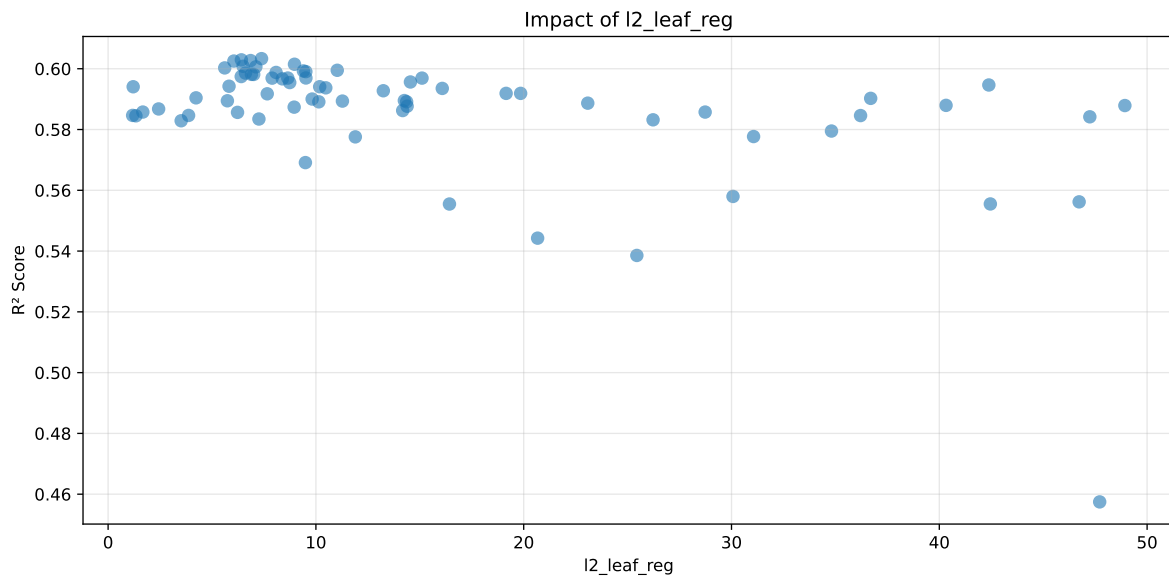
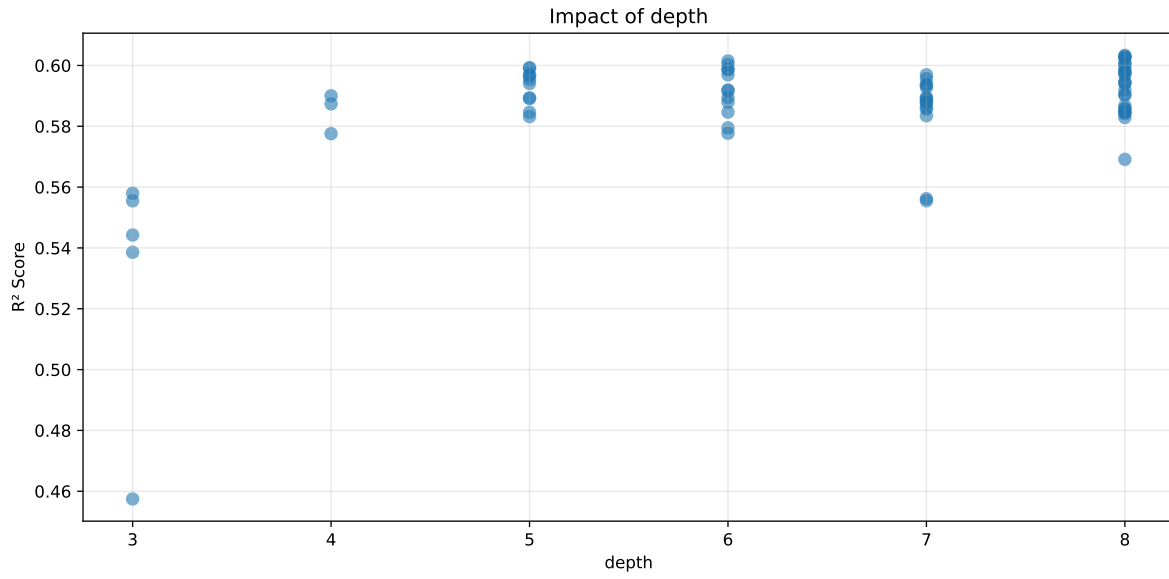
C:\Users\titou\anaconda3\Lib\site-packages\sklearn\utils_tags.py:338: DeprecationWarning:

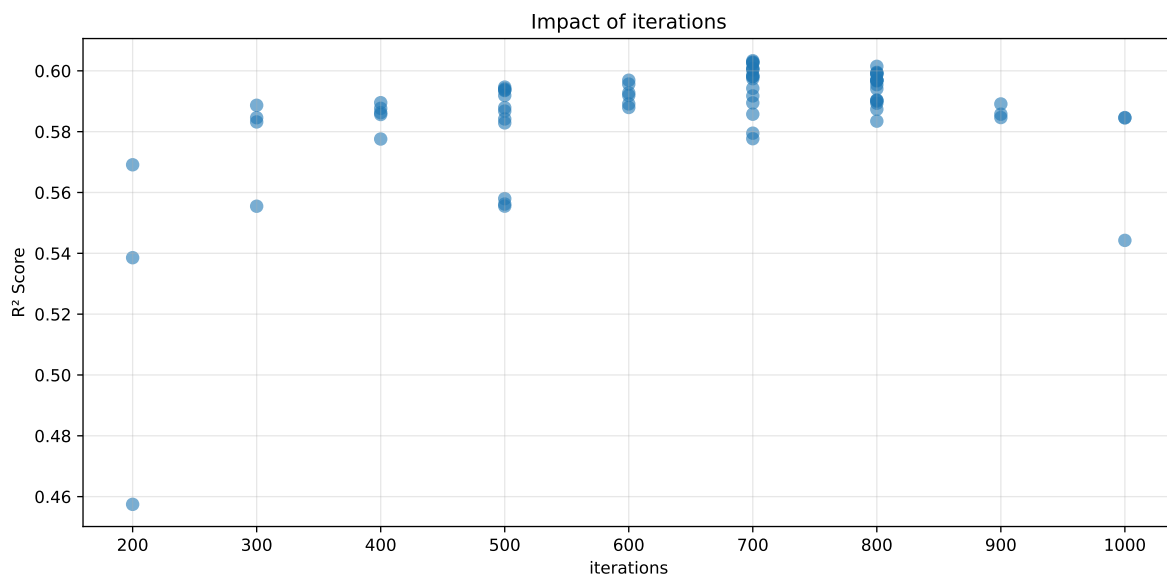
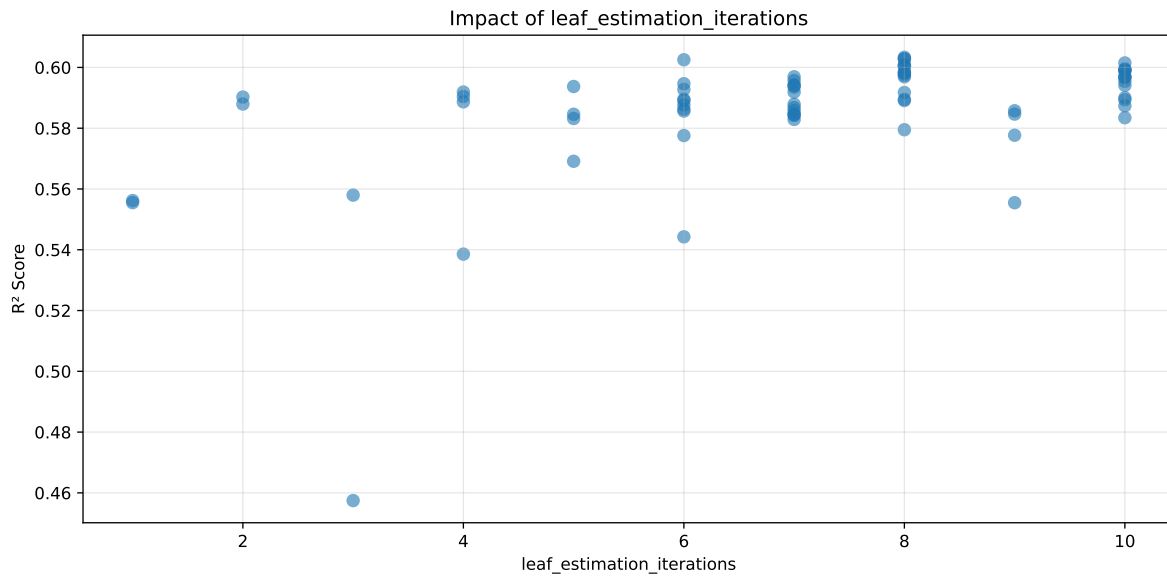
The following error was raised: 'CatBoostRegressor' object has no attribute '__sklearn_tags__'

===== TOP Importance FEATURES =====
feature importance r_umidity 61.510644 r_THI 13.075043 r_NH3 11.398913 r_temperature
7.719498 r_CO2 6.295902



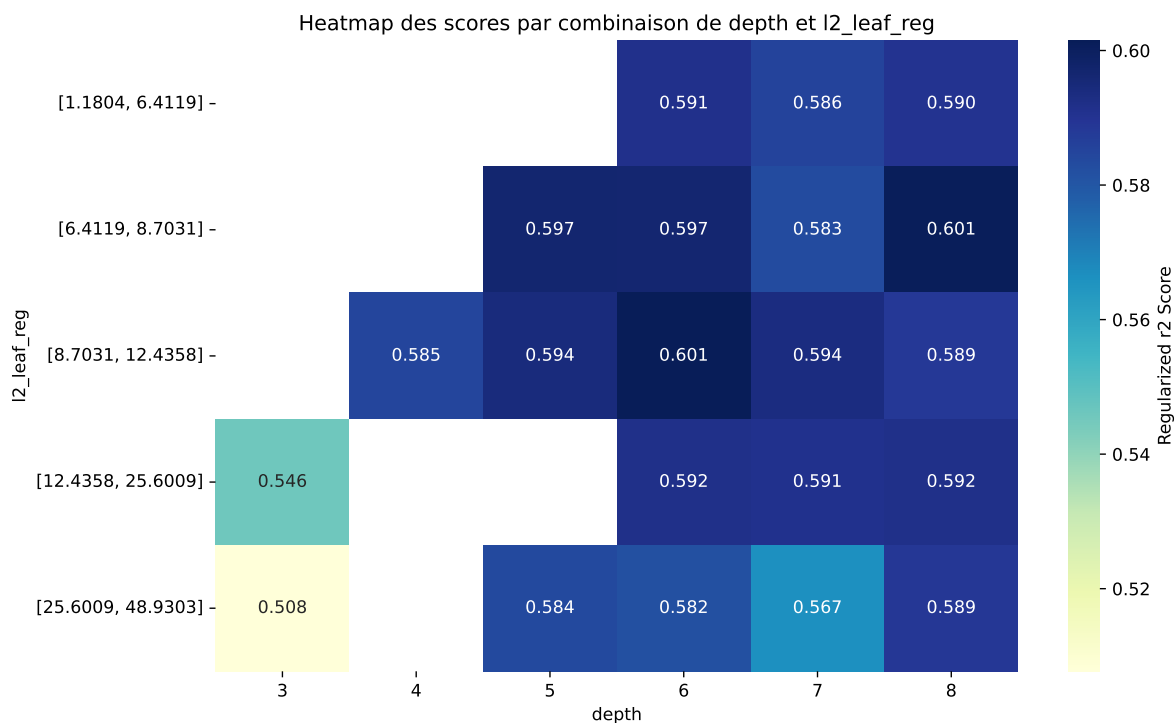
Paramètres avec >5% d'importance : ['depth', 'l2_leaf_reg', 'leaf_estimation_iterations', 'iterations']





C:\Users\titou\Desktop\UNICT\1- Gas sensor calibration\Python\machine_learning\workflow_main

The default value of observed=False is deprecated and will change to observed=True in a future



rank	mean_train_cv	mean_test_cv	test_score_cv	std_test_cv	penalized_score	test	diff	r_test	r_train
1	0.5094	0.4575	0.5682	0.06	0.3018	0.5724	0.1149	1.2512	1.1134
2	0.6828	0.5386	0.6838	0.0722	0.1058	0.6934	0.1549	1.2875	1.2678
3	0.6935	0.5443	0.6892	0.0714	0.0965	0.7004	0.1562	1.2869	1.2742
4	0.7524	0.5691	0.7162	0.0639	0.0194	0.7264	0.1573	1.2763	1.322
5	0.7409	0.5555	0.709	0.0686	-0.0006	0.7301	0.1746	1.3143	1.3337
6	0.7421	0.5562	0.7078	0.0678	-0.0016	0.7327	0.1765	1.3173	1.3343
7	0.7909	0.5795	0.7291	0.0691	-0.0548	0.7449	0.1654	1.2854	1.3648
8	0.7954	0.5777	0.7284	0.0678	-0.0755	0.7435	0.1659	1.2871	1.3769
9	0.8016	0.5776	0.715	0.0706	-0.0943	0.7313	0.1537	1.2662	1.3878
10	0.8099	0.5835	0.731	0.0649	-0.096	0.741	0.1576	1.27	1.3882
---	---	---	---	---	---	---	---	---	---
70	1.0	0.5845	0.7083	0.0541	-0.6619	0.74	0.1554	1.2659	1.7108

Hyperparameters:

Rank 1: {'reg__iterations': 200, 'reg__depth': 3, 'reg__learning_rate': 0.01962824768261938, 'reg__l2_leaf_reg': 47.71796144659072, 'reg__bagging_temperature': 4.663510089783525, 'reg__random_strength': 2.244166614899627, 'reg__subsample': 0.7279878937783963, 'reg__min_data_in_leaf': 31, 'reg__leaf_estimation_iterations': 3}

Rank 2: {'reg__iterations': 200, 'reg__depth': 3, 'reg__learning_rate': 0.04738094030796166,

'reg__l2_leaf_reg': 25.444892975456682, 'reg__bagging_temperature': 2.311377486024384, 'reg__random_strength': 1.343340839909325, 'reg__subsample': 0.5599944919797244, 'reg__min_data_in_leaf': 25, 'reg__leaf_estimation_iterations': 4}

Rank 3: {'reg__iterations': 1000, 'reg__depth': 3, 'reg__learning_rate': 0.011488479121498571, 'reg__l2_leaf_reg': 20.66994695931021, 'reg__bagging_temperature': 4.157548895829988, 'reg__random_strength': 3.827017234011874, 'reg__subsample': 0.8904449096371192, 'reg__min_data_in_leaf': 33, 'reg__leaf_estimation_iterations': 6}

Rank 4: {'reg__iterations': 200, 'reg__depth': 8, 'reg__learning_rate': 0.025716635650485015, 'reg__l2_leaf_reg': 9.494855796082977, 'reg__bagging_temperature': 1.120061068906587, 'reg__random_strength': 3.698546077775074, 'reg__subsample': 0.5497601292459815, 'reg__min_data_in_leaf': 11, 'reg__leaf_estimation_iterations': 5}

Rank 5: {'reg__iterations': 500, 'reg__depth': 7, 'reg__learning_rate': 0.030457030263677288, 'reg__l2_leaf_reg': 42.45627724922543, 'reg__bagging_temperature': 1.708543792703207, 'reg__random_strength': 1.8604461714108296, 'reg__subsample': 0.607173033416835, 'reg__min_data_in_leaf': 24, 'reg__leaf_estimation_iterations': 1}

Rank 6: {'reg__iterations': 500, 'reg__depth': 7, 'reg__learning_rate': 0.031750612992533324, 'reg__l2_leaf_reg': 46.7292944500129, 'reg__bagging_temperature': 1.623503999908889, 'reg__random_strength': 1.575933329733354, 'reg__subsample': 0.5014297245699095, 'reg__min_data_in_leaf': 25, 'reg__leaf_estimation_iterations': 1}

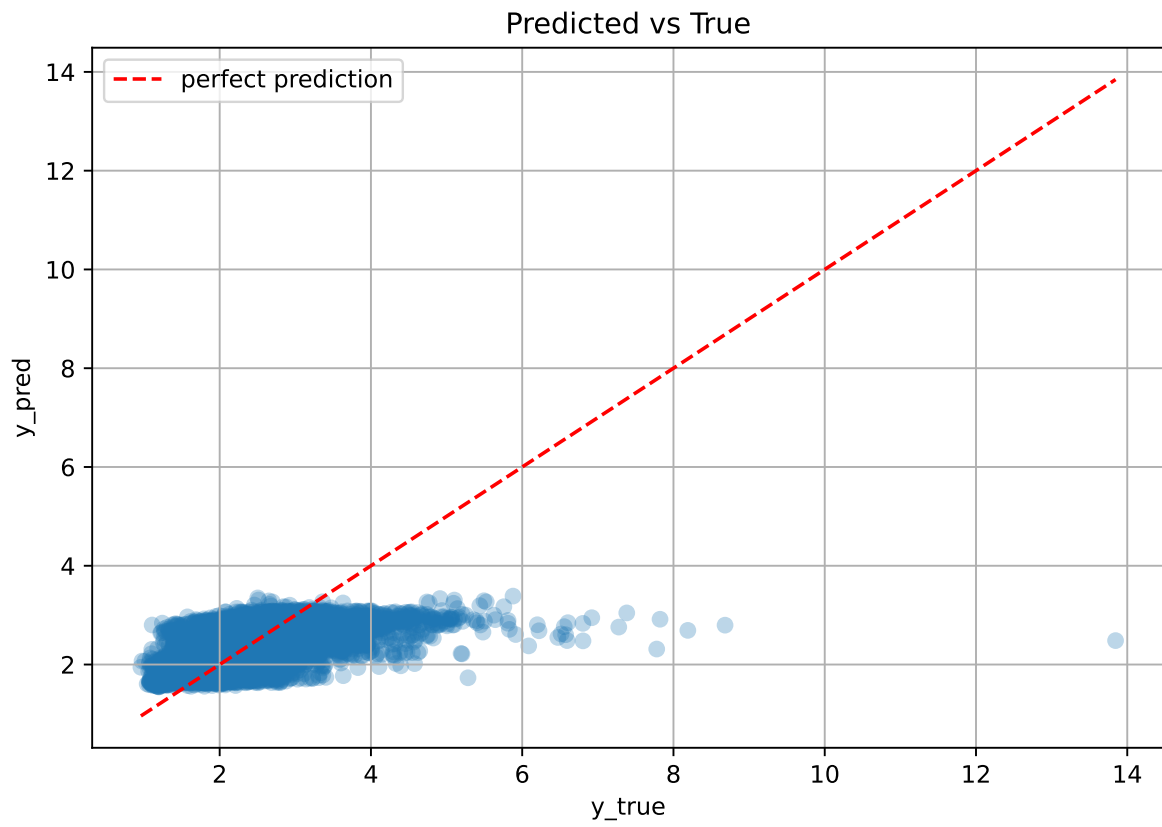
Rank 7: {'reg__iterations': 700, 'reg__depth': 6, 'reg__learning_rate': 0.015974059955237385, 'reg__l2_leaf_reg': 34.81165702307784, 'reg__bagging_temperature': 2.05579552696313, 'reg__random_strength': 4.018931300228304, 'reg__subsample': 0.80798305754077, 'reg__min_data_in_leaf': 19, 'reg__leaf_estimation_iterations': 8}

Rank 8: {'reg__iterations': 700, 'reg__depth': 6, 'reg__learning_rate': 0.014163876060575344, 'reg__l2_leaf_reg': 31.06224078911519, 'reg__bagging_temperature': 2.0740308931468823, 'reg__random_strength': 2.404637362035654, 'reg__subsample': 0.6681091221508337, 'reg__min_data_in_leaf': 49, 'reg__leaf_estimation_iterations': 9}

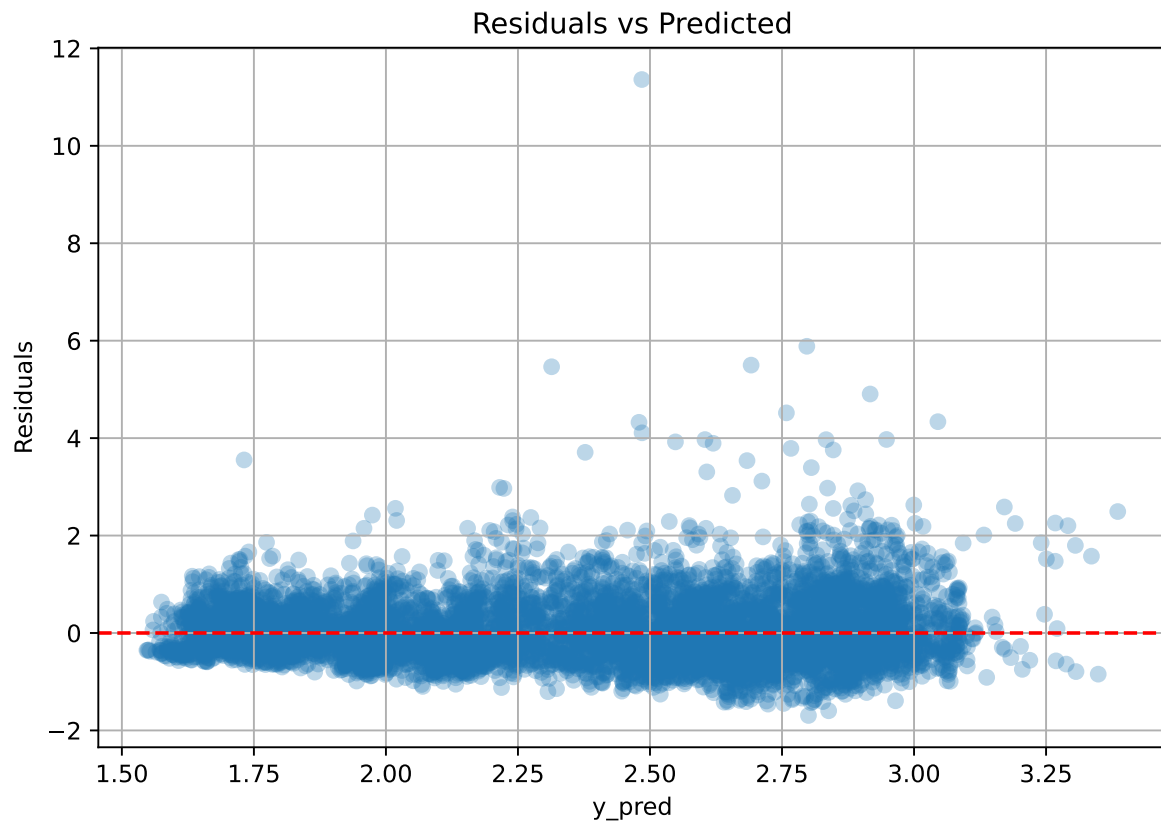
Rank 9: {'reg__iterations': 400, 'reg__depth': 4, 'reg__learning_rate': 0.03767099790217421, 'reg__l2_leaf_reg': 11.89693125104451, 'reg__bagging_temperature': 2.283885768370716, 'reg__random_strength': 2.9049707225617873, 'reg__subsample': 0.5597119663676922, 'reg__min_data_in_leaf': 35, 'reg__leaf_estimation_iterations': 6}

Rank 10: {'reg__iterations': 800, 'reg__depth': 7, 'reg__learning_rate': 0.010192116912999576, 'reg__l2_leaf_reg': 7.257294986639443, 'reg__bagging_temperature': 4.677995355065571, 'reg__random_strength': 4.626625585425949, 'reg__subsample': 0.8860076997022583, 'reg__min_data_in_leaf': 15, 'reg__leaf_estimation_iterations': 10}

2.5.18 Scatter



2.5.19 Scatter residuals



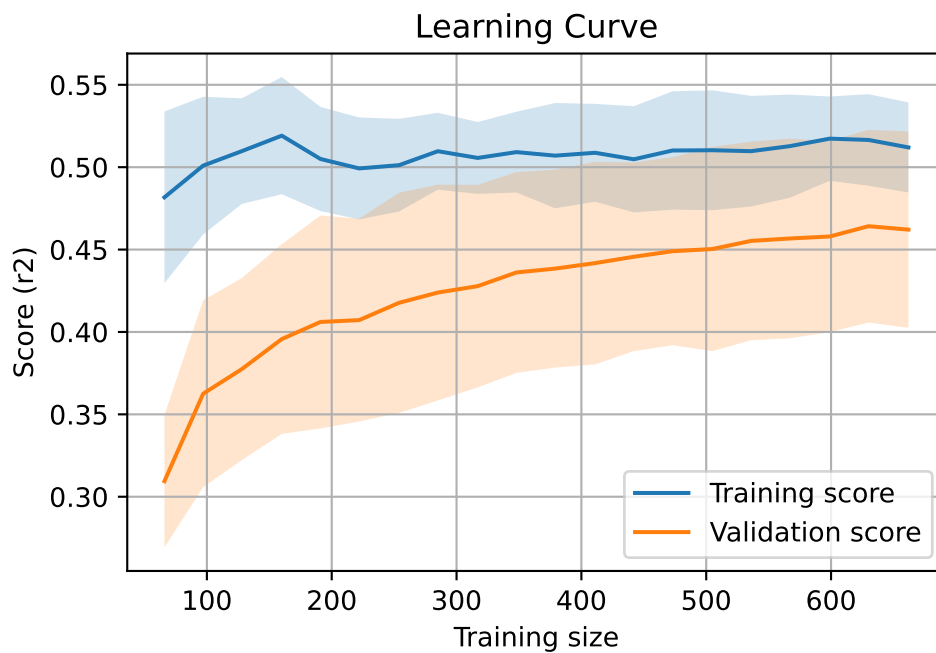
2.5.20 Learning curve — Group: 6

`Len(group)` : 1353, `Len(training)` : 948,

`Len(training_real)` : 664, `Len(test_of_training)` : 284, `Len(test)` : 405

C:\Users\titou\anaconda3\Lib\site-packages\joblib\externals\loky\process_executor.py:782: UserWarning:

A worker stopped while some jobs were given to the executor. This can be caused by a too short timeout or by a too high number of workers.



2.6 Sumup-table

model	group	Len training	Len test	Len total	mean_train_cv	mean_test_cv	std_test_cv	test_cv	test	diff	r_test	r_train	pen_score
Linear	1	557	238	795	0.563	0.551	0.044	0.498	0.5	-0.05	0.908	1.022	nan
Linear	2	1909	817	2726	0.463	0.463	0.017	0.394	0.394	-0.069	0.852	1.0	nan
Linear	3	398	170	568	0.307	0.248	0.036	0.187	0.203	-0.045	0.818	1.241	nan
Linear	4	1909	817	2726	0.431	0.404	0.026	0.457	0.457	0.054	1.133	1.069	nan
Linear	6	948	405	1353	0.361	0.378	0.046	0.413	0.412	0.035	1.092	0.956	nan
Linear	MEAN	5721	2447	8168	0.425	0.409	0.034	0.39	0.393	-0.015	0.961	1.058	nan
LGBM	1	557	238	795	0.611	0.572	0.043	0.54	0.557	-0.015	0.974	1.068	0.4555
LGBM	2	1909	817	2726	0.676	0.648	0.013	0.549	0.563	-0.086	0.868	1.043	0.6484
LGBM	3	398	170	568	0.265	0.23	0.038	0.094	0.133	-0.097	0.577	1.148	0.1278
LGBM	4	1909	817	2726	0.462	0.42	0.026	0.478	0.498	0.078	1.186	1.101	0.2931
LGBM	6	948	405	1353	0.408	0.376	0.053	0.451	0.486	0.11	1.291	1.084	0.282
LGBM	MEAN	5721	2447	8168	0.484	0.449	0.035	0.423	0.447	-0.002	0.979	1.089	0.361
CatBoost	1	557	238	795	0.608	0.563	0.043	0.524	0.555	-0.008	0.986	1.078	0.4307
CatBoost	2	1909	817	2726	0.733	0.699	0.017	0.601	0.604	-0.094	0.865	1.05	0.6986
CatBoost	3	398	170	568	0.618	0.369	0.053	0.219	0.259	-0.11	0.702	1.674	-0.3777
CatBoost	4	1909	817	2726	0.538	0.48	0.023	0.546	0.549	0.068	1.143	1.119	0.3087
CatBoost	6	948	405	1353	0.509	0.458	0.06	0.568	0.572	0.115	1.251	1.113	0.3018
CatBoost	MEAN	5721	2447	8168	0.601	0.514	0.039	0.492	0.508	-0.006	0.989	1.207	0.272

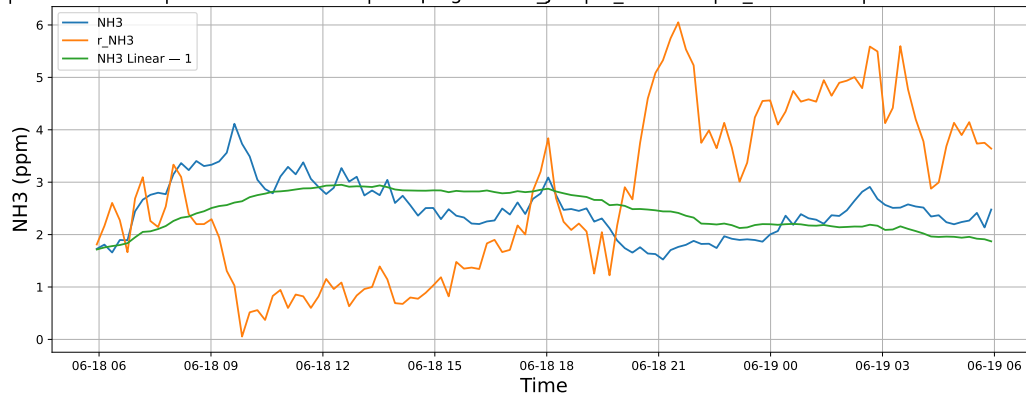
3 Plot

3.1 Standalone plots

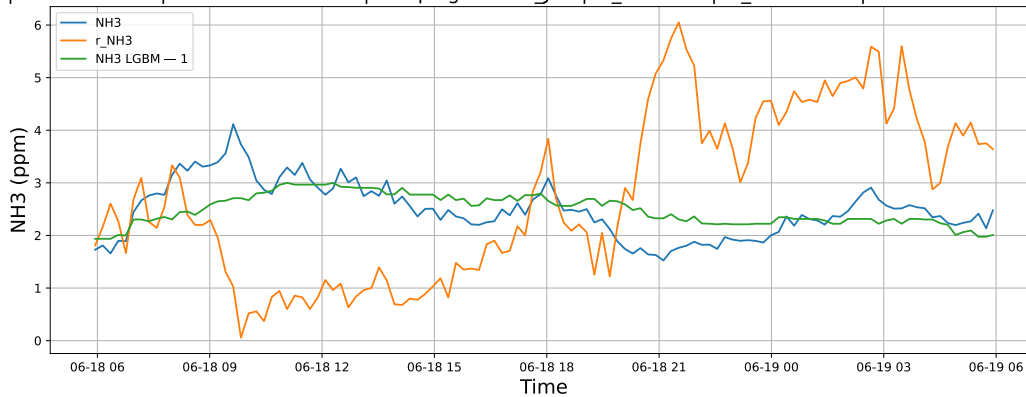
3.1.1 Group: campaign: 2025_Jun,id_rabbit: 1,id_channel: 3

3.1.1.1 Time window : 24

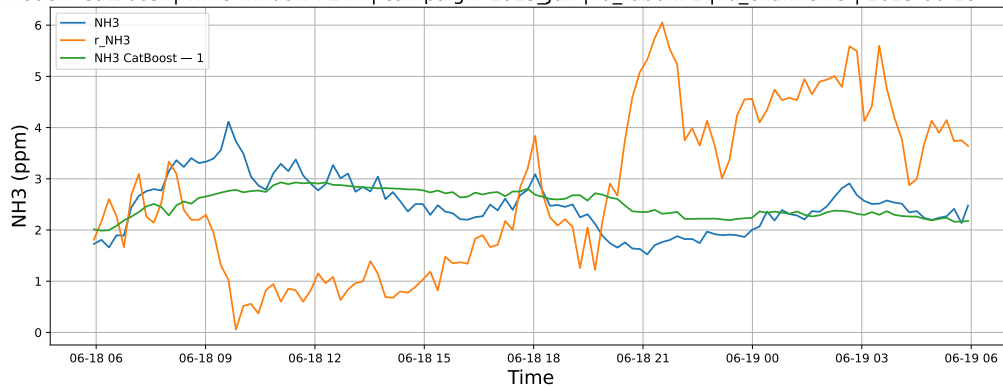
NH3 | Model : Linear | Time window : 24h | campaign: 2025_Jun | id_rabbit: 1 | id_channel: 3 | 2025-06-18 → 2025-06-19



NH3 | Model : LGBM | Time window : 24h | campaign: 2025_Jun | id_rabbit: 1 | id_channel: 3 | 2025-06-18 → 2025-06-19

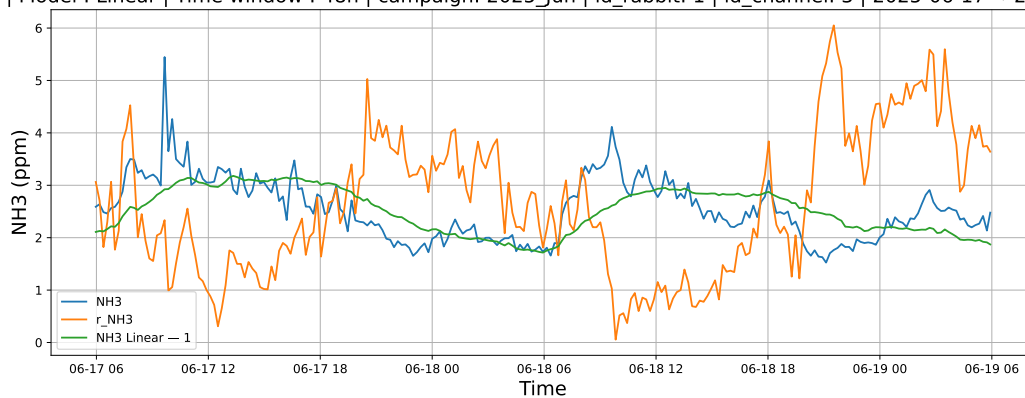


NH3 | Model : CatBoost | Time window : 24h | campaign: 2025_Jun | id_rabbit: 1 | id_channel: 3 | 2025-06-18 → 2025-06-19

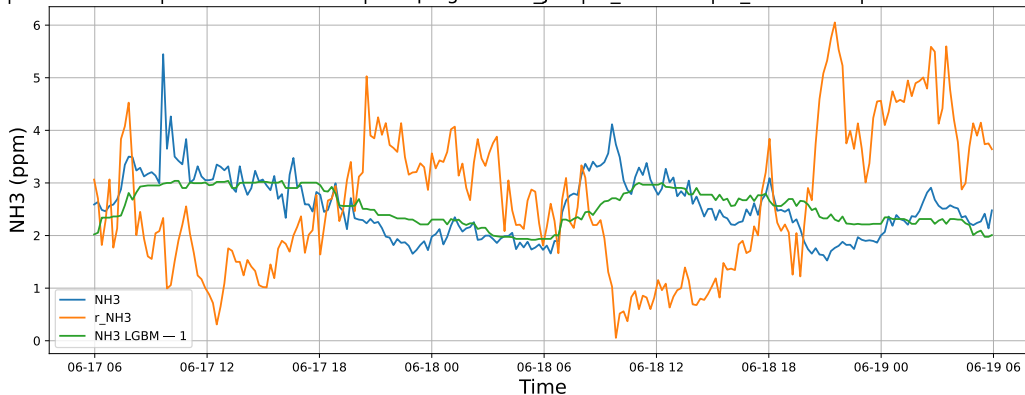


3.1.1.2 Time window : 48

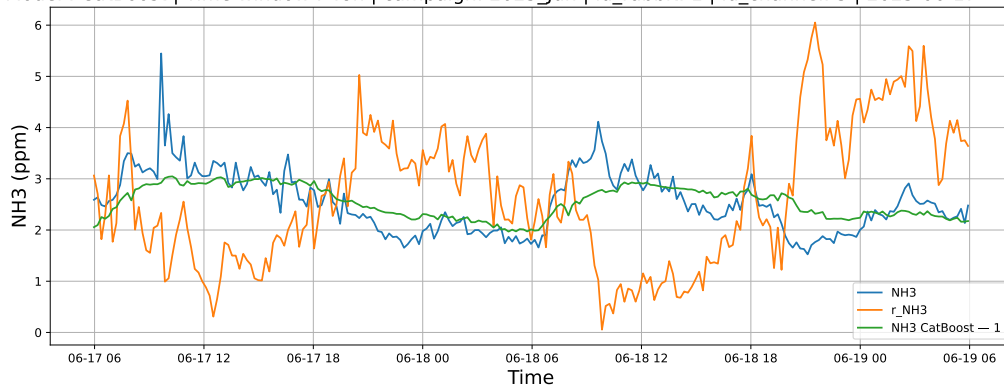
NH3 | Model : Linear | Time window : 48h | campaign: 2025_Jun | id_rabbit: 1 | id_channel: 3 | 2025-06-17 → 2025-06-19



NH3 | Model : LGBM | Time window : 48h | campaign: 2025_Jun | id_rabbit: 1 | id_channel: 3 | 2025-06-17 → 2025-06-19

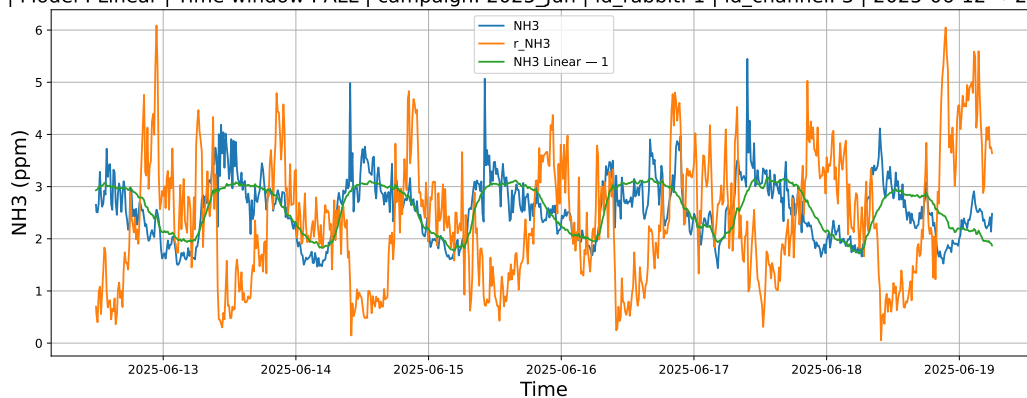


NH3 | Model : CatBoost | Time window : 48h | campaign: 2025_Jun | id_rabbit: 1 | id_channel: 3 | 2025-06-17 → 2025-06-19

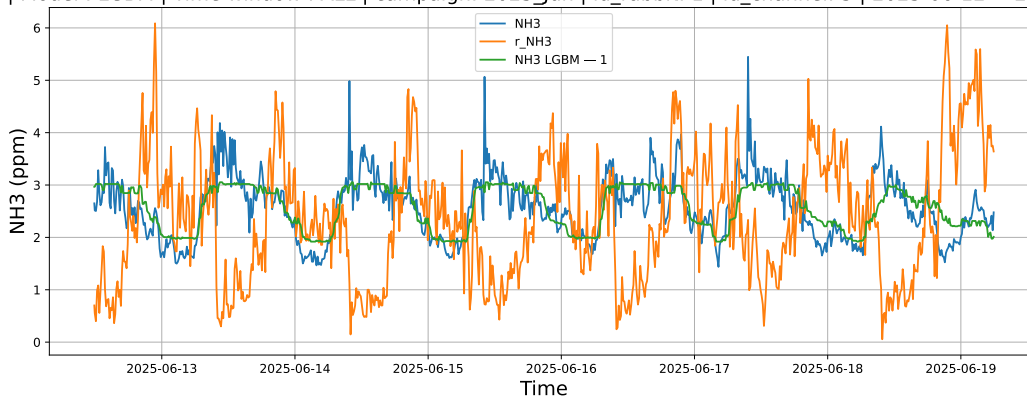


3.1.1.3 Time window : ALL

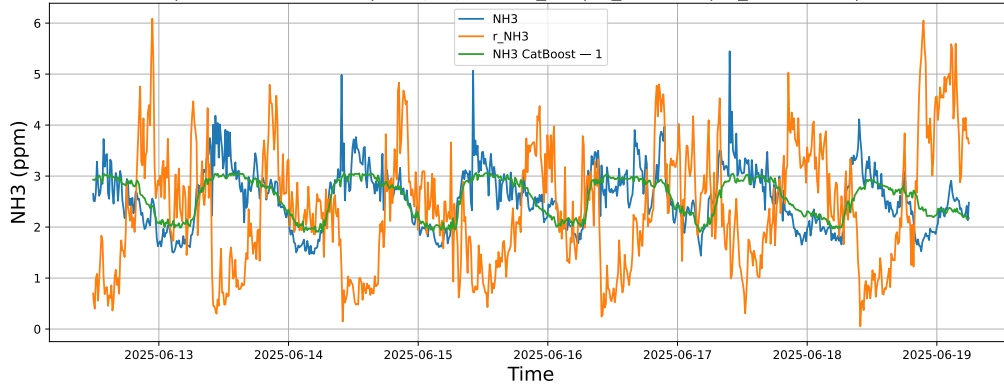
NH3 | Model : Linear | Time window : ALL | campaign: 2025_Jun | id_rabbit: 1 | id_channel: 3 | 2025-06-12 → 2025-06-19



NH3 | Model : LGBM | Time window : ALL | campaign: 2025_Jun | id_rabbit: 1 | id_channel: 3 | 2025-06-12 → 2025-06-19



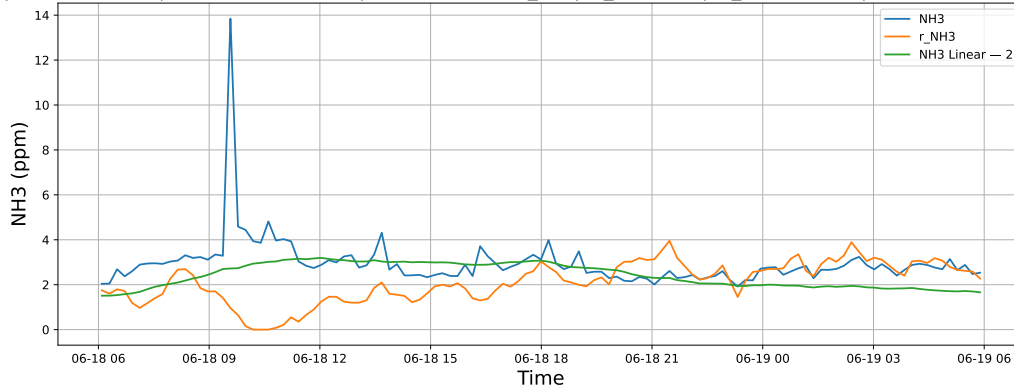
NH3 | Model : CatBoost | Time window : ALL | campaign: 2025_Jun | id_rabbit: 1 | id_channel: 3 | 2025-06-12 → 2025-06-19



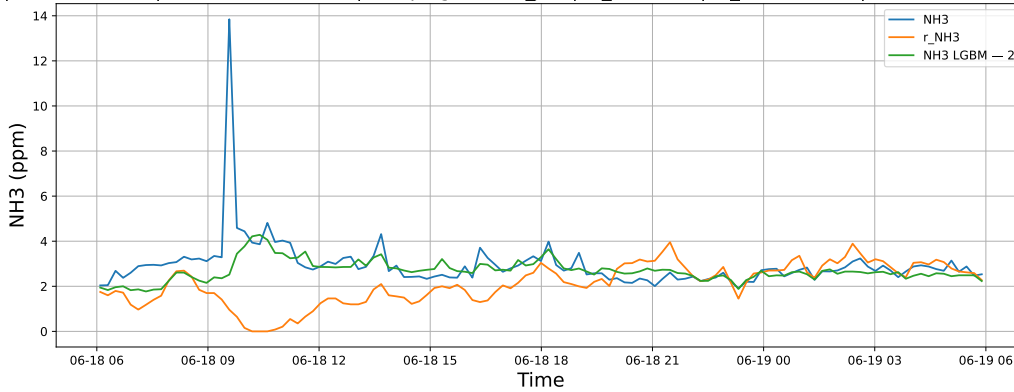
3.1.2 Group: campaign: 2025_Jun,id_rabbit: 2,id_channel: 11

3.1.2.1 Time window : 24

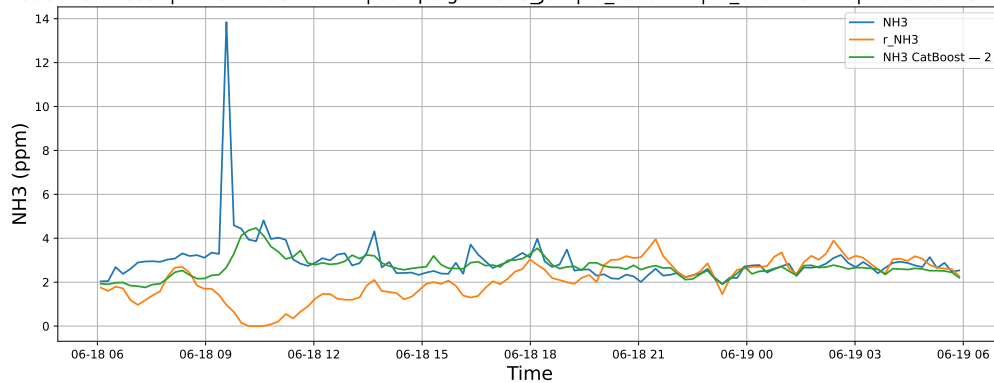
NH3 | Model : Linear | Time window : 24h | campaign: 2025_Jun | id_rabbit: 2 | id_channel: 11 | 2025-06-18 → 2025-06-19



NH3 | Model : LGBM | Time window : 24h | campaign: 2025_Jun | id_rabbit: 2 | id_channel: 11 | 2025-06-18 → 2025-06-19

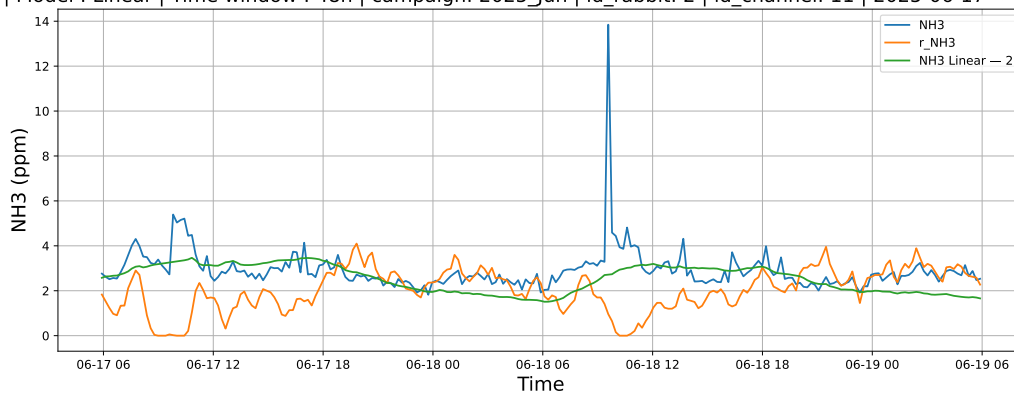


NH3 | Model : CatBoost | Time window : 24h | campaign: 2025_Jun | id_rabbit: 2 | id_channel: 11 | 2025-06-18 → 2025-06-19

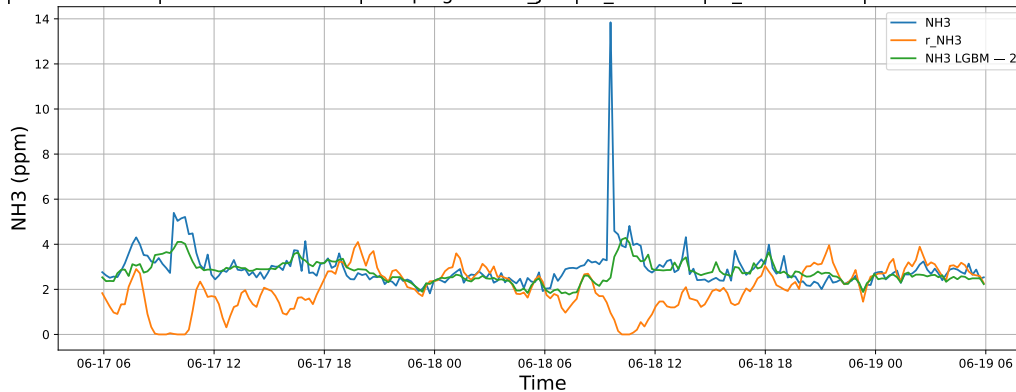


3.1.2.2 Time window : 48

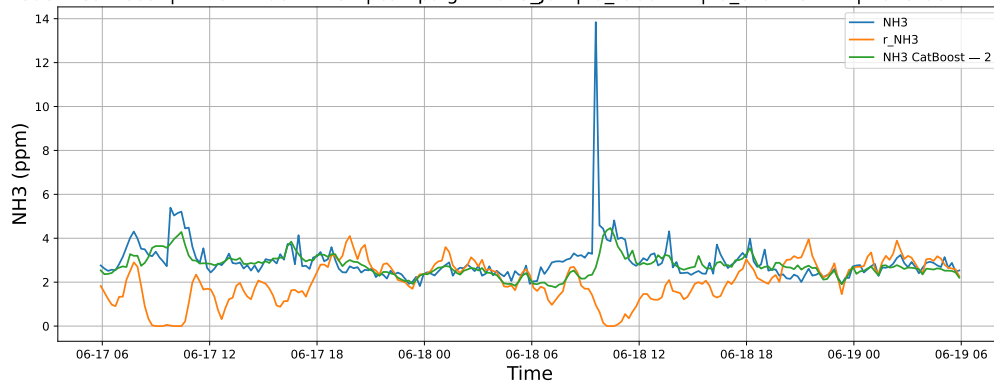
NH3 | Model : Linear | Time window : 48h | campaign: 2025_Jun | id_rabbit: 2 | id_channel: 11 | 2025-06-17 → 2025-06-19



NH3 | Model : LGBM | Time window : 48h | campaign: 2025_Jun | id_rabbit: 2 | id_channel: 11 | 2025-06-17 → 2025-06-19

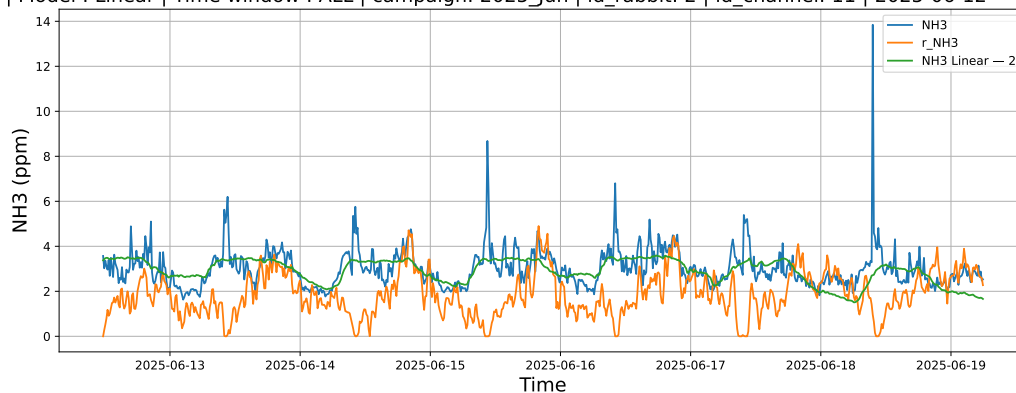


NH3 | Model : CatBoost | Time window : 48h | campaign: 2025_Jun | id_rabbit: 2 | id_channel: 11 | 2025-06-17 → 2025-06-19

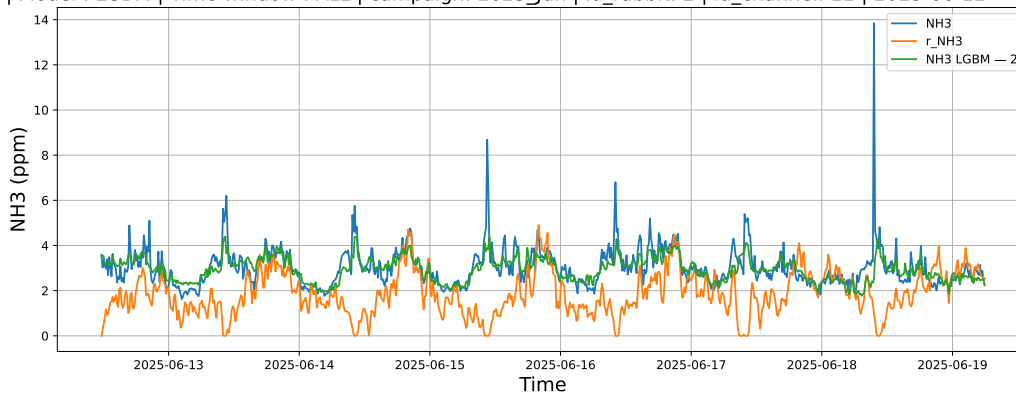


3.1.2.3 Time window : ALL

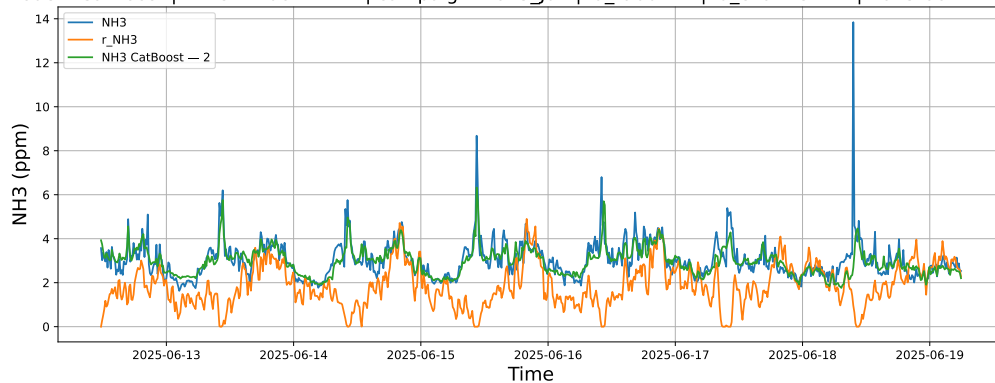
NH3 | Model : Linear | Time window : ALL | campaign: 2025_Jun | id_rabbit: 2 | id_channel: 11 | 2025-06-12 → 2025-06-19



NH3 | Model : LGBM | Time window : ALL | campaign: 2025_Jun | id_rabbit: 2 | id_channel: 11 | 2025-06-12 → 2025-06-19



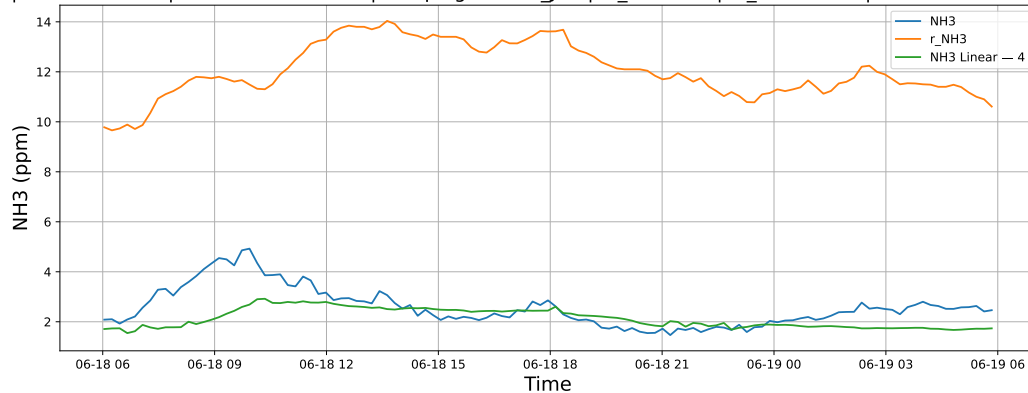
NH3 | Model : CatBoost | Time window : ALL | campaign: 2025_Jun | id_rabbit: 2 | id_channel: 11 | 2025-06-12 → 2025-06-19



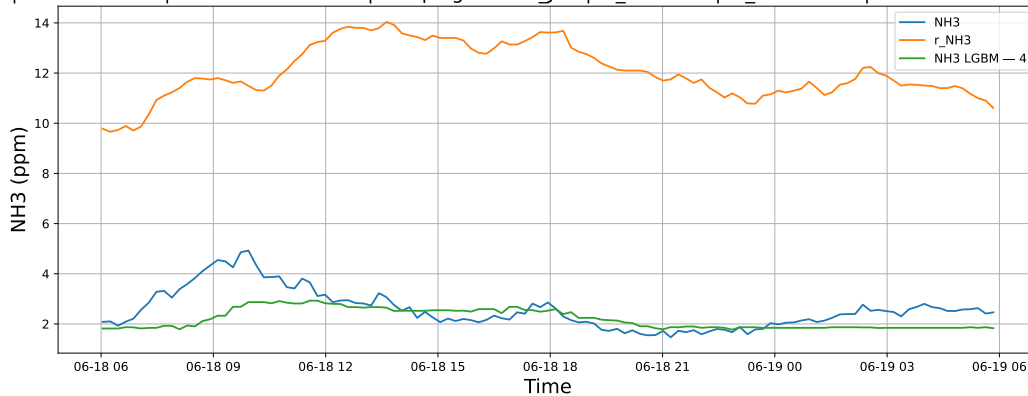
3.1.3 Group: campaign: 2025_Jun,id_rabbit: 4,id_channel: 5

3.1.3.1 Time window : 24

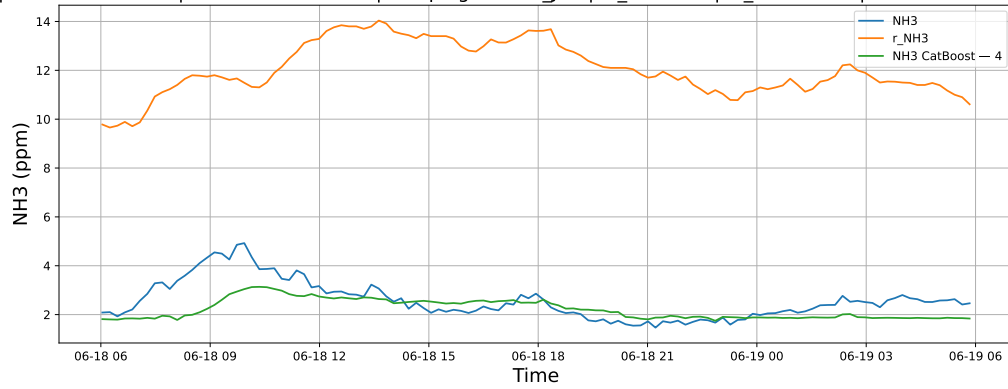
NH3 | Model : Linear | Time window : 24h | campaign: 2025_Jun | id_rabbit: 4 | id_channel: 5 | 2025-06-18 → 2025-06-19



NH3 | Model : LGBM | Time window : 24h | campaign: 2025_Jun | id_rabbit: 4 | id_channel: 5 | 2025-06-18 → 2025-06-19

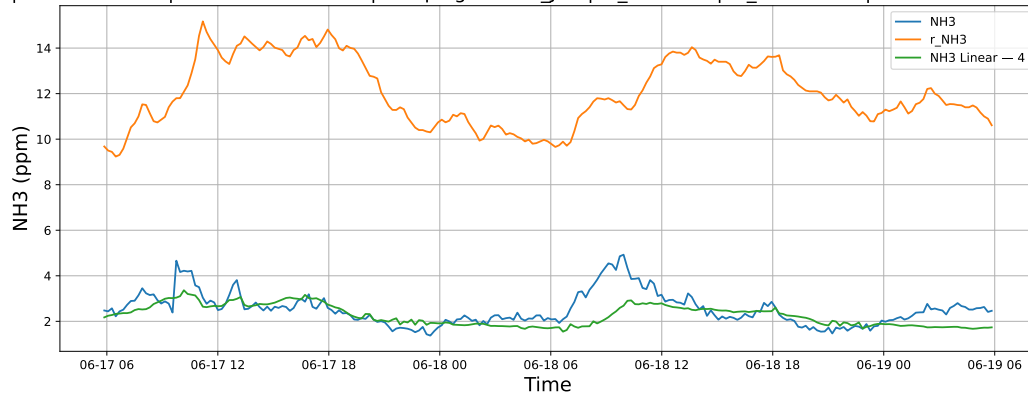


NH3 | Model : CatBoost | Time window : 24h | campaign: 2025_Jun | id_rabbit: 4 | id_channel: 5 | 2025-06-18 → 2025-06-19

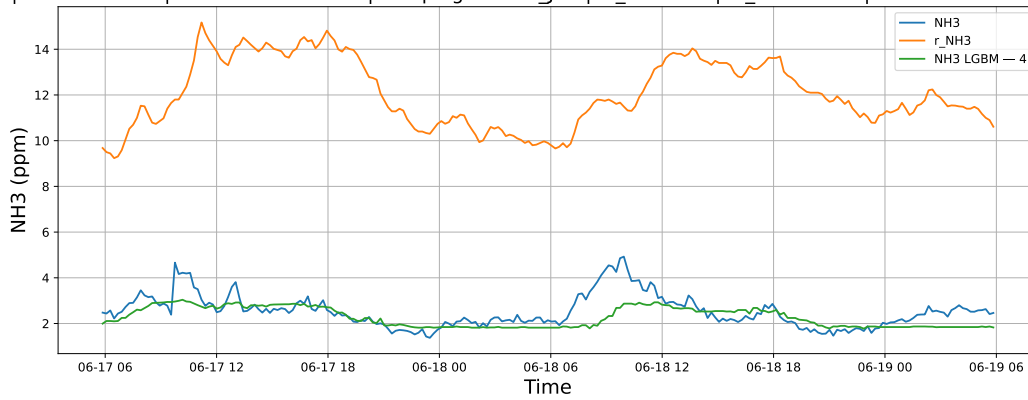


3.1.3.2 Time window : 48

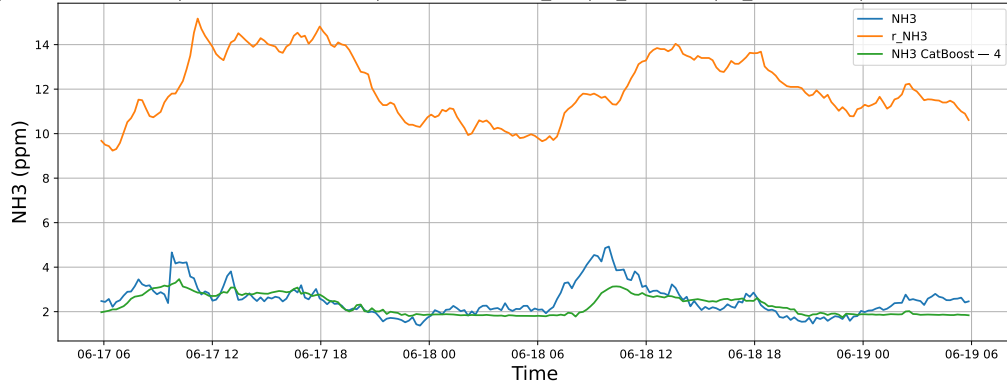
NH3 | Model : Linear | Time window : 48h | campaign: 2025_Jun | id_rabbit: 4 | id_channel: 5 | 2025-06-17 → 2025-06-19



NH3 | Model : LGBM | Time window : 48h | campaign: 2025_Jun | id_rabbit: 4 | id_channel: 5 | 2025-06-17 → 2025-06-19

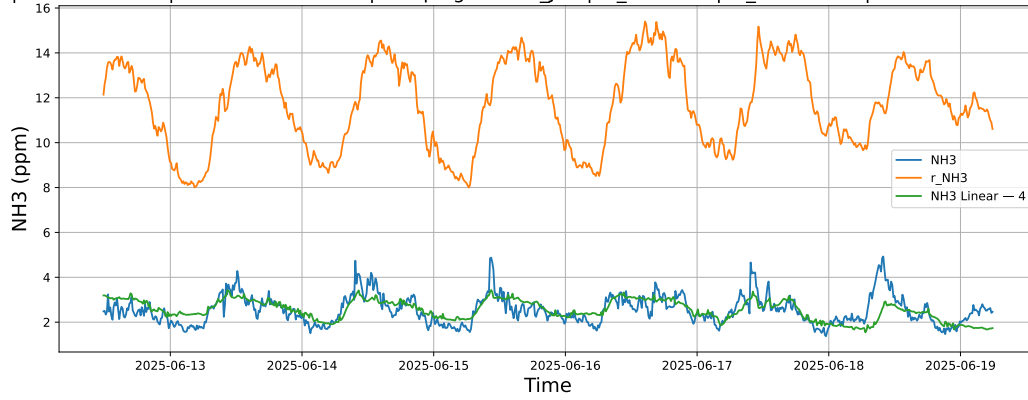


NH3 | Model : CatBoost | Time window : 48h | campaign: 2025_Jun | id_rabbit: 4 | id_channel: 5 | 2025-06-17 → 2025-06-19

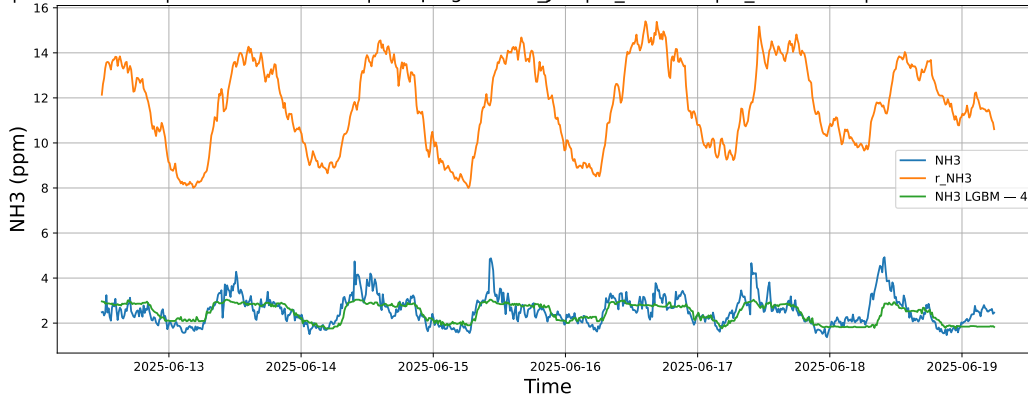


3.1.3.3 Time window : ALL

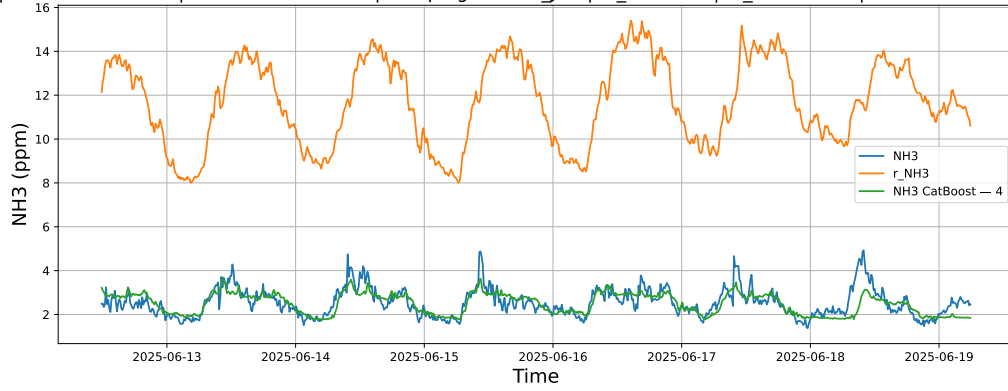
NH3 | Model : Linear | Time window : ALL | campaign: 2025_Jun | id_rabbit: 4 | id_channel: 5 | 2025-06-12 → 2025-06-19



NH3 | Model : LGBM | Time window : ALL | campaign: 2025_Jun | id_rabbit: 4 | id_channel: 5 | 2025-06-12 → 2025-06-19



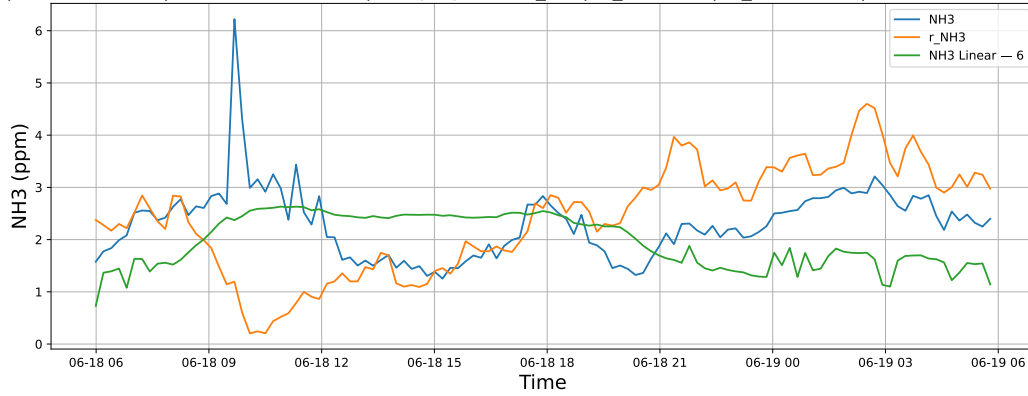
NH3 | Model : CatBoost | Time window : ALL | campaign: 2025_Jun | id_rabbit: 4 | id_channel: 5 | 2025-06-12 → 2025-06-19



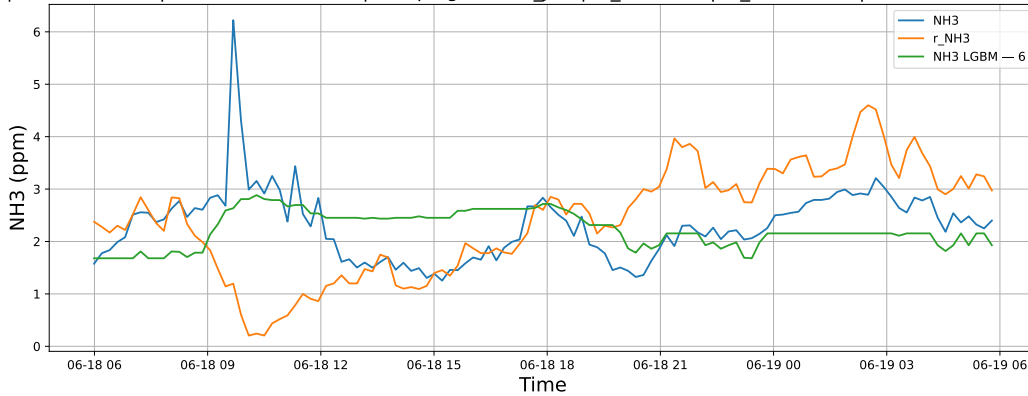
3.1.4 Group: campaign: 2025_Jun,id_rabbit: 6,id_channel: 4

3.1.4.1 Time window : 24

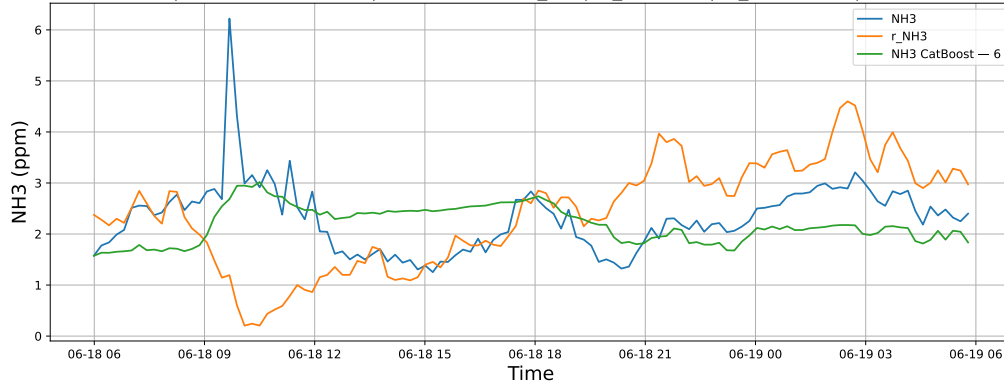
NH3 | Model : Linear | Time window : 24h | campaign: 2025_Jun | id_rabbit: 6 | id_channel: 4 | 2025-06-18 → 2025-06-19



NH3 | Model : LGBM | Time window : 24h | campaign: 2025_Jun | id_rabbit: 6 | id_channel: 4 | 2025-06-18 → 2025-06-19

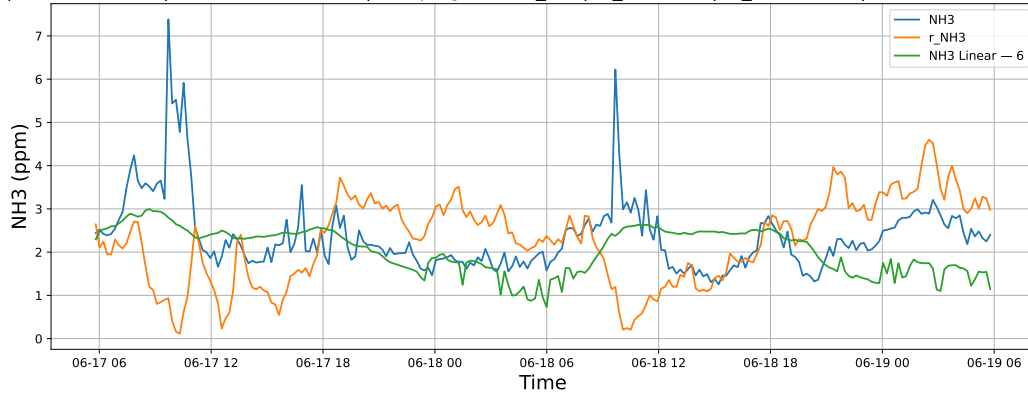


NH3 | Model : CatBoost | Time window : 24h | campaign: 2025_Jun | id_rabbit: 6 | id_channel: 4 | 2025-06-18 → 2025-06-19

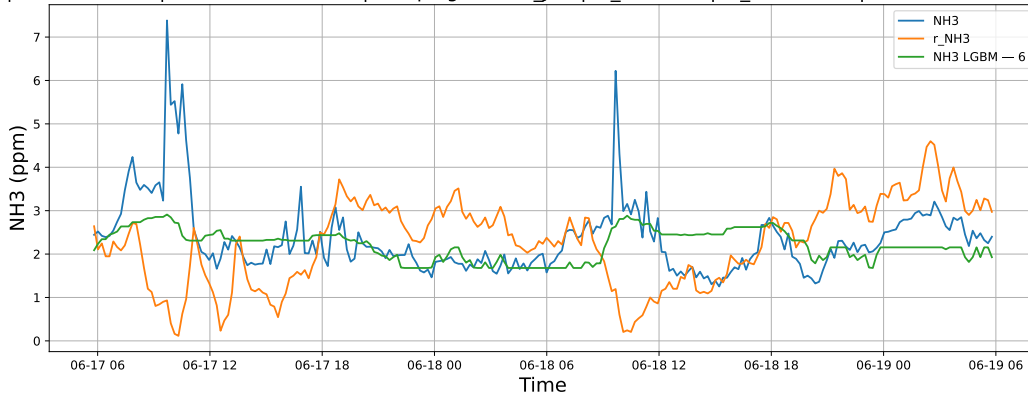


3.1.4.2 Time window : 48

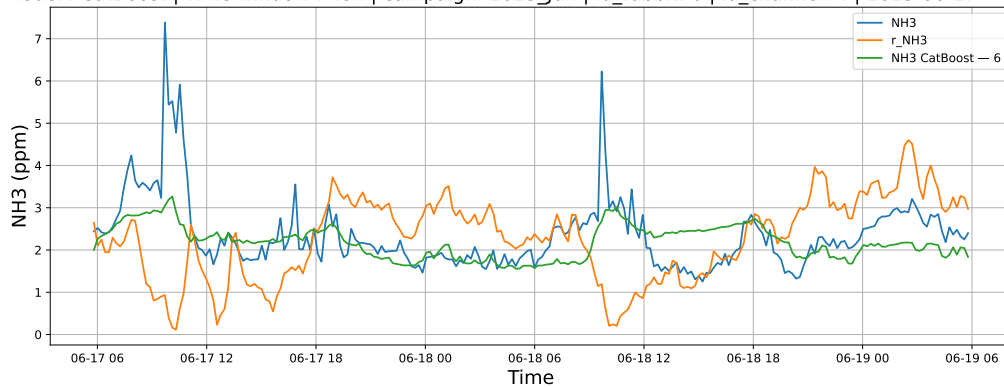
NH3 | Model : Linear | Time window : 48h | campaign: 2025_Jun | id_rabbit: 6 | id_channel: 4 | 2025-06-17 → 2025-06-19



NH3 | Model : LGBM | Time window : 48h | campaign: 2025_Jun | id_rabbit: 6 | id_channel: 4 | 2025-06-17 → 2025-06-19

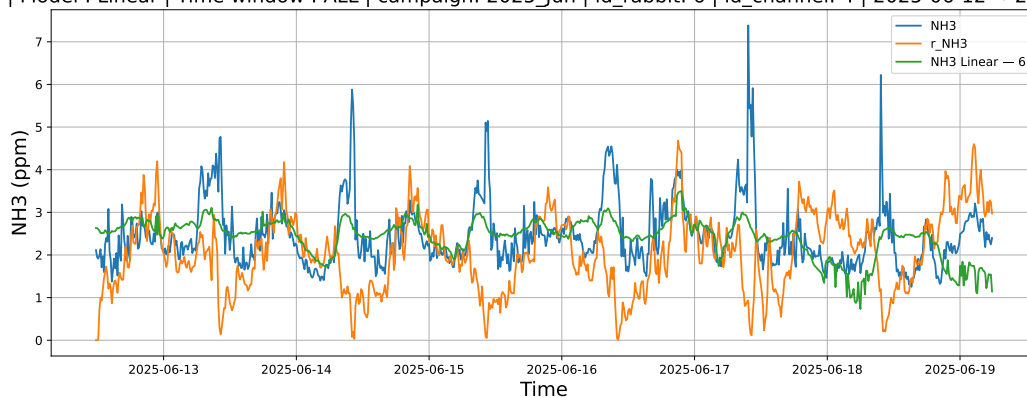


NH3 | Model : CatBoost | Time window : 48h | campaign: 2025_Jun | id_rabbit: 6 | id_channel: 4 | 2025-06-17 → 2025-06-19

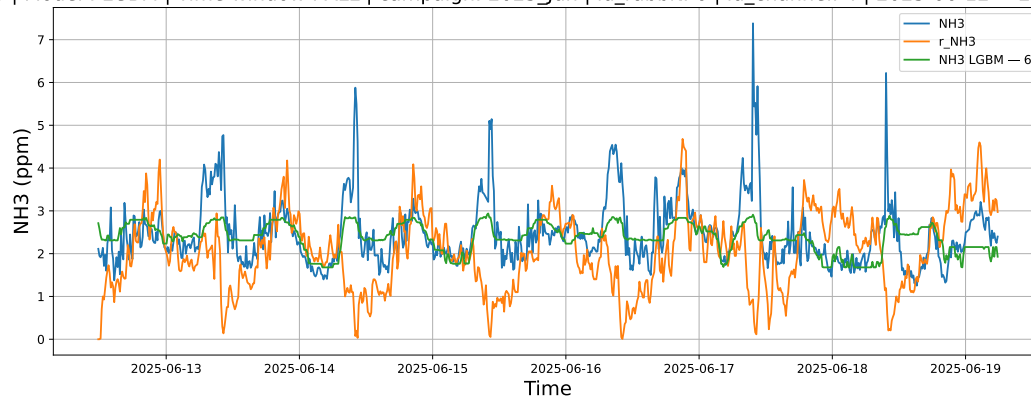


3.1.4.3 Time window : ALL

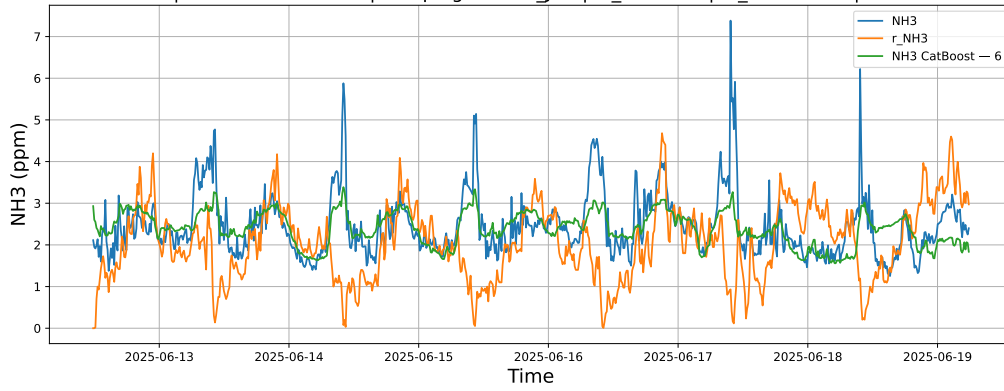
NH3 | Model : Linear | Time window : ALL | campaign: 2025_Jun | id_rabbit: 6 | id_channel: 4 | 2025-06-12 → 2025-06-19



NH3 | Model : LGBM | Time window : ALL | campaign: 2025_Jun | id_rabbit: 6 | id_channel: 4 | 2025-06-12 → 2025-06-19



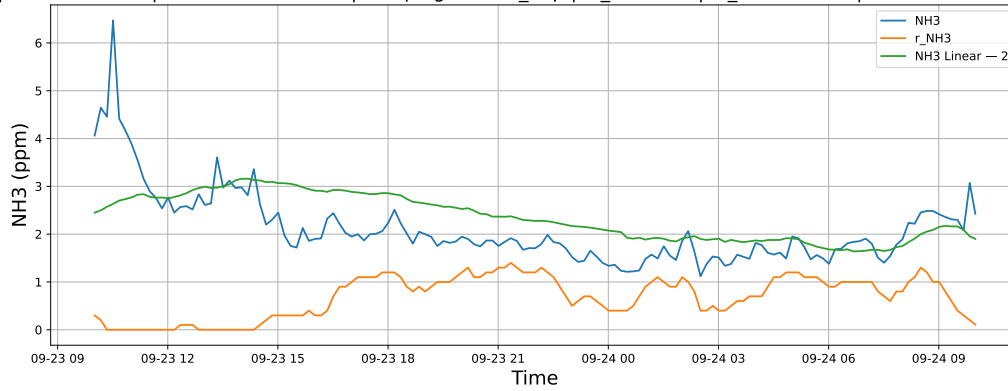
NH3 | Model : CatBoost | Time window : ALL | campaign: 2025_Jun | id_rabbit: 6 | id_channel: 4 | 2025-06-12 → 2025-06-19



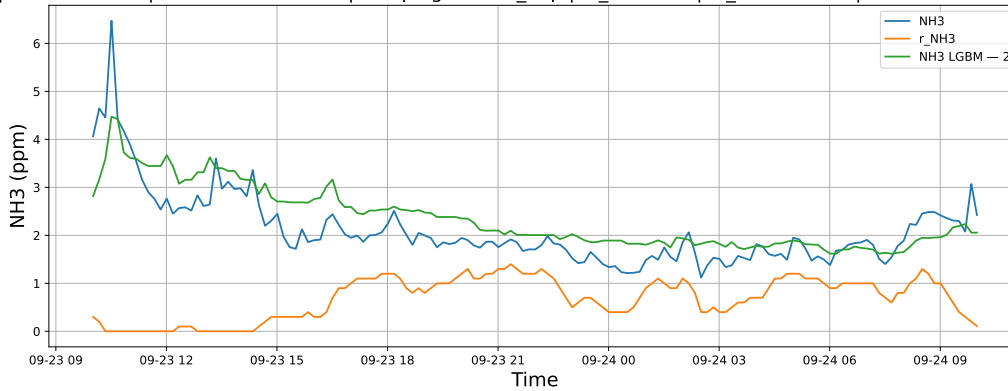
3.1.5 Group: campaign: 2025_Sep,id_rabbit: 2,id_channel: 11

3.1.5.1 Time window : 24

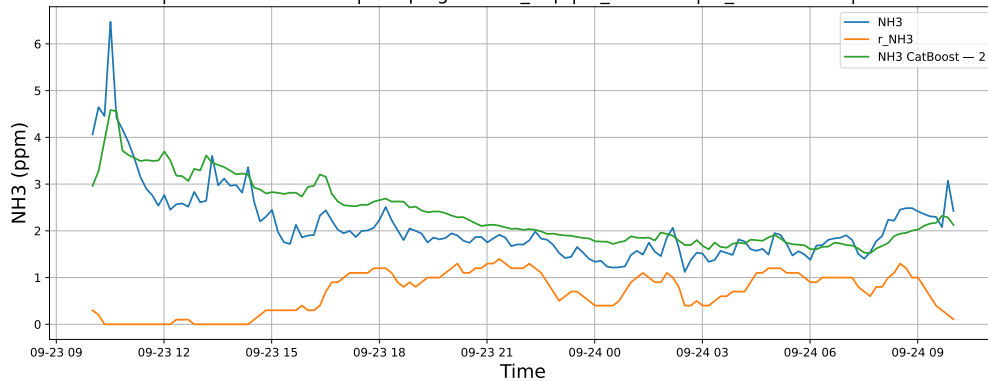
NH3 | Model : Linear | Time window : 24h | campaign: 2025_Sep | id_rabbit: 2 | id_channel: 11 | 2025-09-23 → 2025-09-24



NH3 | Model : LGBM | Time window : 24h | campaign: 2025_Sep | id_rabbit: 2 | id_channel: 11 | 2025-09-23 → 2025-09-24

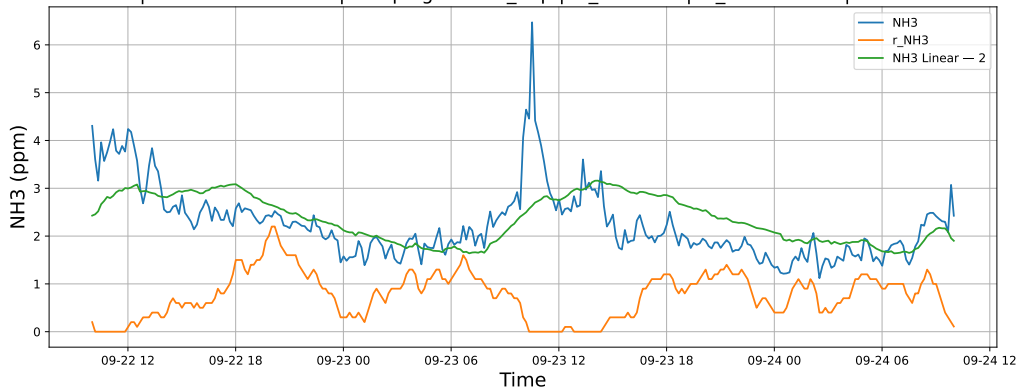


NH3 | Model : CatBoost | Time window : 24h | campaign: 2025_Sep | id_rabbit: 2 | id_channel: 11 | 2025-09-23 → 2025-09-24

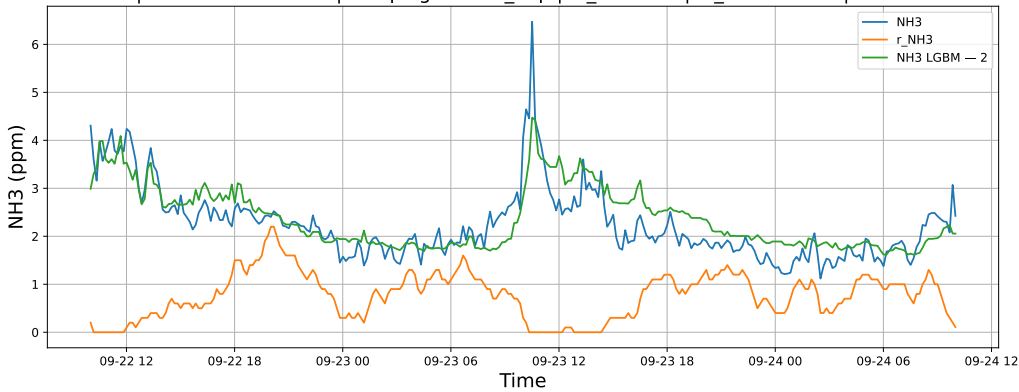


3.1.5.2 Time window : 48

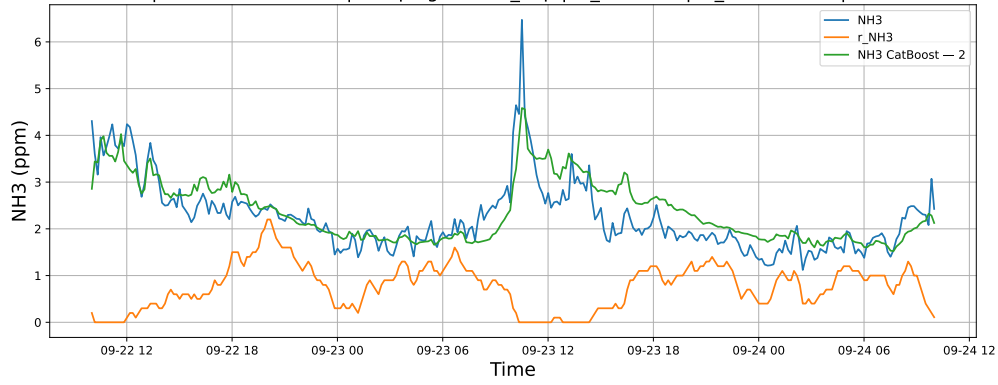
NH3 | Model : Linear | Time window : 48h | campaign: 2025_Sep | id_rabbit: 2 | id_channel: 11 | 2025-09-22 → 2025-09-24



NH3 | Model : LGBM | Time window : 48h | campaign: 2025_Sep | id_rabbit: 2 | id_channel: 11 | 2025-09-22 → 2025-09-24

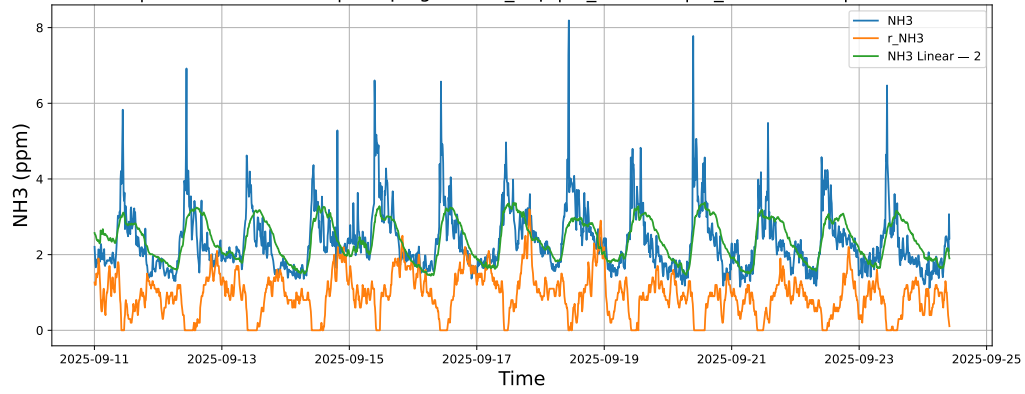


NH3 | Model : CatBoost | Time window : 48h | campaign: 2025_Sep | id_rabbit: 2 | id_channel: 11 | 2025-09-22 → 2025-09-24

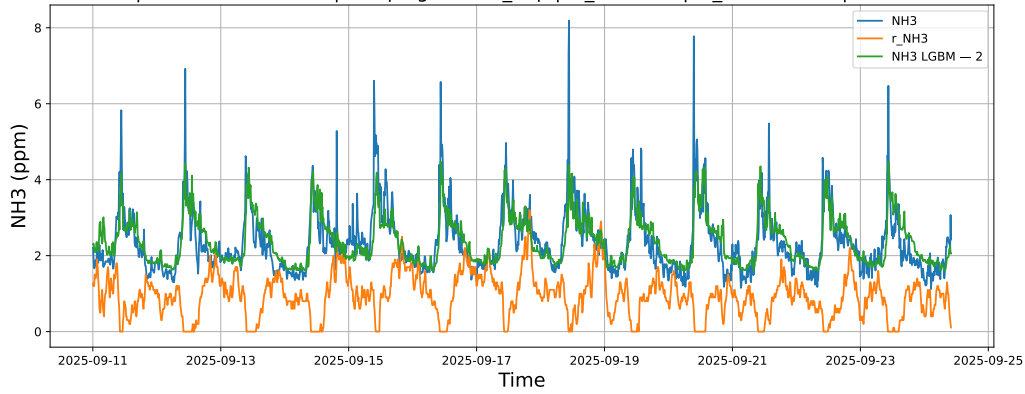


3.1.5.3 Time window : ALL

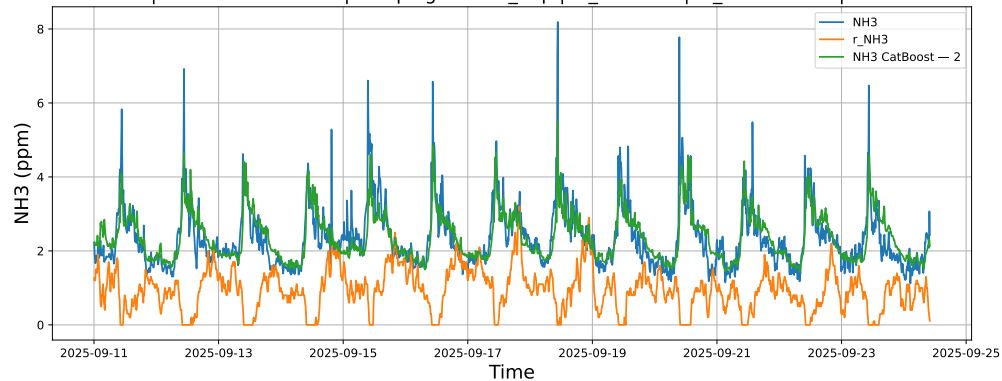
NH3 | Model : Linear | Time window : ALL | campaign: 2025_Sep | id_rabbit: 2 | id_channel: 11 | 2025-09-11 → 2025-09-24



NH3 | Model : LGBM | Time window : ALL | campaign: 2025_Sep | id_rabbit: 2 | id_channel: 11 | 2025-09-11 → 2025-09-24



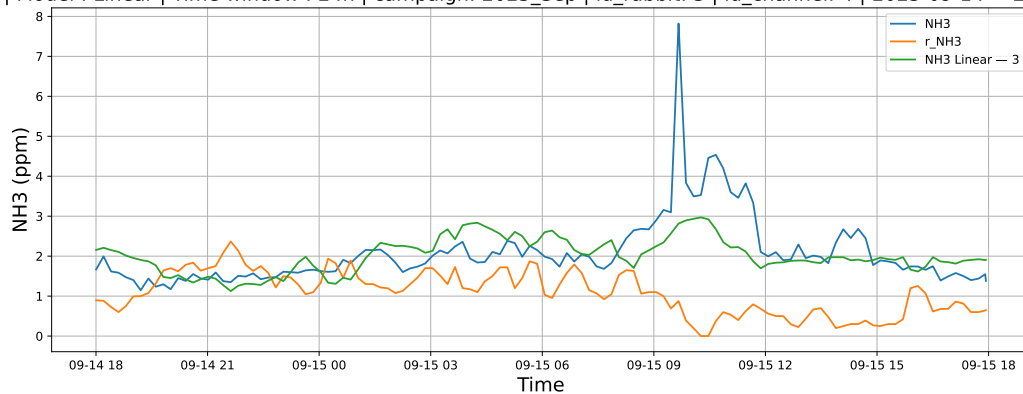
NH3 | Model : CatBoost | Time window : ALL | campaign: 2025_Sep | id_rabbit: 2 | id_channel: 11 | 2025-09-11 → 2025-09-24



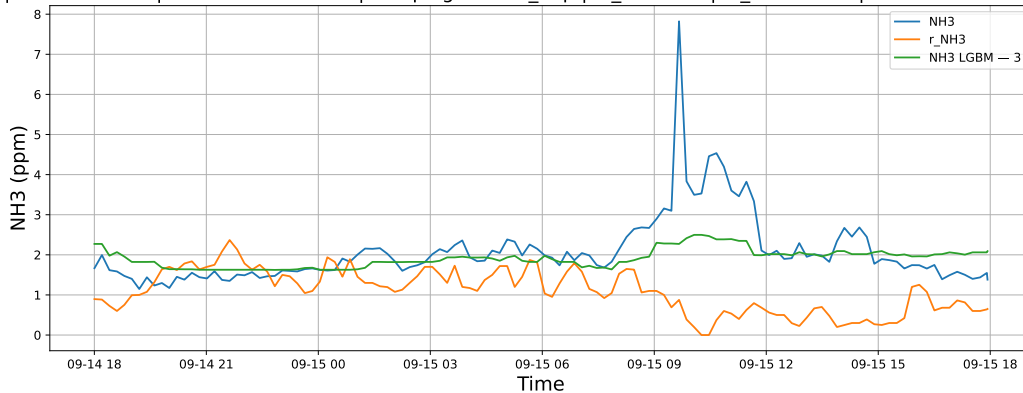
3.1.6 Group: campaign: 2025_Sep,id_rabbit: 3,id_channel: 4

3.1.6.1 Time window : 24

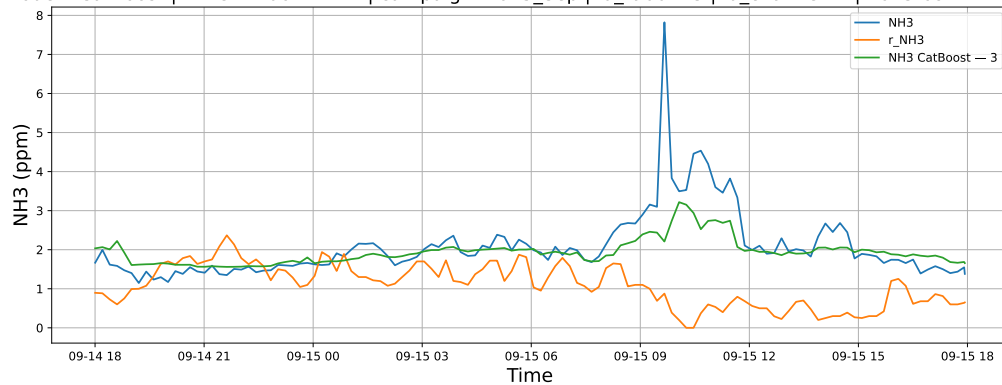
NH3 | Model : Linear | Time window : 24h | campaign: 2025_Sep | id_rabbit: 3 | id_channel: 4 | 2025-09-14 → 2025-09-15



NH3 | Model : LGBM | Time window : 24h | campaign: 2025_Sep | id_rabbit: 3 | id_channel: 4 | 2025-09-14 → 2025-09-15

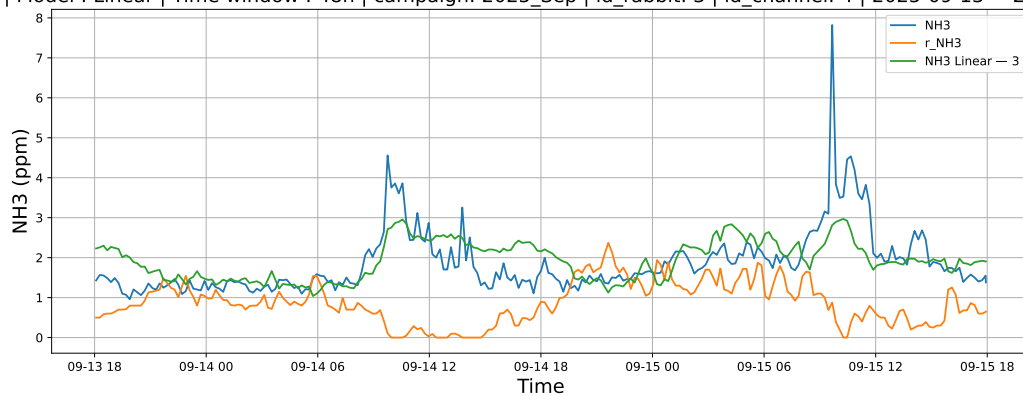


NH3 | Model : CatBoost | Time window : 24h | campaign: 2025_Sep | id_rabbit: 3 | id_channel: 4 | 2025-09-14 → 2025-09-15

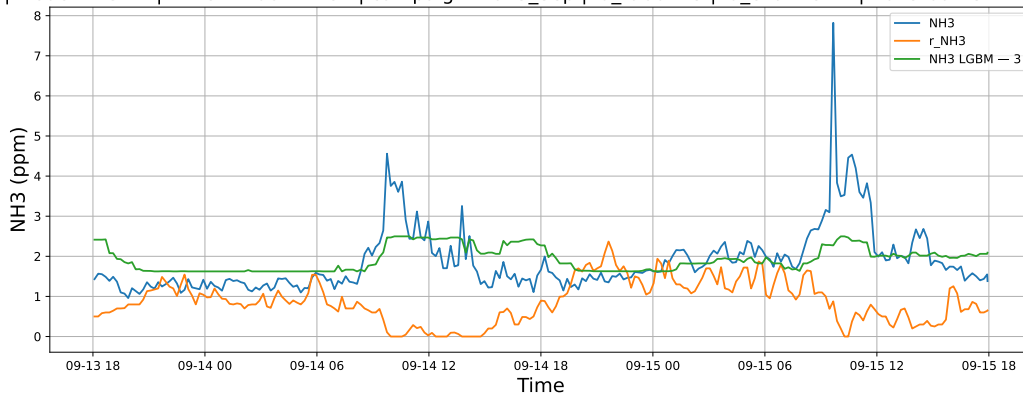


3.1.6.2 Time window : 48

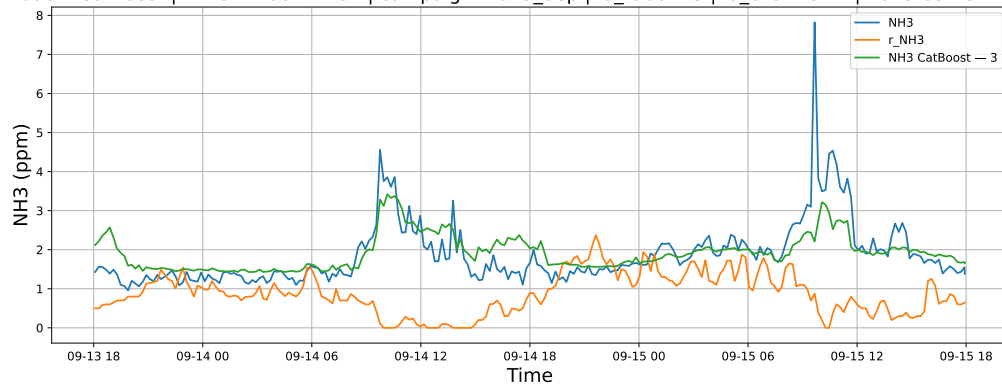
NH3 | Model : Linear | Time window : 48h | campaign: 2025_Sep | id_rabbit: 3 | id_channel: 4 | 2025-09-13 → 2025-09-15



NH3 | Model : LGBM | Time window : 48h | campaign: 2025_Sep | id_rabbit: 3 | id_channel: 4 | 2025-09-13 → 2025-09-15

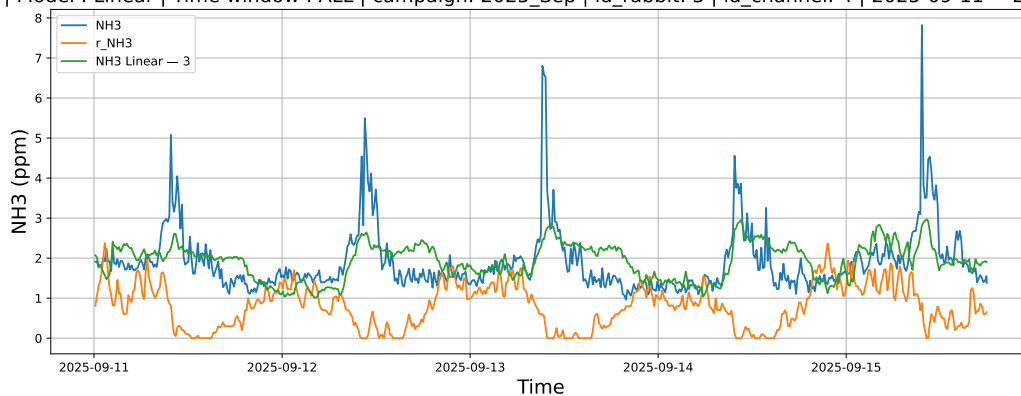


NH3 | Model : CatBoost | Time window : 48h | campaign: 2025_Sep | id_rabbit: 3 | id_channel: 4 | 2025-09-13 → 2025-09-15

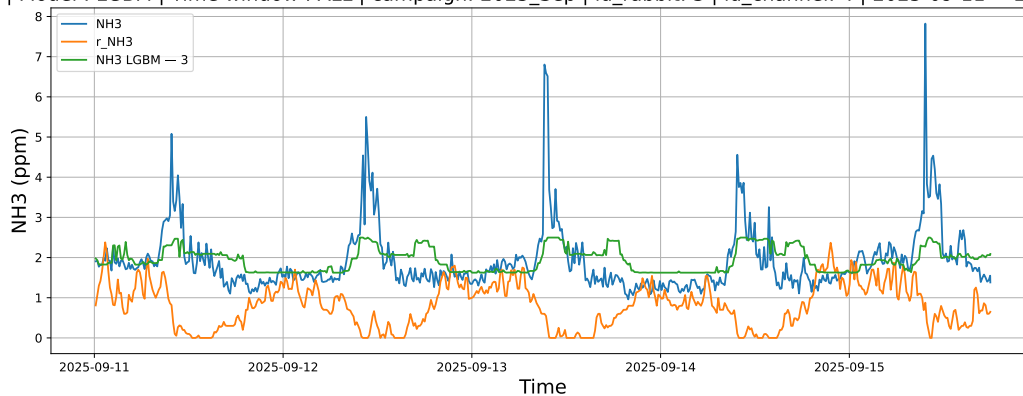


3.1.6.3 Time window : ALL

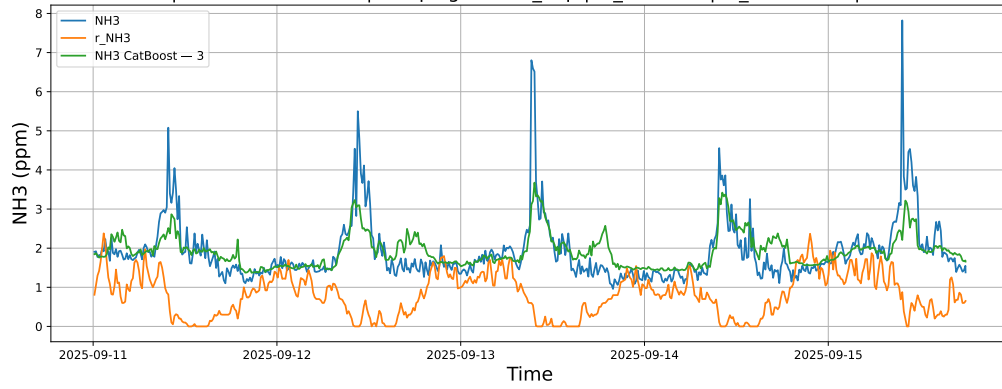
NH3 | Model : Linear | Time window : ALL | campaign: 2025_Sep | id_rabbit: 3 | id_channel: 4 | 2025-09-11 → 2025-09-15



NH3 | Model : LGBM | Time window : ALL | campaign: 2025_Sep | id_rabbit: 3 | id_channel: 4 | 2025-09-11 → 2025-09-15



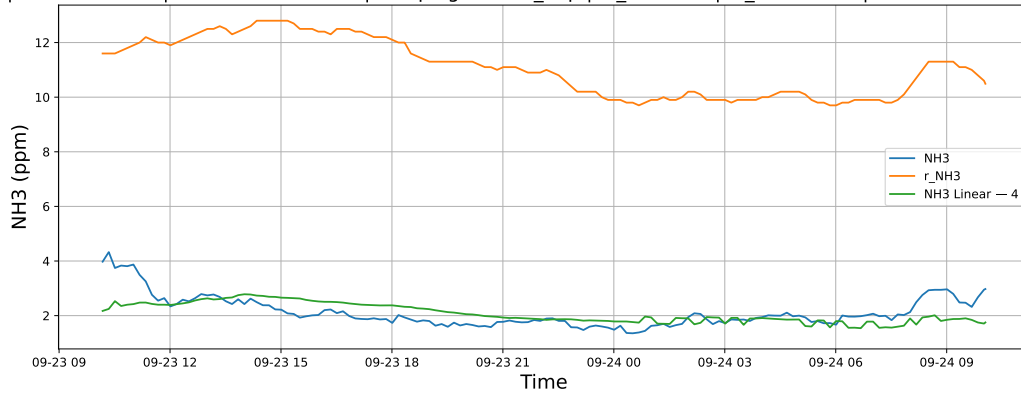
NH3 | Model : CatBoost | Time window : ALL | campaign: 2025_Sep | id_rabbit: 3 | id_channel: 4 | 2025-09-11 → 2025-09-15



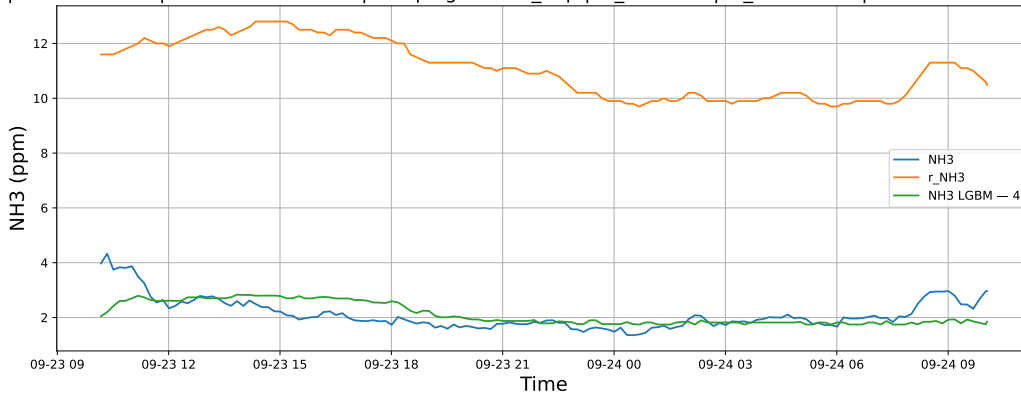
3.1.7 Group: campaign: 2025_Sep,id_rabbit: 4,id_channel: 5

3.1.7.1 Time window : 24

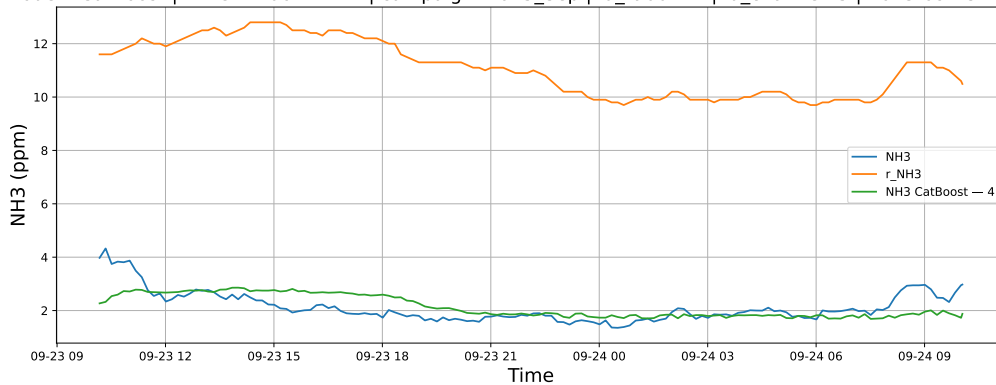
NH3 | Model : Linear | Time window : 24h | campaign: 2025_Sep | id_rabbit: 4 | id_channel: 5 | 2025-09-23 → 2025-09-24



NH3 | Model : LGBM | Time window : 24h | campaign: 2025_Sep | id_rabbit: 4 | id_channel: 5 | 2025-09-23 → 2025-09-24

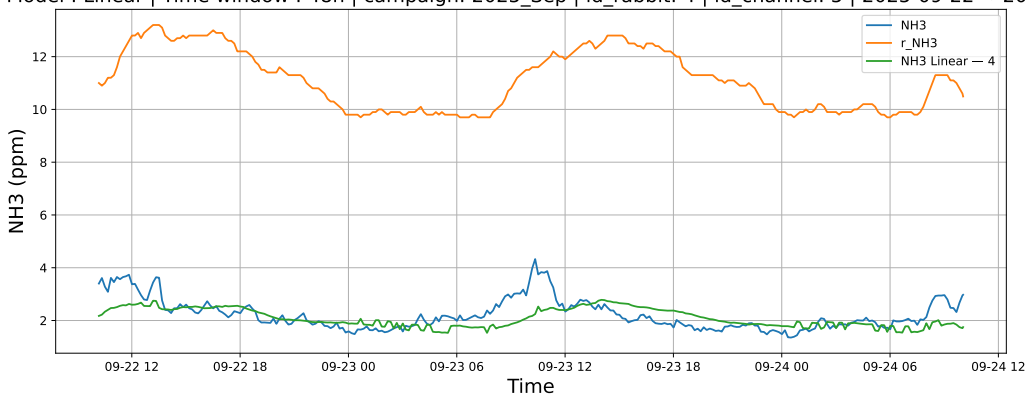


NH3 | Model : CatBoost | Time window : 24h | campaign: 2025_Sep | id_rabbit: 4 | id_channel: 5 | 2025-09-23 → 2025-09-24

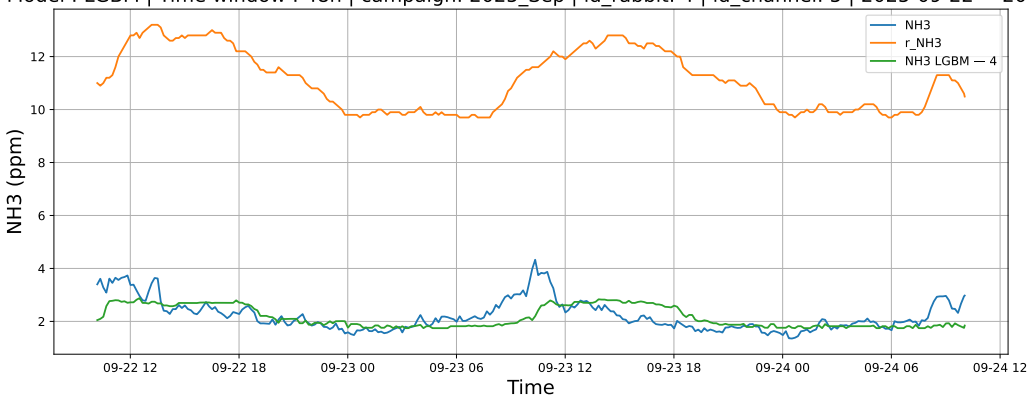


3.1.7.2 Time window : 48

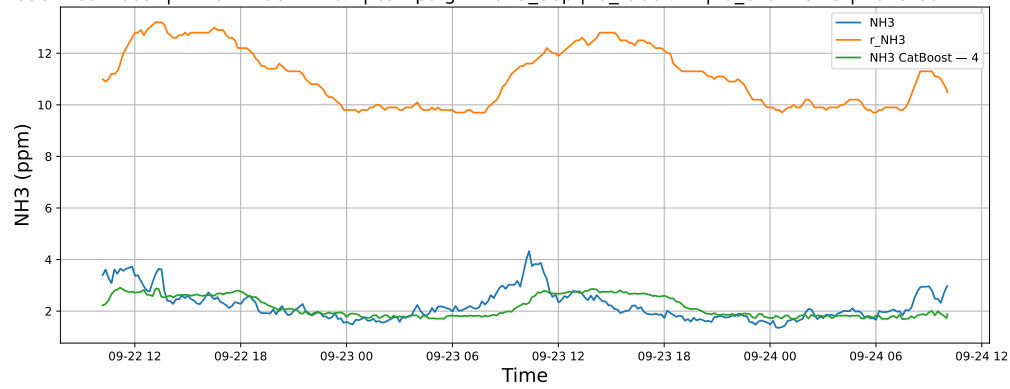
NH3 | Model : Linear | Time window : 48h | campaign: 2025_Sep | id_rabbit: 4 | id_channel: 5 | 2025-09-22 → 2025-09-24



NH3 | Model : LGBM | Time window : 48h | campaign: 2025_Sep | id_rabbit: 4 | id_channel: 5 | 2025-09-22 → 2025-09-24

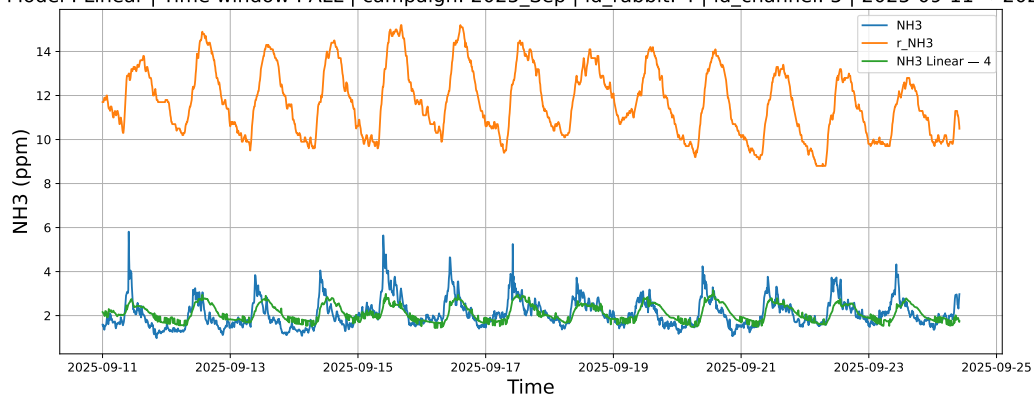


NH3 | Model : CatBoost | Time window : 48h | campaign: 2025_Sep | id_rabbit: 4 | id_channel: 5 | 2025-09-22 → 2025-09-24

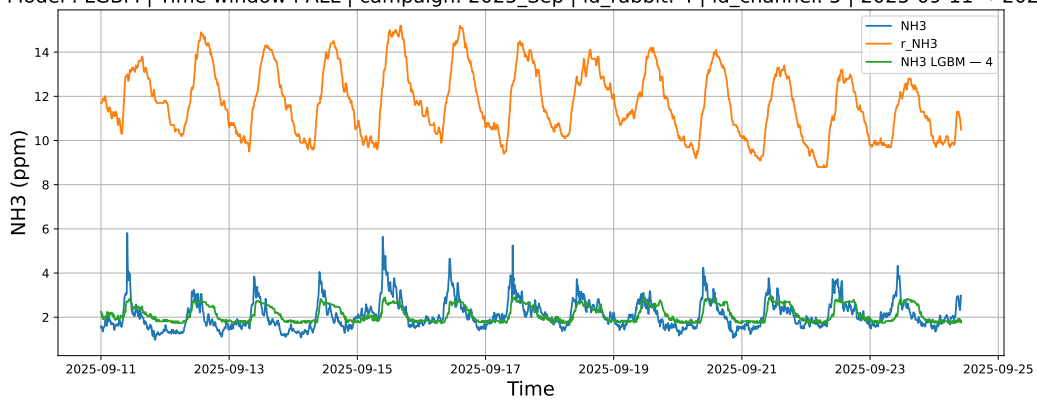


3.1.7.3 Time window : ALL

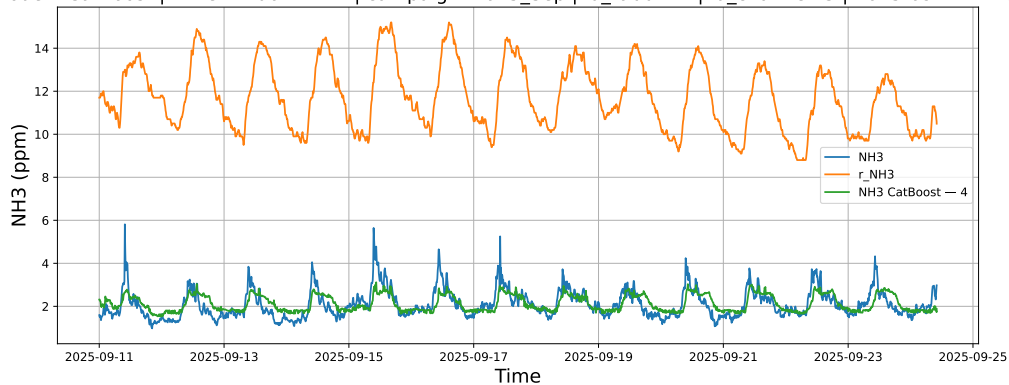
NH3 | Model : Linear | Time window : ALL | campaign: 2025_Sep | id_rabbit: 4 | id_channel: 5 | 2025-09-11 → 2025-09-24



NH3 | Model : LGBM | Time window : ALL | campaign: 2025_Sep | id_rabbit: 4 | id_channel: 5 | 2025-09-11 → 2025-09-24



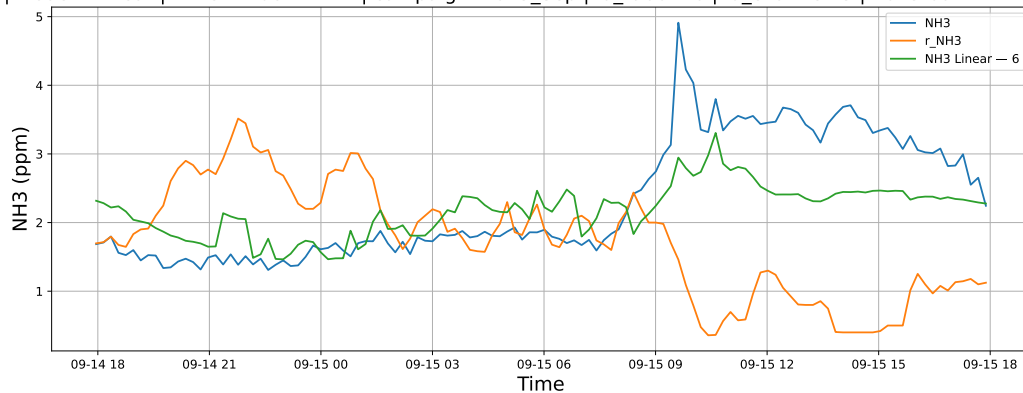
NH3 | Model : CatBoost | Time window : ALL | campaign: 2025_Sep | id_rabbit: 4 | id_channel: 5 | 2025-09-11 → 2025-09-24



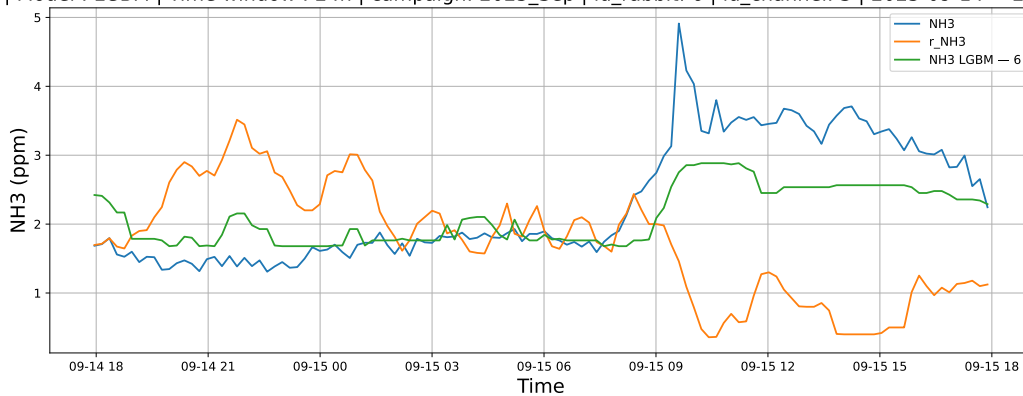
3.1.8 Group: campaign: 2025_Sep,id_rabbit: 6,id_channel: 3

3.1.8.1 Time window : 24

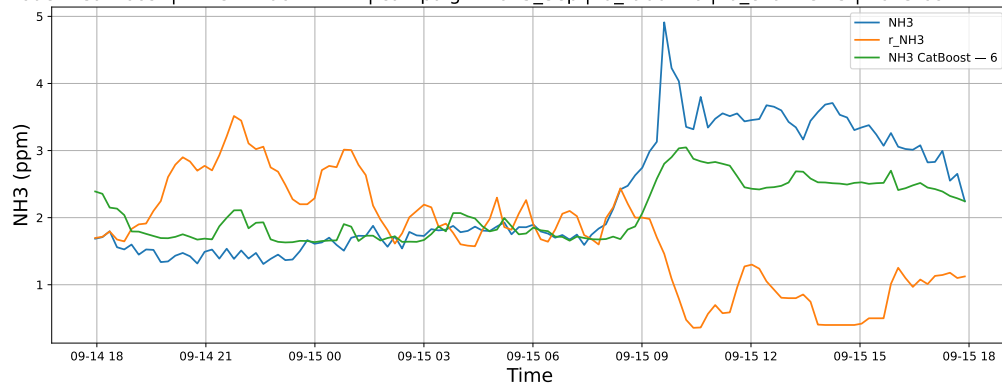
NH3 | Model : Linear | Time window : 24h | campaign: 2025_Sep | id_rabbit: 6 | id_channel: 3 | 2025-09-14 → 2025-09-15



NH3 | Model : LGBM | Time window : 24h | campaign: 2025_Sep | id_rabbit: 6 | id_channel: 3 | 2025-09-14 → 2025-09-15

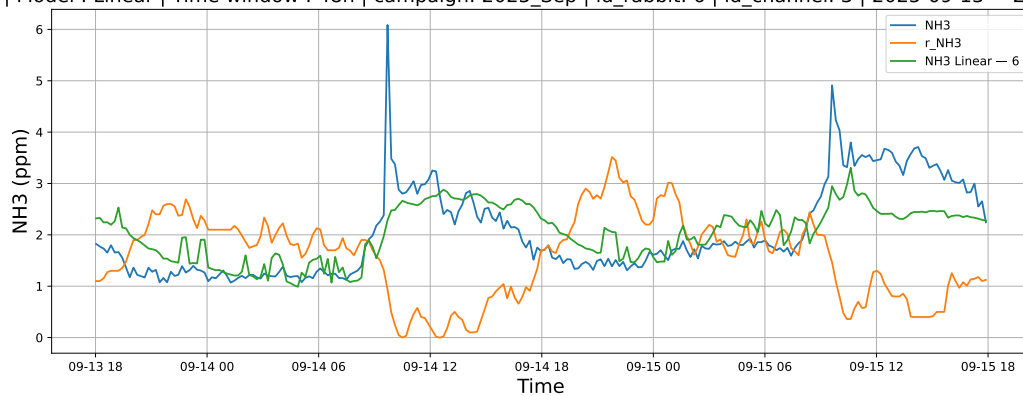


NH3 | Model : CatBoost | Time window : 24h | campaign: 2025_Sep | id_rabbit: 6 | id_channel: 3 | 2025-09-14 → 2025-09-15

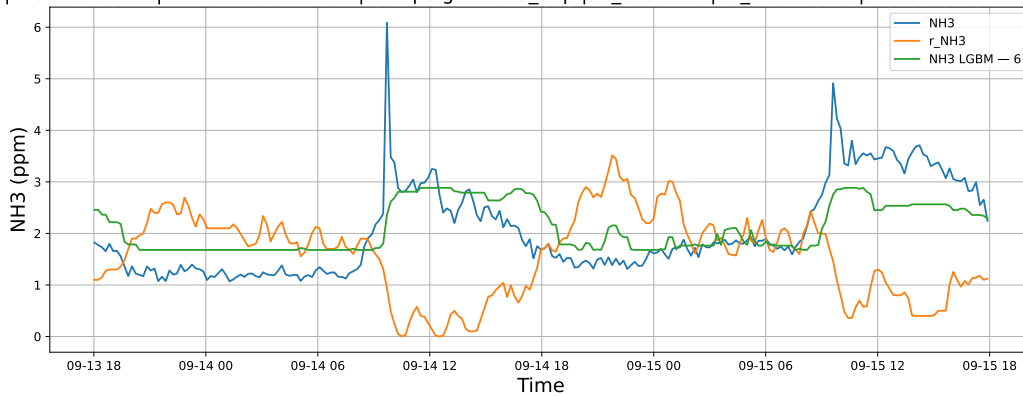


3.1.8.2 Time window : 48

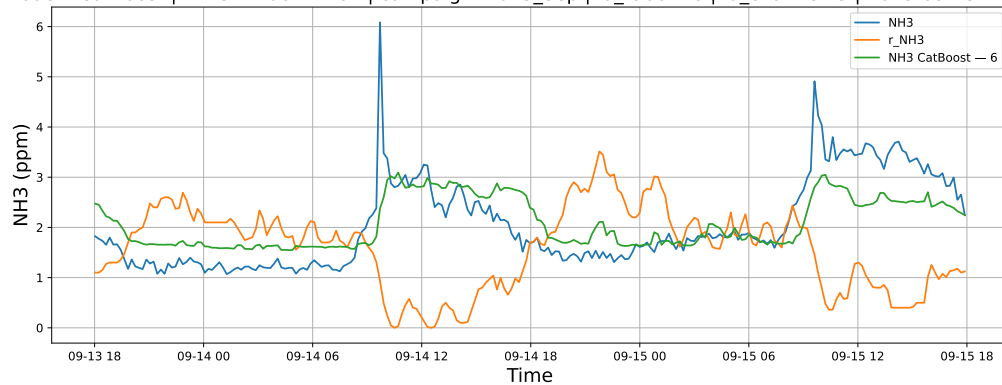
NH3 | Model : Linear | Time window : 48h | campaign: 2025_Sep | id_rabbit: 6 | id_channel: 3 | 2025-09-13 → 2025-09-15



NH3 | Model : LGBM | Time window : 48h | campaign: 2025_Sep | id_rabbit: 6 | id_channel: 3 | 2025-09-13 → 2025-09-15

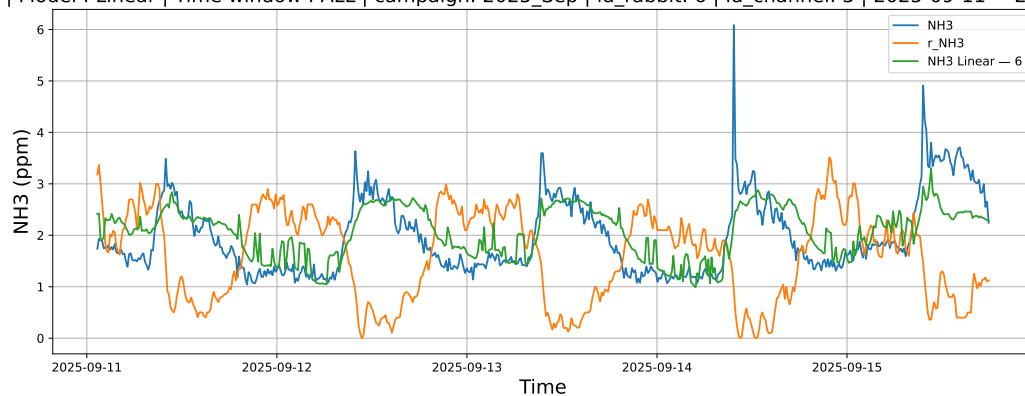


NH3 | Model : CatBoost | Time window : 48h | campaign: 2025_Sep | id_rabbit: 6 | id_channel: 3 | 2025-09-13 → 2025-09-15

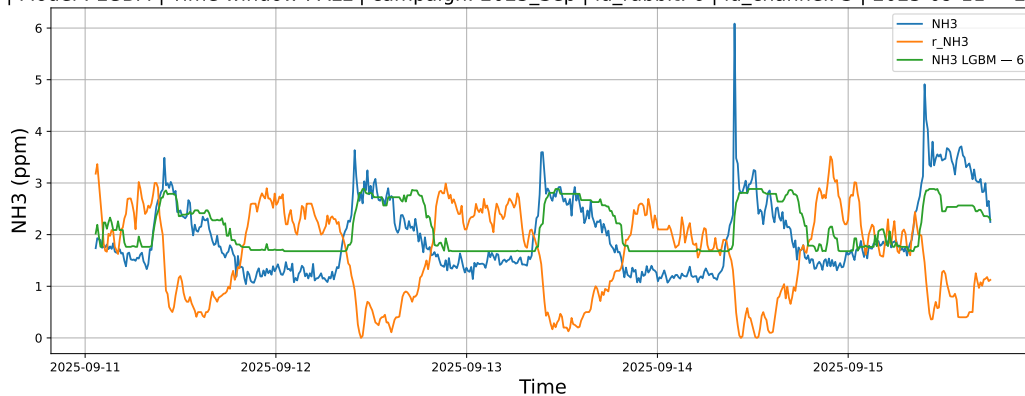


3.1.8.3 Time window : ALL

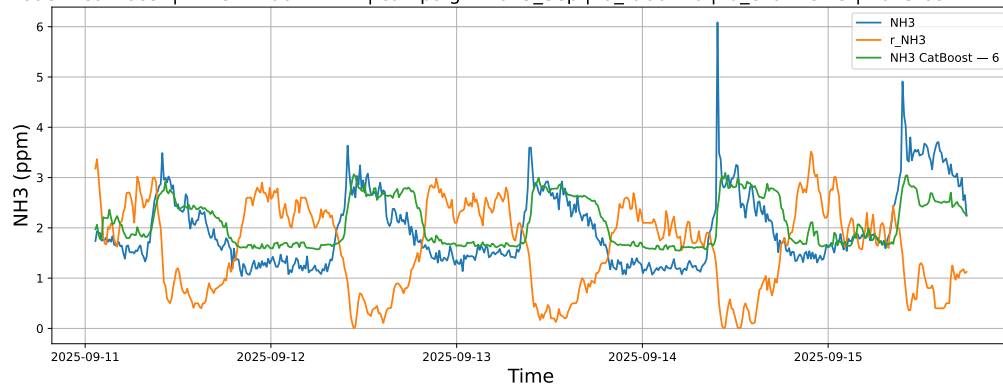
NH3 | Model : Linear | Time window : ALL | campaign: 2025_Sep | id_rabbit: 6 | id_channel: 3 | 2025-09-11 → 2025-09-15



NH3 | Model : LGBM | Time window : ALL | campaign: 2025_Sep | id_rabbit: 6 | id_channel: 3 | 2025-09-11 → 2025-09-15



NH3 | Model : CatBoost | Time window : ALL | campaign: 2025_Sep | id_rabbit: 6 | id_channel: 3 | 2025-09-11 → 2025-09-15

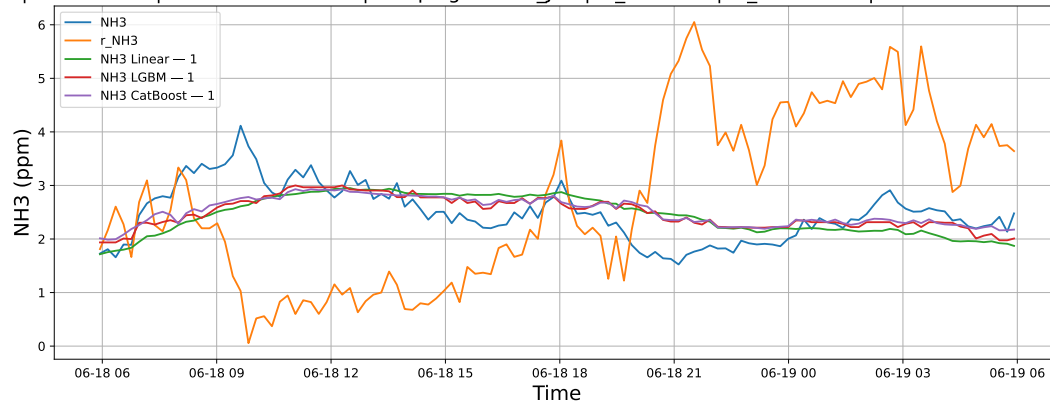


3.2 Compare plots

3.2.1 Group: campaign: 2025_Jun,id_rabbit: 1,id_channel: 3

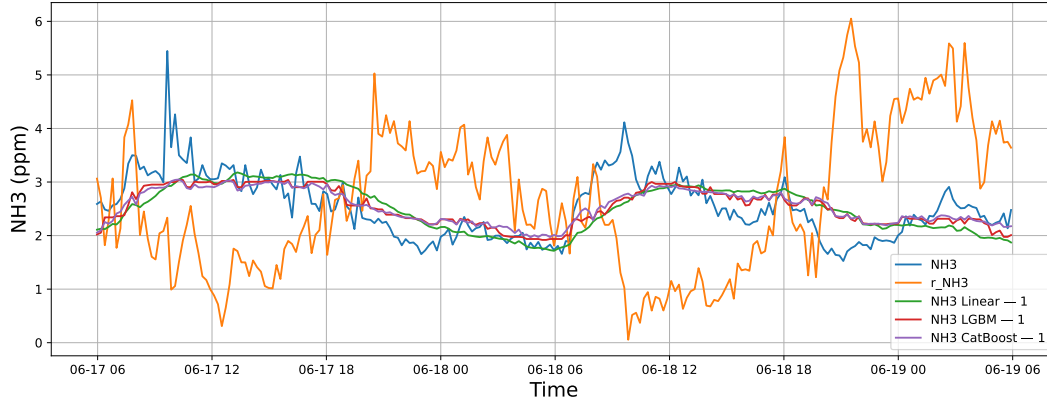
3.2.1.1 Time window : 24

NH3 | Model : ALL | Time window : 24h | campaign: 2025_Jun | id_rabbit: 1 | id_channel: 3 | 2025-06-18 → 2025-06-19



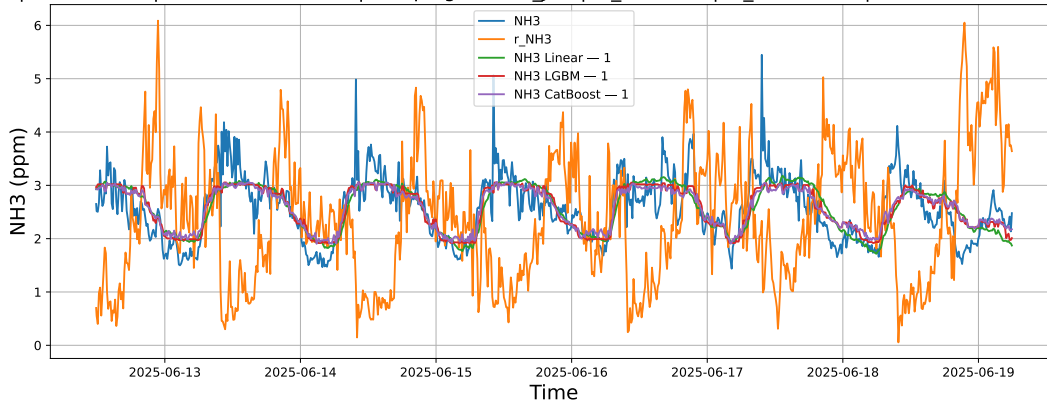
3.2.1.2 Time window : 48

NH3 | Model : ALL | Time window : 48h | campaign: 2025_Jun | id_rabbit: 1 | id_channel: 3 | 2025-06-17 → 2025-06-19



3.2.1.3 Time window : ALL

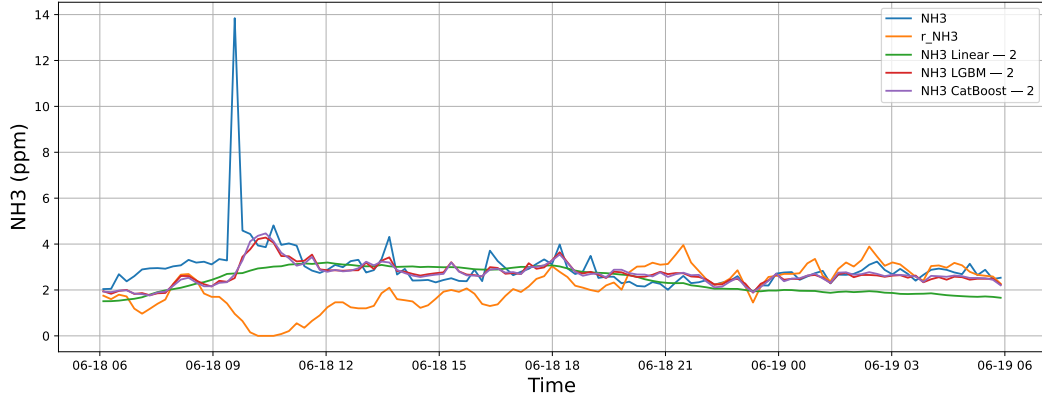
NH3 | Model : ALL | Time window : ALL | campaign: 2025_Jun | id_rabbit: 1 | id_channel: 3 | 2025-06-12 → 2025-06-19



3.2.2 Group: campaign: 2025_Jun,id_rabbit: 2,id_channel: 11

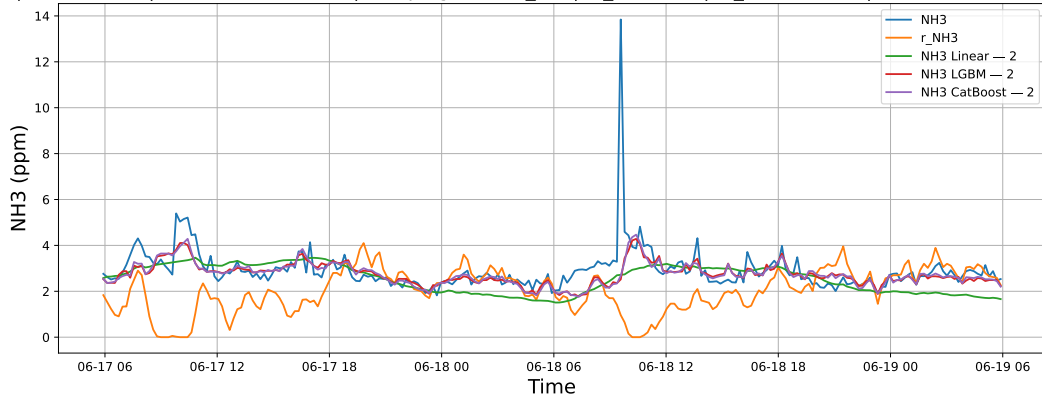
3.2.2.1 Time window : 24

NH3 | Model : ALL | Time window : 24h | campaign: 2025_Jun | id_rabbit: 2 | id_channel: 11 | 2025-06-18 → 2025-06-19



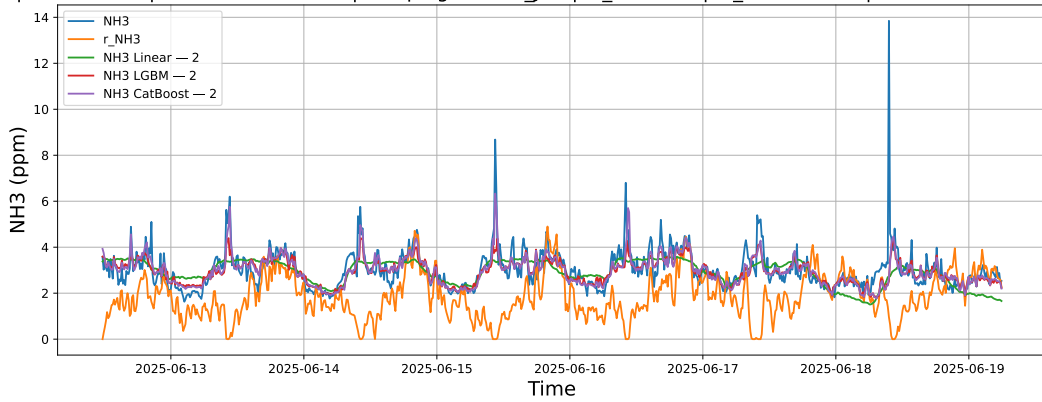
3.2.2.2 Time window : 48

NH3 | Model : ALL | Time window : 48h | campaign: 2025_Jun | id_rabbit: 2 | id_channel: 11 | 2025-06-17 → 2025-06-19



3.2.2.3 Time window : ALL

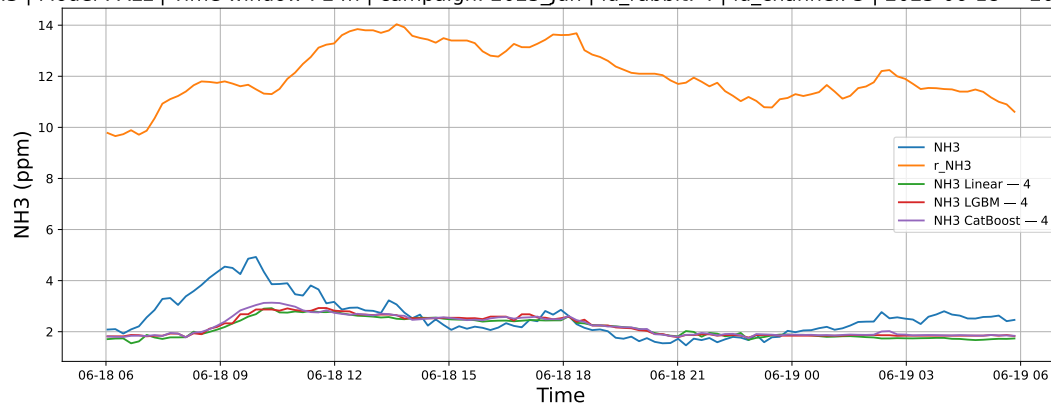
NH3 | Model : ALL | Time window : ALL | campaign: 2025_Jun | id_rabbit: 2 | id_channel: 11 | 2025-06-12 → 2025-06-19



3.2.3 Group: campaign: 2025_Jun,id_rabbit: 4,id_channel: 5

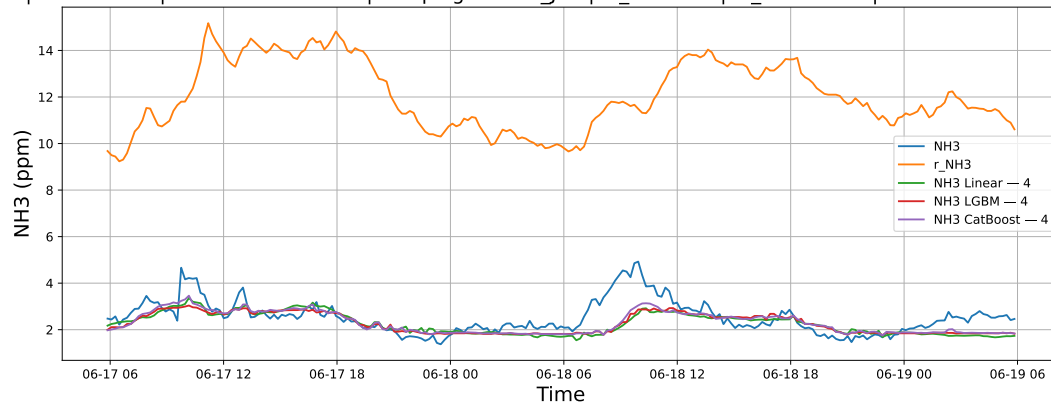
3.2.3.1 Time window : 24

NH3 | Model : ALL | Time window : 24h | campaign: 2025_Jun | id_rabbit: 4 | id_channel: 5 | 2025-06-18 → 2025-06-19



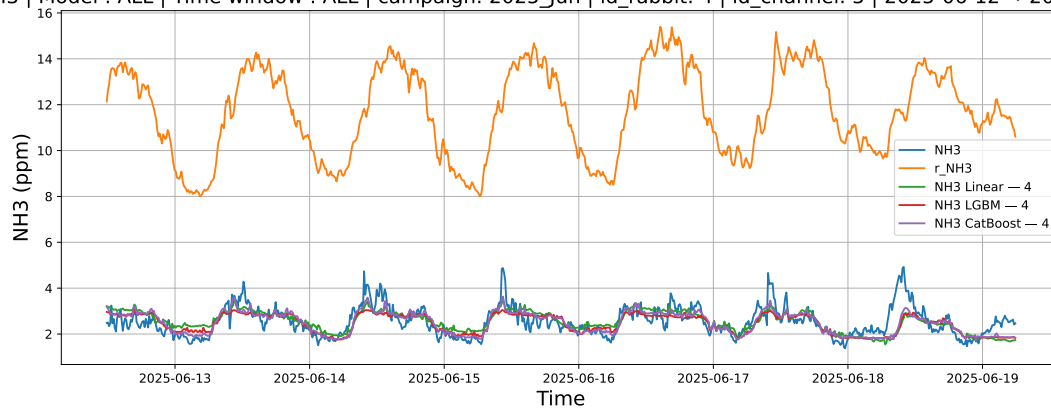
3.2.3.2 Time window : 48

NH3 | Model : ALL | Time window : 48h | campaign: 2025_Jun | id_rabbit: 4 | id_channel: 5 | 2025-06-17 → 2025-06-19



3.2.3.3 Time window : ALL

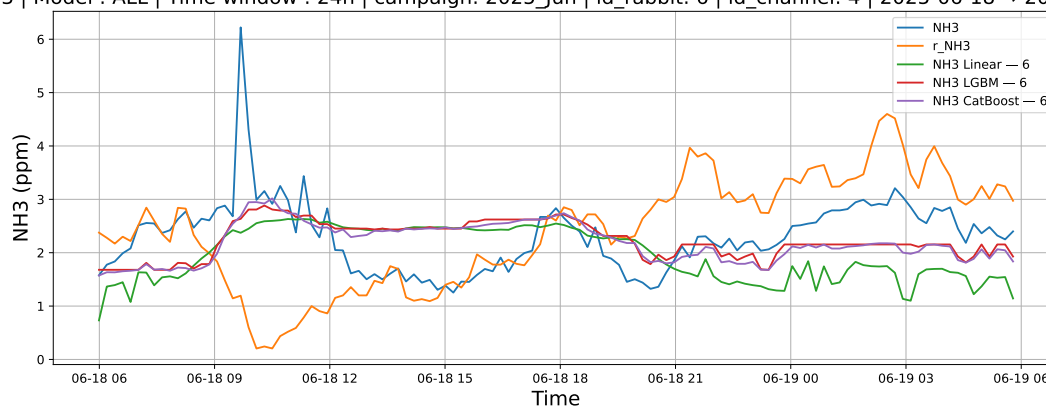
NH3 | Model : ALL | Time window : ALL | campaign: 2025_Jun | id_rabbit: 4 | id_channel: 5 | 2025-06-12 → 2025-06-19



3.2.4 Group: campaign: 2025_Jun,id_rabbit: 6,id_channel: 4

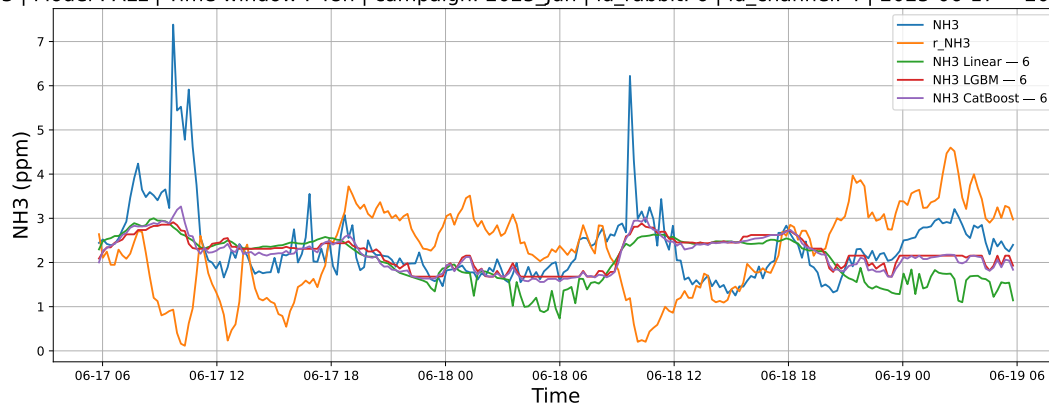
3.2.4.1 Time window : 24

NH3 | Model : ALL | Time window : 24h | campaign: 2025_Jun | id_rabbit: 6 | id_channel: 4 | 2025-06-18 → 2025-06-19



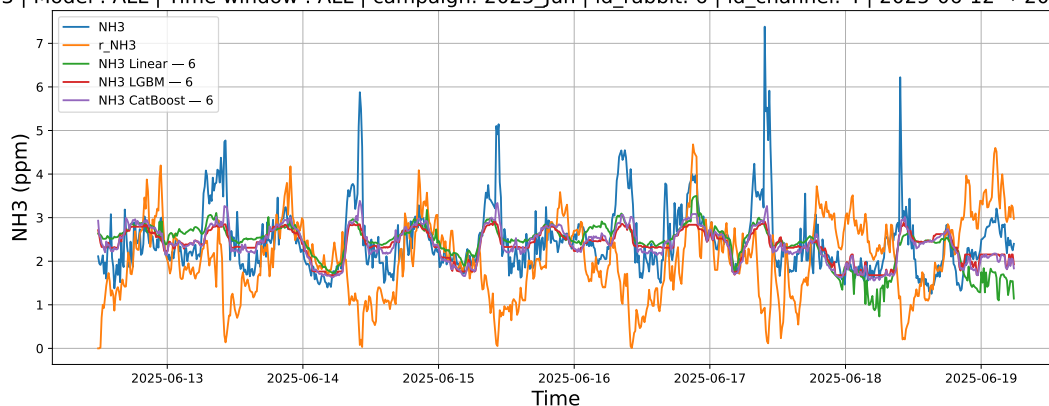
3.2.4.2 Time window : 48

NH3 | Model : ALL | Time window : 48h | campaign: 2025_Jun | id_rabbit: 6 | id_channel: 4 | 2025-06-17 → 2025-06-19



3.2.4.3 Time window : ALL

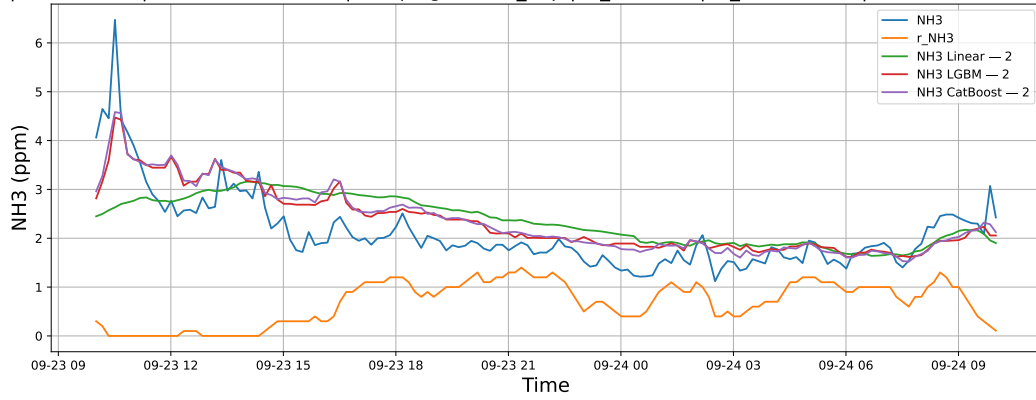
NH3 | Model : ALL | Time window : ALL | campaign: 2025_Jun | id_rabbit: 6 | id_channel: 4 | 2025-06-12 → 2025-06-19



3.2.5 Group: campaign: 2025_Sep,id_rabbit: 2,id_channel: 11

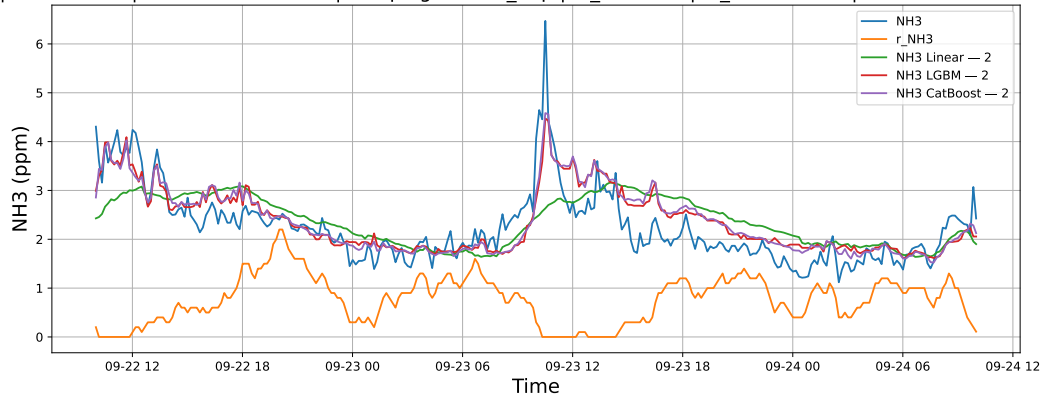
3.2.5.1 Time window : 24

NH3 | Model : ALL | Time window : 24h | campaign: 2025_Sep | id_rabbit: 2 | id_channel: 11 | 2025-09-23 → 2025-09-24



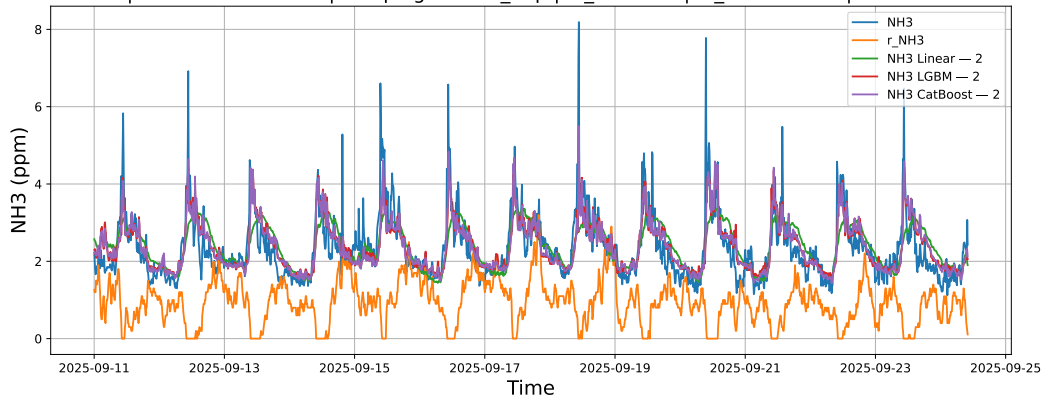
3.2.5.2 Time window : 48

NH3 | Model : ALL | Time window : 48h | campaign: 2025_Sep | id_rabbit: 2 | id_channel: 11 | 2025-09-22 → 2025-09-24



3.2.5.3 Time window : ALL

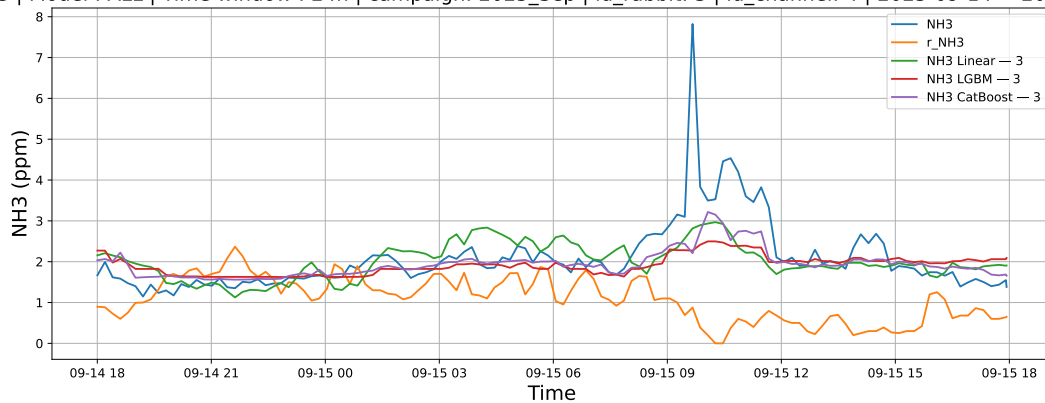
NH3 | Model : ALL | Time window : ALL | campaign: 2025_Sep | id_rabbit: 2 | id_channel: 11 | 2025-09-11 → 2025-09-24



3.2.6 Group: campaign: 2025_Sep,id_rabbit: 3,id_channel: 4

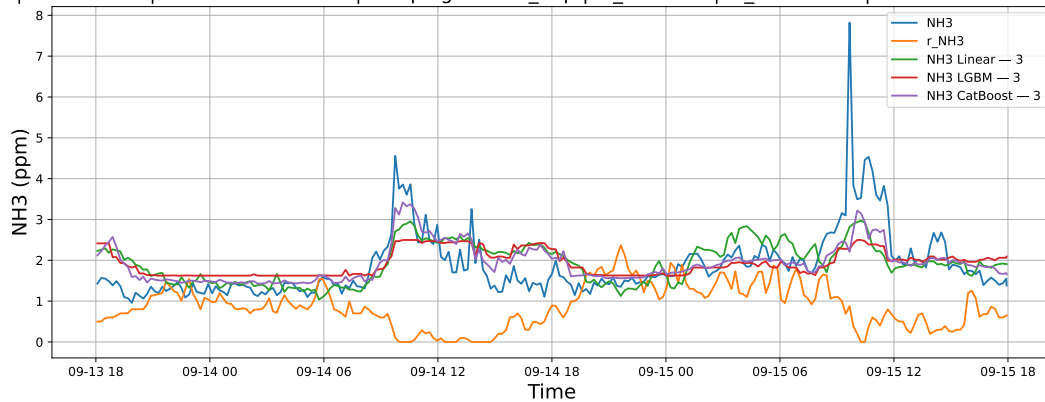
3.2.6.1 Time window : 24

NH3 | Model : ALL | Time window : 24h | campaign: 2025_Sep | id_rabbit: 3 | id_channel: 4 | 2025-09-14 → 2025-09-15



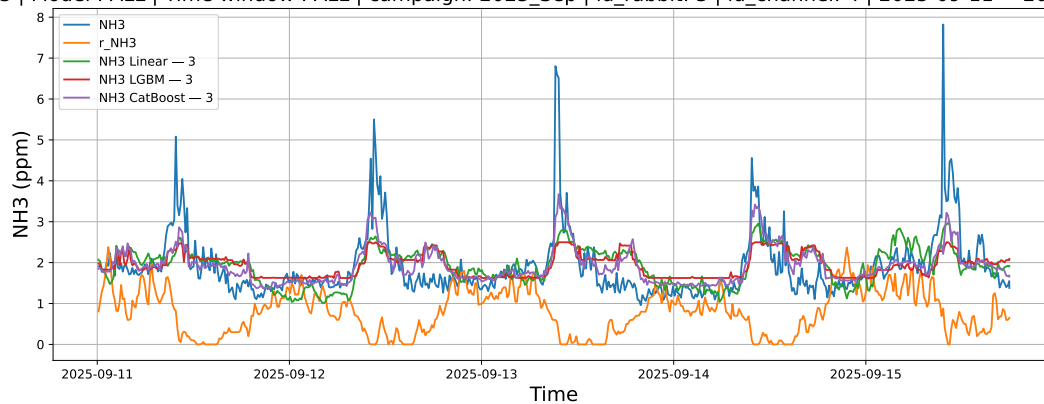
3.2.6.2 Time window : 48

NH3 | Model : ALL | Time window : 48h | campaign: 2025_Sep | id_rabbit: 3 | id_channel: 4 | 2025-09-13 → 2025-09-15



3.2.6.3 Time window : ALL

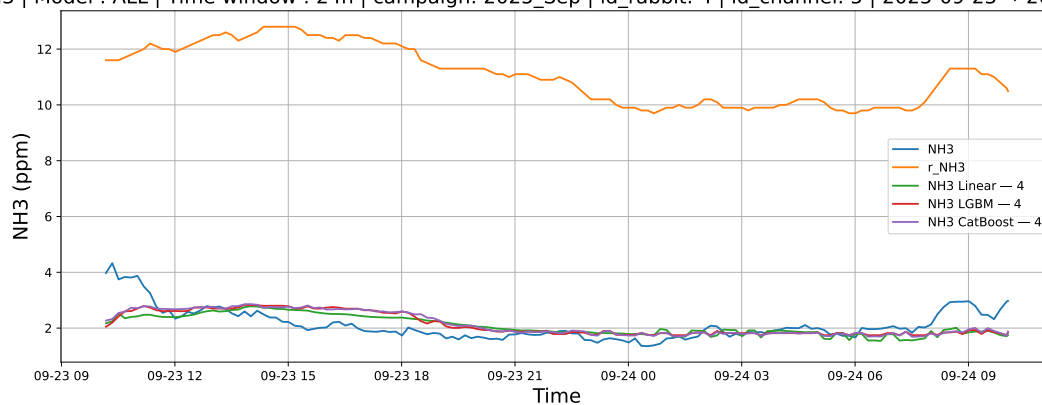
NH3 | Model : ALL | Time window : ALL | campaign: 2025_Sep | id_rabbit: 3 | id_channel: 4 | 2025-09-11 → 2025-09-15



3.2.7 Group: campaign: 2025_Sep,id_rabbit: 4,id_channel: 5

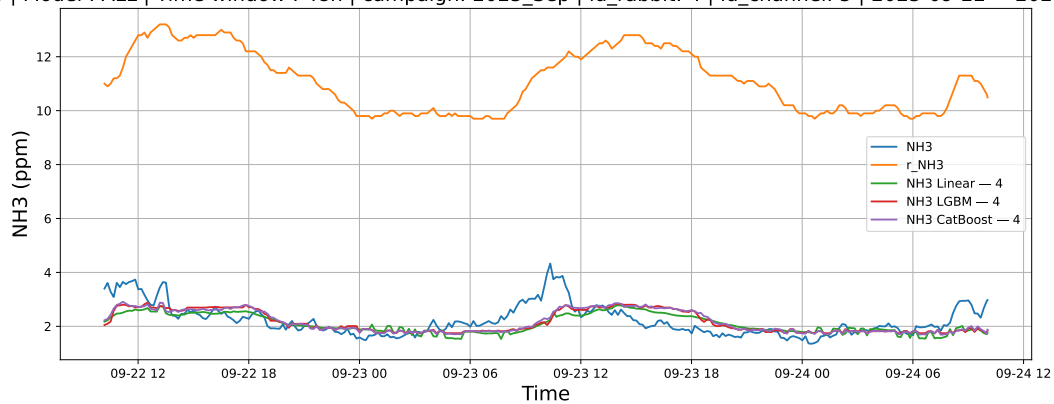
3.2.7.1 Time window : 24

NH3 | Model : ALL | Time window : 24h | campaign: 2025_Sep | id_rabbit: 4 | id_channel: 5 | 2025-09-23 → 2025-09-24



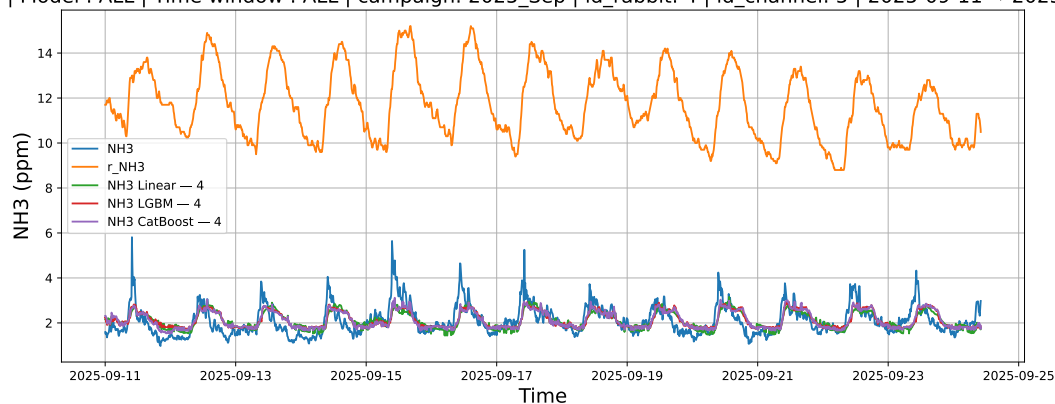
3.2.7.2 Time window : 48

NH3 | Model : ALL | Time window : 48h | campaign: 2025_Sep | id_rabbit: 4 | id_channel: 5 | 2025-09-22 → 2025-09-24



3.2.7.3 Time window : ALL

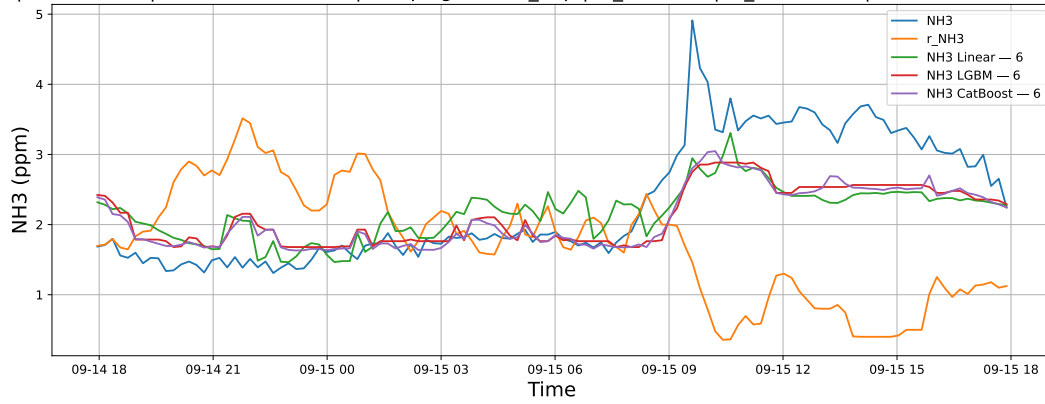
NH3 | Model : ALL | Time window : ALL | campaign: 2025_Sep | id_rabbit: 4 | id_channel: 5 | 2025-09-11 → 2025-09-24



3.2.8 Group: campaign: 2025_Sep,id_rabbit: 6,id_channel: 3

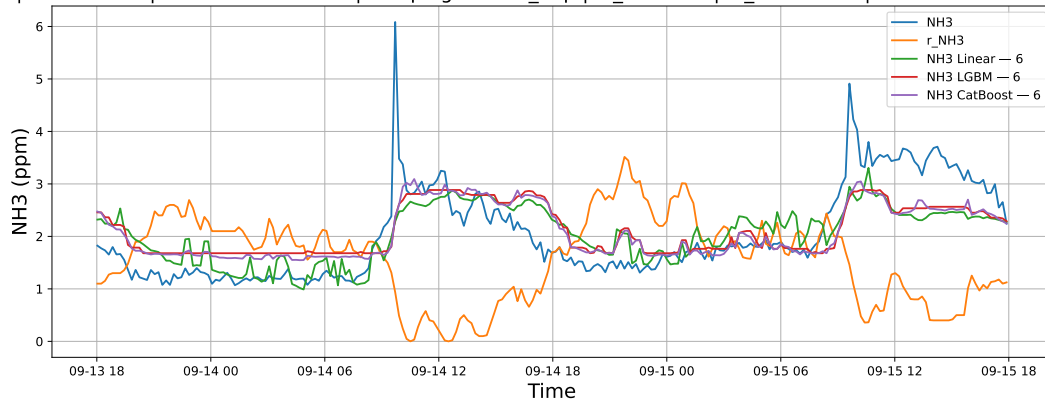
3.2.8.1 Time window : 24

NH3 | Model : ALL | Time window : 24h | campaign: 2025_Sep | id_rabbit: 6 | id_channel: 3 | 2025-09-14 → 2025-09-15



3.2.8.2 Time window : 48

NH3 | Model : ALL | Time window : 48h | campaign: 2025_Sep | id_rabbit: 6 | id_channel: 3 | 2025-09-13 → 2025-09-15



3.2.8.3 Time window : ALL

NH3 | Model : ALL | Time window : ALL | campaign: 2025_Sep | id_rabbit: 6 | id_channel: 3 | 2025-09-11 → 2025-09-15

